ILLUSTRATED BABY NELSON

Special Pediatrics



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Illustrated Baby Nelson

Special Pediatrics

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Foreword

I was very blessed to issue this new edition of Illustrated Baby Nelson with three pioneer figures in pediatrics and neonatology sharing their thoughts, insights and relentless revision of neonatology, hematology and pulmonology chapters. Hats off to ***Dr Ayman Azab, Dr Osama Taha Amer, Dr Laila Sherief****.*

Thank you for the man for whom I myself and this book owe him so much and without his support it would never come to light and persist; ***Mr Sayed Mahmoud****, founder of university book center. Thanks to his soul that exist with and inspire us all*

And finally I end my foreword with these lines to Richard Templer:

"Be prepared to be a little brave every day. If you don't, you'll grow stagnant and mouldy. We all have a comfort zone where we feel safe and warm and dry. But every now and then we need to step outside and be challenged, be frightened, be stimulated. It's this way that we stay young and feel good about ourselves "

Richard Templer (The rules of life)

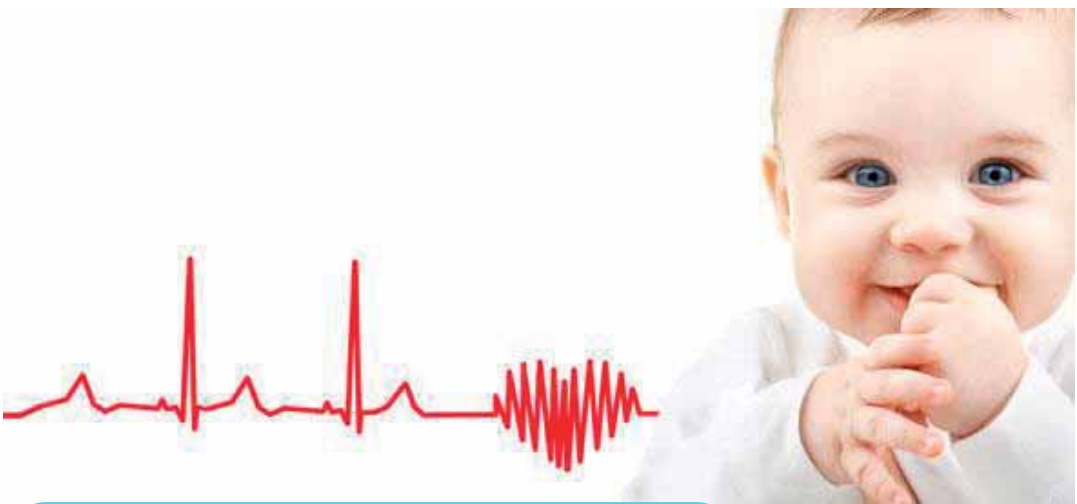
Mohamed El Koumi

October 6 city

7th of March 2017

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Cardiology

List Of Abbreviations

Lt. V. : Left ventricle

Rt. V. : Right ventricle

Lt. A : Left atrium

Rt. A : Right atrium

LVH : Left ventricle hypertrophy.

RVH : Right ventricle hypertrophy

LAD : Left atrium dilatation

RAD : Right atrium dilatation

C.P. angle. : Cardiophernic angle

$P.H^{+}$: Pulmonary hypertension

S_1 : First heart sound

S_2 : Second heart sound

P_2 : Pulmonary component of the second heart sound

A_2 : Aortic component of the second heart sound.

RVF : Right ventricle failure.

LVF : Left ventricle failure

BVF : Biventricular failure

COP : Cardiac output

LPSB : Left parasternal border

CXR : Chest X-ray

SBE : Subacute bacterial endocarditis

RBBB : Right bundle branch block

VMA : Vallinyle mandilic acid

BVH : Biventricular hypertrophy.

PGE1 : Prostaglandin E1

PFO : Patent formen ovale.

TOF : Tetralogy of Fallot

DORV : Double outlet right ventricle

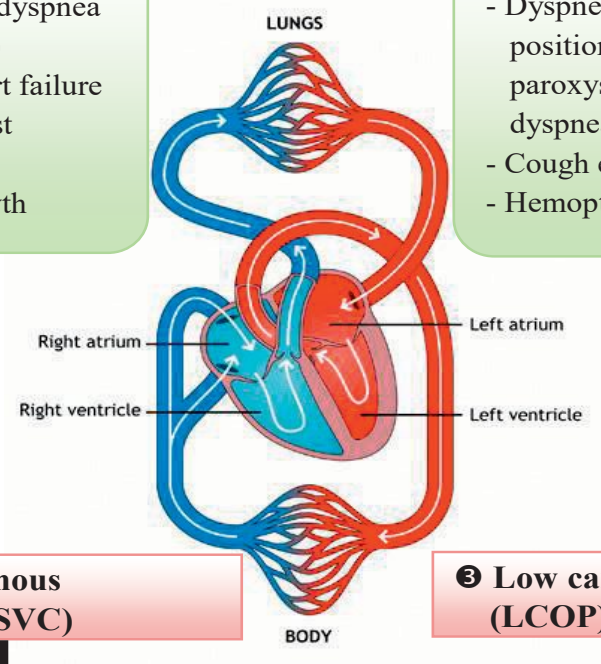
Diagnosis of a cardiac patient

Step 1: History Of

❶ Pulmonary venous congestion (PVC)

In infants
- Poor feeding (dyspnea on suckling)
- Recurrent heart failure
- Recurrent chest infections
- Retarded growth

In older child
- Dyspnea; exertional and positional with or without paroxysmal nocturnal dyspnea
- Cough exertional & positional
- Hemoptysis



❷ Systemic venous congestion (SVC)
- Edema (dependent, bilateral , pitting edema)
- Dyspepsia
- Abdominal pain (enlarged tender liver)

❸ Low cardiac output (LCOP) symptoms
- Dizziness
- Easy fatigability
- Exertional syncopal attacks
- Chest pain
- Retarded growth

❹ Central Cyanosis

Step 2 : General examination for

Suspected congenital heart disease	Suspected rheumatic heart disease
- Cyanosis (in lips & extremities) - Characteristic facies (e.g. Down) - Check patient′s weight and height And plot on appropriate charts - Check blood pressure & pulse - Clubbing of fingers and toes	- Start with measuring blood pressure & checking peripheral signs suggesting AR. - Check for edema in dyspneic patient - Clubbing of fingers and toes and other features suggesting infective endocarditis

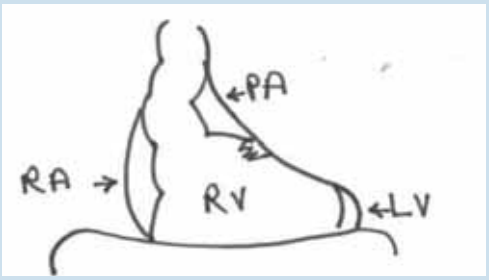
Remember the following notes

A

* Most of the anterior surface of the precordium is made of right ventricle

* Left atrium does not appear in anterior view

* Any chamber enlarges in its longitudinal axis mainly



B

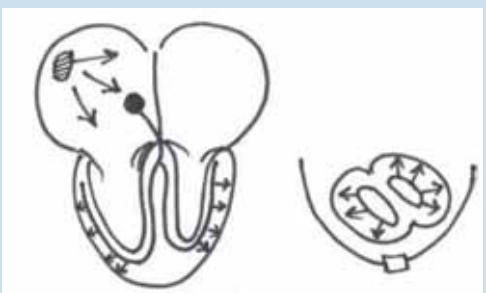
During systole; left ventricle elongate to hit a localized area of chest wall producing the apex beat



C

* Right atrium depolarizes first followed by left atrium
* Both ventricles depolarize at the same time

* Ventricles depolarize from inwards to outwards

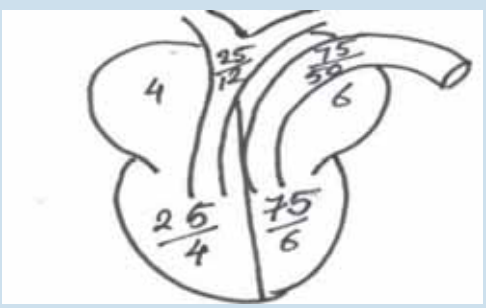


D

* Pressures in the left side exceed that in the right

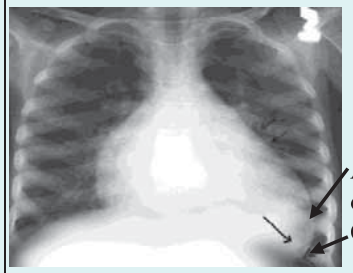
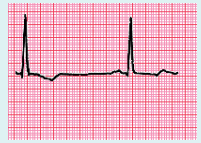
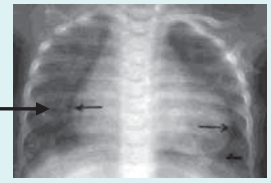
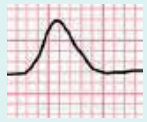
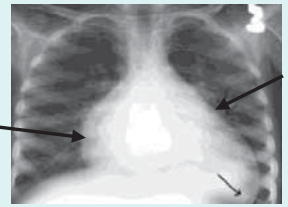
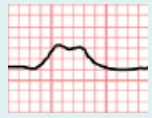
* After birth pressures in the Right side are slightly higher than normal

* Pressure values increase steadily with age



Step 3 : Precordial examination

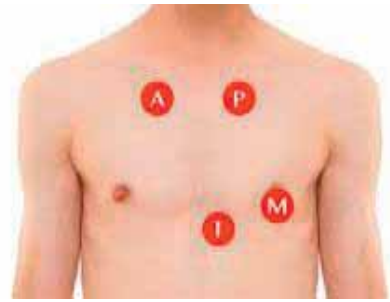
Precordial examination includes inspection, palpation and percussion of the precordium. It detects basically cardiac chamber and or big artery enlargement

	Precordial examination	Chest X Ray	ECG
Left ventricle hypertrophy (LVH) or Enlargement	- Apex beat is shifted down and out - Apex beat is localized		 V_1 → Deep S  V_6 → Tall R
Right ventricle hypertrophy (RVH) or Enlargement	- Apex beat is shifted directly out - Apex beat is diffuse - Precordial bulge - Left para sternal pulsation - Epigastric pulsation		 V_1 →Tall R  V_6 →Deep S
Right atrial dilatation (RAD)	Usually is not detectable clinically	↑ Right cardiac shadow 	Tall ,peaked P wave (P-pulmonale)  
Left atrial dilatation (LAD)	Usually is not detectable clinically	Double contour  Mitralization	wide and bifid P wave (P-mitrale)  

Evidence of pulmonary artery dilatation in cases with pulmonary hypertension(PH)

- Palpation of the pulmonary area: Pulsating pulmonary area (normally silent area)
- Percussion of the pulmonary area: Dull (normally resonant area)

Feel the murmur i.e. Thrills (Thrill is a palpable murmur)



Value of detecting thrill

- Definitely, there will be a murmur to be heard with the same timing of the thrill
- This murmur is organic never functional or innocent
- If the murmur is propagating ; the thrill usually is not propagating (i.e. localizing sign)

If thrill is detected; check where?

- Mitral area (left 5th interspace)
- Left parasternal border area(left 3rd and 4th interspaces)
- Pulmonary area(left 2nd interspace)
- Aortic area(right 2nd interspace)

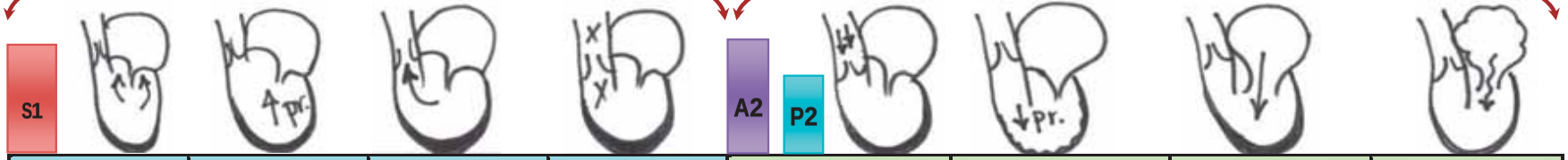
Comment on the apex with (Remember SCART)

	Normal apex	Abnormal apex
Site	Occupies the left 5th interspace, midclavicular line	- Shifted outward & downward (LVH) - Shifted direct outward (RVH) - Absent; impalpable
Character	Normal intensity	- Hyperdynamic (strong and rapid)in volume overload e.g. MR, CM,AR - Heaving(strong and sustained) in pressure overload e.g. AS
Area	Occupies one interspace	- Diffuse (apex beat occupies more than one interspace) in RVH
Rhythm	Regular	- Irregular in irregular arrhythmias
Thrills	Absent	- May be Present: o Systolic thrill (MR, CM,VSD) o Diastolic thrill (MS, CM)

MR: Mitral Regurge; CM: combined mitral lesion (previously double mitral);AR: Aortic Regurge; AS: Aortic stenosis; VSD: ventricular septal defect; MS: Mitral stenosis

Cardiac Cycle

Systole (duration:) **Diastole**



1st heart sound	Isometric contraction phase	Ejection phase	Protodiastolic phase	2nd heart sound	Isometric relaxation phase	Filling phase	Atrial contraction phase
- Due to closure of the atrio ventricular valves at onset of systole - Amplitude depends on amount of blood in ventricles at start of systole - Has 2 components; mitral & tricusped	- Ventricles contract without change in size - Intraventricular pressure is increasing	- Intraventricular pressure exceeds that in big arteries → semilunar valves open	- Pressure in ventricles and large arteries equalize → no net flow of blood	- Due to drop of intraventricular pressure at start of diastole→ blood in big arteries try to backflow→ semilunar valve closure - Amplitude depends on amount of blood in large arteries at start of diastole - Has 2 components; aortic & pulmonary	- Ventricles continue to relax→decline of intraventricular pressure	- Blood flows from atria to the ventricles due to pressure gradient(60%) ± **3rd heart** sound(vibration of myocardial muscle) **N.B:** 3rd heart sound can be heard in normal infants as well as in congestive heart failure	- Atria contract to push blood remaining in the atria(40%) ± **4th heart** sound(atrial contraction against stiff thickened ventricle) **N.B**: 4th heart sound is almost always abnormal

Q1: What is murmur?

- It is abnormal sound due to either blood flowing in an abnormal direction **or** blood flowing across narrow orifice (organic narrowing e.g. stenosis **or** relative)
- Pressure gradient is required for murmur to develop
- Murmurs are graded into 6 grades according to intensity and presence or absence of associated thrills(Grades 1-3 are without thrill and grades 4 - 6 with thrill)
- Murmurs may be:
  1. Organic: due to organic lesion in the heart either congenital or acquired
  2. Relative: due to ↑ blood flow through normal sized valve (soft, non propagating , without thrills and heart sound related is usually accentuated e.g. relative PS in PH)
  3. Innocent murmur: heard over completely normal heart or great vessels. (soft, non propagating , without thrills, *systolic, asymptomatic patient, normal heart sounds*)

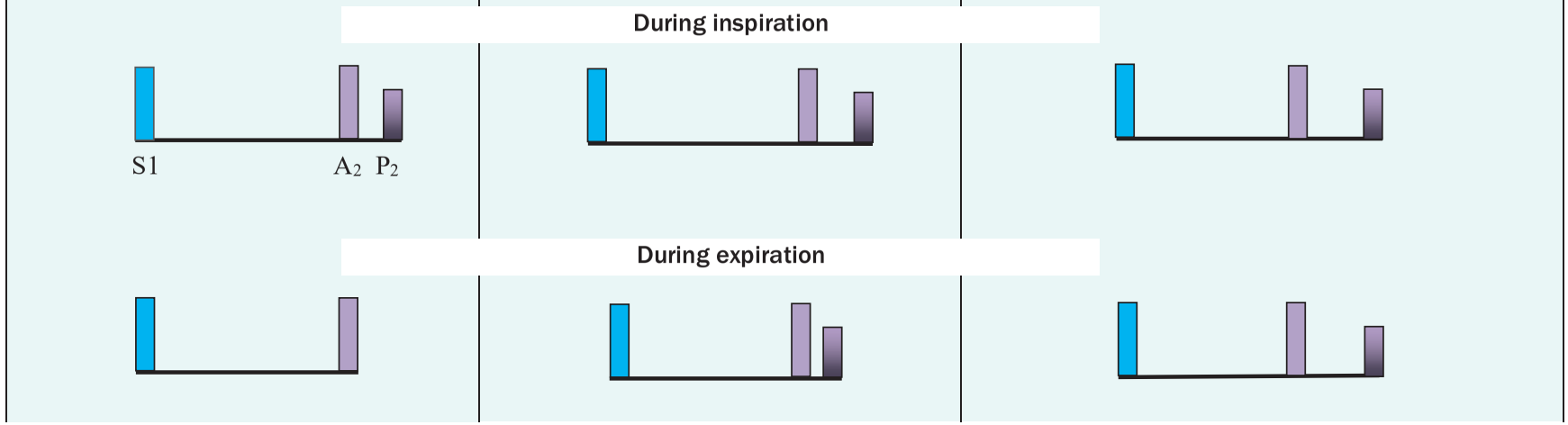
Step 4 : Auscultation

1. Heart sounds

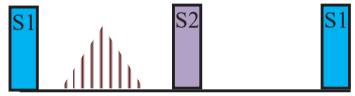
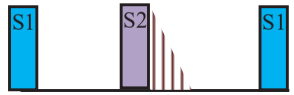
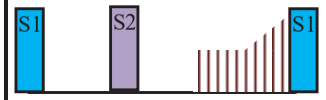
1. First heart sound (S1)	2. Second heart sound (S2)
• Due to closure of Atrio ventricular valves at beginning of systole • Has 2 components →Mitral :best heard over mitral area →Tricusped : best heard over tricusped area • Muffled in →Atrio ventricular valves regurge → Bradycardia • Accentuated in → Atrio ventricular valves stenosis →Tachycardia • **S1 Splits** in right and left Bundle branch block (BBB)	• Due to closure of semilunar valves at beginning of diastole • Has 2 components→ Aortic ;best heard over aortic area → Pulmonary ;best heard over pulmonary area • Muffled in : semilunar valve stenosis: → Aortic stenosis (↓A_2) →Pulmonary stenosis (↓P_2) • Accentuated in hypertension → Systemic hypertension (↑A_2) → Pulmonary hypertension (↑P_2) • **S2 Split** : see below

Splitting of the second heart sound (Detected by auscultating the pulmonary area)

Physiologic splitting	Wide variable splitting	Wide fixed splitting
The two components of S2(A_2: aortic component & P_2:pulmonary component) usually splits in inspiration & unites in expiration	○ S2 splits in whole respiratory cycle but wider splitting in inspiration ○ Occur with prolonged or delayed right ventricular systole	○ Wide splitting ○ Does not vary with the respiratory cycle ○ Occurs with ASD



2. Murmurs

1. Timing	Systolic Murmurs					Diastolic Murmurs	
	Pan systolic 			Ejection systolic 		Early diastolic 	Mid diastolic 
2. Site of maximum Intensity	Mitral area	Left sternal Border(3,4)	Tricusped area	Aortic area	Pulmonary area	Aortic areas	Mitral area
Diagnosis	**MR**	**VSD,ECD**	**TR**	**AS**	**PS**	**AR**	**MS**
Sounds	↓ S1		↓ S1	↓ A_2	↓ P_2		↑ S1
Characters	Blowing	Harsh	Blowing	Harsh	Harsh	Soft	Rumbling
Propagation	Axilla	Precordium		Neck& Apex	Sternal border	Second aortic area	Localized

ECD: Endocardial cushion defect; TR: tricuspid regurge; PS: pulmonary stenosis

N.B: Continuous (machinery or systolic/diastolic) murmurs: See later in PDA

Q2: Auscultatory findings in pulmonary hypertension(PH)?

1. Ejection systolic murmur over pulmonary area (with the relative murmur criteria)
2. Accentuated pulmonary component of the second heart sound

N.B: pulmonary hypertension occur in cases with prolonged pulmonary congestion or prolonged increase pulmonary blood flow

Step 5 : Investigations

1. ECG and chest x ray: Help precordial examination in detecting cardiomegaly (*ECG is invaluable for diagnosis of dysrhythmia and ischemia*)

2. Echocardiography (& preoperative catheterization): **Help auscultation establishing the final diagnosis**

Value: - Describe lesions (site, size, intracardiac pressures, flow across the lesion and ventricular function).

- Detect complications e.g. Infective endocarditis, cardiomegaly

Congenital Heart Diseases (CHD)

Incidence

- 8 per 1000 live born infants have significant cardiac malformation
- The commonest CHD are VSD (30%), PDA (12%), and ASD (7%)
- About 10-15 % have complex lesions with more than one cardiac abnormality

Risk Factors

- Genetic predisposition suggested by family history of CHD
- Exposures during pregnancy e.g.
  a. Drugs e.g. warfarin, anticonvulsants, alcohol
  b. Diseases e.g. maternal rubella (→ PDA), Maternal Diabetes Mellitus
- Chromosomal e.g.
  a. Down syndrome → Atrio ventricular septal defect, VSD
  b. Turner syndrome→ Aortic stenosis, coarctation of aorta
  c. Williams syndrome→ Supra valvular aortic stenosis , pulmonary stenosis



a



b



c

Presentation

- **C**oincidental; accidental discovery of a murmur in an asymptomatic infant
- **C**ardiac failure /Cardiogenic shock e.g. in neonate (critical aortic stenosis and severe coarctation) or in infants(CHD with high pulmonary flow)
- **C**ongestive pulmonary symptoms e.g. CHD with high pulmonary flow
- **C**yanosis e.g. congenital cyanotic heart diseases
- **C**ardiomegaly detectable clinically or during routine chest radiograph
- **C**ertain anatomic diagnosis may not be made by physical examination or chest x ray or ECG alone ,so echocardiography is the mainstay of diagnostic imaging
- **Recently**

1. Antenatal fetal anomaly ultrasound screening
   - Can diagnose up to 70 % of significant lesions ,allowing earlier diagnosis and planning of management
   - Important for any fetus with an increased risk e.g. Down syndrome
2. MRI allows 3 dimensional imaging of complex CHD, assessment of hemodynamics, assists interventional cardiology and reduces the need for cardiac catheterization. ***(Illustrated textbook of Paediatrics, Tom Lissauer)***

Acyanotic Congenital Heart Diseases (ACHD)

ACHD constitutes 80% of all CHD

Classification

i. ACHD with left to right shunt (Potentially cyanotic CHD)

The commonest lesions in this category are

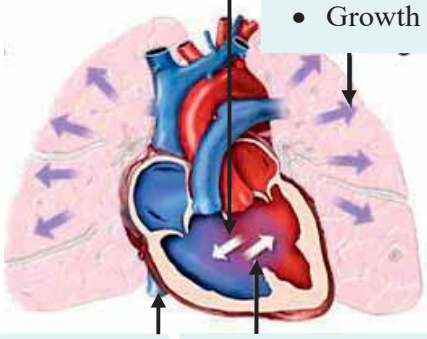
- Ventricular septal defect (VSD)
- Patent ductus arteriosus (PDA)
- Atrial septal defect (ASD)

General clinical features

1. **Degree of the left to right shunt**, and consequently clinical manifestations is dependent on:
   - Size of the defect.
   - Pressure gradient across the defect.

2. Manifestations of high pulmonary blood flow

- Poor feeding (sweating and tachypnea) in babies, exercise intolerance and easy fatigability in children
- Recurrent chest infections & chest wheezes.
- Recurrent heart failure.
- Growth failure (faltering of growth)



3. **Heart failure**
   Usually doesn't occur in full term neonates but can occur in infancy as pulmonary vascular pressure declines.

4. **Eisenmenger syndrome:** Prolonged high pulmonary blood flow→ pulmonary hypertension develops→ with time, right side heart pressures exceeds that of the left → reversal of the shunt → central cyanosis
   Risk is higher with large unrepaired defects

ii. ACHD without shunt

1. Obstructive lesions e.g. Aortic coarctation, Aortic stenosis, Pulmonary stenosis

Common clinical features:

1- Severe obstructive lesions can present early in life with heart failure.

2- Low cardiac output manifestations.

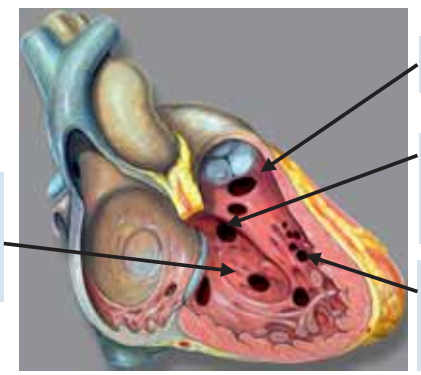
2. Non obstructive lesions e.g. Dextrocardia, Mitral valve prolapse

Ventricular septal defect (VSD)

Definition

Defect anywhere in the interventricular septum

Types of VSDs:



Inlet defect

Outlet defect; (also called infundibular or sub arterial).

Muscular defects: Either single or multiple (Swiss cheese).

Perimembraneous defect: Adjacent to tricuspid valve The commonest type (70%)

Hemodynamics

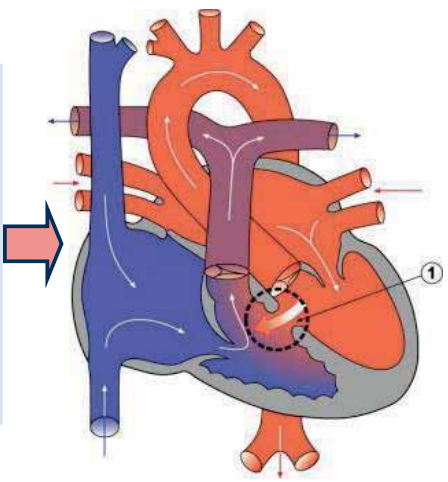
- Blood is shunted from the left ventricle(higher pressure) to the right ventricle(lower pressure)

⇩

- Increased pulmonary blood flow

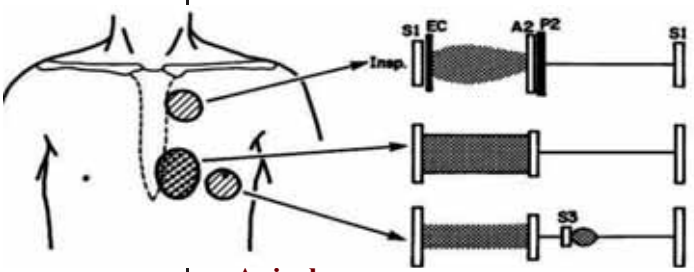
⇩

- Pulmonary congestive symptoms.
- Volume overload over right ventricle ,left atrium and left ventricle



Clinical picture

	Small VSD	Large VSD
General manifestations	- Usually asymptomatic - Discovered accidentally 	History • Breathlessness during suckling • Recurrent chest infections • Recurrent chest wheezes Physical • Breathlessness ,tachypnea, tachycardia and enlarged tender liver (heart failure) • Faltering of growth

	Small VSD	Large VSD
Precordial Examination	- Systolic thrill over the lower left sternal border	- Biventricular enlargement by 2 months of age - Active precordium
Auscultation	• **Murmur of VSD** - Pansystolic. - On lower left sternal border. - Propagate all over the heart. - Harsh ; loud .	• **Murmur of VSD** Same but softer • **Pulmonary area**: Accentuated P_2 & soft systolic murmur indicates pulmonary hypertension.  • **Apical** – Short rumbling mid diastolic murmur – Produced by the increased volume of blood flow across the mitral valve – Usually indicates a Pulmonary-to-systemic blood flow ratio ratio of at least 2 : 1
Complications	• Infective endocarditis	• Heart failure • Infective endocarditis • Higher risk of Eisenmenger syndrome
Investigations		
*Chest X-ray	- Normal	- Cardiomegaly with biventricular enlargement & plethoric lungs.
* ECG	- Normal	- Biventricular enlargement (LVH and RVH
* Echo	- Diagnostic (size smaller than aortic in diameter ;up to 3mm)	- Diagnostic
* Catheter	-	- Pre operative / interventional

Treatment

1. Small defect

- Avoid infective endocarditis by dental hygiene and antibiotic prophylaxis.
- Reassurance ; (surgical intervention is not usually recommended)
- Follow up with ECG & Echo to confirm spontaneous closure

2. Large defect

A. Medical

- Control heart failure (Diuretics, Captopril).
- Antibiotics for chest infections
- Additional calorie input and monitor growth
- Avoid infective endocarditis
- Follow up with ECG & Echo to confirm spontaneous closure.

B. Surgical

- Types
  - i- Palliative: Pulmonary artery banding → reduce pulmonary blood flow
  - ii- Direct closure of the defect
- Indications
  - i- Symptomatic large defects with uncontrollable heart failure or Growth failure
  - ii- Progressive pulmonary hypertension.
- Timing: at 3-6 months of life.

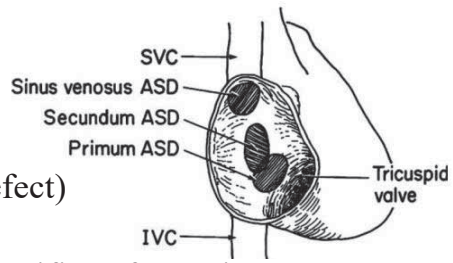
(Illustrated textbook of Paediatrics, Tom Lissauer)

Prognosis

30-50% of small defects (especially muscular) close spontaneously within the first 2-years of life

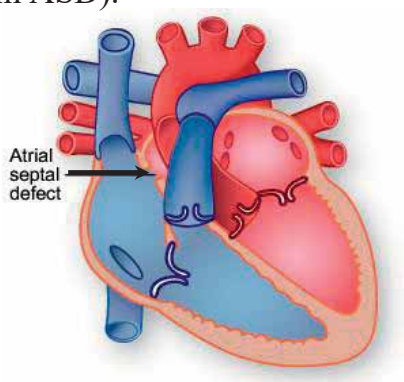
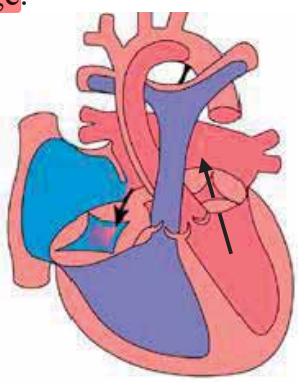
Atrial Septal Defect (ASD)

Definition: Defect in the inter-atrial septum.



Types of ASD

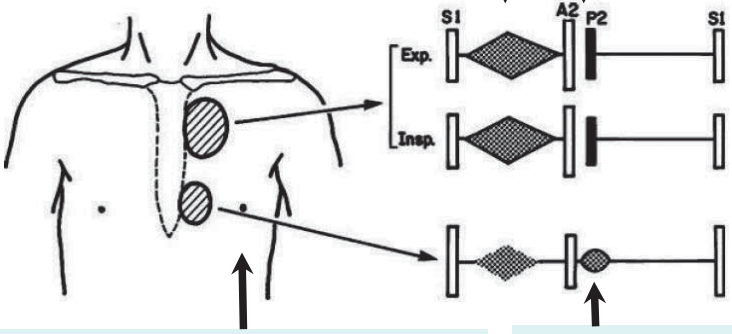
- Secundum ASD
- Primum ASD (Partial atrio ventricular septal defect)
- Sinus venosus defect:
  - Lies in the upper part of the septum near orifice of superior vena cava.
  - Association: usually with partial anomalous pulmonary venous return

Secundum ASD	Primum ASD
• The commonest type (80% of ASD) • Lies in the middle part of the septum • Association May be with **Holt Oram** syndrome (Absent radii, 1st degree heart block, secundum ASD). 	• Less common • Lies in the lower part of the septum • Association Usually associated with cleft of mitral valve leaflet→ mitral regurge. 
Hemodynamics Blood is shunted from left atrium to right atrium → right ventricle →↑ pulmonary blood flow (more with primum ASD)	
General manifestations • Asymptomatic in most cases unless large ASD→ features of increased pulmonary blood flow • Arrhythmias (4th decade onwards)	• Features of increased pulmonary blood flow in infancy including recurrent chest infections & heart failure.
Precordial Examination - Usually normal (may be RVH)	- Cardiomegaly with BVH

Auscultation

Pulmonary area
Murmur of a relative pulmonary stenosis:
- Soft ejection systolic without thrill
- With accentuated P2

Pulmonary area
Wide fixed splitting of S2:
- Wide splitting due to large filling of right ventricle
- Fixed (does not vary with respiration) due to constant filling of right ventricle in all phases of respiration



Apex
Pansystolic murmur of mitral regurge propagating to axilla in Primum ASD

Lower left sternal border
- A short, rumbling mid-diastolic murmur Produced by the increased volume of blood flow across the tricuspid valve
- Usually indicates a Pulmonary-to-systemic blood flow ratio (Qp : Qs) ratio of at least 2 : 1

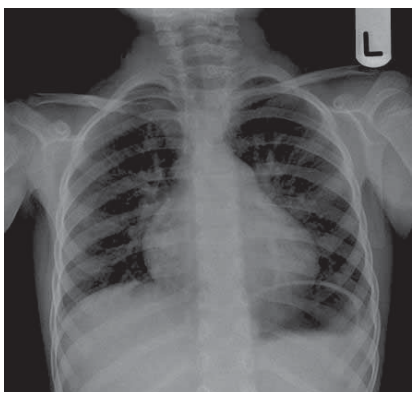
Complications

Very rare; more common with primum ASD:
- Recurrent heart failure may occur with large defects.
- Recurrent pulmonary infections
- Infective endocarditis is extremely rare (can occur in primum defects)
- Reversal of the shunt may occur very late; in adulthood, by the 3rd - 4th decade.

Investigations

1. Chest X-ray

- Cardiomegaly with RVH & RAD.
- BVH in ostium primum defect
- Plethoric lungs (↑ pulmonary vascular markings)



2. ECG:

- RVH & RAD.(BVH in primum ASD)
- Right bundle branch block is common.

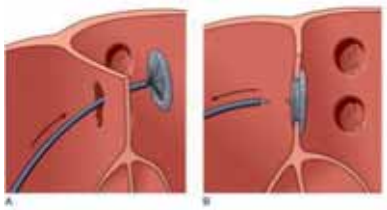


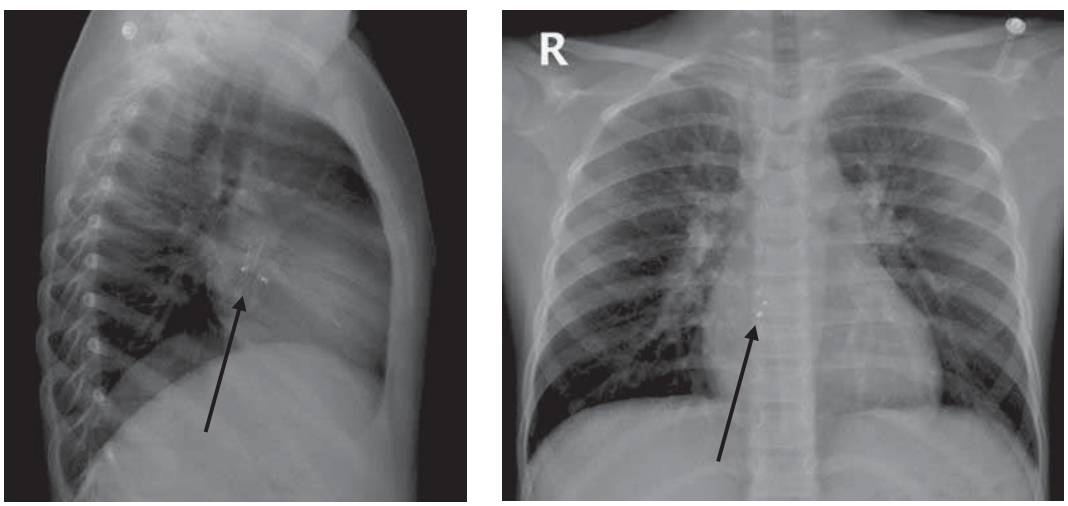
3. Echocardiography is diagnostic

4. Cardiac catheter: - Pre operative/corrective intervention

Treatment

1. Medical
   - Same lines as for VSD
   - Infective endocarditis prophylaxis is needed only for primum ASD with mitral regurge
2. Surgical or transcatheter device closure is advised for
   - All symptomatic patients
   - Asymptomatic patients with a Qp : Qs ratio of at least 2 : 1
   - Those with right ventricular enlargement





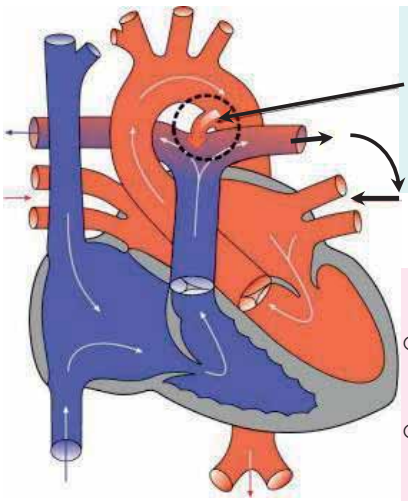
(*Chest x ray after ASD device closure, source: Radiopaedia.com*)

Prognosis

40% of secundum ASD defects close in 1st four years of life spontaneously

Patent ductus arteriosus (PDA)

Definition: Persistent fetal duct connecting the aorta & the pulmonary artery.



Connections

- The aortic end is just distal to left subclavian artery.
- Pulmonary end is at the bifurcation.

Association: Congenital rubella syndrome & prematures

Hemodynamics

- Blood is shunted from the higher pressure of aorta to pulmonary artery → ↑↑ Pulmonary blood flow
- Run off of blood from Aorta to the pulmonary artery mainly during diastole → lower diastolic pressure → Hyperdynamic circulation

General Manifestations

1. Small duct

- Asymptomatic ; discovered accidentally

Precordial examination

- Systolic thrill over the upper left sternal border

2. Big duct

- Symptoms of increased pulmonary blood flow (see before)
- Hyperdynamic circulation e.g. big pulse pressure

Precordial examination

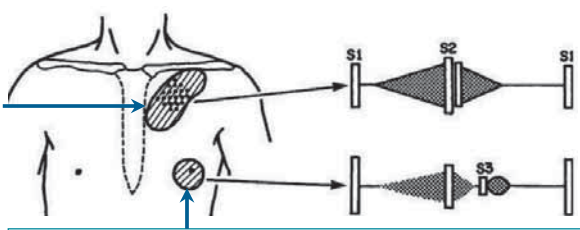
- Systolic thrill over the upper left sternal border
- Evidence of LVH

Auscultation

Left infraclavicular area

Continuous; machinery murmur (Gibson murmur); Murmur may be systolic if the diastolic component is masked:

1. Early in life (due to physiologically raised pulmonary pressure)
2. Very large defects (due to higher pulmonary pressure)
3. Pulmonary hypertension



Apex

- A short, rumbling mid-diastolic murmur
- Produced by the increased volume of blood flow across the mitral valve in big defects

Investigations

1. Chest X ray and ECG

- Small /asymptomatic PDA→ Both are normal
- Large and symptomatic PDA → Indistinguishable from large VSD
  - Chest X-ray: Cardiomegaly and Plethoric lungs.
  - ECG: → Left ventricle hypertrophy with large left to right shunt
    → Right ventricle hypertrophy with pulmonary hypertension

2. Echocardiography is readily diagnostic

Complications

- As in large VSD plus Aneurismal dilatation and rupture of the duct

Treatment

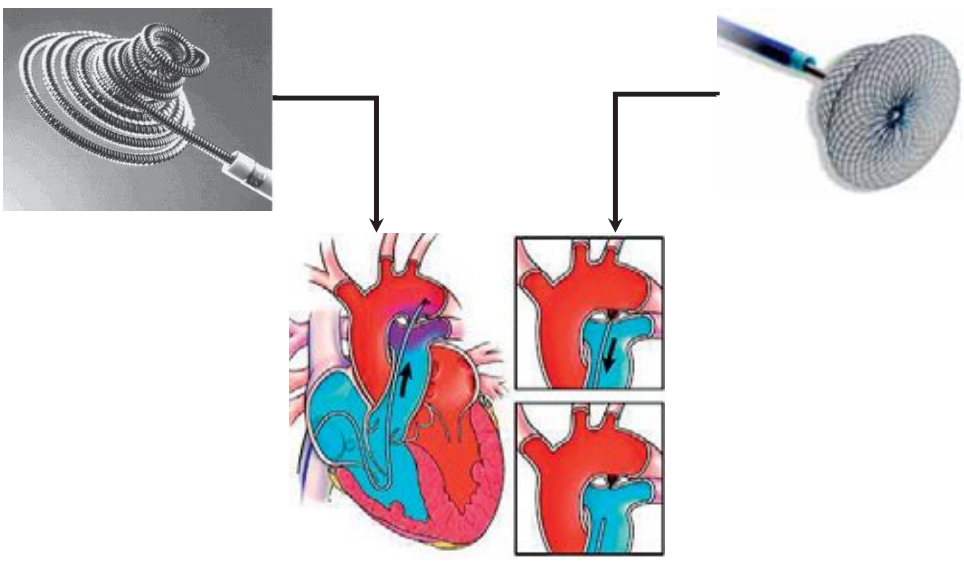
A. Medical

- Control heart failure & prevent infective endocarditis (Infective endarteritis)
- Medical closure in preterm by I.V. indomethacin in the 1st week of life .

B. Transcatheter or Surgical closure

Irrespective of age, size or symptoms ,all PDAs should be closed, preferably before 1 yr of age.

- **Small PDAs** are closed with intravascular coils
- **Moderate to large PDAs** are closed with an umbrella-like device



Differential diagnosis of continuous murmurs:

- An aorticopulmonary window defect
- A sinus of Valsalva aneurysm that has ruptured into the right side of the heart or pulmonary artery, Coronary arteriovenous fistulas
- Truncus arteriosus
- VSD with aortic insufficiency, and combined aortic and mitral insufficiency (to-and-fro murmur rather than continuous nature)

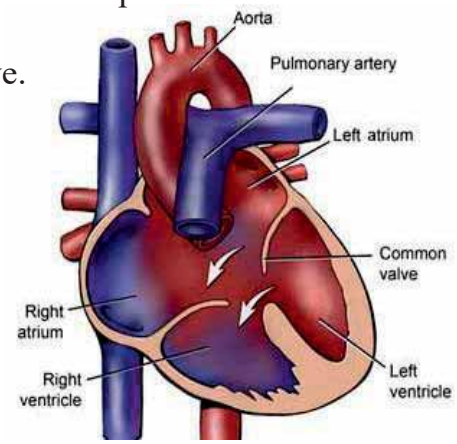
Atrio ventricular septal defect (AVSD)
(Endocardial cushion defect; ECD)

Definition

Defect in atrioventricular septum; complete defect is composed of:

1- Large ASD
2- Common & incompetent atrio ventricular valve.
3- Large VSD.

Association: common with Down syndrome.



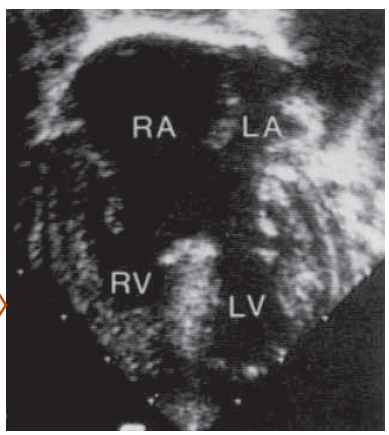
Manifestations

- Features of increased pulmonary blood flow and intractable heart failure develop early in infancy.
- Evidence of cardiomegaly and hyperactive precordium (RVH mainly)
- Systolic thrill on lower left sternal border.

Auscultation

1- Lower left sternal border→Pansystolic murmur propagating all over the precordium
2- Pulmonary area→ Systolic murmur of a relative pulmonary stenosis or pulmonary hypertension

Clinical picture is very similar to large VSD and cases are only diagnosable by Echocardiography



Treatment

1. Medical: - As for large VSD
2. Surgical
   - Because of the risk of pulmonary vascular disease developing as early as 6-12 mo of age, Early surgical repair is mandatory to avoid early pulmonary hypertension and intractable heart failure
   - Complication :surgically induced heart block requiring placement of a permanent pacemaker
   - Pulmonary artery banding is an option as a palliative surgery

Coarctation of Aorta (CoA)

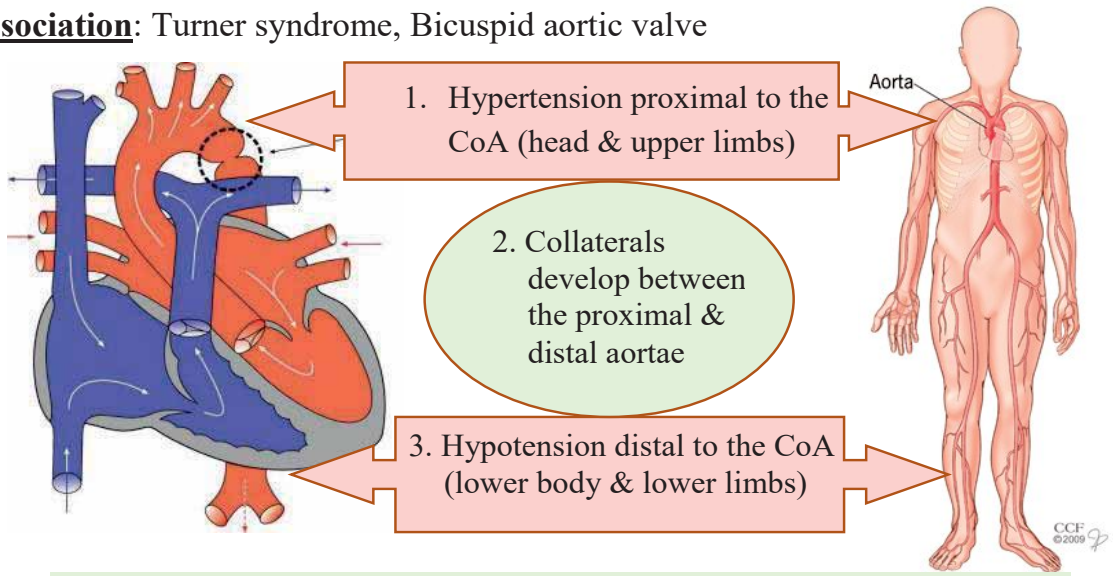
Constriction of the aorta to anywhere from aortic arch to aortic bifurcation.

Types

Ductal CoA: adjacent to ductus arteriosus; below origin of left subclavian artery

Preductal or postductal CoA : far uncommon

Association: Turner syndrome, Bicuspid aortic valve



4. In severe coarctation , blood is shunted from the pulmonary artery to the descending aorta via a patent arterial ductus with subsequent:
   - Perfusion of the lower body is dependent on the patent ductus(duct dependent systemic flow)
   - May be differential cyanosis (lower limbs blue, upper limbs pink).

Clinical Picture

1. Severe coarctation

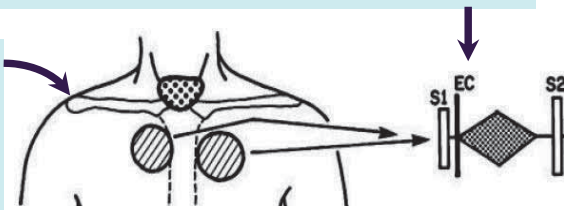
- Usually no symptoms are apparent at birth, but can develop within a week of birth with the closure of the ductus arteriosus
- Presentation: poor feeding, congestive heart failure, cardiomegaly and acidosis

2. Milder cases: Usually asymptomatic; may present in older child or adult with:

- **Pulse**
  - Bounding in upper limbs and carotids , weak or absent in lower limbs
  - Femoral pulse is delayed than radial pulse (unlike normal).
- **Blood pressure** :Higher in upper limbs than lower limb(unlike normal)
  The upper-to-lower extremity pressure gradient will increase with exercise

- **Murmur**
  - Systolic murmur with ejection click
  - Over the base of the heart
  - Transmitted to the left infrascapular area

In older patients with well-developed collaterals:
A continuous murmur may be heard over chest laterally and posteriorly



3. Complicated cases: May present with:

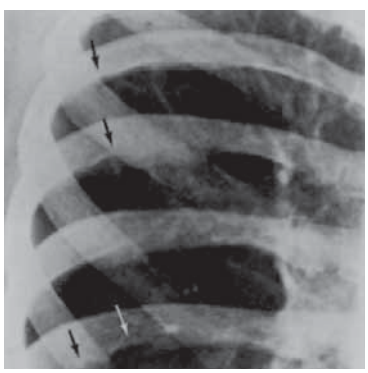
* Systemic hypertension due to renal hypoperfusion.

* Intracranial hemorrhage due to hypertension or associated aneurysm of circle of Willis.

Investigations

1- Chest X-ray:

- Cardiomegaly (LVH)
- Notching of the inferior border of the ribs from pressure erosion by enlarged collaterals is common by late childhood(Rosler sign)



2- ECG: - LVH in older patients

3- Echocardiography is Diagnostic

4- When echocardiogram is equivocal: cardiac CT , MRI and catheter are alternative diagnostics

Treatment

1. In neonates with severe coarctation of the aorta
   - Infusion of prostaglandin E1 to reopen the ductus and re-establish adequate lower extremity blood flow.
   - Once a diagnosis has been confirmed and the patient is stabilized, surgical repair should be performed
2. Older cases require:
   - Control of hypertension and prophylaxis against infective endocarditis.
   - Surgical repair

Cyanotic Congenital Heart Disease (CCHD)

Classification

CCHD with decreased pulmonary blood flow	CCHD with increased pulmonary blood flow
With RVH - Fallot tetralogy - Pulmonary atresia with VSD **With LVH** - Tricusped atresia - Ebstein anomaly - Pulmonary atresia with VSD	**With RVH** - **T**ransposition of great arteries (TGA) - **T**otal anomalous pulmonary venous return - Eisenmenger Syndrome **With BVH** - **T**runcus arteriosus - Single ventricle
Criteria - Hypercyanotic spells (attacks of increasing cyanosis) - Heart failure is very rare - P_2 (pulmonary component of S_2) → ↓ - Chest X-ray → lung oligaemia	- Poor feeding "dyspnea on suckling". - Recurrent chest infections is common - Recurrent heart failure is common - P_2 → ↑ - Chest X-ray → plethora.
- Growth retardation occurs in long standing, symptomatic, uncorrected lesions.	

Central cyanosis

Definition

Bluish discoloration of skin and mucous membranes due to presence of > 5 gm /ml reduced hemoglobin in the capillary blood.

Causes

1. Congenital heart diseases are the main cause of chronic central cyanosis
   - The commonest cyanotic congenital heart disease is Fallot tetralogy.
   - The commonest cyanotic congenital heart disease presenting at birth is TGA.
2. Respiratory failure (usually acute cyanosis)

Presentation

- Early it may be overlooked and become evident only during crying and feeding
- Then it becomes apparent at rest; noted mainly in inner lips and tongue

Consequences

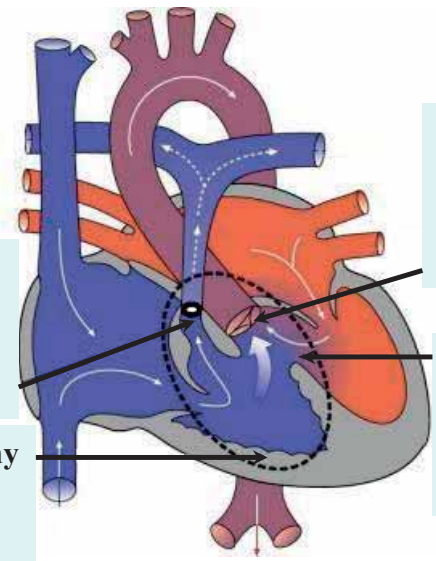
- Cyanotic clubbing, dusky blue skin, gray sclerae with engorged blood vessels
- Polycythemia
- Increased risk of Relative iron deficiency and Cerebral stroke

CCHD with decreased pulmonary blood flow

Fallot Tetralogy

Definition

Commonest cyanotic congenital heart disease with decreased pulmonary blood flow ; Composed of



Aortic dextroposition
- Aorta overrides the ventricular septal defect
- Receives mixed blood

Pulmonary stenosis
Mainly infundibular (muscular) but may be valvular

VSD
- Usually large
- Lies just below the aortic valve

Right ventricle hypertrophy
- Usually mild RVH
- Due to right ventricle out flow obstruction

Hemodynamics

A. Degree of pulmonary stenosis (PS) determines degree of right to left shunt:

1. Severe PS

- Early right to left shunt → cyanosis appear in the neonatal period
- Pulmonary blood flow is mainly dependent on flow through the ductus arteriosus
- When the ductus begins to close in the 1st few hours or days of life, severe cyanosis and circulatory collapse may occur

2. Mild to moderate PS

Patient is initially pink; over time, pulmonary stenosis gradually increases → right to left shunting → cyanosis appear within months

B. Pulmonary blood flow is maintained

- In neonate via ductus arteriosus
- In older child via Multiple aortopulmonary collateral arteries arising from the ascending and descending aorta

Clinical picture

1. Central Cyanosis



- Often, cyanosis is not present at birth; but with increasing hypertrophy of the right ventricular infundibulum as the patient grows, cyanosis occurs later in the 1st yearr of life
- Infants with severe degrees of PS, neonatal cyanosis is noted immediately
- Infants with mild degrees of PS may initially be seen with heart failure caused by a ventricular-level left-to-right shunt
- Cases who aren't initially visibly cyanotic are termed Acyanotic or pink Fallot .

2. Cyanotic Clubbing of fingers and toes



3. Growth retardation (Stunting) in unrepaired cases

4. Squatting position



- Older children ,with significant cyanosis at rest, have dyspnea on exertion
- After physical effort → dyspnea increases → the child assumes squatting position for the relief of dyspnea
- Theory: Squatting →kink of femoral arteries →↑ systemic vascular resistance →↑ aortic pressure →↑ pulmonary blood flow →↑ blood oxygenation.

5. Paroxysmal hypercyanotic Spells (Hypoxic, **"blue,"** or **"tet" spells)**

with collapse

dehydration

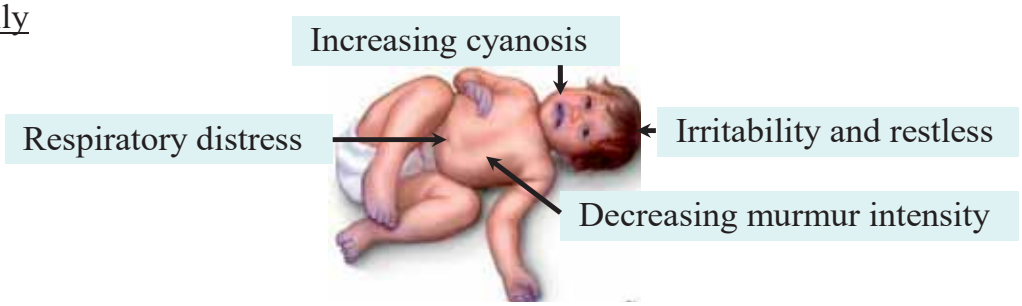
Incidence

Mainly in the 1st 2 years of life; triggered by crying, feeding or infection

Mechanism

Infundibular spasm → reduction of an already reduced pulmonary blood flow→ if prolonged→ severe systemic hypoxia and metabolic acidosis

Clinically



- Spell lasts from a few minutes to a few hours
- Severe spell may lead to convulsions, cerebrovascular stroke or even death

6. Cardiac examination

Precordial

- Normal heart size (may be mild RVH)
- Systolic thrill over left sternal border.

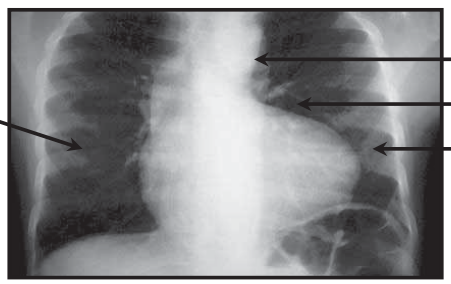
Auscultation

- S_2: single (A_2 only is heard)
- Murmur: systolic (organic PS) on the upper and mid left sternal border.

Investigations

1. Chest X-ray

Oligaemic lung fields

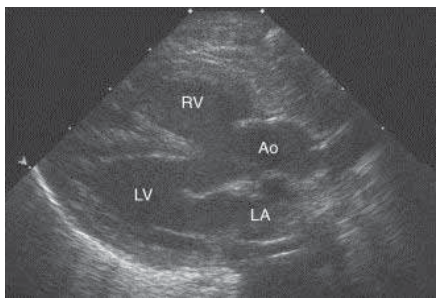


Coeur en sabot or boot shaped heart

- -Narrow base.
- -Exaggerated waist
- -Rounded uplifted apex.
- Heart size is normal or mild RVH

2. Echocardiography

Two-dimensional Echo is Diagnostic



3. Others

- ECG shows RVH
- Catheter → Pre-operative
- CBC show polycythemia

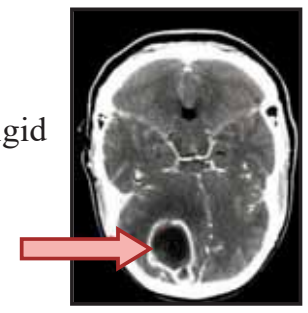
Complications

Early corrective surgery in infancy made it rare

1. Polycythaemia due to
   - Hypoxemia → ↑ erythropiotine → polycythaemia
   - Relative iron deficiency increase polycythaemia

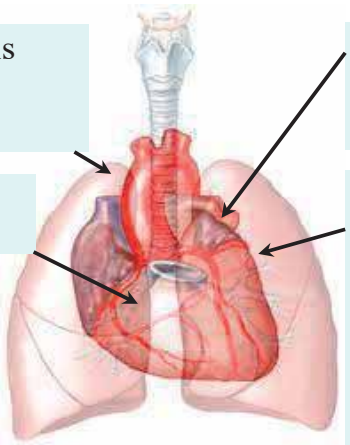
2. Cerebral thrombosis due to:
   - Extreme polycythemia → sluggish blood flow
   - Microcytosis due to relative iron deficiency →rigid non deformable RBCs

3. Brain abscess due to septic emboli and lack of pulmonary filtration



4. Pulmonary tuberculosis due to pulmonary oligaemia

5. Infective endocarditis



6. Heart failure is rare; may be iatrogenic with big shunt

7. Associated anomalies e.g. *DiGeorge syndrome or CATCH 22* (cardiac defects, abnormal facies, thymic hypoplasia, cleft palate, hypocalcemia).

Treatment

1. Medical

A. Treatment of hypoxic spells (*in sequence*)



- Don't Panic
- Calm and hold the infant in Knee-chest (frog) position ± pressure to femoral pulses.
- Facial oxygen (if not distressing the child)
- Avoid premature attempts to obtain blood samples to avoid further agitation

⇩

- Morphine 0.1mg/kg SC/IM/IV
  - Suppress the respiratory center
  - Caution in infants less than 3 months of age
- Fluid bolus 10ml/kg of NaCl 0.9% or Colloid IV
- Sodium bicarbonate 1mEq/kg IV (= 1ml/kg of 8.4%)
- Improve right ventricle out flow and reduce right to left shunt by either:

  i. Propranolol 0.1mg/kg IV over 5 minutes.
  - Value : Reduces infundibular spasm
  - Contraindicated if already on maximal dose of oral beta blockers.

  OR

  ii. Phenylephrine 5-10mcg/kg (slow IV push)
  - Value: Increases systemic vascular resistance

⇩

- PICU Intervention : Intubation and ventilation for spells resistant to the above management

- After the spell is over:
  - Oral propranolol prophylaxis 0.5-1 mg/kg/6hours
  - Avoid digitalis as it may induce infundibular spasm
  - Arrange for surgery

B. Avoid cerebral thrombosis by

- Treatment of relative iron deficiency
- For severe polycythemia
  - Adequate hydration
  - Phlebotomy
  - Partial exchange transfusion with albumin or saline.

C. Prophylaxis & treatment of infective endocarditis.

2. Surgical

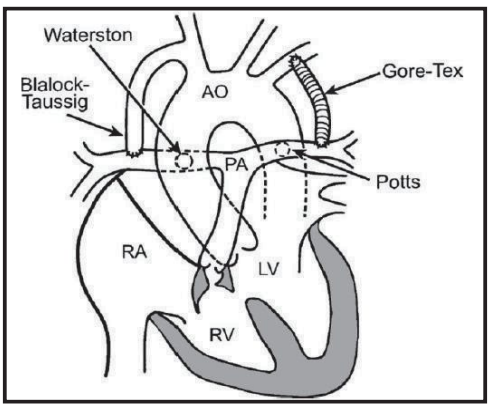
Indicated as soon as the spells begin

A. Palliative shunts

* Idea: Anastomosis between aorta and pulmonary artery to allow ↑ pulmonary blood flow

* Modified Blalock Taussig operation:

Anastomosis between subclavian artery & ipsilateral pulmonary artery using Gore Tex conduit (Potts and Waterston operations are obsolete)



B. Total repair:

Can be done between 4 months to 2 years according to severity and available cardiac center

Other causes of CCHD with decreased pulmonary blood flow

- All are Fallot tet. like in clinical presentation and management
- Differentiated from Fallot tet by Echocardiography and diagnostic catheter

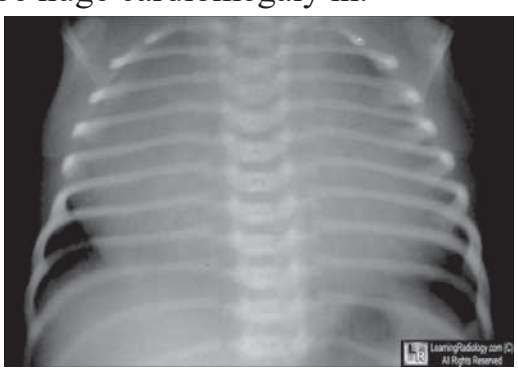
Ebstein's Anomaly

Composed of

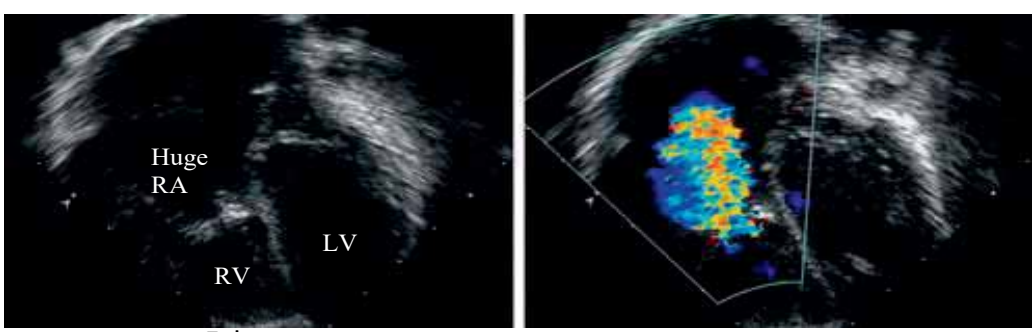
- Huge right atrium
- Downward displacement of tricusped valve leaflets.
- Tricusped regurge is common.
- Small right ventricle

Clinically

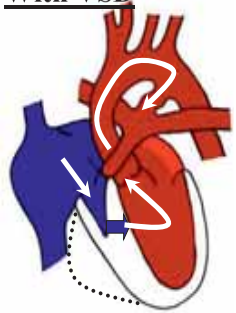
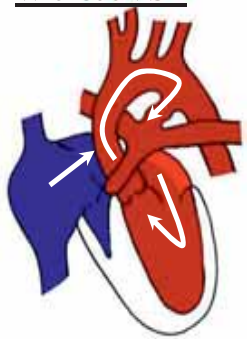
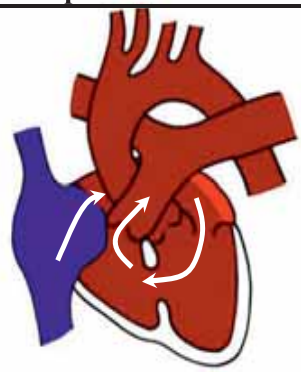
- May be asymptomatic
- Splitting of S1 and S2
- May be
  - Mild cyanosis
  - Atrial arrythmias
  - Pansystolic murmer (tricusped regurge)
- May be heart failure.
- Chest X-ray: May be huge cardiomegaly in.



- ECG → Right bundle branch block (RBBB) and RAD
- Echocardiography and color Doppler : diagnostic



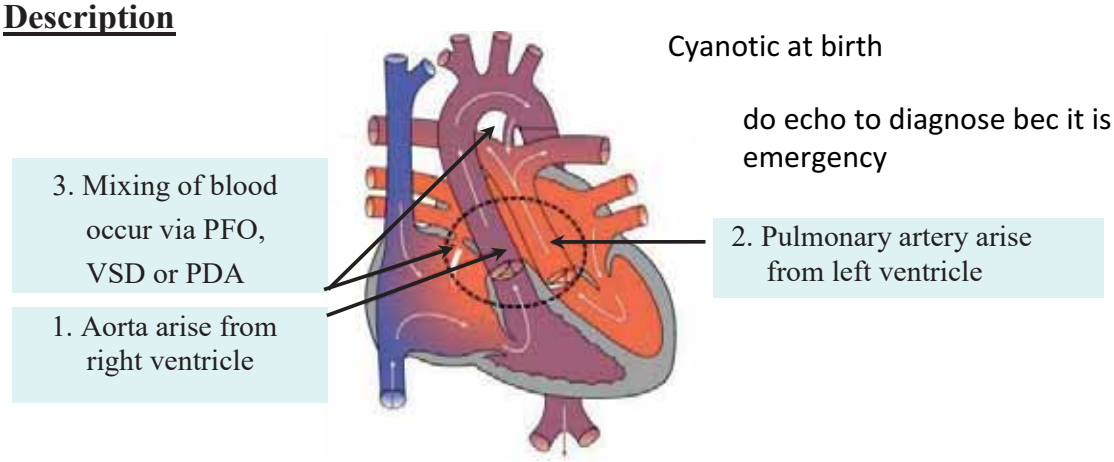
Echo Doppler

<table>
<tr><th colspan="2">3. Pulmonary atresia</th><th>4. Tricusped atresia</th></tr>
<tr>
<td><u>With VSD</u>
 
• Blood in right. ventricle pass to left ventricle via VSD so, Left ventricle contain mixed Blood → cyanosis.
• Pulmonary blood flow depends on PDA or collaterals between Aorta & pulmonary artery.</td>
<td><u>Without VSD</u>
 
• Blood in right atrium pass via patent foramen ovale (PFO) → left atrium → left ventricle → cyanosis.
• PFO <u>&</u> PDA are essential for life.</td>
<td> 
• Atretic tricusped valve → Blood in right atrium pass via PFO → left atrium → left ventricle → cyanosis.
• Pulmonary blood flow is dependent on VSD <u>or</u> PDA.
• Right ventricle is hypoplastic.</td>
</tr>
<tr><td colspan="3"><u>Presentation</u>
- Cyanosis is evident at birth → increase in intensity with ductus arteriosus closure.
- Single second heart sound (only A_2 is heard).</td></tr>
<tr><td colspan="2">- No murmurs. (may be machinery of PDA or collaterals).</td><td>- Murmur of VSD (± PDA).</td></tr>
<tr><td colspan="3"><u>Diagnosis</u>
- Echocardiography can differentiate it from Fallot tetralogy.</td></tr>
<tr><td colspan="3"><u>Treatment</u>
- Once suspected after birth: PGE_1 infusion to keep the ductus open
- Operative: Palliative shunts/ Total correction
- Medical: measures for polycythemia and infective endocarditis(as before)</td></tr>
</table>

CCHD with increased pulmonary blood flow

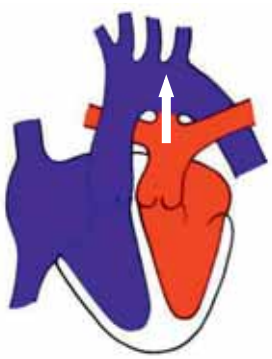
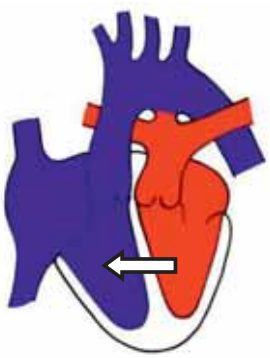
1. Transposition of Great arteries (TGA)

Description



Incidence: Common in infant of diabetic mother.

Types: - Isolated TGA
- TGA With VSD
- TGA With VSD and pulmonary stenosis
- Corrected TGA

Isolated (intact ventricular septum)	TGA With VSD
 Medical emergency * Severe cyanosis at birth * With ductus closure → marked cyanosis with acidosis and hypoglycemia * No murmur * Single accentuated S_2 (anteriorly placed aorta)	 - Milder cyanosis - Manifestations of increased pulmonary blood flow - VSD murmur.

Investigations

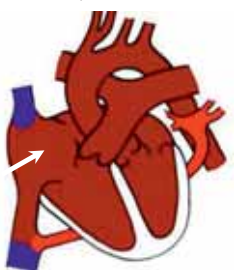
- Chest x ray
  - Cardiomegaly (egg on side)
  - Plethoric lung



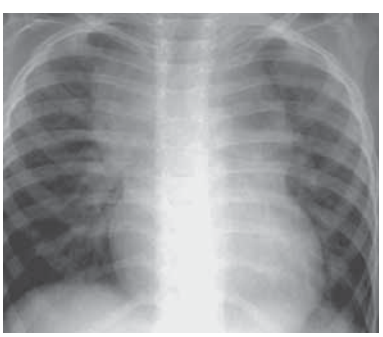
- ECG: RVH.
- Echo.: Diagnostic
- Cardiac catheter: Preoperative

Treatment	
1- Maintain PDA → PGE_1 infusion & avoid O_2.	- Treatment of heart failure (3D).
2- Palliative operation: Rashkind balloon atrial septostomy → create large ASD → free intra cardiac mixing.	
3- Total correction: either	
- Arterial switch operation (anatomical correction)	- Arterial switch operation
- Atrial switch operation.	
4- Other lines of treatment: - Avoid cerebral thrombosis (see Fallot tet) - Precautions against infective endocarditis	

2. Total anomalous pulmonary venous return (TAPVR)



- Pulmonary veins drain into right side of the heart either:
  - In superior vena cava (supra cardiac)
  - In coronary sinus (cardiac)
  - In inferior vena cava (infra cardiac)
- ASD allow blood in right atrium to pass to left atrium → blood mixing → cyanosis.

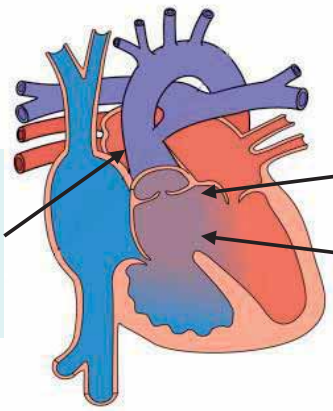


- Chest X-ray
  Snowman in snowstorm **or**
  Figure 8 shaped heart.
- Echo is diagnostic

3. Truncus arteriosus (TA)

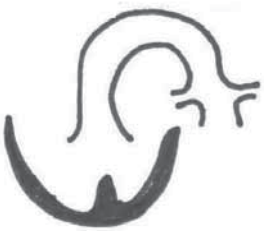
Description

1. One arterial trunk leave the heart giving rise to both aorta & pulmonary artery



2. One semilunar valve (Truncal valve)
3. Large VSD below the trunk

Types

Type I	Type II	Type III	Type IV "pseudo truncus"
			
Single pulmonary artery from left. Side	Two pulmonary arteries from the posterior wall	Two pulmonary arteries from the lateral wall	Arteries arising from the descending aorta → supply the lungs.

Clinical picture

- Cyanosis → variable onset (usually minimal esp. in neonate & infants).
- Features of increased pulmonary blood flow
- S_2 → single.
- VSD murmur.
- Chest X-ray → Right sided aortic arch in 50% of cases.

Treatment

1. Treatment of heart failure.
2. Surgical correction.

4. Single ventricle

- *Absent interventricular septum* → both Aorta & pulmonary artery arise from common ventricle → free mixing of blood → cyanosis.
- Degree of cyanosis depends on whether pulmonary valve is stenotic or not which determine pulmonary blood flow.

Valvular lesions

1. Mitral Stenosis (MS)

Causes

1- Rheumatic (mainly)

2- Congenital (rarely)

Hemodynamics

- MS → Blood accumulate in left atrium → left atrial dilatation → pulmonary congestion and pulmonary congestive symptoms
- Prolonged pulmonary congestion → pulmonary hypertension with low cardiac output symptoms
- Prolonged pulmonary hypertension →Right ventricular hypertrophy & right ventricle failure→ systemic congestive symptoms

Symptoms

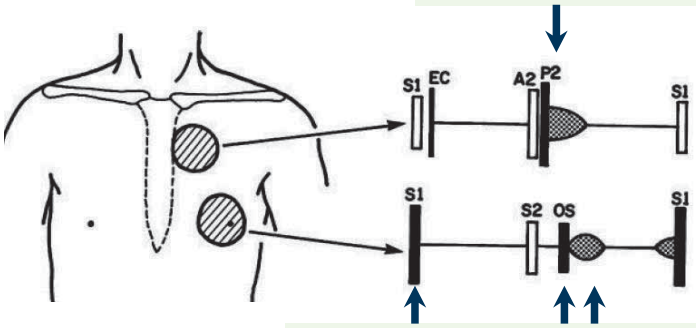
- Mild cases may be asymptomatic ; discovered accidentally
- Pulmonary venous congestive manifestations with or without systemic venous congestive manifestations.

Precordial examination

1. Weak apex beat; Slapping apex due to palpable first heart sound.
2. Pulmonary pulsation (palpable second heart sound) and dull pulmonary area in cases with pulmonary hypertension.

Auscultation

Pulmonary area (in cases with pulmonary hypertension):
- Accentuated S_2.
- Ejection systolic murmur



Apex
- Accentuated S1
- Opening snap.
- Murmur: localized mid diastolic rumbling with pre systolic accentuation

Complications

* Valve	- Rheumatic activity
	- Bacterial endocarditis
	- Calcification
* Left atrium	- Dilatation→ compression manifestations
	- Dysrhythmias
	- Intra atrial thrombi
* Ventricle	- Right ventricular failure

2. Aortic Regurge (AR)

Causes

1- Rheumatic (mainly)

2- Syphilitic, Marfan syndrome or post-operative (rare).

Hemodynamics

1- Incompetent aortic valve → In diastole blood returns to the heart leading to:
   - Decreased diastolic pressure.
   - Large end diastolic volume of left ventricle → increased systolic pressure.

2- High systolic pressure and low diastolic pressure result in hyperdynamic circulation.

Symptoms

1- Manifestations of low cardiac output.

2- Palpitation with exertion.

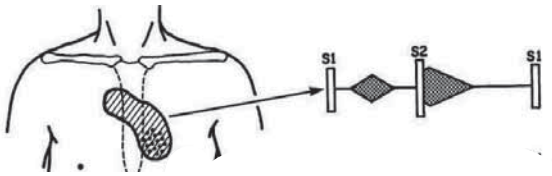
General examination: (peripheral signs suggesting A.R.)

1. Corrigan sign = visible arterial pulsations in carotid arteries.
2. De Musset sign = head nodding with each heartbeat.
3. Wide pulse pressure.
4. Water hammer pulse.
5. Capillary pulsations (in nail beds and lips).
6. Hill's sign: Lower limb's systolic pressure is higher than upper limb by > 20 mmHg.
7. Pistol shot due to push of blood in an empty artery.
8. Duroisier sign: diastolic murmur on pressing femoral artery by distal edge of stethoscope.

Precordial examination

- Hyper dynamic apex (forcible, non-sustained).
- Left ventricular enlargement (LVH).

Auscultation



Aortic area

1. Early diastolic murmur
   - On the 1st & 2nd aortic areas
   - Increases by leaning forward & in expiration
   - Propagates to the apex
2. May be ejection systolic murmur due to functional aortic stenosis

3. Mitral regurge (MR)

Causes

1- Rheumatic (mainly)
2- Congenital (rarely)
3- Mitral valve prolapse

Hemodynamics

1- Incompetent mitral valve → in systole blood returns to the left atrium → left atrial dilatation and pulmonary congestion → pulmonary hypertension.
2- Large end diastolic volume of left ventricle → left ventricle enlargement.

Symptoms

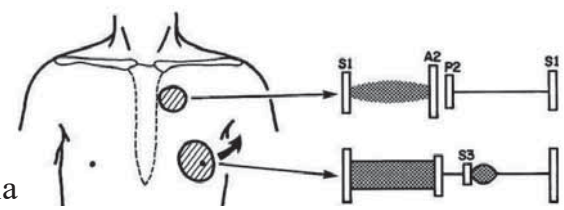
- May be asymptomatic in early cases
- Palpitation with exertion.
- Pulmonary venous congestive manifestations.

Precordial examination

- Hyper dynamic apex. (forcible, non-sustained).
- Left ventricular enlargement (LVH).
- May be apical systolic thrill.
- May be evidence of pulmonary hypertension(dull, pulsating pulmonary area)

Auscultation

1. Apex



- Muffled S_1.
- Murmur → Pansystolic.
  → Propagates to the axilla

2. Pulmonary area

In cases with pulmonary hypertension:
- Accentuated S_2.
- Ejection systolic murmur.

Aortic stenosis (AS)

Causes

1. Rheumatic	2. Congenital
Valvular stenosis - The commonest - About 70%	a. Valvular: bicusped aortic valve b. Subvalvular: a membrane or septal hyperterophy (hypertrophic obstructive cardiomyopathy c. Supravalvular (With *Williams syndrome* : Elfin facies, hypercalcemia, Cocktail party chatter).

Hemodynamics

- Left ventricle outflow obstruction
  → Low cardiac output.
  → Left ventricle hypertrophy & left ventricle failure.

Symptoms

1. Critical stenosis can present early in life with heart failure
2. Milder cases
   - May be asymptomatic
   - Manifestations of low cardiac output in older child.
   - Manifestations of complications

Precordial examination

- Heaving apex (forcible, sustained).
- Left ventricular enlargement (LVH).
- Systolic thrill over aortic area and the neck.

Auscultation

Aortic area: - Muffled, delayed aortic component of S_2.
- Murmur → Harsh ejection systolic murmur
→ Best heard on 1st aortic area
→ Propagate to the neck & the apex.

Complications

1. Left ventricular failure
2. Bacterial endocarditis
3. Sudden cardiac death

Treatment

- **Medical**
  - Exercise restriction for severe stenosis
  - Prophylaxis against endocarditis
  - Avoid vasodilators
- **Interventional / surgical** *(Balloon valvoplasty or valvotomy or valve replacement)*
  For: Symptomatic cases and/ or Pressure gradient across the valve > 40mmHg

Pulmonary stenosis (PS)

Causes

- Congenital mainly.

Symptoms

1- Asymptomatic in mild cases
2- Severe obstructive lesions can present early in life with heart failure.
3- Manifestations of low cardiac output in older child.

Precordial examination

- Right ventricular enlargement (RVH).
- Systolic thrill over pulmonary area.

Auscultation

Pulmonary area

Muffled pulmonary component of S_2.

Murmur → Ejection systolic on pulmonary area propagate to the left parasternal area.

Investigations of valvular lesions

Investigations	Value
Chest x ray	- Detect cardiomegaly - Pulmonary vascular markings: - Oligemic in pulmonary stenosis - Prominent in left sided failure - Specific configuration
ECG	- Detect chamber enlargement - Myocardial ischemia
Echocardiography	- Diagnostic ; see before
Catheterization	- Diagnostic ; done pre-operative - Interventional

Treatment of valvular lesions

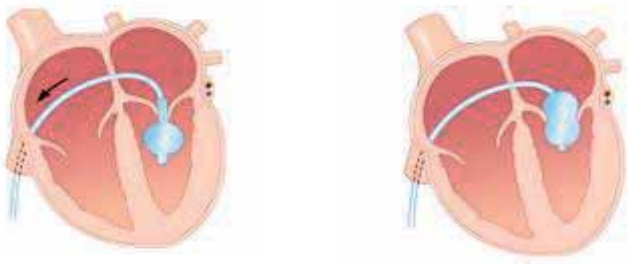
A. Medical

1. Afterload reducers e.g. captopril for regurgitant lesions.
2. Exercise restriction for severe stenotic lesions.
3. Treat complications:
   - Heart failure
   - Infective endocarditis.
   - Arrythmias

4- Prophylaxis against
- Infective endocarditis.
- Rheumatic fever (in rheumatic cases).

B. Interventional

Balloon valvoplasty for symptomatic, non-calcific, stenotic lesions e.g.
- AS
- PS
- MS



C. Surgical

1- Valvotomy for stenosis
2- Valve repair for regurge
3- Valve replacement.

Acute Rheumatic Fever (ARF)

Definition

Immunologic disease affecting mainly the heart and joints& less frequently central nervous system, skin and subcutaneous tissue

Risk factors

- Peak of onset between 5-15 year (rare before 5 years)
- Genetic predisposition
- Moderate and High risk populations : Developing countries
- Low risk populations : USA, Canada, western Europe

Pathogenesis

Group A Streptococcal (GAS) pharyngitis (M serotypes 1, 3, 5, 6,18, 24)



Antibodies formed against streptococcal cross react against host connective tissue

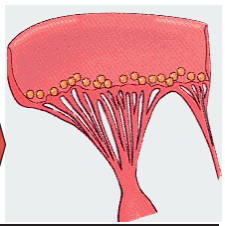
Streptococcal infection alter host connective tissue antigenicity

Inflammation

Proliferative with Aschoff nodules as in the heart and subcutaneous nodules

Exudative (in joints) with inflammatory edema → resolve without residual

Valvulitis due to endocardial inflammation may destroy valves resulting in valve regurge (in acute stage) OR end up in valve fibrosis and stenosis OR a mix of both



Clinical Picture

A latent period of 1-3 weeks usually exist between pharyngitis and clinical symptoms of acute rheumatic fever

Major Criteria of Rheumatic Fever

1. Arthritis

- The commonest presenting sign 75%
- Distribution
  - Usually affect big joints (e.g. knee, ankles, wrist, elbow).
  - Polyarticular arthritis; either simultaneous or successive.
  - Migratory arthitis →Fleet form one joint to another.
- Joint examination
  - Inspection→ Red ,hot ,swollen (inflamed)
  - Passive movement→Severely tender
  - Active movement→ Absolute limitation of movement
- Course :
  - Dramatic response to salicylates within 48 hours.
  - Resolve without residuals, even without treatment, over one week.
- Arthritis now refers to polyarthritis in low-risk populations, but to monoarthritis or polyarthralgia in moderate/high-risk populations

2. Carditis

- The 2nd common(50%) and most serious manifestation of ARF
- Pancarditis→ inflammation of endocarium,myocardium and pericardium
- Carditis may be silent **or** late onset appearing after 6 weeks – 6 months
- Carditis is now defined as clinical and/or subclinical (echocardiographic valvulitis)

i. Endocarditis: Valvulitis

Affect commonly the mitral valve with or without aortic valve:

a. Mitral valve

- Valve edema → transient mitral stenosis → mid diastolic rumbling murmur (*Carey Combs* murmur)
- Valve destruction → mitral regurge → apical holosystolic murmur propagating to the axilla with muffled 1st heart sound.

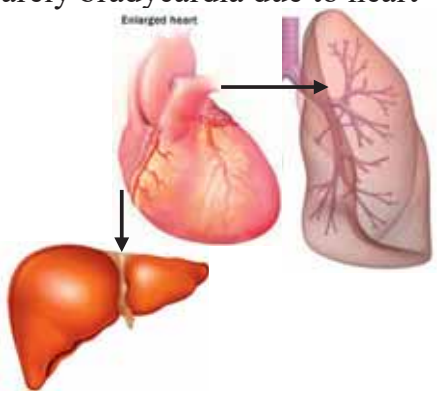
b. Aortic valve

- Aortic regurge→ left sternal border early diastolic murmur.



2. Myocarditis

- Tachycardia out of proportion to age & fever(rarely bradycardia due to heart block)
- Heart failure indicates severe carditis with
  - Marked dyspnea
  - Enlarged tender liver
  - Gallop rhythm, muffled heart sounds
  - Cardiomegaly



3. Pericarditis

a. Dry pericarditis
- Stitching chest pain .
- Pericardial rub (on the bare area of the heart, unrelated to respiration).

b. Pericardial effusion
- Dull aching pain.
- Distant heart sounds.

Investigations for carditis

- ECG may show Low voltage ECG in pericardial effusion.
- Chest x ray may show cardiomegaly
- Echocardiographic findings include mitral and/or aortic regurgitation, pericardial effusion, and decreased ventricular contractility

3. Rheumatic chorea "*Sydnham Chorea*" (10%)

Incidence
- More in girls 8-12 years (school age).
- Occur weeks or months after pharyngitis so, other criteria are usually lacking.
- May coexist with other rheumatic fever manifestations in 10%

Due to: Dysfunction of the basal ganglia.

Manifestations

1. Emotional lability and personality changes
2. Involuntary movements
   - Sudden ,jerky ,spontaneous pseudo purposeful movements of limbs
   - Facial grimace.
   - Distribution: Proximal more than distal.
   - Increase with emotional stress and decrease by sleep.
3. Hypotonia



Tests for rheumatic chorea

- Cannot maintain arms extended → spooning & pronation of hands (*Choreic hand*).
- Milk maid's grip: irregular contraction & relaxations on squeezing examiner hand
- Wormian movements of tongue upon protrusion (*Darting tongue*).
- Examination of handwriting to evaluate fine motor movements

Outcome

- Self-limited ; usually resolve within 4-8 weeks
- Acute phase reactant are usually normal in isolated chorea

4. Erythema Marginatum (< 5%)



Site

- The trunk
- Proximal parts of the limbs

Criteria

- Large erythematous macules.
- With pale centers and serpiginous borders(map like appearance)
- Evanescent.(may come and go for several months)
- Not pruritic

5. Subcutaneous Nodules (< 1%)

Appear several weeks after the attack



Site

- Over the extensor surfaces of tendons near bony prominences.

Criteria

- Size is about 1 cm.
- Firm, freely mobile,and painless.
- Usually associated with severe carditis.

N.B: Both sub cutaneous nodules and erythema marginatum are rare in Egypt

Minor criteria of Rheumatic Fever

A. Clinical

1. Fever: of > 38° C (>38.5° C in Low-Risk populations)

2. Arthralgia

- Moderate/High-Risk populations only include monoarthralgia (polyarthralgia for Low-Risk populations)
- Can't be used as minor manifestation in presence of arthritis

B. Investigations

1. ECG: Prolonged P-R interval

- Can't be used as minor manifestation in presence carditis

2. Elevated acute phase reactants

- ↑ ESR; >30 mm/hr (>60 mm/hr in Low-Risk populations).
- ↑ C reactive protein

Evidence of recent Group A Streptococcal (GAS) infection

- Positive throat culture Or
- Elevated or increasing streptococcal antibody titers : ASO , Anti Deoxyribonuclase (DNase) β , Anti hyaluronidase

American heart association's revised Jones criteria (2015)

- **For diagnosis of initial attack of ARF**
  - 2 major manifestations, **Or**
  - 1 major and 2 minor manifestations
    Plus
  - Evidence of recent GAS infection
- **For diagnosis of recurrent attacks of ARF**
  - 2 major, **Or**
  - 1 major and 2 minor, **Or**
  - 3 minor manifestations (only in the Moderate/High-Risk population)
    Plus
  - Evidence of recent GAS infection
- Carditis Redefinition (see before)
- Arthritis Redefinition (see before)
- Minor criteria Redefinition (see before)

Notes on Jones criteria

There are 3 circumstances in which the diagnosis of acute rheumatic fever can be made without strict adherence to the Jones criteria:

- When chorea occurs as the only major manifestation of acute rheumatic fever
- When indolent carditis is the only manifestation
- Patients with recurrences of acute rheumatic fever in particularly high-risk populations.

Prognosis and complications

1. Arthritis sub side within days to weeks even without treatment.
2. Chorea subsides within few months without residuals.
3. Only carditis can cause permanent damage especially in recurrences which :

* Suggested by: - Appearance of new murmurs.
  - Change in character of already existing murmur
* Result in organic valve lesion e.g. MS,MR, AS, and/or combined valve lesion.
* Carditis and chronic valve lesions may be complicated by:
  - Heart failure
  - Infective endocarditis
  - Embolic manifestations
  - Arrhythmias
  - Pulmonary hypertension

Differential diagnosis

1. Other causes of arthritis

- **Post streptococcal arthritis**
  - May follow infection with either group A or group G streptococcus
  - Typically oligoarticular, affecting lower extremity joints, and mild symptoms can persists for months
  - Some clinicians consider it to be an incomplete form of acute rheumatic fever
- **Transient synovitis (toxic synovitis)**
  - Post-infectious arthritis, typically affects the hip, often after an upper respiratory tract infection
  - Acute onset of severe pain in the hip, with referred pain to the thigh or knee, lasting approximately 1 wk
- **Rheumatoid arthritis:**
  - Chronic deforming arthritis; last ≥ 6 weeks.
  - Non migratory
  - Some involve small peripheral joints.
  - Absent dramatic response to salicylates within 48 hours.
  - Associations e.g. Spiking fevers, lymphadenopathy, and splenomegaly

- Reactive arthritis related to gastrointestinal infections (e.g., **S**higella, Salmonella, Yersinia)
- **S**eptic arthritis (usually monoarthritis)
- **S**erum sickness
- Hematologic e.g. **S**ickle cell disease, Acute leukemia , hemophelia.
- Immunologic e.g. **S**ystemic lupus erythematosis& anaphylacteod purpura

2. Other causes of carditis e.g.

- Vial carditis
- Infective endocarditis
- Drug induced.

3. Other causes of chorea e.g

- Wilson disease
- Cerebral palsy

Treatment of Acute Rheumatic Fever

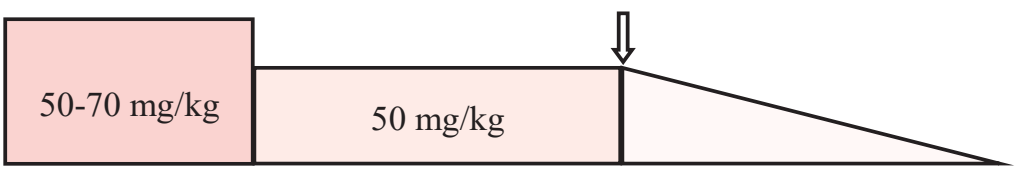
A. Treatment of Acute Attack

1. Eradicate group A Streptoccoci from the upper respiratory tract

- Oral penicillin or erythromycin for 10 days or a single intramuscular injection of benzathine penicillin
- Long-term antibiotic prophylaxis after this initial course of antibiotic therapy

2. Typical migratory Arthritis

1. Bed rest for 2 weeks
2. Anti-inflammatory drug: Salicylates



- Dose: 50-70 mg/kg/day (max = 6 gram /day) For 3-5 days
- Then 50 mg/kg/d for 3 weeks.
- Then gradual withdrawal monitored by decline in ESR & CRP
- Side effects: Gastritis, GI bleeding(R/Gastriprotection),Reye syndrome

3. Daily examination is vital to pick carditis that can present within 2 weeks of the onset

3. Carditis

1. Bed rest

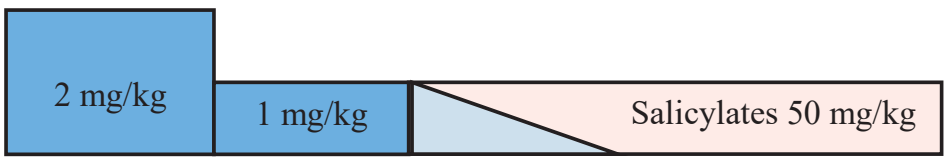
- For cases with severe carditis and heart failure
- Rest for 3 months and gradual ambulation for a similar period



2. Anti-inflammatory drugs

a. Corticosteroids (Prednisone)

* Indications:
  - Moderate to severe carditis(with cardiomegaly)
  - Heart failure.
* Dose:



- 2 mg/kg/d for 2-3 weeks
  - Followed by half the dose for 2-3 wk
  - And then tapering of the dose by 5 mg/24 hr every 2-3 days.
- At the beginning of tapering; salicylates is started at dose of 50 mg/kg/d and continued for 6 weeks to avoid rebound phenomenon.

b. Salicylates can be used in mild carditis without cardiomegaly nor heart failure

3. Heart failure with respiratory distress

a. Oxygen inhalation
b. Restriction of salt and fluid intake.
c. Steroids
d. Other anti-failure drugs:
   - Diuretic: Furosemide 1mg/kg every 6-12 hours
   - VasoDilators: e.g. captopril
   - Digoxin: used cautiously; begin with half the usual recommended dose

(Nelson Textbook of Pediatrics, 2016)

- Anti-inflammatory agents (e.g., salicylates, corticosteroids) should be withheld if arthralgia or atypical arthritis is the only clinical manifestation of presumed acute rheumatic fever.
- Premature treatment with one of these agents may interfere with the development of the characteristic migratory polyarthritis and thus obscure the diagnosis of acute rheumatic fever.
- Acetaminophen can be used to control pain and fever while the patient is being observed for more definite signs of acute rheumatic fever or for evidence of another disease

(Nelson Textbook of Pediatrics)

4. Treatment of Rheumatic Chorea

- Avoid emotional stress.
- Control abnormal movements:
  - Phenobarbitone 16-32 mg / 8 hours oral Or
  - Haloperidole (Safinase tablet) 0.01-0.03 mg/kg.
- Long acting penicillin prophylaxis.

5. Treatment of complications

- Treatment of heart failure; see before.
- Treatment of infective endocarditis
- Treatment of valve lesions

B. Prophylactic

1. Primary prevention: prevent 1st attack by:

- Hygienic housing.
- Proper treatment of Strept. infection: penicillin **or** erythromycin for 10 days.

2. Secondary prevention : Prevent recurrence of Rheumatic fever by:

- Long acting penicillin (Benzathine penicillin)
- Dose : 600.000 units for those < 27 kg and 1.2 million unit for those > 27 kg
- Route: Single injection, I.M every **3-4** weeks.
- Alternatives: Oral penicillin V **or** Macrolide (Erythromycin)
- Duration

Rheumatic fever category	Duration
– Without carditis	– 5 yr or until 21 yr of age, whichever is longer
– With carditis but without residuals	– 10 yr or until 21 yr of age, whichever is longer
– With carditis and residual heart disease	– 10 yr or until 40 yr of age, whichever is longer – Sometimes lifelong prophylaxis

(Nelson Textbook of Pediatrics, 2016)

3. Infective Endocarditis (IE) prophylaxis

- American heart association (AHA) no longer recommends routine IE prophylaxis for rheumatic heart diseases
- Prophylaxis is still recommended only for those with prosthetic valves or prosthetic material used in valve repair (use antibiotic other than penicillin)

Infective endocarditis

Definition

- Infection of the valvular & mural endocardium
- Infection can be bacterial, viral, or fungal

Pathogenesis

Two factors are essential

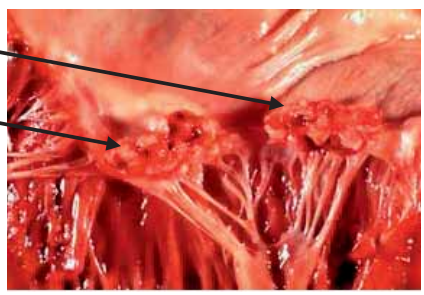
1. Presence of cardiac structural abnormality with significant pressure gradient
2. Bacteremia; even transient

Commonest causative organisms

1. Streptococcus viridans (50%):
   - Follows dental surgery ,dental caries, tonsillectomy, dental extraction
2. Staphylococcus aureus and epidermidis
   - Mainly postoperative
   - Risk is high with prosthetic valve and central venous catheter
3. Group D enterococci
   - More often after lower bowel or genitourinary procedures
4. HACEK group : Haemophilus, Actinobacillus, Cardiobacterium, Eikenella, and Kingella species
4. Pseudomonas aeruginosa or Serratia marcescens
   - Seen more frequent in intravenous drug users
5. Fungal: Seen more frequent in immunodeficient & post open heart surgery.

Pathology

Implantation of the organism in the diseased endocardium → Local inflammation & formation of friable *vegetations* composed of platelets, fibrin, inflammatory cells, and organisms



Clinical picture

A. History: Suggestive of a risk factor or bacteremia

B. General manifestations

1. Fever (**p**yrexia)
2. **P**oor appetite → weight loss & malaise.
3. **P**alpable spleen (tender splenomegaly)
4. **P**ale clubbing
5. **P**allor
6. **P**urpura



7. Rare manifestations due to immune complexes induced vasculitis:

Osler nodules
- Transient tender nodules
- Pea-sized
- Intradermal in the pads of the fingers and toes



Janeway macules
- Irregular, painless macules
- In palms and soles



Splinter hemorrhage
- Linear hemorrhagic streaks
- Beneath the nails



B. Cardiac manifestations

- Appearance of new murmurs
- Change in the character of previous murmurs
- Sea gull murmur (musical) → due to rupture of valve leaflets.
- Heart failure or arrhythmias.

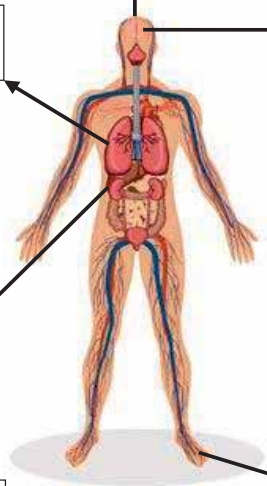
C. Embolic manifestations

1. Neurologic
- Embolic stroke (seizures, hemiparesis)
- Cerebral abscess
- Mycotic aneurysm → intracranial hemorrhage



2. Pulmonary embolism

3. Retinal hemorrhages; (Roth spots = oval with pale centers), blindness







4. Renal infarction → hematuria & renal failure

5. Skin infarcts and gangrene

Investigations

A. For diagnosis

1. Blood culture
   - 3-5 blood samples before the start of antibiotics (contact the lab staff !!!)
   - The causative agent is recovered from the 1st 2 blood cultures in 90% of cases
2. Echocardiography: transthoracic and transesophageal

   Value
   - Detects vegetations (early & minute vegetation < 3 mm^3 may be missed)
   - May predict embolization (fungating vegetation > 1 cm3).
   - Detects underlying cardiac lesions
   - Detects complication e.g. valve dysfunction or leak , myocardial abscess
3. Culture of other specimens may include scrapings from skin lesions, urine, synovial fluid, abscesses, and, CSF in case of meningitis
4. With unusual microorganisms
   - PCR of surgical material e.g. resected valve tissues
   - Specific serology

B. For monitoring/effect

1. Acute phase reactant (ESR, CRP, Procalcitonin and Leucocytosis)
2. Complete blood count: for anemia of chronic illness
3. Renal function tests, urinalysis

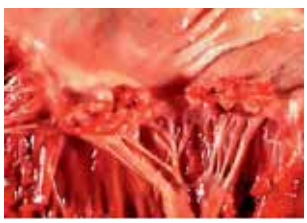
Prevention

A. Oral and dental care;the important

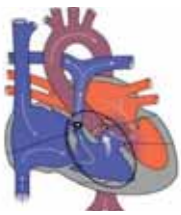
B. Antibiotic prophylaxis; Required for cases with**:**



1. Prosthetic cardiac valve or prosthetic material used for cardiac valve repair



2. Previous infective endocarditis



3. Congenital heart disease (except ASD)





4. Repaired Congenital heart disease with prosthetic material or device

5. Palliative shunts and conduits



6. Permanently damaged valves due to Rheumatic heart disease

Procedures requiring prophylaxis ?

- Dental /oral
- Invasive respiratory tract procedures
- Cardiac surgery/catheterization



- Prophylaxis for gastrointestinal or genitourinary procedures is no longer recommended in the majority of cases
- Direct consultation with the child's cardiologist is still the best method for determining a specific patient's ongoing need for prophylaxis

Prophylactic antibiotics

- Oral amoxicillin (50 mg/kg) or Ampicillin Or
- Ceftriaxone (50 mg/kg) IM or IV Or
- If penicillin allergic: Azithromycin or clindamycin

3. Vigorous treatment of sepsis and local infections

Treatment

Medical

1- Hospitalization, Bed rest and treat heart failure

2- Antibiotic therapy

- Start immediate parenteral antibiotic combinations while waiting for culture results.
- For 4-6 weeks (modify in view of clinical and laboratory response)
- Best empirical therapy in patients without a prosthetic valve is vancomycin plus gentamicin
- Other antibiotics include crystalline penicillin G, ceftriaxone, nafcillin or oxacillin.
- Amphotricin B and 5 florocytosine for fungal endocarditis.

Surgical

- Removal of vegetation with or without valve replacement
- Indications
  - Treatment failure e.g. increasing size of vegetations while under therapy
  - Complications: Severe valve involvement with intractable heart failure, myocardial abscess, recurrent emboli

(American Heart Association 2007, Nelson 2016)

Prognosis: Despite the use of antibiotic agents, mortality is at 20-25%. Serious morbidity occurs in 50-60%

Heart failure

Definition

Clinical syndrome in which the heart is unable to pump enough blood to meet body needs.

Causes

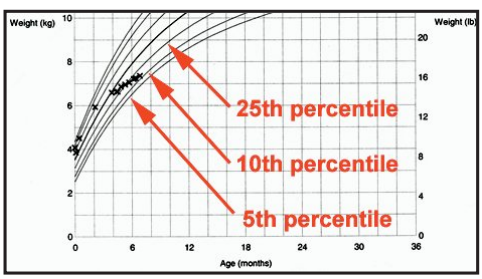
Congenital heart diseases are the commonest causes of heart failure in infancy	Rheumatic heart diseases (in school age)
▪ Myocarditis - Viral e.g. Coxachie A, B & Echo viruses - Toxic e.g. drugs, diphtheria . ▪ Dilated cardiomyopathy. ▪ Infective endocarditis	▪ Acute hypertension ▪ Severe anemia
Dysrhythmia ▪ Supraventricular tachycardia ▪ Complete heart block	▪ Acute cor pulmonale ▪ Broncho pulmonary dysplasia
▪ Nutritional e.g. Beri Beri, Kwashiorker, Keshan disease(selenium deficiency)	

Clinical features

A. Symptoms

Infants

- Poor feeding; takes less volume per feeding, becomes dyspneic while sucking, and may perspire profusely
- Poor weight gain.



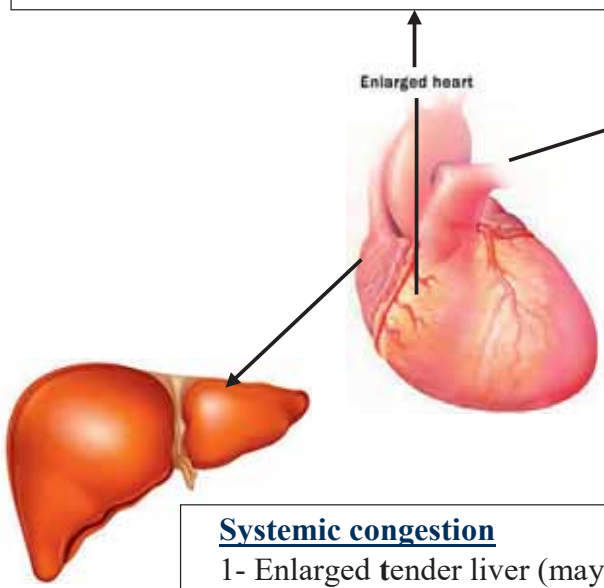


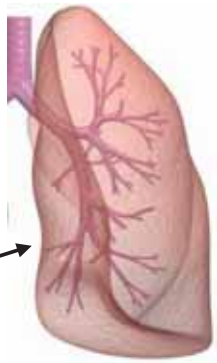
Older child

- Dyspnea on exertion.
- Effort intolerance.
- Ankle edema.

B. Signs

Compensatory response to heart failure
1- **T**achycardia, gallop rhythm & weak pulse.
2- Cardiomegaly is almost always present.
3- Cold, sweaty skin (increased sympathetic derive)





Pulmonary congestion
1- **T**achypnea
2- Exertional dyspnea
- Infant → poor feeding.
- Child → dyspnea & orthopnea
3- Chest wheezes & fine crepitation.

Systemic congestion
1- Enlarged **t**ender liver (may be absent in early left sided failure).
2- Congested neck veins; hard to detect in infants due to short neck.
3- Edema → generalized start in ankles (sacral in bed ridden)
4- Edema in infants usually involve eye lids and the sacrum

Investigations (*Heart failure is a clinical diagnosis*)

1. Chest X-ray



- Cardiomegaly
- Fluffy perihilar pulmonary markings

2. Echocardiography



- Confirm ventricular dysfunction
  - Using fractional shortening (difference between end-systolic and end-diastolic diameter divided by end-diastolic diameter)
  - Normal value is between 28% and 42%
- Doppler can estimate cardiac output
- May detect the cause of failure.

3. Magnetic resonance angiography (MRA)

- Quantifies left and right ventricular function, volume and mass.
- Quantifies the regurgitant fraction in valvar regurgitation

4. ECG: - Detect arrhythmias.

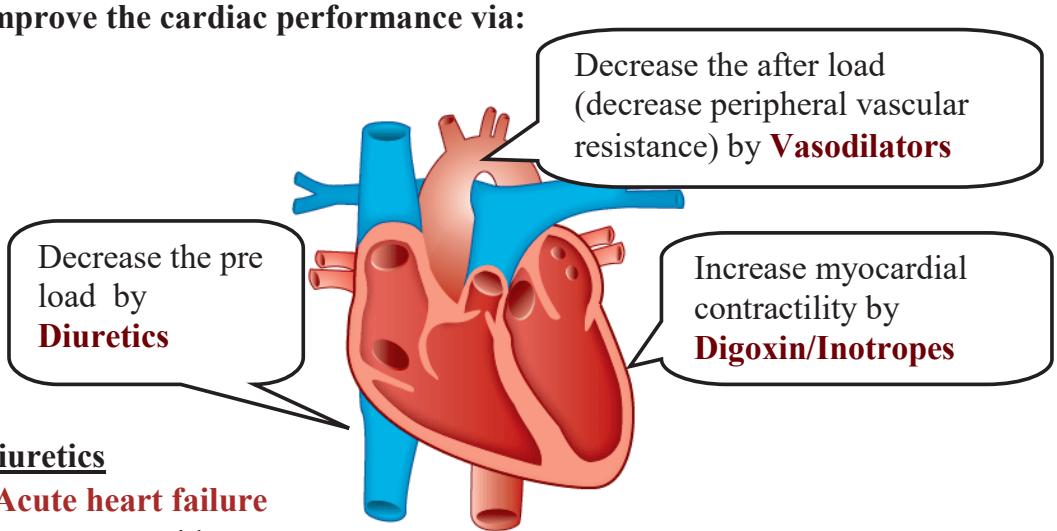
Treatment

1. Hospitalization

- Bed rest as needed in a semi-upright position(infant chair for infants)
- O2 inhalation ± positive pressure ventilation can significantly reduce total body oxygen consumption
- Low salt diet → avoid further salt and water retention.
- Increasing daily calories (nasogastric feedings may be helpful)
- Sedation

2. Treat the cause

3. Improve the cardiac performance via:



a. Diuretics

Acute heart failure

- Furosemide
  - I.V. (0.5-2 mg/kg/dose) or oral (1-4 mg/kg/day)
  - Precautions : monitor serum electrolytes and acid base(Hypokalemia and alkalosis → may increase digitalis toxicity)
- Nesiritide (B-type natriuretic peptide) IV infusion

Chronic heart failure

- Spironolactone (potassium sparing diuretic)
- Thiazide duiretics

b. Vasodilators

Used if blood pressure allows and in absence of obstructive lesions

Actions

- After load reducers
- Cardiac remodeling

Commonly used drugs

- Angiotensin converting enzyme (ACE) inhibitors e.g. Captopril, Enalapril
- Angiotensin receptor blockers [ARBs]

Indicated mainly in:

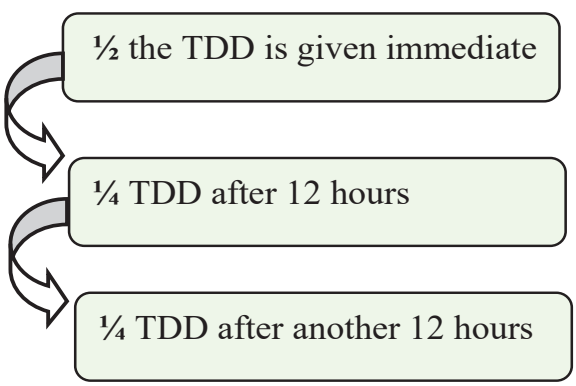
- Dilated cardiomyopathy/myocarditis
- CHD with large left to right shunt.
- Severe MR and AR
- Hypertensive heart failure

c. Digoxin

Digitalization

- Total digitalizing dose (TDD)

	Oral TDD (microg/kg)	I.V. TDD
Premature	20	75% of oral T.DD.
Newborn	30	
Infants < **2** y	40-50	
Child > **2** y	30-40	





- Precautions
  - Obtain ECG before each of the 3 digitalizing doses
  - Measure baseline serum electrolyte before and after digitalization
- Maintenance dose
  - Oral:5-10 µg/kg/day, divided q12h
  - IV dose is 75% of oral dose
  - Trough serum level: 1.5-3.0 ng/mL <6 mo old; 1-2 ng/mL >6 mo old

Absolute contraindications to digitalis

- Cardiac outlet obstruction e.g. Hypertrophic cardiomyopathy.
- Fallot's tetralogy
- Heart block.

Other inotropes

- Used for short term in ICU setting ;given by intravenous infusion
- Dobutamine ± Dopamine
- Milrinone(Phosphodiesterase inhibitor)

Digitalis toxicity

1. Causes
   - Accidental over dose.
   - Renal impairment.
   - Increased myocardial sensitivity e.g.: hypokalemia & active myocarditis
   - Drug interactions.
2. Signs
   - Anorexia, vomiting
   - Drowsiness & visual disturbance in older child.
   - Bradycardia
   - Worsening of heart failure.
   - Arrhythmias (supraventricular arrythmia & heart block).
3. Treatment
   - Continuous ECG monitoring.
   - Stop digitalis
   - Correct hypokalemia
   - Correct arrhythmias by
     a- Atropine 0.01 mg/kg/6 hours for heart block.
     b- lidocaine for ventricular arrhythmia
   - Increase excretion of digoxin by Digoxin immune Fab (*Digibind*), slow I.V.

Systemic Hypertension

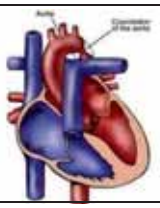
Definition

- Average systolic blood pressure (SBP) and/or diastolic blood pressure (DBP) that is ≥ 95th percentile for age, sex, and height on ≥ 3 occasions
- Prehypertension was defined as average SBP or DBP that are ≥90th percentile but <95th percentile
- In adolescents beginning at age 12 yr, prehypertension is defined as BP between 120/80 mm Hg and the 95th percentile

Causes

1. Essential (primary) hypertension
   - Rare in children; common in adults.
   - Associations → obesity, hereditary, increased sensitivity to salt intake.

2. Secondary

Etiology	Acute	Chronic
Renal 	- Acute glomerulonephritis. - Acute renal failure. - Hemolytic uremic syndrome.	- Renal tumors, hypoplasia, dysplasia. - Chronic pyelonephritis - Hydronephrosis/reflux nephropathy. - Renovascular: o Renal artery stenosis, thrombosis, o Polyarteritis. o Renal vein thrombosis.
Endocrine 		- Cushing syndrome - Hyperaldosteronism - Congenital adrenal hyperplasia. - Hyperparathyroidism(hypercalcemia)
Tumors 		- Neuroblastoma - Wilm's tumor. - Pheochromocytoma
Cardiac 		- Coarctation of aorta 
Neurologic 	- Acute ↑ intra cranial tension. - Guillian Barre syndrome - Poliomyelitis.	
Drugs	- Sympathomimitics.	- Steroids - NSAIDs

Possible mechanisms in 2ry hypertension

- Stimulation of Renin-Angiotensin-Aldosterone system e.g. renal hypertension.
- Salt & water retention e.g. Cushing & hyperaldosteronism.
- Stimulation of vasomotor center e.g. neurologic hypertension.
- Vasoconstriction due to:
  - ↑ Release of catecholamines e.g. pheochromocytoma
  - Sympathomimitc drugs.

Presentation

1- Usually asymptomatic.

2- May be → headache, irritability, blurr of vision (in severe cases)

3- Complications

* Hypertensive heart failure.
* Acute pulmonary oedema.
* Hypertensive encephalopathy :
  - Severe bursting headache, vomiting,
  - Irritability, convulsions and coma.
  - Fundus examination
    - Vasospasm
    - Papilloedema
    - Retinal hemorrhage.



Investigations



- 24 hour Ambulatory Blood Pressure Monitoring (ABPM).
- Calculate mean daytime BP, and sleep BP over 24 hr.
- ABPM is useful in the evaluation for:
  - White coat hypertension
  - Risk of hypertensive target organ damage
  - Response to pharmacologic therapy (confirms physiologic nocturnal dipping of blood pressure essential for kidney vitality)

Workup for 2^{ry} causes

1. Renal

- Urine analysis, urine culture, renal function tests.
- Abdominal ultrasound.
- Renal Doppler.
- **If blood pressure is not controlled on ≥ 2 drugs**
- Pre and post captopril renal scintigraphy and or
- CT or MRI angiography
- **If strong suspicion of renovascular hypertension**
- Digital Subtraction Angiography
- Selective renal vein rennin level

2. Cardiac

- Chest X-ray
- Echocardiography (For a cause or effect of hypertension).

3. Endocrinal

- Electrolytes (potassium & sodium).
- Night time blood or salivary cortisole level →↑ in Cushing.

4.Tumors

- 24hr urine vallynile mandilic acid (VMA); (metabolite of catecholamines) →↑ in pheochromocytoma & Neuroblastoma.
- Abdominal ultrasound, CT, MRI.

Treatment

- Children with BP between the 95th and 99th percentile plus 5 mm Hg are categorized as *stage 1 hypertension*
- children with BP above the 99th percentile plus 5 mm Hg have *stage 2 hypertension.*
- Stage 1 hypertension, if asymptomatic and without target organ damage, allows time for evaluation before starting treatment
- Stage 2 hypertension calls for more prompt evaluation and pharmacologic therapy

Goals

- Reduce BP below the 95th percentile
- In the presence of chronic kidney disease, diabetes, or target organ damage, the goal should be to reduce BP to less than the 90th percentile
- If blood pressure between 90 -95th percentile continue monitoring

A. Chronic hypertension

I. Primary hypertension

1. Non pharmacologic

- Weight reduction may result in a 5-10 mmHg reduction in systolic pressure
- Low salt, potassium rich diet
- Dynamic aerobic exercises
- Physical fitness

2. Drug therapy

Indications

- Family history of early complications of hypertension
- Target organ damage (ocular, cardiac, renal, neurologic)
- Symptomatic hypertension

Stepped care approach

- Step 1: A small dose of single antihypertensive drug either diuretic or an adrenergic inhibitor
- Step 2: If the first drug ineffective a second drug is added to or substitute the initial drug starting with small dose then proceed to a full dose.
- Step 3: If blood pressure is still high a third drug; usually a vasodilator, is added

Drugs

* Diuretics e.g. hydrochlorothiazide, chlorothiazide, spironolactone
* Adrenergic inhibitor e.g. atenolol, prazocin
* Vasodilator e.g.
  - Hydralazine
  - ACE inhibitors e.g. captopril , enalapril
  - Calcium channel blockers e.g. nifedipine, amelodipine

II. Secondary hypertension

- Treat the cause whenever possible
- Drug therapy as in essential hypertension

B. Acute hypertension

- The blood pressure should be reduced by 10% in the 1st hour, and 15% more in the next 3-12 hr, but not to normal during the acute phase of treatment
- Hypertensive urgencies, usually accompanied by few serious symptoms such as severe headache or vomiting, can be treated either orally or intravenously.

Action plan

1. Ensure safe airway, breathing and circulation (ABC)
2. Slow reduction of blood pressure is mandatory.
3. Drugs useful in acute hypertension:

With severe symptoms

- Hydralazine 4 hourly IM,IV
- Labetalol IV infusion
- Esmolol IV infusion
- Nicardipine IV infusion
- Sodium nitroprusside IV infusion



With less severe symptoms PO

- Clonidine, hydralazine, or isradipine.
- Minoxidil is the most potent oral vasodilator; long-acting

4.Treat the cause (in 2^{ry} hypertension).

5.After adequate control of acute hypertension shift to oral antihypertensives



Hematology

Introduction to Pediatric Hematology

Intrauterine hematopoiesis passes into 3 stages:

Yolk sac hematopoiesis	Visceral hematopeiosis	Medullary hematopeiosis
In the 1st 8 weeks	8th week → 6th month	6th month → onwards
In the yolk sac	In the liver, spleen, lymph nodes & thymus	Settles in the bone marrow

Exposure to hematologic stress (e.g. chronic hemolysis) → ++ bone marrow then ++ extramedullary hematopoiesis in the spleen and the liver.

Bone marrow contain pleuripotent stem cells which give colony forming unit (CFU):

- Erythroid→Erythroblasts→Normoblasts→Reticulocytes→RBCs
- Myeloid→Myeloblast→Promyelocyte→Myelocyte→Metamyelocyte→ mature WBCs
- Megakaryocytic →Megakaryoblast→Megakaryocyte →Platelets

•Normal erythropoiesis requires

Regulatory hormones
- Erythropoietin
- Androgen
- Thyroxin
- ACTH
- Cortisol
- Growth hormone



• Normal stem cells in bone marrow

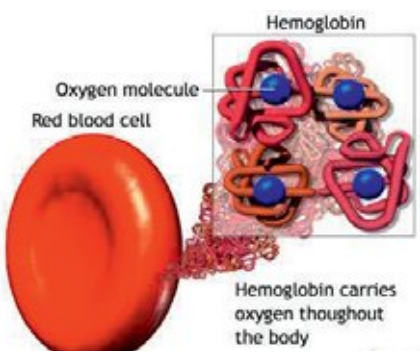
Essential nutritional elements
- Proteins
- Iron
- Folic acid
- Vitamin B12
- Copper

Hemoglobin (Hb) composition

Hb molecule is composed of Heme groups (ferrous iron containing) attached to 4 polypeptide chains which define the type of Hb.

Types of normal hemoglobins

1. **Emberyonic hemoglobin** :
   - Gower 1, 2 and Portland
   - Disappear by the 3rd month
2. **Foetal hemoglobin**
   - Hb F (α_2, γ_2)
   - Has high affinity to O_2
3. **Adult hemoglobin:** Hb A (α_2, β_2),Hb A_2 (α_2, Δ_2)



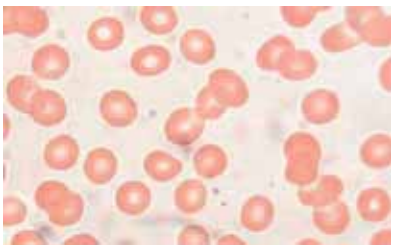
Switch mechanism in hemoglobin synthesis

	Emberyonic life	6th month	At Birth	6-12 month postnatal
Emberyoic	Dominant	--	--	--
Hb F	--	90 %	70 %	< 1 %
Hb A	--	10 %	30 %	97 %
Hb A_2	--	--	Trace	2 %

* At the 3rd – 6th month → normal switch from γ to β chain production occurs

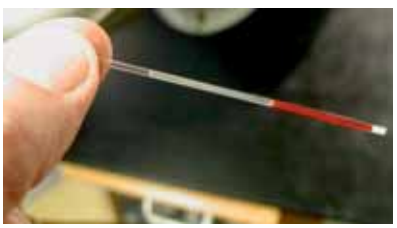
Blood indices

- **Hemoglobin content**:
  - In 1st 2 weeks→ 16-20 gm/dl (intrauterine hypoxia →↑erythropoietin).
  - In infancy → 10-14 gm/dl
  - Adult male → 13.5- 17.5 gam/dl
  - Adult female → 11.5 – 15.5 gm/dl



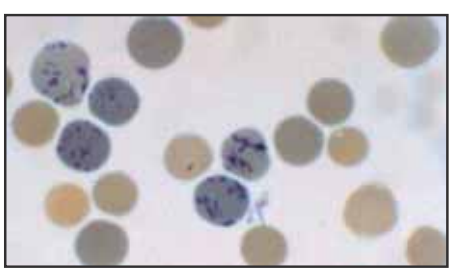
- **RBCs count:**
  - In newborn → 6 million / mm^3.
  - Adult male → 4.5- 6 million / mm^3
  - Adult female → 4 - 5.5 million / mm^3
- **Hematocrite value(Ht. value)** ; packed red cell volume

  * Percent of RBCs volume in 100 ml blood ≈ 40-50%
  * Increased in
    - Polycythemia
    - Hemoconcentration(dehydration)
  * Reduced in
    - Anemia
    - Hemodilution



- **Mean corpuscular volume (MCV)**
  - Normal: 72 – 79 femto liter.
  - If < 70 → RBCs are small (Microcytes).
  - If > 85 → RBCs are big (Macrocytes)
- **Mean corpuscular hemoglobin (MCH)**
  - Normal: 27- 34 pg.
  - If < 27 pg → RBCs are hypochromic.
- **Mean corpuscular hemoglobin concentration (MCHC)**
  - Concentration of Hb. in an erythrocyte
  - Normal: 33%
  - If < 30 % → RBCs are hypochromic
- **Reticulocytic count (RC)**

  * In neonatal period < 5 %
  * Later on 0.5 – 1.5 %
  * <u>Reticulocytosis occur in</u>:
    - Hemolytic anemia.
    - Hemorrhage
    - Response to hematinic e.g. (iron, folic acid)
    - Recovery of bone marrow from suppression.
  * Reticulocytopenia occur in bone marrow failure

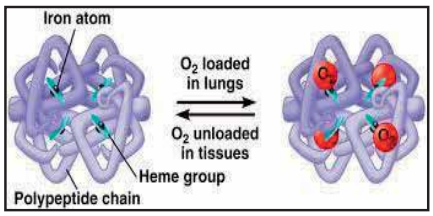


- **White blood cells count**
  - In neonatal period =15.000 – 20.000 / mm^3
  - Later on = 4.000 – 11.000 / mm^3
- Platelet count 150.000 – 450.000 / mm^3

Anemia

Definition

Reduction of hemoglobin and/or RBCs count
Below the average value for age and sex
Interfering with oxygen carrying capacity of blood



General features of anemia

Symptoms (all are non-specific)	**Signs** (all are non-specific)
– Fatigue, headaches and faintness are all very common – Palpitations – Breathlessness – Angina – Intermittent claudication	– Pallor – Tachycardia (palpitation) – Hemic murmurs (functional, systolic). – Heart Failure in severe anemia (with hemoglobin < 4 gm/dl)

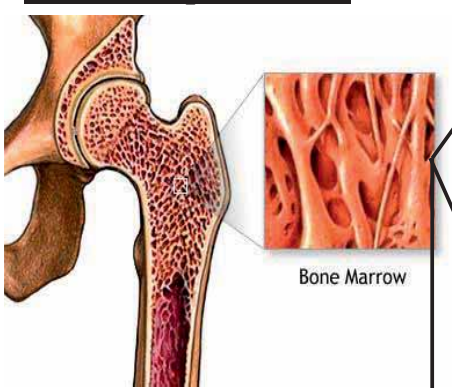
Classification of anemia

A. Morphologic Classification

	Microcytic anemia	**Normocytic anemia**	**Macrocytic anemia**
MCV	< 70 fl	72-79 fl	>85 fl

B. Etiologic Classification

I. Decreased production



a. Decreased erythroid cells in bone marrow (Bone marrow failure)
- Pure red cell anemia.
- Aplastic anemia
- Marrow infiltration e.g. Leukemia

b. Decreased red cells production despite normal RBCs precursors

* Anemia of chronic disease:
- Chronic inflammation
- Chronic infection
- Chronic renal failure

c. Specific factor deficiency (Dyshemopiotic anemia) e.g. deficiency of

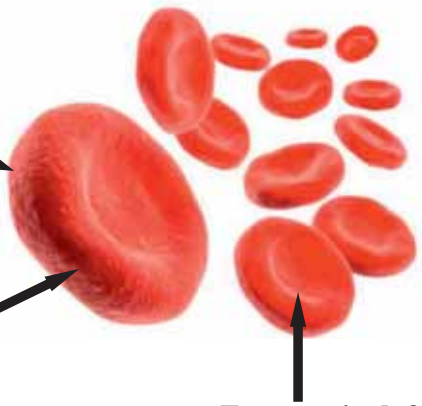
* Minerals: Iron, copper * Vitamins: B1, B6, Folic acid and B12. * Protein

II. Increased destruction (hemolytic Anemia)

A. Intra corpuscular causes

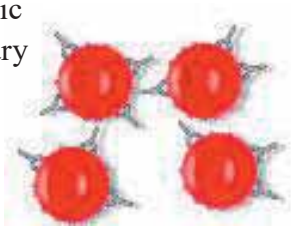
a. Membrane defects
- Hereditary spherocytosis
- Hereditary elliptocytosis.
- Paroxysmal nocturnal hemoglobinuria

b. Hemoglobinopathy
- α thalassemia
- β thalassemia.
- Sickle cell disease



c. Enzymatic defects
- G6PD deficiency
- Pyruvate kinase deficiency.

B. Extra corpuscular causes

Immunologic (Coombs test Positive)	Non immunologic (Coombs test Negative)
1. Iso immune hemolytic anemia (passively acquired antibodies) - Hemolytic disease of newborn - Incompatible blood transfusion 2. Autoimmune hemolytic anemia (actively formed antibodies): - Idiopathic - Secondary 	1.**M**icro**a**ngiopathic **H**emolytic **A**nemia(MAHA) - DIC - Hemolytic Uremic Syndrome (HUS). - Renal Vein Thrombosis (RVT). - Artificial & Calcified Cardiac Valves - Disseminated Cancer - Thrombotic Thrombocytopenic Purpura 2. Septicemia (E.G. Clostridia Welchii) 3. Malaria Due To: - Direct Effect - Drug Induced 4. Drugs, Heavy Metals, Snake Venom 5. March Hemoglobinuria 6. Paroxysmal Nocturnal Hemoglobinuria 7. Hypersplenism

III. Hemorrhagic anemia
- Acute hemorrhage
- Chronic hemorrhage.

Bone Marrow Failure

A. Trilineage failure

Criteria: - Failure of the 3 cell lines with pancytopenia
- No organomegaly nor lymphadenopathy.

Causes

a. Congenital

1. Fanconi anemia
2. Familial aplastic anemia
3. Dyskeratosis congenital; Ectodermal dysplasia with:
   - Skin pigmentation
   - Mucous membrane leukoplakia
   - Nail dystrophy
   - Others : short stature ,cataract, mental retardation

b. Acquired

1. Idiopathic
2. Secondary

B. One cell line failure

1. Red cells (hypoplastic anemia;pure red cell anemia)

a. Congenital

- Diamond Blackfan anemia
- Congenital dyserythropoietic anemia
- Pearson's Syndrome

b. Acquired

* Idiopathic: transient erythroblastopenia of childhood
* Secondary: to Drugs, Infections, Parvo B19 , Malnutrition

2. White cells

1. Schwashman Diamond syndrome: pancreatic insufficiency, metaphyseal dysplasia
2. Kostmann disease (severe congenital neutropenia)
3. Reticular dysgenesis

3. Platelets

1. Congenital amegakaryocytic thrombocytopenia
2. TAR syndrome (thrombocytopenia absent radii syndrome)

N.B : Dyserythropoiesis (ineffective erythropoiesis).

i. Primary (congenital dyserythropoietic anemia).

ii. Secondary dyserythropoiesis:

- Megaloblastic anemia (folic acid, or vit B_{12} deficiency).
- Thalassemia syndromes.
- Sideroblastic anemia.
- Paroxysmal nocturnal hemoglobinuria

Aplastic anemia

Definition

Marked decrease **or** absence of blood forming elements in the bone marrow (BM) with peripheral pancytopenia (decreased RBCs, WBCs and platelets).

Etiology

1. Congenital aplastic anemia: e.g. Fanconi anemia
2. Acquired aplastic anemia.
   - Idiopathic (50%)
   - Secondary
     - Irradiation
     - Insecticides
     - Infections e.g. Ebstein Barr virus, hepatitis viruses , HIV
     - Drugs → Dose dependent bone marrow depression e.g. chlorambucil
       → Idiosyncratic non dose dependent BM depression by:
       - Antibiotics → Chloramphenicol , sulphonamides
       - Anticonvulsants → Carbamazepine, phenytoin
       - Antithyroid → Carbimazole, thiouracil
       - Antimalarial → Chloroquin
       - Antirheumatic → Indomethacin, phenylbutazone
     - Diseases → SLE, Paroxysmal nocturnal hemoglobinuria (PNH)

Pathogenesis Of acquired aplastic anemia:

- Theory: Altered bone marrow stem cells antigen proteins e.g. by drugs or infection → Activated T lymphocytes → ↑ TNF and Interferon γ → accelerated stem cells apoptosis → pancytopenia.
- Aplastic anemia is unstable condition; may progress to myelodysplastic syndrome and leukemia

Clinical picture





- Thrombocytopenia →**p**urpura
- Anemia → **p**allor (+ weakness, fatigue etc……)
- Lecuopenia→ frequent, **p**ersistent infections
- No organomegaly (*no hepatosplenomegaly nor lymphadenopathy*).

Fanconi anemia

Inheritance: Mainly autosomal recessive

Clinical features

Skeletal anomalies (60%)
- Abnormal thumb (absent or hypoplastic)
- Absent radii

Short stature with short trunk

Skin pigmentation (café au lait spots) mainly over the trunk





Microcephaly / mental retardation (17 %)
Triangular facies

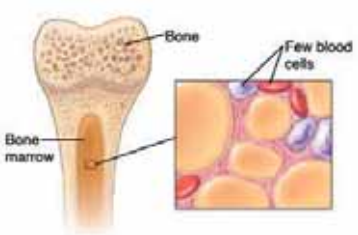
Congenital anomalies e.g.
- Eye
- Ears
- Cardiac
- Renal
- Genital

Hematological
- High risk of malignancy e.g. acute myeloid leukemia and solid cancers
- Pancytopenia :
  - Typically starts with thrombocytopenia or leukopenia
  - Usually between 4th – 12th year (earlier or later presentation do occur)
  - Bone marrow is aplastic

Investigations

1. CBC
- Pancytopenia (with macrocytic anemia)
- Reticulocytopenia

2. BM
- Hypo cellular BM
- Replaced by fibro fatty tissue.



3. In Fanconi anemia :
- Cytogenetic of blood lymphocytes shows increased chromosomal breakages and exchanges induced by mutagen (e.g. *Mitomycin C*)
- Mutation analysis



4. Investigations for a cause e.g. viral serology

Differential diagnosis

1- From other causes of pancytopenia

2- From other causes of purpura e.g. Idiopathic thrombocytopenic purpura.

Treatment

1. Supportive care; see leukemia
2. Specific treatment: Indicated in severe aplastic anemia

a. For Fanconi anemia

1. Androgen therapy (Oxymetholone) effective in 50 %
2. Hematopoietic stem cell transplantation (HSCT) from HLA matched donor

b. For acquired aplastic anemia

1. Hematopoeitic stem cell transplantation from HLA matched donor
2. If HSCT was unavailable use (alone or in combination):
   - Anti thymocyte globulin
   - Cyclosporine
   - Methyl prednisolone
   - Granulocyte colony stimulating factor (G-CSF)

Causes of pancytopenia:

* Bone marrow failure e.g.
  - Aplastic anemia
  - Advanced megaloblastic anemia
  - Myelophthisis: Bone marrow infiltration
* Hypersplenism (Pancytopenia with compensatory bone marrow hyperplasia)
* Fulminant sepsis
* Auto immune (Evans syndrome)

Congenital Pure red cell anemia

(Diamond Blackfan Anemia)

Definition

Familial disease due to decreased sensitivity of erythroid cells to erythropoietin → hypoplasia of RBCs precursors.

Clinical picture

- Pallor: usually evident by the 2nd – 6th months.
- Associated anomalies
  - Abnormal thumbs; (*triphalangeal thumb*)
  - Congenital heart diseases.
  - Short stature
- Hepatosplenomegaly, if present, is due to chronic transfusion therapy.
- High risk of acute myeloid leukemia.



Investigations

1. CBC

- Macrocytic anemia &reticulocytopenia
- Normal platelets and WBCs.
- Increased erythrocyte adenosine deaminase activity (ADA)& hemoglobin F

2. BM

- Decreased erythroid cells.
- Normal myloid and megakaryocytic cells.

Treatment

1- Steroids: Give remission in 80%.
2- Chronic transfusion therapy **with** iron chelation for steroid resistant cases
3- Bone marrow transplantation.

Differential Diagnosis

Form acquired pure red cell anemia

*** Idiopathic: Transient erythroblastopenia of childhood;**

- Due to transient immunologic suppression of RBCs synthesis
- No anomalies
- Normocytic normochromic red cells
- Usually need no treatment (recover within 1-2 months).

*** Secondary:** to Drugs, Infections, Parvo B19, Malnutrition

Iron Deficiency Anemia (IDA)

Incidence: The most common cause of anemia in pediatrics.

Iron metabolism

- Daily requirements: 8-15 mg/day
- Most of dietary iron present in ferric state.

⇩

- Iron changes to ferrous by combined action of HCL & vitamin C.
- Only **5-10%** of dietary iron is absorbed from the duodenum and proximal jejnum

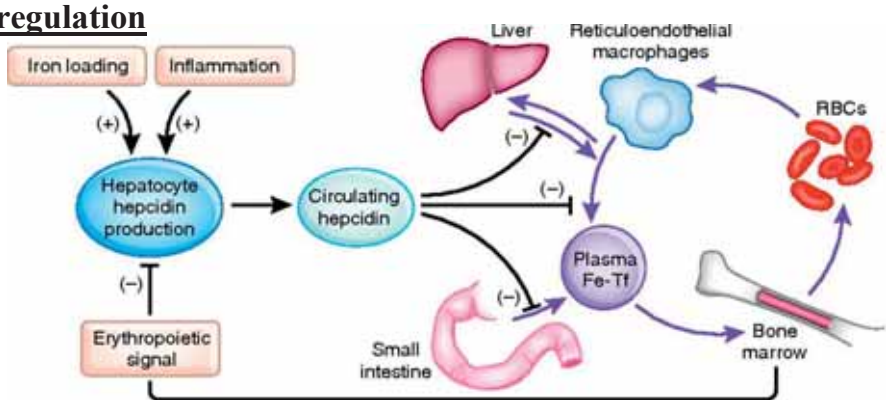
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- Factors enhancing absorption e.g. decreased hipcidin, vitamin C, ↑ erythropoiesis
- Factors impairing absorption e.g. increased hepcidin, hypochlorhydra, tannate

⇩

- Absorbed iron is bound to serum **transferrin** and stored as **ferritin** to be used in
  a. In bone marrow → RBCs
  b. In cell enzymes e.g. Catalase, peroxidase, mono amine oxidase (MAO).

Iron homeostasis regulation



- Hepcidin as the main regulator of systemic iron homeostasis
- Hepcidin synthesis is *induced* by iron loading and inflammation and *suppressed* by erythropoiesis.
- Increased hepcidin levels limit further enteral iron absorption and release of iron from the liver and the reticuloendothelial system to normalize plasma iron levels.
- With increased inflammation, elevated hepcidin levels cause the same sequence of events, leading to reticuloendothelial blockade and the anemia of inflammation.

Causes

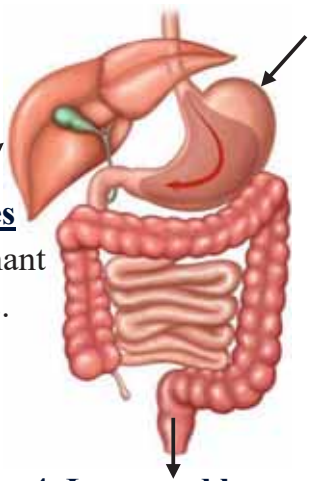
1. Decreased intake In infants 6-24 months due to:



a. Delayed weaning (Prolonged breast-feeding without supplementation) due to:
- Depletion of the baby's iron store by the 6th month
- Iron in breast milk is no more enough beyond 6th month



b. Unfortified animal (Cow) milk feeder Due to:
- Low iron content
- Low lactoferrin content
- Presence of heat labile protein which induces occult blood loss



2. Decreased iron absorption
- Excess tea, phytate & antiacids
- Achlorhydra e.g. atrophic gastritis
- Malabsorption syndrome

3. Decreased iron stores
- Iron deficient pregnant
- Perinatal blood loss.
- Preterm.



4. Increased loss

a. Occult blood loss due to
- Ankylostoma
- Cow milk protein allergy
- Drug induced gastritis.
- Peptic ulcer, polyps, GERD
- Meckle's diverticulum
- Esophageal varices

b. Overt blood loss e.g
Epistaxis, hematuria, hemodialysis



5. Increased requirements due to:

- Accelerated rate of growth in preterm, infants, adolescence.
- Congenital cyanotic heart diseases (due to polycythemia)



Clinical picture

1. Mild anemia is asymptomatic.

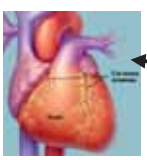
2. Manifestations of anemia (anorexia, pallor,…) vary with severity of iron deficiency.

3. Systemic manifestations:

- Decreased Alertness, learning & concentration span (due to ↓ iron containing cellular enzymes).



- Atrophic glossitis : tongue is pale ,glazed (smooth)
- Angular stomatitis



- Palpitation (Tachy cardia) on exertion
- Cardiomegaly in severe deficiency





- Nails ⇒ Brittle , longitudinal ridges, flattening and spooning (koilonychia)
- Palpable spleen in about 10 % of cases



- Pica (Geophagia) desire to ingest unusual substances e.g. dirts , mud ,chalk (increases the risk of concomitant lead poisoning)

4. Clinical signs of an underlying cause e.g. Ankylostoma anemia

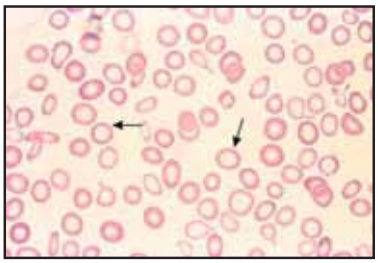
Iron deficiency anemia plus

- Parotid enlargement (endemic parotitis)
- Recurrent abdominal Pain
- Bouts of diarrhea and constipation
- B complex vitamins deficiency: pellagra , nutritional edema
- Eosinophelia in the CBC and ova in stool analysis

Investigations

A. For diagnosis

1. CBC



- Anemia: ↓ Hb%, ↓ RBCs count
- Hypochromia: - MCH < 27 Pg
  - MCHC < 30%
- Microcytosis MCV < 70 fl

* Blood film shows:
  - Abnormally large central pallor (hypochromic cells)
  - Target cells
  - *Wide red cell distribution width (RDW)* and anisocytosis
* Normal white blood cells
* Thrombocytosis is often present

2. Iron indices

Index	Iron deficiency	normal range
○ Serum Iron. ○ Transferrin saturation. ○ Total iron binding capacity (TIBC) ○ Soluble serum transferrin receptors	**< 30** µg / dl **< 16 %** Increased Increased	**60 - 140** µg / dl **30**% **250-400**µg / dl Variable cutoff
▪ Serum ferritin (index of iron stores) ▪ Reticulocyte hemoglobin content * ▪ Marrow iron stores ▪ Hepcidin**	**< 15** ng / ml **< 27.5** pg Absent Usually ≤10 ng/mL	**15-300** ng/ml

* A sensitive indicator that falls within days of onset of iron-deficient erythropoiesis and is unaffected by inflammation

** Extremely elevated in anemia of inflammation and suppressed in iron deficiency anemia

B. For the cause

- Stool analysis for parasites, ova and occult blood tests
- GIT barium study, endoscopy,and tests for achlorhydra
- Workup for malabsorption
- Workup for hemorrhagic diseases

Treatment

1. Treat the cause

2. Diet

- Excessive intake of milk, particularly bovine milk, should be limited
- infants who are breast and cow milk feeders require prophylactic oral iron given at 4th – 6th months (2mg/kg/d)
- in weaned; encourage intake of vitamin C, meat, fish

3. Iron preparation

Oral Iron

- Dose:3- 6 mg/kg/d. elemental iron
- In-between meals
- Continued for 8 wk after blood values normalize to re-establish iron stores
- Side effects: GIT upset, constipation and dark stool.
- Preparations:
  - Ferrous sulphate (The best; contain 20 % elemental iron)
  - Ferrous gluconate
  - Ferrous lactate

Parentral Iron

- Preparation
  - Iron Dextran (Imferon) or Iron sucrose (Venofer) IV
  - Iron sorbitol (Jectofer) deep IM
- Indications
  - Intolerance to oral iron
  - GIT disorders aggravated by oral iron e.g. IBD
  - Malabsorption
  - Rapid loss of iron
- Side effects
  - With IM→ staining, abscess
  - With IV → anaphylaxis (start with an observed test dose)

4. Packed red cell transfusion: is considered in

- Severe anemia (Hb < 7- 8 gm/dl).
- Anemic heart failure.
- Infection interfering with iron therapy.

Response to iron therapy

Time after iron administration	Response
By the 1st day	Replaced cellular enzymes → decreased irritability and improved appetite.
By the 2nd day	Erythroid hyperplasia in bone marrow.
By the 3rd day	Reticulocytosis peaking at 5-7days
By the 1st month	Increase hemoglobin at rate of 0.25 – 0.5 gm/dl/day
By the 3rd- 6th months	Repletion of stores.

Mentzer index (MI) *= Packed cell volume / RBC count in millions*

** Help differentiate microcytic anemia of iron deficiency from β thalassemia trait*

** MI In iron deficiency anemia >13 * MI In β thalassemia trait <13*

Differential Diagnosis of microcytic anemia that fails to respond to oral iron

1. Poor compliance / incorrect dose or medication
2. Malabsorption of administered iron
3. Ongoing blood loss
4. Concurrent infection or inflammatory disorder inhibiting the response to iron
5. Concurrent vitamin B_{12} or folate deficiency
6. **Diagnosis other than iron deficiency**
   a) **Hemoglobinopathies**: Thalassemia , Sickle thalassemia , Hemoglobin C and hemoglobin SC disease
   b) **Anemia of chronic disease**
      – In any disease with chronic inflammation e.g. Chronic infections
      – Anemia is usually normocytic ; occasionally microcytic
      – Inflammatory mediators and raised Hepcidin are the main etiologies
   c) **Lead poisoning**
      – History of exposure or iron deficiency with Pica
      – Blue staining of gums, abdominal pain
      – Lab: Basophilic stippling of RBCs ,high serum lead level, free erythrocyte protoporphyrin
   d) **Siderblastic anemia** (X-linked disorder)
      – Due to abnormal erythrocytic 5-aminolevulinic acid synthetase (ALAS), the rate-limiting enzyme reaction in heme synthesis. An important cofactor for ALAS is pyridoxal phosphate.
      – Treatment: Stem cell transplantation for transfusion dependent cases

	IDA	Anemia Of Chronic Disease	Thalassemia traits	Sideroblastic Anemia
Serum Ferritin	– Decreased	– Increased	– Normal	– Increased
Serum iron	– Decreased	– Decreased	– Normal	– Increased
TIBC	– Increased	– Decreased	– Normal	– Normal
Transferrin saturation	– Decreased	– Decreased	– Normal	– Increased
Bone Marrow	– Absent iron	– Increased iron	– Normal iron	– Increased iron – Ringed sideroblasts
Specific	– ↑↑ Soluble transferrin receptors – ↓↓ Hepcidin	– Normal soluble transferrin receptors – ↑↑ Hepcidin – High CRP	– Electrophoresis	– HSM, icterus – May respond to B6

(Nelson textbook of pediatric)

Megaloblastic anemia

Definition: Anemia with megaloblasts in BM and macrocytes in peripheral blood.

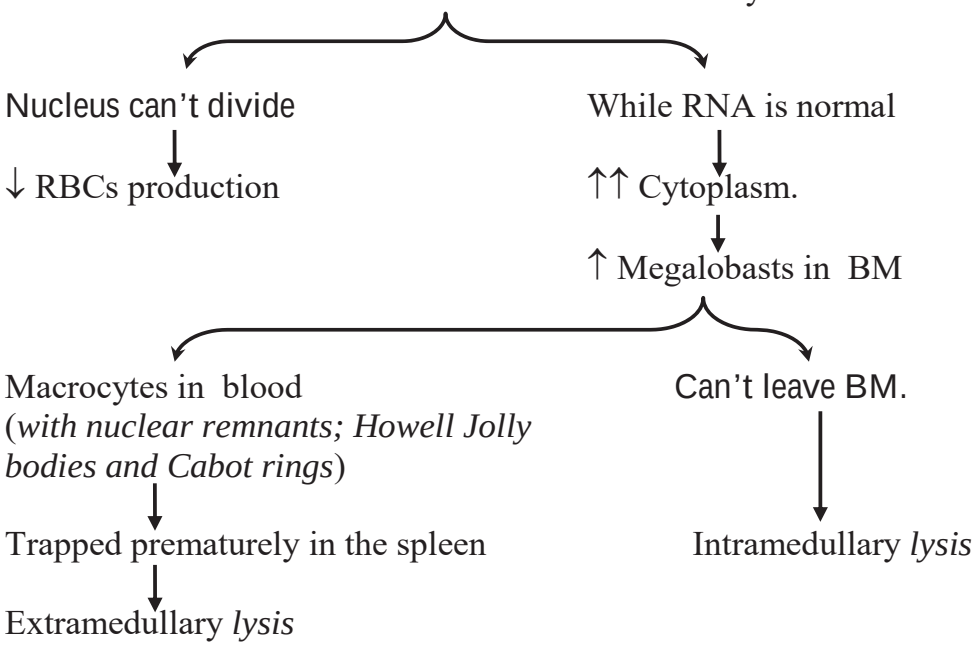
Causes

i- Vitamin B_{12} (cobalamin) deficiency

ii- Folic Acid deficiency.

Pathogenesis

Folic acid & B_{12} are essential for DNA synthesis in stem cells of RBCs, platelets and WBCs. **So** in folic acid or B12 deficiency



Metabolism

Vitamin B_{12}	**Folic acid**
Sources Animal origin only e.g. milk, meat.	- Animal & plant (green leaves, fruits)
Requirements 5 – 20 µg /day	20 – 50 µg /day
Absorption Gastric parietal cells release intrinsic factor (IF) which binds to B_{12} → B_{12}/IF complex → absorbed from the terminal ileum	Absorbed from the proximal intestine (duodenum and jejunum) Requires vitamin C for absorption
Stores enough for 2-4 years	Stores enough for 2-4 months

Causes of B_{12} and folic acid deficiency

	Vitamin B_{12}	**Folic acid**
Decreased intake	* Infants breast fed to vegetarian mums	* Infants fed on: Goat's milk
Decreased absorption	* Malabsorption syndrome * Intrinsic factor defect - Pernicious anemia - Gastrectomy * B_{12} /IF consumption by Diphyllobothrium latum **or** bacterial overgrowth. * Ileal disease **Or** resection	* Malabsorption syndrome * Vitamin C deficiency
Impaired metabolism	Defective transport: Transcobalamin II deficiency	* Cytotoxic drugs → methotrexate * Anticonvulsant → phenytoin → valproate
Others	.	* **Increase requirements: e.g.** - Prematures (↓stores) - Pregnancy - Chronic Hemolytic anemia * Reduced stores: liver cirrhosis * Increased loss: hemodialysis

Clinical picture

A. Hematologic

* Anemia (Anorexia, pallor, *tiredness* , ……..) with slight jaundice.
* Advanced megaloblastic anemia → thrombocytopenic purpura and leucopenia
* Mild hepatosplenomegaly due to intramedullary hemolysis

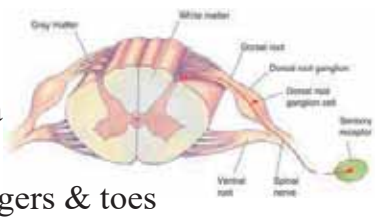


B. GIT manifestations esp. in folate deficiency:

* Atrophic glossitis → Beefy red glazed tongue in *25 %*
* Atrophic gastritis → Dyspepsia, vomiting, risk of cancer stomach
* Atrophy of intestinal mucosa → Abdominal pain and chronic diarrhea.

C. Neurologic manifestations: Sub acute Combined Degeneration (SCD)

Only with severe vitamin B_{12} deficiency (? irreversible)



* Degeneration of
  - Posterior column → deep sensory loss, sensory ataxia
  - Pyramidal tract → Progressive *weakness*, *Paraplegia*
  - Peripheral Nerve → Symmetrical paraesthesiae in fingers & toes

Diagnosis

1. Is it megaloblastic anemia?

CBC

Macrocytic anemia

- Low Hb% & Ht value
- MCV > 100 fl
- MCHC = normal

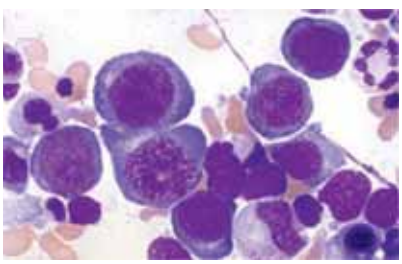
Hyper segmented (hyper mature) neutrophils (contain 4-5 lobes).



Thrombocytopenia, leucopenia and reticulocytopenia in advanced cases.

BM

- Erythroid hyperplasia
- Megaloblastic changes



2. What is the cause?

A. Vitamin B12 deficiency

i. Is there Vitamin B_{12} deficiency?

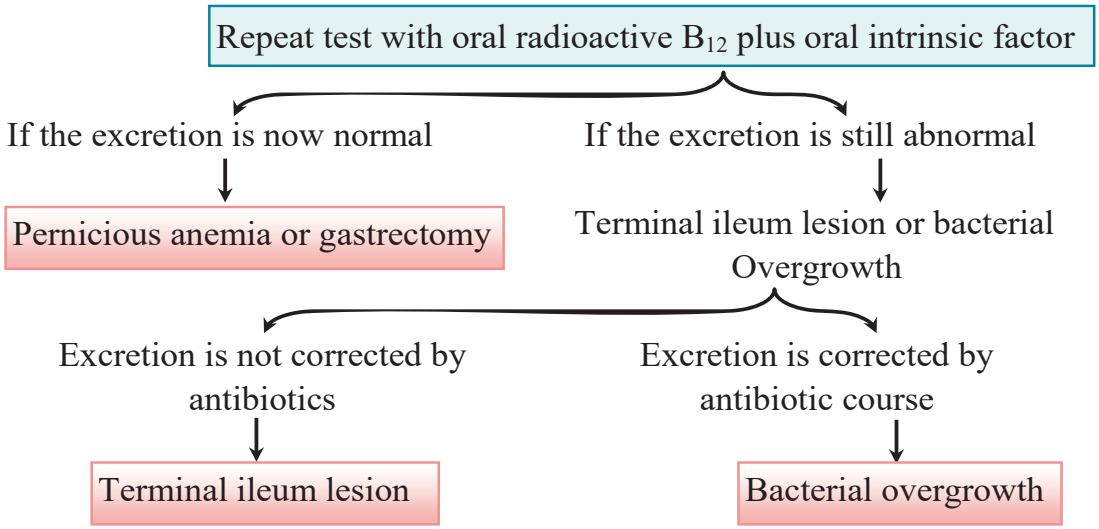
- Low serum vitamin B_{12} (Diagnostic)
- Therapeutic test: **1** µg B_{12} → reticulocytosis at 6th day

ii. What is the cause of Vitamin B_{12} deficiency?

a. Schilling test: done if the etiology was unclear

Radioactive B_{12} is given orally followed by an I.M. injection of non-radioactive B12 to saturate B12 binding proteins

Normally	If this is abnormal
>**10** % of oral B_{12} excreted in urine	The test is repeated with the addition of oral intrinsic factor capsules



b. Gastric function tests: Histamine or pentagatrin test confirm achlorhydra

c. Serological tests: Anti-parietal and anti-IF antibodies

B. Folic acid deficiency

* Low serum folate; RBC level is better measure of tissue folate (Diagnostic)
* Therapeutic test: 0.2 mg Folate → look for reticulocytosis at 6th day

Treatment

A. B12 deficiency

a. Pernicious anemia
- Hydroxocobalamin IM 1000 µg to a total of 6 mg over the course of 3 weeks Then 1000 µg every 3 months for the rest of the patient's life.
- Recent alternatives: high oral or sublingual dose 2 mg per day

b. Gastrectomy or ileal disease:
Monitor serum B_{12}→whenever low → give prophylactic vitamin B_{12}

c. Treat the underlying disease

B. Folic acid deficiency

a. Treat the underlying disease

b. Exclude subclinical B_{12} deficiency which ca be aggravated by folic acid therapy

c. Folic acid 5 mg/day continued for 4 months to replace body stores.

d. Prophylaxis
- Chronic hemolytic anemia 5 mg weekly
- Premature<1500gm require 1 mg daily for 6 weeks

N.B:

Folic acid may produce a hematological response in vitamin B12 deficiency but may aggravate the neuropathy. So Large doses of folic acid alone should not be used to treat megaloblastic anemia unless the serum vitamin B12 level is known to be normal

Hemolytic Anemia

Definition

- Anemia resulting from increased RBCs destruction exceeding bone marrow capacity for compensation (reduced RBCs survival).
- Normal RBCs life span 120 days.

Acute Hemolytic Anemia
(Intra vascular hemolytic anemia)

Causes

A. Intra corpuscular causes: - Enzymatic defects e.g. G6PD deficiency

B. Extra corpuscular causes

Immunologic
(*Coombs positive*)

- Isoimmune
  - Hemolytic disease of newborn.
  - Mismatched blood Transfusion.
- Autoimmune

Non immunologic
(*Coombs negative*)

1. **M**icro**a**ngiopathic **h**emolytic **A**nemia (MAHA) Can be acute or chronic hemolytic anemia)
   - DIC
   - Hemolytic uremic syndrome (HUS).
   - Renal vein thrombosis (RVT).
   - Artificial & calcified cardiac valves
   - Disseminated cancer
   - Thrombotic thrombocytopenic purpura
2. Septicaemia (e.g. Clostridia Welchii)
3. Malaria due to: - Direct effect
   - Drug induced
4. Drugs, heavy metals, snake venom
5. March hemoglobinuria
6. Paroxysmal nocturnal hemoglobinuria

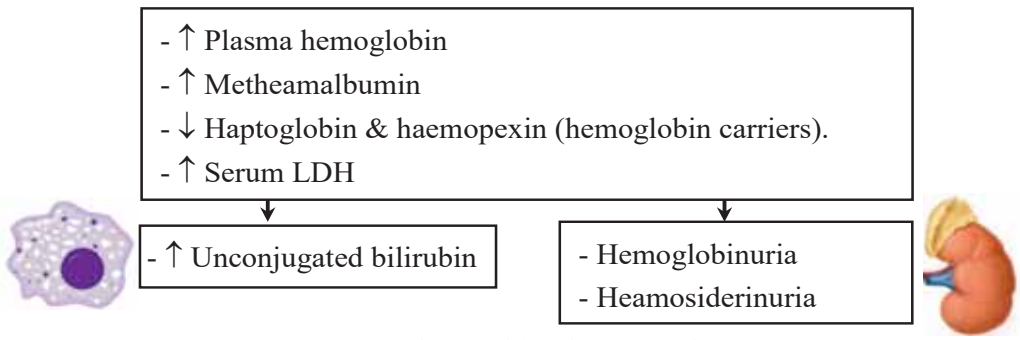
General clinical manifestations

- Acute pallor.
- HyperBilirubinemia: - Indirect; lemon yellow color
  - Dark colored urine (*hemoglobinuria*)
- Pyrexia and rigors.
- Loin Pain.
- In Profound anemia: rapid decompensation with blurred vision, lassitude and anemic heart failure even death



General workup

1. **Anemia**: Usually normocytic normochromic , low Hb% and Ht. value.
2. **Decreased RBCs Survival** indicated by:

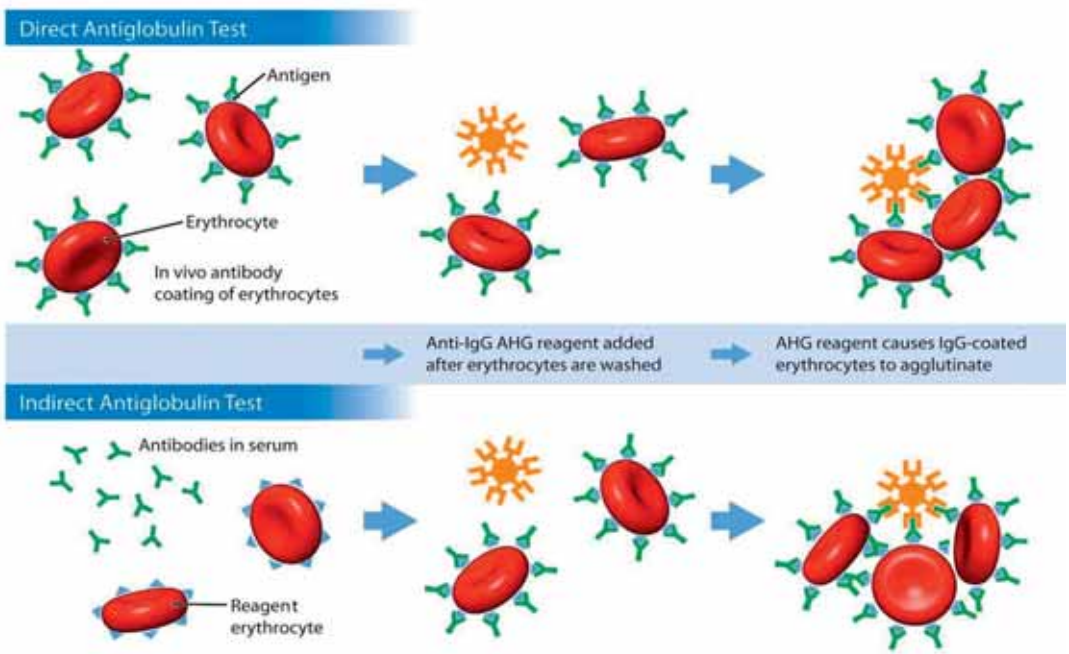


3. **Increased erythropioesis** indicated in the CBC by:
   - Reticulocytosis: Reticulocytic count > **8%**
   - ↑ Normoblasts (immature nucleated RBCs).

Specific investigations

1. Coombs test (Anti Globulin Test): Diagnose Immunologic hemolytic anemia.

- Direct Coombs test detect antibody coated RBCs
- Indirect Coombs test detect free antibodies in serum



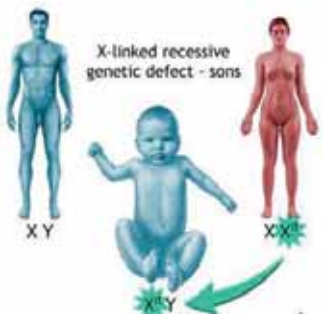
2. Blood film for:

- Heinz bodies in G6PD deficiency.
- Microangiopathic hemolytic anemia: - Fragmented RBCs (schistocytes).
  - Thrombocytopenia.
- Malaria.

Glucose - 6 - Phosphate Dehydrogenase Deficiency

(G6PD Deficiency)

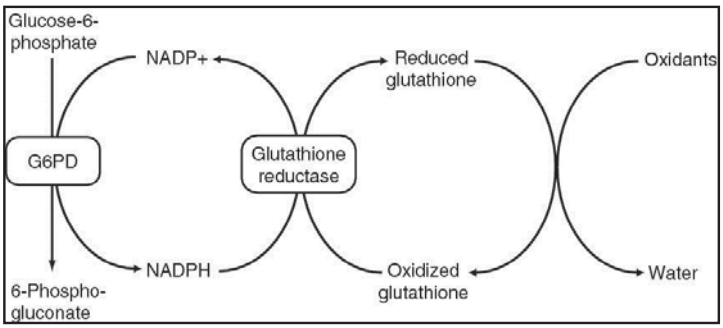
Etiology



* Hemolytic anemia due to *age labile* Glucose - 6 - Phosphate Dehydrogenase enzyme
* Sex linked recessive disorder which seen mainly in males
* Occasionally seen in females if in homozygous state or heterozygous state with random inactivation of the other normal X chromosome (Lyon hypothesis).

Pathogenesis

* G6PD enzyme is key enzyme of Hexose Mono Phosphate (HMP) shunt which produce reduced glutathione.
* Reduced glutathione protects the red cells against oxidizing agents.



* In G6PD deficiency → ↓ reduced glutathione → impaired elimination of oxidants → oxidation of Hb → methemoglobin → precipitate inside RBCs → acute hemolysis.

Oxidizing agents include



- Food → Fava beans (favism)
  contain *vicine* and *convicine* oxidants
- Infections → Viral or bacterial
- Drugs → Anti pyretics e.g. - Acetyle salicylic acid
  - Phenacetin
  → Anti-microbial e.g.- Sulpha ,Chloramphenical ,Nitrofurantion
  → Anti-malarial e.g. Primaquine
  → Anti tuberculous e.g. Para amino salsylic acid , isoniazid

Genetic variants

- Types A^+ & B^+ → Normal variants.
- Type A^- → American type (enzyme activity = 5-15%).
- Mediterranean type → Severe deficiency (enzyme activity < 5%).
- Canton type.

Clinical picture

A. Acute hemolytic anemia

- Features of acute hemolysis without organomegaly (*see before*).
- Occur 2- 3 days after exposure to the oxidant trigger.
- Degree of hemolysis varies with the triggering agent and the severity of G6PD enzyme deficiency
- Episodes usually brief as newly produced young RBCs have a higher enzyme activity that can withstand oxidant stress
- May occur in the neonatal period → neonatal anemia & jaundice.

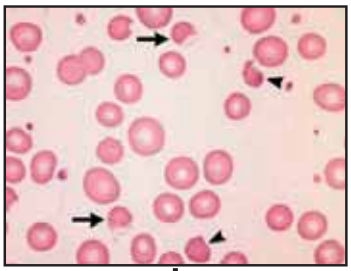
B. Chronic non spherocytic hemolytic anemia

- Extremely rare.
- Presents with pallor, tinge of jaundice and mild splenomegaly.

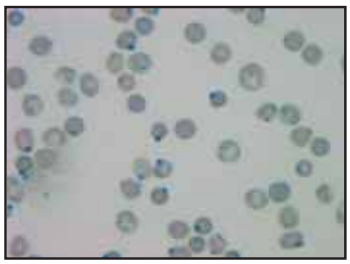
Investigations

1. For **anemia** → Low Hb% and Ht value.
2. For **acute hemolysis** → Evidence of ↓ RBCs survival and ↑ Erythropiosis.
3. For **the cause**:

i. Blood film



- Fragmented RBCs
- Reticulocytosis



Heinz bodies

- Denatured hemoglobin
- Appear as purple blue intracellular inclusion bodies visible with supravital stains

ii. G6PD Enzyme assay

- Done few weeks (2-4 weeks) after the acute attack
- During the acute attack, the bone marrow produces young RBCs with higher enzymatic activity which may give false normal results.

Treatment



A. During the attack

- Resuscitation: Ensure ABC
- Packed RBCs transfusion (5-10 ml/kg) for severe attack (Hb < 7 gm/dl or < 9 gm/dl with persistent hemoglobinuria).Can be repeated
- Control triggers e.g. infection, stop oxidant agent
- In a neonate: treatment of associated neonatal jaundice

B. After the attack

- Avoid oxidant food & drugs.
- Anti oxidant → vitamin E.
- Offer screening of other sibling boys
- Offer a *card* with diagnosis that include forbidden and alternative drugs (e.g. Anti pyretics in fever → paracetamol **or** ibubrufen) and avoided foods

Chronic Hemolytic Anemia

Causes

A. Intra corpuscular causes

1. Membrane defects	- Hereditary spherocytosis
	- Hereditary elliptocytosis (ovalocytosis)
2. Hemoglobinopathy	- α thalassemia
	- β thalassemia.
	- Sickle cell disease
3. Enzymatic defects	- Pyruvate kinase deficiency

B. Extra corpuscular causes

1. Immunologic (Coombs +ve): Autoimmune hemolytic anemia (Warm type)
2. Non immunologic (Coombs –ve): Hypersplenism

General clinical features

a. Initially

1. Features of progressive **A**nemia (pallor, fatigue,...)
2. Tinge of jaundice :
   - Indirect hyper**B**ilirubinemia ; lemon yellow color
   - Normal colored urine (acholuric jaundice)



b. In advanced and neglected cases

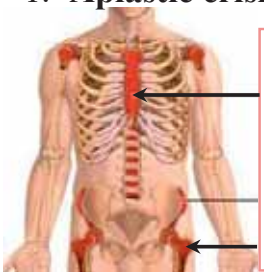
1. **S**keletal changes due to bone marrow expansion:
   * Head (Mongloid features):
     - Macrocephaly.
     - Depressed nasal bridge.
     - Prominent maxillae.
     - Prognathism.
   * Generalized osteoporosis.



2. **S**plenohepatomegaly due to extramedullary hematopioesis & hemosiderosis
3. Gall bladder **S**tones (Calcium bilirubinate) in chronic hemolysis for > 4 years.

c. Hematologic crises

1. **Aplastic crisis** (*Erythroblastopenic crisis*)



- Transient bone marrow hypoplasia
- Due to parvo-B19 infection (infect erythroid cells).
- Clinically:
  - Increased pallor <u>without</u> deepening of jaundice
  - Reticulocytopenia

2. **Megaloblastic crisis** due to folate deficiency→ aggravation of anemia
3. **Hemolytic cri**sis
   - Increased rate of hemolysis precipitated by infection
   - Increasing pallor, jaundice , reticulocytic count
   - Hemoglobinuria
4. **Hyperhemolytic crisis** due to associated G6PD deficiency

General Investigations

1. Anemia

* Low Hb% and Ht value
* Usually normocytic normochromic but may be:
  - Macrocytic due to associated folate deficiency or marked reticulocytosis
  - Microcytic in thalassemia and chronic hemoglobinuria

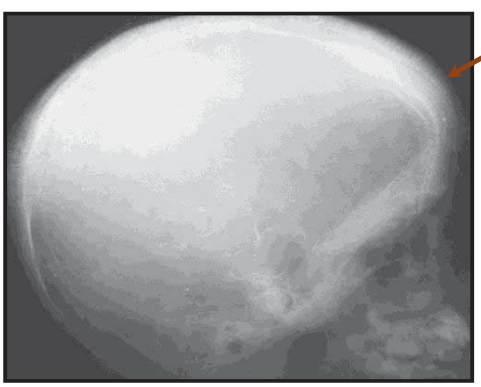
2. Decreased RBCs survival

* ↑Unconjugated bilirubin (usually < 5 mg/dl)
* Iron buildup: ↑ Serum iron and ferritin

3. Increased erythropoiesis

* Modest reticulocytosis ;peaks in hemolytic and hyperhemolytic crises
* **Skull X-ray**: shows

(*Hair on end appearance*).



- Marrow space expansion
- Wide diploic space
- Macrocephally

* **Bone marrow aspirate:** shows



- Erythroid hyperplasia
- May be megaloblastic in associated folate deficiency
- May be aplastic in aplastic crisis or PNH

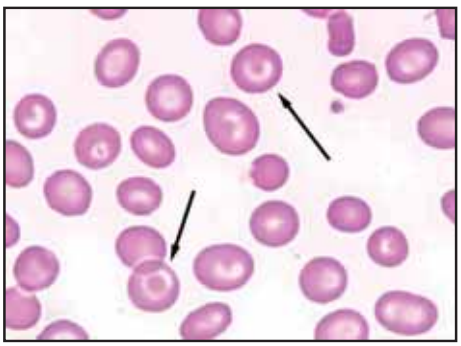
4. ↓ Chromium (^{51}Cr) labelled RBCs survival:

Direct evidence of short RBCs survival (research tool only)

Specific investigation

1. Coombs test: Diagnose Immunologic hemolytic anemia.

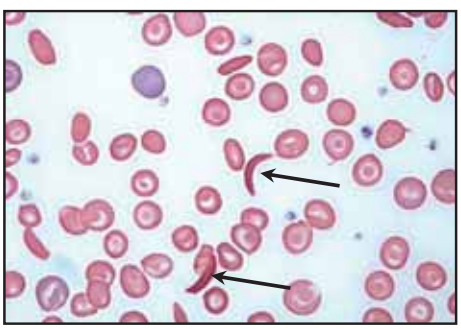
2. Blood film for abnormal RBCs morphology e.g.



a. Target cells

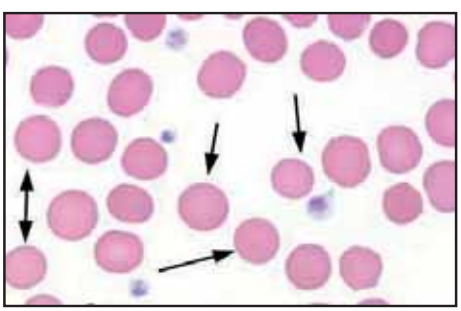
RBCs with central staining, a ring of pallor, and an outer rim of staining seen in

- Thalassemia
- Sickle cell disease
- Hyposplenism
- Obstructive liver disease
- Iron-deficiency anemia



b. Sickle cells

Seen in sickle cell anemia



c. Microspherocytes

Small spherical cells with loss of central pallor , Seen in

- Hereditary sherocytosis
- Immune hemolytic anemia
- Hypersplenism
- Burn
- Sickle Cell disease

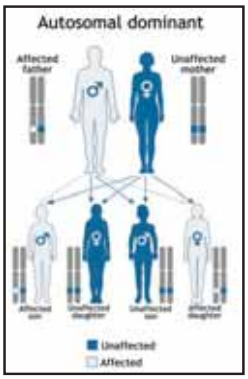
3. Osmotic fragility test/auto hemolysis test for Hereditary spherocytosis

4. Flow cytometry for PNH (Ham test is no longer used)

5. Hb electrophoresis for hemoglobinopathies

Hereditary Spherocytosis

(Familial acholuric jaundice)



Pathogenesis

- The most common inherited hemolytic anemia in northern Europeans
- Autosomal dominant disorder
- Due to deficiency of red cell cytoskeleton proteins(Ankyrin or Spectrin or protein 4.2)

↑ Cell wall permeability to Na^+ → ↑ Intracellular Na^+ influx → ↑ Water content inside RBCs

Release of micro vesicles

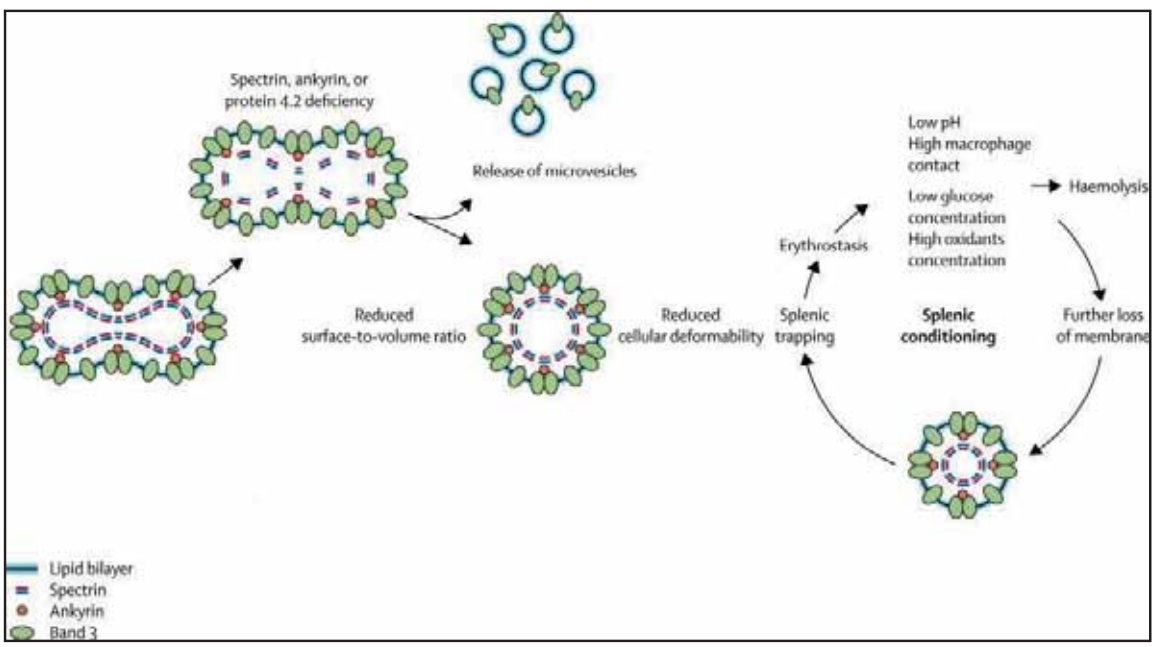
- Cells become spherical with more rigid and less deformable
- Premature RBCs aging (over work of Na/K pump)

↓

Exposure to stressful splenic circulation (low pH, low glucose, high oxidants, high macrophage contact)

↓

RBCs are trapped and destroyed in the spleen.



Clinical picture

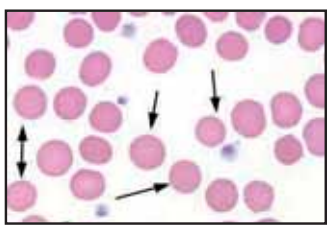
* Some patients may go through life with no symptoms
* Positive family history is present in **75**% of symptomatic cases

1- Features of anemia
2- Features of chronic hemolysis
} starting early in life; **50%** present by neonatal anemia and jaundice.

3- Gall stones are seen in **50%** of unsplenctomized cases by 4-5 years

Investigations

1. For **anemia** → ↓Hb% and ↓Ht value (usually normocytic anemia)
2. For **chronic hemolysis** → ↓ RBCs survival & ↑ erythropoiesis.
3. For **the cause**:
   a. Blood film → RBCs are small, rounded without central pallor i.e. spherocytes.



   b. Negative Coombs' test rules out autoimmune hemolytic anemia as a cause of spherocytes in blood film.
   c. Incubated osmotic fragility test:
      - Test is non-specific ;miss up to 20% of cases
      - Normally when red cells are placed in solutions of increasing hypotonicity, it takes in water, swell, and eventually lyses.
      - Spherocytes (already swollen cells) lyse more readily than normal biconcave cells
      - In equivocal results ; perform *incubated osmotic fragility test* (incubate for 24 hours at 37 ^{0}C)
      - Hemolysis is partially corrected by addition of glucose
   d. Flow cytometric EMA (Eosin-5- Maleimide) binding test **and** the cryohemolysis test are much more sensitive and specific

The diagnosis of HS can be established from:

1. A positive family history and
2. The presence of typical clinical and laboratory features of the disease: splenomegaly, spherocytes on the blood smear, reticulocytosis, and an elevated mean corpuscular hemoglobin concentration.

- If these are present, no additional testing is necessary to confirm the diagnosis.
- If the diagnosis is less certain, the recommended tests that have a high predictive value for HS are the flow cytometric EMA (eosin-5- maleimide) binding test and the cryohemolysis test **(Nelson textbook)**

Medical treatment

1- Supportive → folic acid 1mg/day (till splenectomy is done)

2- Slight anemia (Hb > 10 gm/dl & reticulocytic count < 10%) → follow up

3- Severe anemia requires packed red cells transfusions.

Elective splenectomy:

Indications → Moderate to Severe anemia
→ Frequent crises
→ Poor growth
→ Cardiomegaly

Value : Clinical cure → prevent hemolysis, crises and gall bladder stones

Timing : It is best to postpone splenectomy until after 6 years to avoid overwhelming fatal infections

Risks : Increased risk of overwhelming infections, particularly encapsulated organisms

Precautions A. Vaccinate 2–3 weeks before splenectomy for:
- Pneumococcal polysaccharide vaccine; repeated every 5 years
- Meningococcal group C vaccine
- Influenza vaccine; repeated annually
- Haemophilus influenzae type B (Hib) vaccine

B. Long-term penicillin V 250mg 12-hourly (or erythromycin)

C. Aggressive treatment for any febrile illness

N.B: Often, concomitant cholecystectomy should be performed if there are gallstones detected in pre-operative abdominal ultrasound.

Paroxysmal Nocturnal Hemoglobinuria

Etiology

* Acquired hemolytic anemia where RBCs are susceptible for complement damage
* Hematopoietic stems cells membranes lack decay accelerating factor (CD55) and CD59 which are involved in complement degradation
* Absence of CD55 and CD59 → uncontrolled hemolytic action of complement
* Anemia is chronic with acute exacerbations

Clinical picture

* Features of intravascular hemolytic anemia with hemoglobinuria especially after sleep by night
* Attacks of abdominal and back pain due to micro thrombi
* Complications
  - Venous thrombosis
  - Iron deficiency
  - Aplastic anemia

Investigations

A. General investigations for intravascular hemolysis

B. Specific: Flow cytometric analysis of red cells with anti-CD55 and anti-CD59

Treatment

1. Blood transfusions (Leucocyte-depleted blood to prevent transfusion reactions)
2. Eculizumab
   - A recombinant humanized monoclonal antibody that prevents the cleavage of C5 (and therefore the formation of the membrane attack complex).
   - It reduces intravascular hemolysis, hemoglobinuria, the need for transfusion
   - Given IV every 2weeks
   - Very expensive
3. Long-term anti-coagulant prophylaxis for patients with deep venous thrombosis
4. Folic acid 3mg daily
5. Bone marrow transplantation

Thalassemia

Definition: Autosomal recessive disorders due to defective globin chain production

A. α thalassemia syndromes

✧ Impaired α chain production

✧ Due to deletion of one or more of the 4 α globin genes on chromosome 16

1. One gene missing → Silent carrier (Asymptomatic)

2. Two gene missing → α thalassemia trait

- Familial microcytic anemia ; commonly mistaken as iron deficiency
- Normal iron indices
- Normal Electrophoresis for age.
- Diagnosed by DNA analysis.

3. Three genes missing → Hemoglobin H disease

- Mild to moderate microcytic hemolytic anemia at birth
- Evidence of chronic hemolysis
- Electrophoresis shows: Hb H (4β chains).

4. Four genes missing → Fetal hydropes

- Severe intra uterine anemia and anemic heart failure
- Resulting in death in-utero or short after birth
- Electrophoresis shows: Dominant hemoglobin Bart (4 γ globin chains) with complete absence of normal fetal and adult hemoglobin

B. β-thalassemia syndromes

− Impaired β chains production

− Due to mutation of one or more of the 2 β globin genes on chromosome 11

A. One gene mutation → **β thalassemia trait** (Heterozygous β-thalassemia)

- − Microcytic anemia with no evidence of overt hemolysis
- − Differentiated from iron deficiency anemia by
  - o Normal iron indices and Mentzer index <13
  - o Characteristic hemoglobin electrophoresis:
    - ➢ Hb A_2 up to 3-7% in over 90% of cases.(diagnostic)
    - ➢ HbF up to 1-3% in only 50% of cases.

B. Two genes mutation → β thalassemia major

C. Thalassemia intermedia

- Due to a combination of homozygous mild β and α thalassemia
- Moderate anemia (Hb 7–10 g/dL) doesn't require regular transfusions

β Thalassemia Major

(Cooley's anemia)

The commonest chronic hemolytic anemia in Egypt & Mediterranean areas.

Pathophysiology

A. Impaired β chain production

↓ Production of HbA (α_2 β_2)

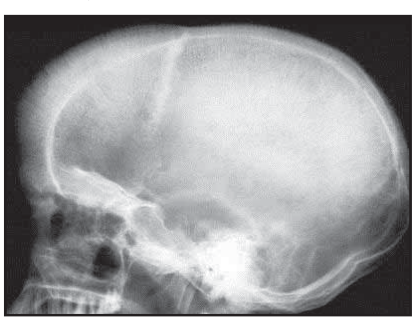
Deposition of excess unmatched α chain inside the RBCs → hemolysis → Anemia

Compensatory production of other Hb containing non Beta chains especially Hb F ($\alpha_2\gamma_2$) → ↑ HbF (↑O_2 affinity)

Tissue hypoxia → Compensatory increased RBCs production

Medullary hematopoiesis → Bone marrow expansion (Skeletal changes)





Extra medullary hematopoiesis → Splenomegaly (± hepatomegaly) ± hypersplenism



B. Iron overload due to

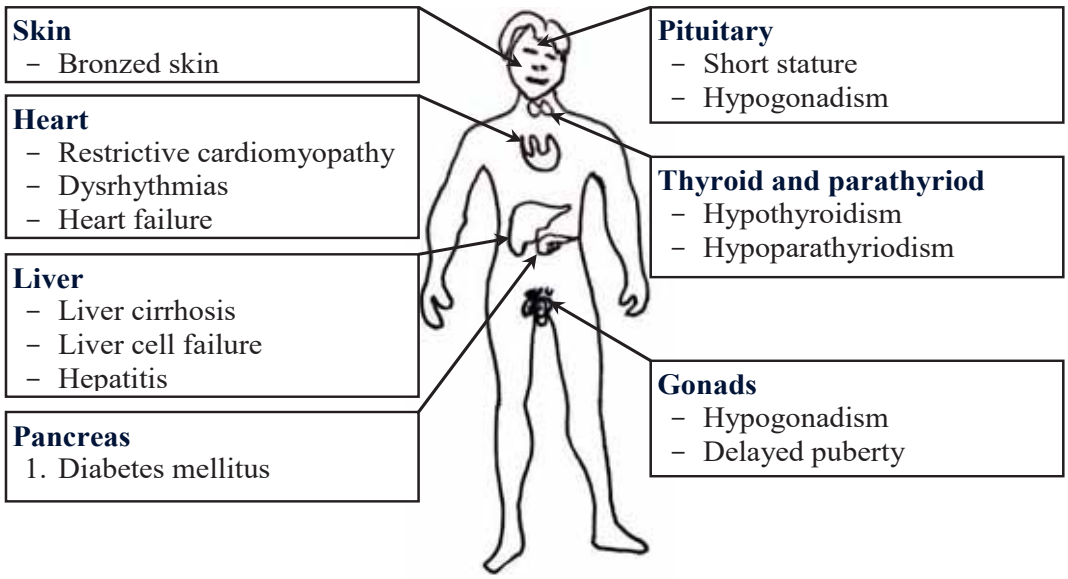
- Chronic hemolysis.
- Enhanced iron absorption.
- Repeated blood transfusion.

Clinical picture

A. General features of ***chronic hemolytic anemia*** (see before)

- Manifestations start insidiously after the 6th month of age when switch from γ to β chain production usually normally occur.
- Classic features: Progressive anemia, jaundice ,thalassemic facies,and organomegaly

B. Hemosiderosis → Iron deposition in:

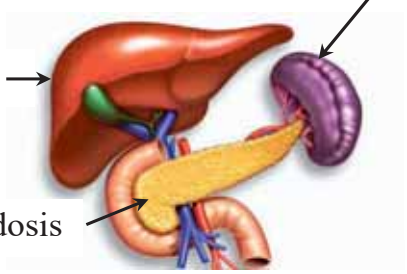


C. Complications and causes of death

1. Heart failure → 

2. Liver cell failure →

3. Diabetic keto acidosis



4. Hypersplenism :
   - Pallor, purpura, pyrexia, and organomegaly
   - CBC: pancytopenia
   - Bone marrow: marrow hyperplasia , no malignant cells

5. Frequent blood transfusions carry risk of blood bone infections e.g. HBV, HCV, AIDS, and CMV →

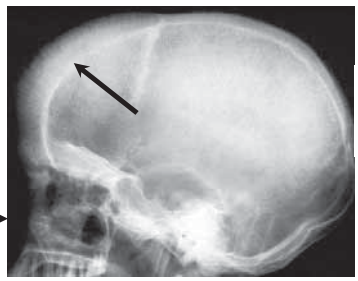


Investigations

a. For anemia → Low Hb% and Ht. value.

b. For chronic hemolysis

- Unconjugated hyperbilirubinemia
- Reticulocytosis (commonly <8%)
- Skull X-ray in children >3 years →



"*Hair on end" appearance*

c. For the cause

1- Blood film → hypochromic, microcytic anemia with target cells.

2- Alkaline denaturation (Apt) test → Hb F resist denaturation by alkali

3- Hemoglobin electrophoresis:

- Markedly raised Hb F (80-90%)
- Slightly raised Hb A_2
- Absent or near absent Hb A

4- Prenatal diagnosis is possible by **c**horionic **v**illous **s**ampling (CVS)

d. For diagnosis of iron overload

- Serum iron and ferritin
- Cardiac / hepatic MRI*
- Liver biopsy*
- Liver iron by Superconducting Quantum Interference Device (SQUID)

Treatment

1. Chronic transfusion therapy

- Indications for initiation of regular red cell transfusions include:
  - Hemoglobin level ,7 g/dl (on at least 2 measurements)
  - Poor growth
  - Facial bone changes
- Aim: To keep pre transfusion Hb > 9.5- 10.5 gm/dl
- Dose: 10-15 ml/kg packed RBCs monthly.
- Benefits
  - Allow normal growth and activity
  - Decreases bone marrow activity → ↓ skeletal changes.
  - Decreases extra medullary hematopoiesis → ↓ organomegaly.

2. Iron chelation therapy

- Often deferred until age 3 to 4 years.
- Indications:
  - Cumulative transfusion load of 120 ml/kg or greater
  - Serum ferritin level persistently >1,000 ng/ml
  - Liver iron concentration .5–7 mg/g dry weight

- **Drugs used**

A. Desferroxamine (Desferal)

- Dose: 25-50 mg/kg/day
- Route: IV or by continuous SC pump for 10 hours, 5-6 nights per week.
- Side effect: anaphylaxis, deafness, cataract, retinal damage , yerssinia sepsis and skeletal changes
- Ascorbic acid 200 mg daily enhance the chelating effect of Desferal

B. Recent Oral drugs

1. Deferiprone (Ferriprox)

- Dose : 75-100 mg/kg
- Value: Effective in Reducing cardiac iron overload.
- Side effects: Gastric upset ,neutropenia and agranylocytosis

2. Defrasirox (Exjade)

- As effective as desferal but oral with longer half life
- Once daily, 20-40 mg /kg
- Monitoring of liver enzymes and serum creatinine is essential

3. Supportive treatment

- Low iron diet
- Folic acid 1mg/day
- Endorcine support as necessary.
- Hepatitis A and B vaccine

4. Splenectomy

- Indications

A. Hypersplenism suggested by:

- Increasing need for transfusion by ≥ 50% than usual for > 6 months.
- Annual PRBCs > 250 ml/kg/year in face of uncontrolled iron overload
- Severe leucopenia and / or thrombocytopenia (Pancytopenia)

B. Huge spleen with pain or pressure symptoms.

- When: Preferably after the 5th – 6th year
- Risk: Overwhelming sepsis (especially if done < 5 years)
- Precautions: See before

5. Other lines of treatment

- Hydroxyurea → Induction of Hb F → ↓Unmatched α chain accumulation → ↓ hemolysis (of limited value due to serious side effects).
- Stem cell transplantation → best for patient less than 17 years
- Gene therapy is under research

6. Genetic counseling

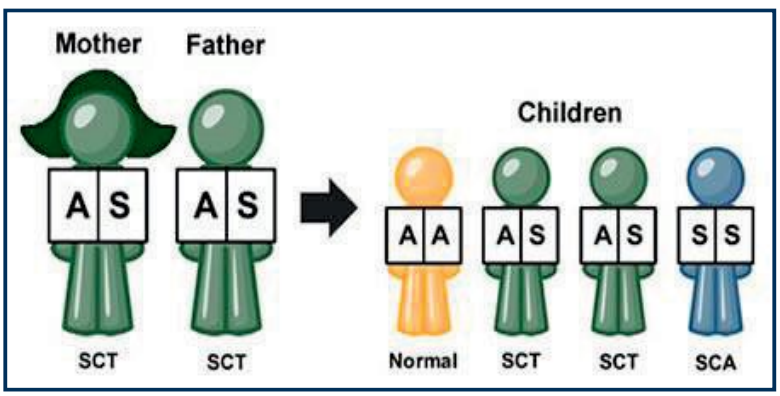
Sickle Cell Disease

Etiology

- Autosomal recessive disorder.
- Due to single amino acid substitution in the number **6** position of the **β**-chains (valine for glutamic) resulting in new Hb → HbS **(**$\alpha_2\beta_2^{6glut.} \rightarrow \alpha_2\beta_2^{6val.}$ = HbS**)**.

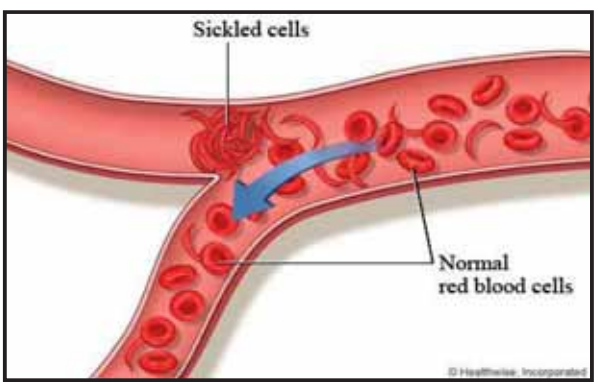
Forms

- Sickle cell Anemia; SCA → Hb SS ; homozygous
- Sickle cell trait ; SCT → Hb AS ; heterozygous



Pathogenesis

HbS can't withstand hypoxia
↓
If exposed to low O_2 tensions
↓
HbS polymerize
↓
RBCs distortion
↓
Intra vascular sickling with subsequent
↓



1. Aggregation → vascular occlusion
2. Trapping and hemolysis in reticulo endothelial system in the spleen & liver

Clinical picture

- Common in negroes
- Features of anemia
- Features of chronic hemolysis } Starting after the 6th month of age
- Renal disorders → proteinuria, nephrotic syndrome, chronic renal failure.

Crisises

1. Aplastic
2. Hemolytic
3. Megaloblastic
4. Hyperhemolytic (as before)
5. **Vaso occlusive crisis (painful crisis)**

- Mechanism
  - In vivo sickling → vascular occlusion → ischemia ± infarction.
  - Reduced Nitric Oxide bioavailability → vasoconstriction and platelet activation
  - WBCs counts are often elevated, adhere to endothelial cells and may further trap sickled red cells, contributing to stasis
- Precipitating factors

  Fever, Acidosis, Dehydration, Infection & Hypoxia, exposure to cold
- Clinically

CNS

→ Cerebrovascular stroke

→ Retinopathy

Myocardial infarction

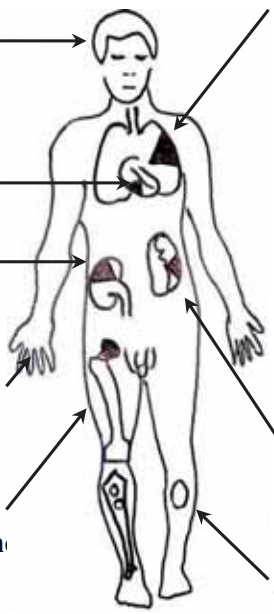
Renal infarction

→ Hematuria

→ Chronic renal failure

Hand and foot syndrome: Ischemia of metacrapal & metatarsal bones

Avascular necrosis of bon e.g. femoral head



Acute chest syndrome (ACS):

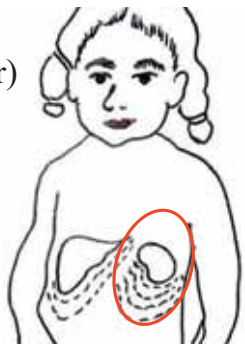
- Due to pulmonary emboli of necrotic bone marrow (fat emboli) infection or pulmonary infarction
- Clinical Presentation :
  - Gradual or catastrophic.
  - Severe respiratory distress
  - Chest pain, fever
  - Hypoxemia
  - CBC: Leucocytosis
  - CXR: Lung consolidation.

Splenic infarctions→ fibrosis → shrinking→ autosplenectomy & hyposplenism

Others e.g. leg ulcers, priapism

6. **Splenic sequestration crisis**

- Sudden pooling of the blood in the spleen (± the liver)
- Precipitated by dehydration
- Occurs primarily in infants



Clinically

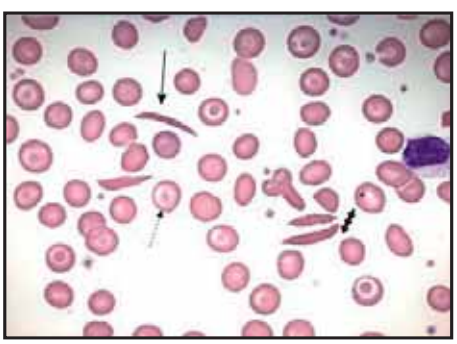
- Acute pallor and acute abdominal pain
- Massive splenomegaly
- Hypovolemic shock.
- Decline in hemoglobin of ≥2 g/dL from the patient's baseline hemoglobin; reticulocytosis and a decrease in the platelet count may be present

7. Infectious crisis

- Due to hyposplenism (↓opsonization)
- Common organisms: usually encapsulated bacteria
- Site
  - Meningitis (Pneumococci & H. influenza)
  - Pneumonia (Pneumococci)
  - Osteomyelitis (Salmonella)

Investigations

1. For **anemia** → Low Hb% & Ht value.
2. For **chronic hemolysis** →↓ RBCs survival & ↑ erythropoiesis.
3. For **the cause**
   a) Blood film:
      - Detect sickle cells in peripheral blood. If not detected, sickling can be enhanced by adding sodium metabisulfite (Sickling test).
      - Howell–Jolly bodies (nuclear remnants) and Sub membranous pits in RBCs may be seen indicating hyposplenism.



   b) Hemoglobin electrophoresis: Show HbSS (90%) & Hb F (2-10%).
   c) Neonatal screening allows early detection , adequate care and longer survival

Treatment

1. Avoid factors precipitating painful crisis e.g.

- Vigorous hydration during exposure to extreme stress
- Vigorous treatment of infections

2. Chronic transfusion therapy & iron chelation (Equivocal)

Indications:

- Stroke
- Recurrent painful crisis.
- Recurrent acute chest syndrome

3. Treatment of crises:

A. Vaso occlusive

- Oxygen
- Hydration ; oral and IV to maintain euvolemia
- Pain control e.g. ibuprofen, acetaminophen, codeine, or morphine
- Empirical antibiotics (cephalosporin and macrolide) for infections
- Simple transfusion target post transfusion hemoglobin ±10 g/dL
- Exchange transfusion program for recurrent cases

B. Sequestration

- Blood transfusion, typically 5 mL/kg of packed red blood cells
- Exchange transfusion
- Prophylactic splenectomy is the only effective strategy for preventing future life-threatening episodes

C**. Aplastic / hemolytic crisis** → Blood transfusion

4. Control infection:

- Prophylactic penicillin for life
- Immunize against: Pneumococci & H. influenza

5. Alternative treatment

* Hydroxyurea

Value

- Induction of Hb F (takes time so not suitable for acute therapy)
- Improve RBCs hydration
- The only effective drug proved to reduce the frequency of painful episodes

Dose: typical starting daily dose is 15-20 mg/kg

* Hematopoietic stem cell transplantation.

Sickle cell trait (Hb AS)

- Hemoglobin analysis shows Hb A, typically >50% and (Hb S <50%)
- Asymptomatic but complications do exist e.g.
  - Sudden death during rigorous exercise
  - Splenic infarcts at high altitude
  - Hematuria
  - Bacteriuria
  - Susceptibility to eye injury with formation of a hyphema
  - Susceptibility to renal medullary cancer
- Patients are resistant to lethal effects of falciparum malaria

Auto Immune Hemolytic Anemia (AIHA)

Definition

Hemolytic anemia due to circulating antibodies against patients own RBCs.

Explanation

- Altered immune response (not recognize self-antigens).
- Altered RBCs antigenicity by infection **or** drugs.

Clinical types: According to the type of auto antibodies

Causes	Warm reactive autoantibodies	Cold reactive autoantibodies
Primary	– 50 % are idiopathic	– Idiopathic
Secondary		
▪ Infection	– CMV, HBV	– Mycoplasma pneumonia – Infectious mononucleosis
▪ Vaccination		– MMR
▪ Disease	– Leukemia/ Lymphoma. – SLE	– Lymphoma
▪ Drugs	– Methyle Dopa, penicillin, interferon	
Criteria of Antibodies	- IgG. - Active at 37°C	- IgM - Active at below 37°C

Clinical picture

1. AIHA due to warm reactive autoantibodies

A. Acute transient type	B. Chronic type
– Acute hemolytic anemia with splenomegaly.	– Chronic hemolytic anemia.
– In child 2-12 year.	– In child < 2 or > 12 years
– May follow respiratory infection.	– May be underlying systemic disease e.g. lymphoma.
– Responsive to steroids	– Variable response to steroids.
– Lasting 3-6 months	– Last months and years

2. AIHA due to cold reactive autoantibodies (*cryopathic hemolytic syndromes*)

A. Cold agglutinin disease

- Cold reacting auto antibodies may present in low titer
  - Exposure to triggering agent e.g. Infection with mycoplasma
    - Increased auto antibody titer
      - Eventual chronic hemolytic anemia

B. Paroxysmal cold hemoglobinuria

- Due to Donath Landsteiner antibody (IgG which can activate complement)
- Antibodies are biphasic, reacting with red cells in the cold in the peripheral circulation, with lysis occurring due to complement activation when the cells return to the central circulation.
- The lytic reaction is demonstrated in vitro by incubating the patient's red cells & Serum at 4°C and then warming the mixture to 37°C (*Donath–Landsteiner test*)
- Association:
  - Infection with mycoplasma, Ebstein Barr virus, Cytomegalo virus, measles, mumps or chickenpox
  - Congenital or acquired syphilis
- Course: Mild, resolve with infection resolution.

Investigations

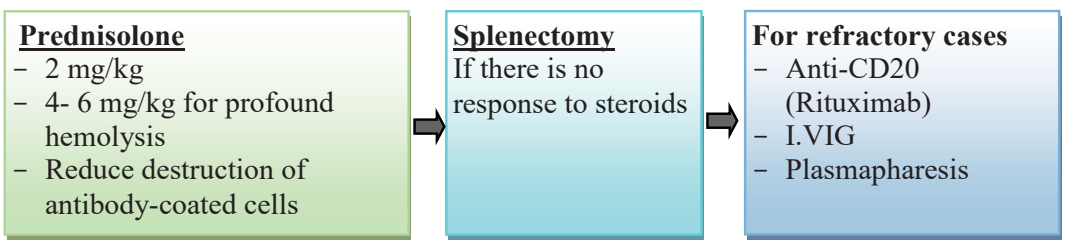
- For **anemia** → Low Hb % & ↓ Ht value.
- For **acute hemolysis** (see before)
- For **chronic hemolysis** (see before)
- For **the cause**.

1. CBC :
   - Micro spherocytes.
   - AIHA plus autoimmune thrombocytopenia → **Evan's Syndrome.**
2. Positive Coombs test (*Antiglobulin test*)
   - Direct: Detects high titer of autoantibodies coating the RBCs
   - Indirect: Detects the free autoantibodies in patient serum

Treatment

A. AIHA due to warm antibodies

1. Mild and asymptomatic anemia needs only follow up
2. Treatment of symptomatic anemia



B. Cold agglutinin disease

* Patients should avoid exposure to cold.
* Steroids and splenectomy are usually ineffective.

* Anti-CD20 (Rituximab) has been successful in some cases.
* Plasmapharesis: for severe cases.

C. Supportive care

* Treat the underlying cause
* Blood transfusion:

▪ **Value**	– **Life saving for severe anemia**
▪ **Problem**	– Compatibility testing is complicated by the presence of unspecific red cell autoantibodies
▪ **Use**	– Small volume of packed RBCs starting with a test dose – Use the least incompatible blood
▪ **Disadvantage**	– Hard to find totally compatible blood. – Transient effect

Approach to a Case of Anemia

A. History

1. Onset

- G6PD, spherocytosis can present from neonatal period
- Thalassemia, sickle cell anemia present > 6 month



2. Past history

- Hemorrhage
- Drugs which may induce (aplasia, acute hemolysis in G6PD deficiency).
- Infection or fava beans → G6PD deficiency.

3. Family history →for similar cases and consanguinity for inherited causes.

B. Clinical approach : Search for clinical clues of :

1. Bone marrow failure (Pallor, Purpura, Pyrexia ± organomegaly)


+

+


2. Acute hemolytic anemia


OR

→

+


Triggers | Acute pallor/jaundice | Dark urine

3. Chronic hemolytic anemia


+

→


Gradual pallor/jaundice | Organomegaly | Complications

4. Anemia with congenital anomalies:



Fanconi anemia.

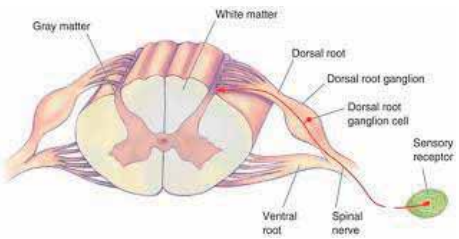


Diamond Blackfan anemia.

5. Anemia with systemic associations



GIT upset (Folic acid deficiency)



Neuropathy (B_{12} deficiency)

C. Laboratory approach

1. Is it anemia? → ↓ Hb% & ↓ Ht value.

2. Is there bone marrow failure?

Features	Consider	Workup
– Anemia – Thrombocytopenia – Leucopenia or increased WBCs	- Aplastic anemia - Leukemia - Bone marrow infiltration - Don't forget hypersplenism	- Blood film for abnormal cells - BM examination

3. Is it microcytic anemia?

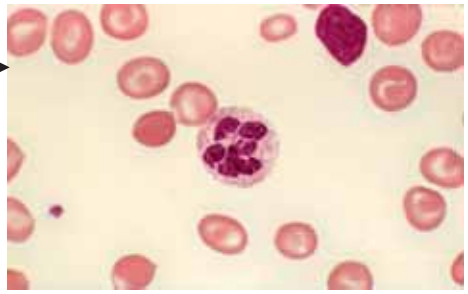
Features	Workup
– MCV < 70 fl – MCH < 27 pg – MCHC < 30%	- Iron indices (diagnose iron deficiency and anemia of chronic illness) - Blood film for basophilic stippling - Hemoglobin electrophoresis for thalassemia - Bone marrow examination for ringed sideroblasts - Serum lead level

4. Is it macrocytic anemia?

Features : MCV > 85 fl

Causes

- Folic acid and B12 deficiency →
- Other causes:
  - Aplastic or hypoplastic anemia
  - Hepatic disease
  - Hypothyroidism
  - Drugs: antifolate, cytotoxics



Workup: Consider Serum folic acid and B12 assays and Schilling tests

5. Is it hemolytic anemia?

- **Is there increased red cell breakdown?**
  - Anemia
  - ↑Bilirubin: unconjugated, from hem breakdown
  - ↑Urinary urobilinogen (no urinary conjugated bilirubin).
  - ↑Serum lactic dehydrogenase (LDH), as released from the RBC
- **Is there increased red cell production?**
  - Reticulocytosis
- **Is the hemolysis mainly extra- or intravascular?**
  - Extravascular hemolysis may lead to splenic hypertrophy and splenomegaly
  - Features of intravascular hemolysis are:
    - ↑ Plasma hemoglobin
    - ↓ Haptoglobin & haemopexin(hemoglobin carriers).
    - ↑ Metheamalbumin
    - ↑ Unconjugated bilirubin, fecal and urinary urobilinogen
    - Hemoglobinuria (and heamosiderinuria)
    - ↑ Serum LDH

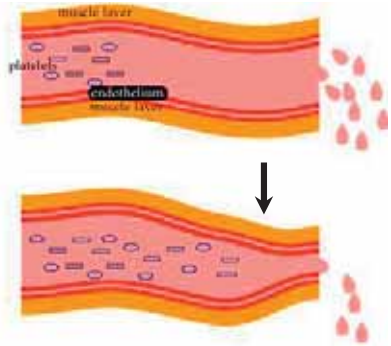
Causes : See classification of anemia

Workup:

- Positive Coomb's test: Immune hemolytic anemia
- Negative Coomb's test : Other causes of hemolytic anemia: Consider
  - Osmotic fragility and autohemolysis tests
  - Enzyme assays e.g. G6PD deficiency
  - Blood film and sickling tests
  - Electrophoresis

Hemorrhagic disorders

Hemostasis is the mechanisms of stoppage of bleeding after injury of a blood vessel



I. 1ry Hemostatic Mechanisms

a. Vascular factor

Role

- Reflex vasoconstriction at the site of bleeding.
- Damaged endothelium release serotonin and activate F XII

Assessment

- Bleeding time.
- Hess test

b. Platelets

Criteria

- Life span: 8-10 day
- Count: 150- 400.000 platelets per ml
- Mean platelet diameter: 1- 4 μM
- Distribution: one third (30%) in the spleen; two-thirds in the bloodstream
- Controlled by thrombopeiotin

Role

1. Adhesion to exposed collagen fibers (Von Willbrand factor is essential).
2. Aggregation; platelets accumulate at injured site helped by adenosine diphosphate (ADP) & thromboxane A_2(TXA_2)
3. Release of:
   - Thromoxane A2 →↑ platelet aggregation.
   - Serotinin →↑ vasoconstriction
   - Platelet factor 3 (PF_3) → enhance clotting
   - Thrombasthinine →↑ clot retraction.
4. Platelet plug formation

Assessment

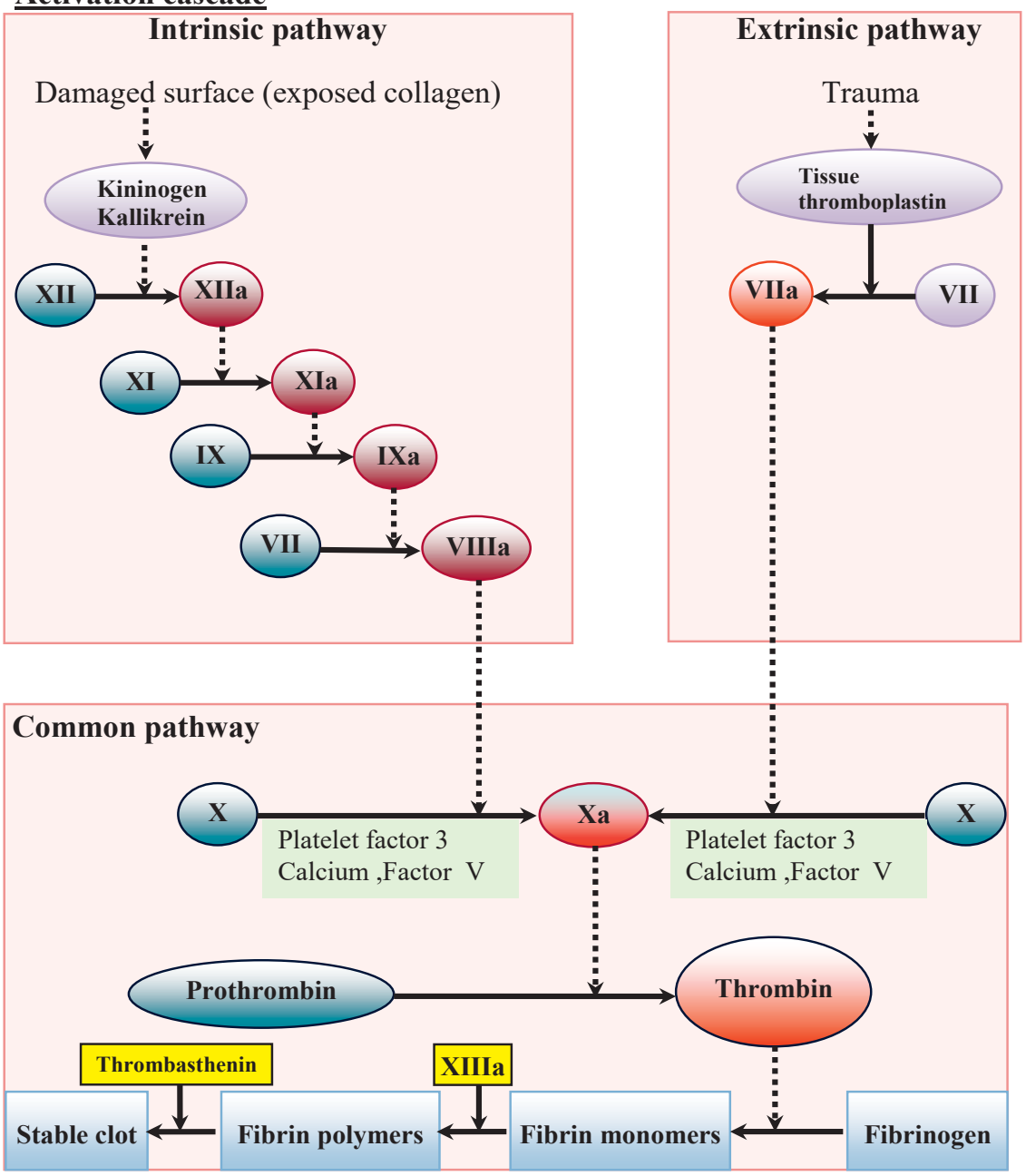
- Bleeding time→ normal = 4-8 min.
- Hess test (capillary fragility test) → cuff of sphygmomanometer is inflated between systole & diastole for 5 min.→ if > 5 petechiae appear within 5 cm circle in the forearm → +ve test.
- Platelet count (N = 150 – 400.000 /mm^3)
- Platelet function tests whenever thromocytopathy is suspected
  - Assess platelet adhesiveness and aggregation.
  - Assay of PF3 level.
  - Clot retraction test.

II. Coagulation factors

Criteria

- Most coagulation factors are formed in the liver and circulate in an inactive form (*Pro coagulants*) which are activated in a cascade manner
- Vitamin K dependent factors → II, VII, IX, X.
- Most activation reactions occurred on the surface of the platelets
- Calcium is a cofactor for many steps of activation cascade ;particularly the common pathway

Activation cascade



Coagulation factors names

I	Fibrinogen	**II**	Prothrombin
III	Tissue thromboplastin	**IV**	Calcium
V	Labile factor	**VII**	Stable factor
VIII	Anti-hemophilic factor	**IX**	Chrismas factor
X	Stuart Prower factor	**XI**	Plasma thromboplastin antecedent
XII	Hageman factor	**XIII**	Fibrin stabilizing factor

Others : High Molecular Weight Kinonogen , Kallekrein

Assessment:

1. Clotting **t**ime → normal = 8-12 **min**.

rough test → prolonged with defects in any phase

2. Thrombin **t**ime (TT) → normal = 15-20 Sec.

- Time needed to plasma to clot after addition of bovine thrombin
- Prolonged in fibrinogen deficiency.

3. Prothrombin **t**ime (PT) → normal = 12-14 Sec.

- Time needed to plasma to clot after addition of thromboplastin & Calcium.
- Test extrinsic & common pathways.

4. Partial **t**hromboplastin **t**ime (aPTT) → normal = 25-40 Sec.

- Time needed for plasma to clot after addition of kaolin, Calcium & platelets.
- Test intrinsic & common pathways.

Interpretation

Defect in	**PT**	**PTT**	**Specific**
Common pathway (X,V,II,I)	Prolonged	Prolonged	Specific factor assay
Extrinsic pathway (VII)	Prolonged	Normal	
Intrinsic pathway (XII,XI,IX,VIII)	Normal	Prolonged	

N.B: Prolonged both PT & PTT also occur in multiple factors deficiency e.g.

- Liver cell failure
- Vitamin K deficiency
- Disseminated Intra vascular Coagulation (DIC)

5. Specific tests e.g. Von Willbrand disease:

- Prolonged both bleeding time and clotting time
- Decreased both F VIIIc & F VIIIa
- Von Willbrand factor assay

Control of coagulation factors

1. Natural coagulation inhibitors

Value: localizes thrombosis to the site of injury.

a. Antithrombin (AT)

- Potent inhibitor of factors XIa, IXa, Xa, and thrombin.
- Its action is greatly potentiated by heparin.

b. Activated protein C

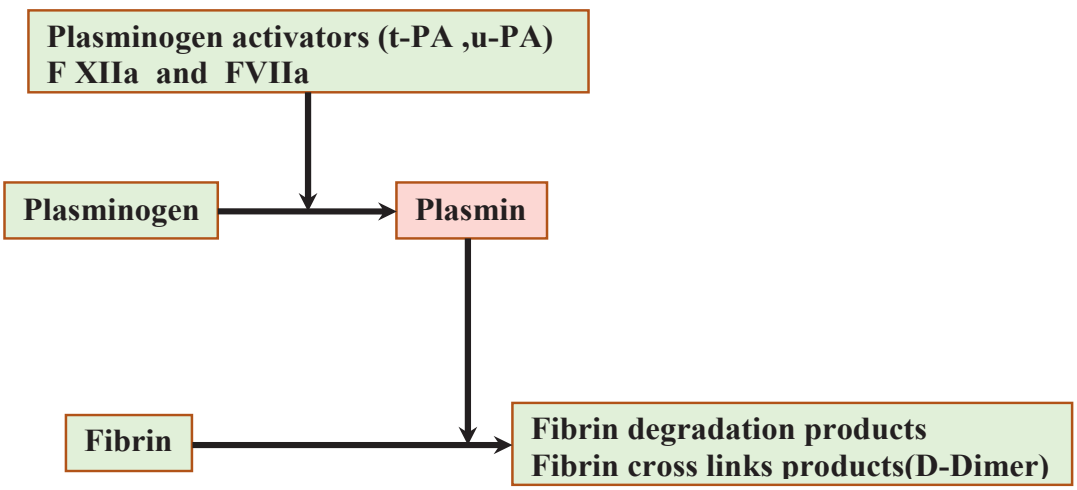
- Vitamin K dependent protein
- Activated by thrombin
- Activated protein C inactivates factor V and factor VIII

c. Protein S

- This is a cofactor for protein C

2. Fibrinolytic system: (Fibrinolysis)

- Plasmin is generated from its inactive precursor plasminogen
- This is achieved principally via tissue plasminogen activator (t-PA) released from endothelial cells.
- Some plasminogen activation may also be promoted by urokinase, produced in the kidneys (u-PA).
- Plasmin breaks down fibrinogen and fibrin into fragments X, Y, D and E, collectively known as fibrin (and fibrinogen) degradation products (FDPs).
- D-dimer is produced when cross-linked fibrin is degraded.
- Fibrinolysis helps to restore vessel patency after vascular damage.



Purpura

Definition: Group of disorders characterized by:

1- Multiple, spontaneous hemorrhages in the skin and mucous membranes.

2- Range from pin point (petechiae) to several centimeters (ecchymosis)

3- Petechiae are purple in color, not elevated, do not blanch on pressure, not pruritic

Possible bleeding sites:

- Skin: petechiae, ecchymosis
- Mucous membranes: oral, gingival, conjunctiva,..
- Orificial: epistaxis, hemoptysis, hematuria, melena..
- Internal: intracranial, retinal, pleural, pericardial,..

Causes

I. Non thrombocytopenic purpura: (Normal platelet count)

1. Vascular purpura

a. Hereditary: e.g.

Ehler Danlos Syndrome

- Defect in type III collagen in connective tissue.
- Clinical picture
  - Hypermobile joints
  - Purpura



b. Acquired

Vasculitis

- Henoch schonlein purpura
- Sepsis (meningeococcal).
- Infective endocarditis

Weak blood vessels

- Scurvy → defective connective tissue collagen.
- Cushing syndrome

2. Thrombasthenia (Platelet dysfunction; Thrombocytopathy)

a. Hereditary thrombocytopathy

- Defective adhesion: Von Willbrand disease , Bernard Soulier disease
- Defective aggregation: Glanzmann disease.

b. Acquired

- Uremia (Renal failure)
- Cholemia (Liver failure)
- Drug induced: Aspirin (↓ ADP & Thromboxan A2),Heparin
- Autoimmune: SLE

II. Thrombocytopenic purpura

Defined as platelet count <100.000/mm^3

A. Decreased production

With decreased marrow megakaryocytes

I. Congenital

1. Fanconi anemia

2. Thrombocytopenia absent radii (TAR)syndrome

- Thrombocytopenia
- Absent or hypoplastic radii
- Other associated anomalies e.g. congenital heart disease





3. Wiskott – Aldrich syndrome

- Sex linked recessive disorder
- Microthrombocytopenia
- Infantile eczema and bloody diarrhea are the commonest presentation
- Combined immunodeficiency


Petechiae due to thrombocytopenia


Eczema


Pneumonia and other infections

II. Acquired

1. Aplastic anemia
2. Bone marrow infiltration (e.g. Leukemia).
3. Advanced megaloblastic anemia

B. Increased destruction

With compensatory increased marrow megakaryocytes

I. Immunologic

- Primary : Idiopathic thrombocytopenic purpura
- Secondary to
  - Immunologic diseases e.g. SLE
  - Post transfusion purpura
  - Drug induced e.g. Heparin, Phenytion
  - Malignancy e.g. Lymphoma

II. Non immunologic

- Microangiopathic hemolytic anemia e.g. DIC, hemolytic uremic syndrome.
- Hypersplenism.
- Acute infections, Sepsis.
- Thrombotic thrombocytopenic purpura
- **Kasabach-Merritt syndrome (KMS):**



  - Combination of giant cavernous hemangioma, thrombocytopenia, and coagulopathy
  - Platelets are trapped and destroyed in the giant cavernous hemangioma
  - Frequently associated with disseminated intra vascular coagulation
  - Treatment : steroids ,interferon, surgical ligation of feeding vessels

Idiopathic Thrombocytopenic Purpura (ITP)

(Immune Thrombocytopenic Purpura)

Definition: Purpura characterized by:

- Shortened platelets survival due to antiplatelet antibodies
- Thrombocytopenia (< 100.000 /mm^3).
- Increased bone marrow megakaryocytes.
- Absent other identifiable thrombocytopenic disorders (a diagnosis of exclusion).

Pathophysiology

- Anti-platelets antibodies triggered by preceding viral infection
- Antibodies coated platelets is trapped in the spleen by macrophage Fc receptors
- Spleen is the primary site for antibody production and platelets destruction.

Clinical picture

Incidence

- Age: peak between 1 – 4 years
- Sex incidence: Equal

Acute purpura

- The classic presentation of ITP is a previously healthy 1-4 yr old child who has sudden onset of generalized petechiae and purpura.
- The parents often state that the child was fine yesterday and now is covered with bruises and purple dots.
- Often there is bleeding from the gums and mucous membranes, particularly with profound thrombocytopenia (platelet count <10 × 10^9/L).
- There is a history of a preceding viral infection 1-4 wk before the onset of thrombocytopenia in 50-60% of patients.



Important exclusions

1. No significant organomegaly ;however tip of the spleen is palpable in 10 % of cases
2. No significant pallor except with severe bleeding or associated autoimmune hemolytic anemia (Evans syndrome)

Clinical classification of bleeding severity

1	**No symptoms**
2	**Mild symptoms**: – Bruising and petechiae – Occasional minor epistaxis – Very little interference with daily living
3	**Moderate**: – More severe skin and mucosal lesions – More troublesome epistaxis and menorrhagia
4	**Severe**: – Bleeding episodes: menorrhagia, epistaxis, melena—requiring transfusion or hospitalization – Symptoms interfering seriously with the quality of life

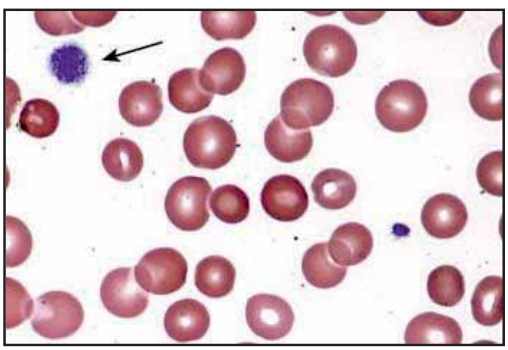
(Nelson textbook of pediatrics, 2016)

Diagnosis

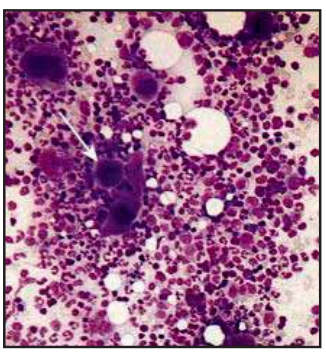
- Isolated thrombocytopenia in healthy child with normal blood smear
- **Blood smear:**
  - Platelet count → always < 100.000 / mm^3; platelets usually large sized
  - Normal WBCs and RBCs (except in Evans and severe bleeds); no abnormal cells
- **Bone marrow examination**
  - Increased numbers of megakaryocytes, many of which appear immature
  - Normal myeloid and erythroid cells

  Indications for bone marrow examination include

  1. Abnormal WBC count or differential
  2. Unexplained anemia
  3. History and physical examination suggestive of a bone marrow failure syndrome or malignancy.
  4. No response to supportive therapy and steroid therapy needs to be started

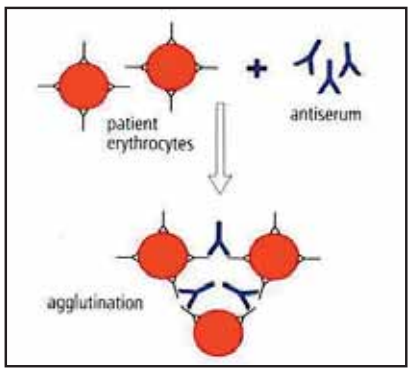


Blood smear



Bone marrow smear

- **Other tests**
  - A direct Coombs should be done if there is unexplained anemia to rule out Evans syndrome or before instituting therapy with IV anti-D.



  - In adolescents, an antinuclear antibody test should be done to evaluate for SLE.
  - HIV studies should be done in at-risk populations, especially sexually active teens.
  - Platelet antibody testing is seldom useful in acute ITP.

Prognosis

- Spontaneous resolution occurs in **80** % of children with acute ITP within 6 mo.
- About **20**% of children with acute ITP go on to have chronic ITP.
- Fewer than **1**% of patients develop an intracranial hemorrhage
- Therapy does not appear to affect the natural history of the illness. There is no evidence that therapy prevents serious bleeding.
- The prognosis may be related more to age, as ITP in younger children is more likely to resolve whereas the development of chronic ITP in adolescents approaches 50%.

Differential Diagnosis

From other causes of **purpura** :

	ITP	BM aplasia	BM infiltration	Hypersplenism
Clinical	– Isolated purpura in clinically well child	– Purpura – Pallor – Pyrexia – No organomegaly	– Purpura – Pallor – Pyrexia – Organomegaly	– Purpura – Pallor – Pyrexia – Splenomegaly
Blood film	– ↓Platelets	– Pancytopenia	– Pancytopenia	– Pancytopenia
Bone Marrow	– Megakaryocyte hyperplasia	– Hypoplastic	– Infiltrated	– Hyperplastic

Treatment of ITP

I. No therapy (Observation)

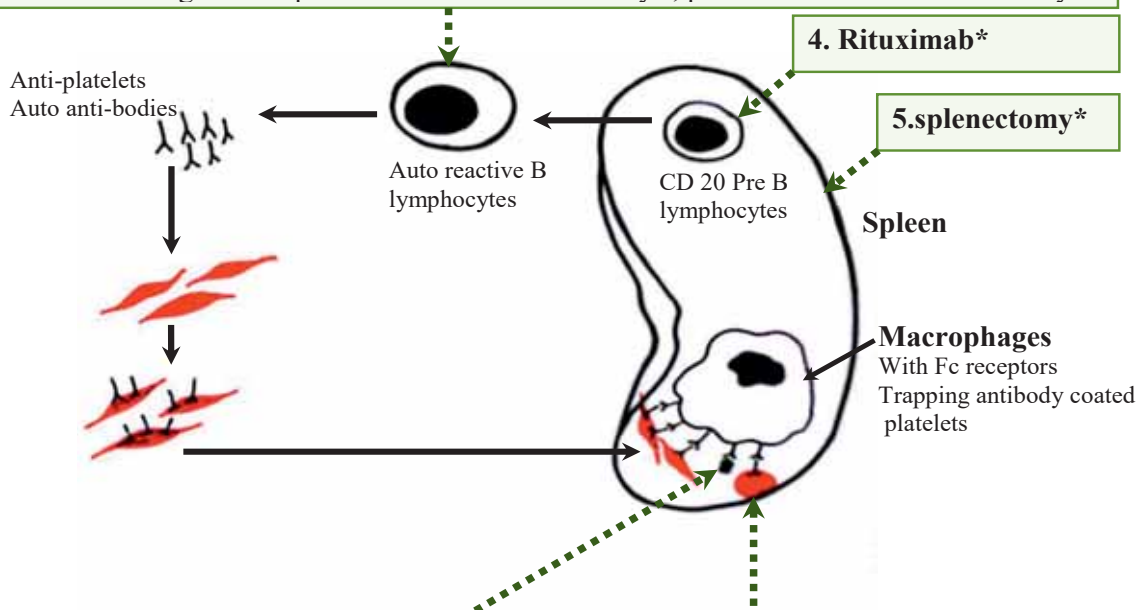
- For patients with minimal, mild, and moderate symptoms
- Education and counseling of the family about benign nature of ITP
- Avoid trauma, aspirin, contact sports.

II. Drug therapy

- Indicated for significant bleeding with platelet counts below 20×10^9/L
- Aim : to raise platelets above theoretically safe level $>20 \times 10^9$/L

1. Prednisone

- Decrease platelet antibodies production and phagocytosis of antibody coated platelets
- Standard dose: 2 mg /kg/day for 21 days → gradually tapered
- High dose :4 mg / kg/day for 4 days → less steriod side effects
- Two thirds will respond but relapse is common
  - Standard dose : response in 1 week
  - High dose : platelets > 20×10^9/L in 2 days , platelets > 50×10^9/L in 3-4 days



2. Intravenous Immunoglobulin (IVIG)

- Block macrophage Fc receptors → protect platelets from destruction
- Dose : **0.8-1** gm/kg for 1- 2 days
- Induces a *rapid* rise in platelet count (usually $>20 \times 10^9$/L) in 95% of patients within 2 days
- Very useful in serious bleeding or urgent surgery

3. intravenous anti D (Win Rho)

- Coat Rh positive RBCs → RBC-antibody complexes bind to macrophage Fc receptors → platelet escape destruction with transient mild hemolysis (Patient must be unsplenctomized, Rh positive , with Hb > 9 gm/dl)
- Should not be used with IVIG
- Dose : **50-75** µg/kg
- Causes a rise in platelet count to $>20 \times 10^9$/L in 85% of patients within 48-72 hr

*** Splenectomy**

Indications - Severe ITP with life threatening bleeding unresponsive to steroids and intravenous immunoglobulin
- A second line therapy for Chronic ITP uncontrolled medically

Response - Two-thirds will achieve a normal platelet count

*** Rituximab**

Nature Anti CD20 monoclonal antibody

Role Deplete auto reactive B cells (mainly for chronic ITP)

Treatment of life threatening bleeding (e.g. intracranial hemorrhage):

1. Resuscitation (including blood transfusion if needed)
2. Platelet transfusion (especially if platelets count < 20.000/mm^3).
   * Have shorter life span but can be life saving in life threatening hemorrhage or in case of emergency operation e.g. splenectomy
   * Dose: 0.2 platelet units per kg as a bolus followed by continuous infusion if required
3. I.V. immunoglobulin 1gm/kg/day for 2-3 days.
4. I.V. methylprednisolone 30 mg / kg/day for three days
5. Emergency splenectomy (not usually recommended).
6. Recombinant human factor VIIa (Rhu VIIa)

Take home messages

- There are no data showing that treatment affects either short- or long-term clinical outcome of ITP
- Platelet transfusion in ITP is usually contraindicated unless life-threatening bleeding is present

(Nelson Textbook Of Pediatrics, 2016)



Anaphylactoid Purpura (Henoch Schonlein Purpura)

Definition

- The commonest vasculitis of childhood
- Characterized by immunoglobulin (Ig) A deposition in the small vessels mainly in the skin, joints, gastrointestinal tract, and kidney.
- Peak age incidence between 2-8 years ; more in males
- May follow drugs or upper respiratory viral or bacterial (sterptoccoci) infection

Clinical picture

1. Skin rash

- In 100 % of cases
- Skin lesions are usually symmetric and occur in gravity-dependent areas (lower extremities) or on pressure points (buttocks and extensor surfaces)
- Start as palpable erythematous maculopapular rash then become purpuric (petechiae).
- May be pruritic.
- Association: Non pitting angioedema of hands, feet, lips, scalp



Typical HSP rash



Angio edema of dorsa of hands

2. Arthritis/arthralgia

- In 75 % of cases.
- Oligoarticular usually in large joints e.g. ankle & knee.
- Swollen, hot, tender, with limitation of movement.
- Usually resolves within 2 wk but can recur without residuals



Swollen ankles with typical HSP rash, and angio edema of dorsa of feet

3. Gastro intestinal tract manifestations



- In 80% of cases
- Presentations
  - Vomiting, diarrhea, paralytic ileus
  - Bowel angina (postprandial abdominal pain, bloody diarrhea)
- Complications
  - Intussusception (? due to submucosal hematoma)
  - Bowel infarction and perforation

4. Renal

Mainly	Rarely
- Hematuria	- Proteinuria
- Acute nephritis	- Nephrotic syndrome.
- Normal renal function	- Chronic renal disease develops in 1-2% of children with HSP, and approximately 8% of those with HSP nephritis go on to have end-stage renal disease.

5. Others vascultic manifestations

- Neurological → seizures, paresis.
- Testis → Hemorrhage.

Diagnosis

- ➢ The diagnosis of HSP is a clinical one
- ➢ No laboratory finding is diagnostic of HSP

1. For purpura :
- Normal platelet count & function (may be thrombocytosis) .
- Normal coagulation profile.

2. For renal manifestations
- Urine analysis for → RBCs, RBCs casts, proteinuria
- Renal function tests
- Renal biopsy for renal insufficiency or heavy proteinuria

3. For gastro intestinal manifestations
- Stool analysis for gross or occult blood
- Abdominal ultrasound and/or abdominal CT for intussusception

4. Others
- Increased ESR and serum IgA
- Biopsies of skin and kidney
  - Can give diagnostic clue, particularly in atypical or severe cases
  - Shows characteristic leukocytoclastic vasculitis with intramural granulocytes in small arterioles and/or venules with deposits of IgA

Treatment

1. Symptomatic and supportive care e.g. bed rest for joint pain

2. Medications :

- Anti-inflammatory medicines e.g. ibuprofen or painkillers e.g. paracetamol may help relieve some of the joint pain
- Empiric use of prednisone (1 mg/kg/day for 1 to 2 wk, followed by taper)
- Evidence exist to support use of steroids for bowel angina(Reduces abdominal pain) and neurological manifestations
- It does not alter overall prognosis nor prevent renal disease

3. Follow up

It is recommended that children with HSP undergo serial monitoring of blood pressure and urinalyses for 6 mo after diagnosis, especially those who presented with hypertension or urinary abnormalities.

Coagulation Disorders

A. Hereditary

1. Intrinsic pathway disorders:

- Factor VIII deficiency (Hemophilia A)
- Factor IX deficiency (Hemophilia B or Christmas disease).
- Factor XI deficiency (Hemophilia C).
- Factor XII deficiency
- Von Willbrand disease (Vascular hemophilia) ⇒ commonest disorder.

2. Extrinsic pathway disorders

- Factor VII deficiency

3. Common pathway disorders:

- Factors II, X deficiency
- Factors V (Para Hemophilia)
  - Fibrinogen deficiency:
    - Congenital afibrinogenaemia.
    - Congenital dysfibrinogenaemia
- Factor XIII deficiency

B. Acquired

1- Vitamin K deficiency
* Hemorrhagic disease of newborn: see neonatology
* Vitamin k malabsorption due to:
  - Biliary atresia / obstruction
  - Fat malabsorption e.g. celiac disease,
* Vitamin K antagonists

2- Liver cell failure

3- Disseminated intravascular coagulation (DIC).

4- Others:
- Massive transfusion syndrome
- Medications: heparin , thrombolytics , L-Asparginase
- Inhibitors of coagulation due to:
  a. Hemphelia inhibitors
  b. Auto immunes diseases : SLE , rheumatoid arthritis

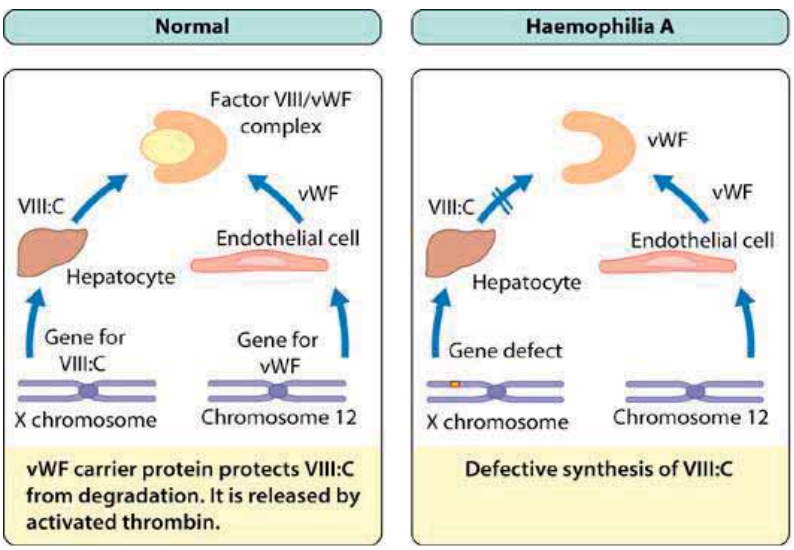
Hemophelia A (classic hemophelia)

Definition

- Sex-linked recessive coagulation defect due to deficiency of factor VIII-C
- 20% of cases are new mutations.
- Hemophilia A represents 80% of all hemophiliacs

Pathophysiology

- Factor VIII-C is synthesized by the liver and reticuloendothelial system
- In plasma, factor VIII-C is stabilized and protected from degradation by Von Willebrand factor(vWF) protein
- In the presence of normal vWF, the half-life of factor VIII-C is approximately 12 hours, whereas in the absence of vWF, the half-life of factor VIII-C is reduced to 2 hours



(Illustrated peadiatrics ,Tom Lissauer)

Pattern of deficiency

- Hemostatic level of F VIII-C is >30- 40U/L (30-40%); below which bleeding occur
- Plasma level of Factor VIII-C in carrier females is between **30-50** %

Clinical picture

Severity of bleeding episode is dependent on the plasma level of factor VIII-C and the severity of trauma

	Severe disease	Moderate disease	Mild disease
F VIII-C	< 1%	1-5%	5-25%
Bleeding trigger	Bleed spontaneously	Bleed with minor trauma	Bleed with severe trauma

1. In neonate
Unusual bleeding from the circumcision site or umbilical stump



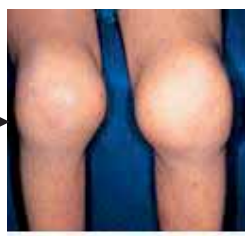
2. External bleeding
e.g.
- Epistaxis
- Dental /mouth bleeding
- GIT bleeding
- Hematuria



3. Skin bleeding
- Easy bruising
- Ecchymotic patches
- Hematomas
- No petechaie



4. Hemarthrosis
- The hallmark of hemophilia A and B
- Affects mainly the big joints of the lower limbs
- Affected joint become swollen, red, hot, and tender with limited mobility
- Tend to be recurrent
- Recurrent hemarthrosis→ joint fibrosis / ankylosis & muscle atrophy



5. Internal bleeding
e.g.
- Muscle hematoma
- Intracranial
- Retroperitoneal
- Hemothorax



Investigations

1. For diagnosis

- Prolonged Clotting time
  - Prolonged aPTT ⇓ **F VIII-C assay** → Low (Diagnostic)
  - Normal PT
- Normal Bleeding time

2. For carrier detection

- Direct F VIII-C gene mutation analysis.

3. For prenatal diagnosis

- Chorionic villous sample or amniocentesis and gene mutation analysis.
- Fetal blood sampling → immunoradiometric assay of F VIII-C antigen

Treatment

1. Primary prophylaxis

- Regular F VIII replacement 20 unit/kg 3 times a week
- The National Hemophilia Foundation recommends prophylaxis for children with severe hemophilia to reduce frequency of bleeding episodes. Usually, such programs are initiated with the first joint hemorrhage.
- Hepatitis B and A vaccination.
- Avoid trauma, I.M. injections & aspirin.



2. During bleeding episodes/peri operative

a. Factor VIII replacement

1. IV infusion recombinant factor VIII ⇒ dose is according to site & severity
   - Minor bleeding: the factor VIII level should be raised to 20–30 IU/dL.
   - Severe bleeding: the factor VIII should be raised to at least 50 IU/dL.
   - Major surgery: the factor VIII should be raised to 100 IU/dL preoperatively and maintained above 50 IU/dL until healing occurred.
2. Others e.g.
   - Cryopreciptate
   - Fresh frozen plasma





b. Ancillary therapy

a. Desmopressin



(D - amino- D - Arginine VasoPressin; DDAVP)

- Role: Increases F VIII level by 4 folds
- Alternative to recombinant F VIII in mild bleeding in mild hemophelia
- Intra nasal or Intravenous (0.3 μ gm /kg in 50 ml saline over 20 min)

b. Antifibrinolytics

- Role: inhibit fibrinolysis → stabilizes the clot
- Indication: - Adjuvant therapy to recombinant factor VIII in mucosal bleeding (oral bleeding, or epistaxis) due to high fibrinolysins in saliva
- Avoided: - In urinary tract hemorrhages to avoid risk of intra renal clot formation and obstructive uropathy
- Examples: Epsilon Amino Caproic Acid (EACA),Tranexamic acid

c. Special situations e.g.

	F VIII dose (IU/kg)	Other lines
Intra cranial hemorrhage	50 IU/kg / 8 hours Then / 12 hours for 8 days Then / 24 hours for 7 days	Urgent hospitalization
Epistaxis and oral bleeding	25 IU/kg / day	EACA 50 mg/kg/6hrs For 1 week
Hematuria and Hemarthrosis*	25- 50 IU/kg / 12 hours for 1-3 days	Followed by prednisone short course for 3-5 days

*Hemarthrosis care include rest, immobilization, cold compresses and elevation

Complications of hemophilia

A. Due to bleeding

- Hemophilic arthropathy→joint stiffness & Muscle atrophy
- Severe intracranial hemorrhage.
- Severe blood loss→ hypovolemic shock

B. Complications of transfusion (See chronic transfusion in thalassemia)

C. Complications of factor VIII therapy

1. Hypersensitivity reactions
2. Factor VIII inhibitors (Antibodies)
   - Develop in about 5-10 %
   - Hemorrhage become refractory to treatment
   - Inhibitors (measured by Bethesda Units) should be screened for annually

- Treatment options

Low responders (< 5 Bethesda Units)	High dose factor VIII
High responders (>10 Bethesda Units)	1. Prothrombin complex concentrate (contain prothrombin, F VII, F IX, F X) 2. Recombinant activated factor VII (*NovoSeven*) 3. Immune tolerance induction: daily infusion of the missing protein till inhibitors disappears 4. Rituximab (Anti CD -20)

Other hemophilias

Hemophelia B (Christmas disease)

- Factor IX deficiency.
- Incidence: 1 / 50.000(in contrast to hemophelia A which is 1 / 10.000)
- Sex-linked recessive disorder.
- As hemophelia A **but** milder.
- Treated by: Recombinant factor IX or factor IX concentrate given/24 hours.
- Prophylactic treatment: Recombinant factor IX twice a week.

Hemophelia C

- Factor XI deficiency.
- Autosomal recessive disorder so can affect both sexes.
- Very mild disease.
- Treated by fresh frozen plasma given / 48 hours.

Von Willebrand Disease; VWD (Vascular Hemophelia)

- Von Willebrand factor (VWF) is a large multimeric glycoprotein
- Synthesized in megakaryocytes and endothelial cells
- Stored in platelet and endothelial cell

Functions :VWF is essential for

- Platelet adhesion to sub endothelial matrix after vascular damage
- Carrier for F VIII protecting it from proteolysis

Deficiency

- **VWF** deficiency results in: Platelet dysfunction and decrease plasma F VIII.
- Pattern of deficiency: low (type 1) **or** abnormal (type 2) **or** absent (type 3).
- Inherited mainly as autosomal dominant disorder
- Diagnosed more in women than in men
- The commonest hereditary bleeding disorder , and some reports suggest that it is present in 1-2% of the general population

Clinical picture

- Mild bleeding: commonly mucocutaneous hemorrhage including skin bruising, epistaxis, gum bleeding and menorrhagia.
- May be post operative bleeding
- Very rarely severe bleeding episodes or hemarthrosis (type 3)

Because VWF is an acute-phase protein, stress will increase its level. Thus, patients may not bleed with procedures that incur major stress, such as appendectomy and childbirth, but may bleed excessively at the time of cosmetic or mucosal surgery (**Nelson Textbook Of Pediatrics, 2016**)

Investigations

A. Screening

- Long bleeding time + long PTT + Normal PT
- Normal results on screening tests do not preclude the diagnosis of VWD

B. Diagnostic

Note: if the history is suggestive of a mucocutaneous bleeding disorder, VWD testing should be undertaken

Tests

1. Quantitative assay for VWF antigen
2. Testing for VWF activity (Ristocetin cofactor activity)
3. Determination of VWF structure (VWF multimers)
4. F VIII assay (F VIII decreases in some subtypes ;Type 2N vWD)
5. Platelet count And platelet adhesion test

Treatment

1. Minor bleeds require no treatment
2. Desmopressin is effective especially in type 1(mainstay of treatment)
3. Antifibrinolytics (e.g EACA) → as adjuvant treatment in oral bleeding.
4. VWF replacement therapy

   Indications
   - For severe bleeding episodes.
   - The only effective treatment in type 3.

   Use
   - Cryoprecipitate
   - Fresh frozen plasma.
   - Purified or recombinant VWF concentrates (containing no factor VIII) may become available in the near future

Disseminated Intravascular Coagulation (DIC)

Wide spread activation of clotting factors all over the body resulting in:

- Thrombosis and ischemia of different vessels.
- Consumption of coagulation factors → bleeding.
- Microangiopathic hemolytic anemia & thrombocytopenia.

Predisposing factors

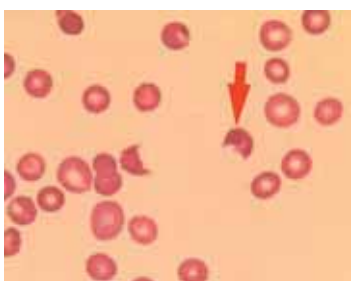
- Severe tissue damage e.g. in shock, dehydration, burn, hyperthermia, crush injury,ashyxia.
- Sepsis: meningeococcal, pneumococcal ,rickettsia, malaria, viral
- Snake bites
- Tumors e.g. AML, disseminated malignancy e.g. neuroblastoma
- Incompatible blood Transfusion
- Protein C deficiency; congenital or acquired with purpura fulminans
- Gastro intestinal causes e.g. fulminant hepatitis, pancreatitis, severe inflammatory bowel disease

Clinical picture

- **Manifestations of the cause.**
- Bleeding first occurs from sites of venipuncture, ecchymosis , purpura.
- Thrombotic manifestations: gangrene in the skin, subcutaneous tissues, extremities or renal infarction.
- Severe anemia → Shock
- May be multi organ system failure

Investigations(3C)

1. Investigations for the **C**ause
2. **C**oagulation profile:
   - Defective all phases of coagulation ; prolonged PT , aPTT, thrombin time, and low fibrinogen.
   - Prolonged bleeding time
   - Fibrinogen degradation products (FDPs, D-dimers) appear in the blood. The D-dimer assay is as sensitive as the FDP test and more specific for activation of coagulation and fibrinolysis.
3. **C**BC/blood smear:
   - Anemia
   - Fragmented and burr- and helmet-shaped red blood cells (schistocytes)
   - Thrombocytopenia



Treatment

The 1st 2 steps in the treatment of DIC are the most critical:

1. Aggressive treatment the trigger that caused DIC and
2. Restore normal homeostasis by correcting the shock (fluids/blood transfusion), acidosis, and hypoxia that usually complicate DIC.

3. Control of bleeding
   - Platelet infusions for marked thrombocytopenia .
   - Cryoprecipitate for hypofibrinogenemia.
   - Fresh frozen plasma for replacement of other coagulation factors and natural inhibitors.







4. Heparin use is controversial ; used only in:
   - Purpura fulminans
   - Chronic low grade DIC with giant heamangioma
   - Severe ischemic manifestations.

Prognosis

Generally bad , primarily dependent on the outcome of the treatment of the primary disease and prevention of end-organ damage

Evaluation of a bleeding child

Is there an emergency?

* Does the patient need immediate resuscitation or senior help?
* Is the patient about to exsanguinate (shock, coma)?
* Is there hypovolemia (postural hypotension, oliguria)?
* Is there CNS bleeding (meningism, CNS, and retinal signs)?
* Is there an underlying condition that escalates this apparently minor bleeding into an evolving catastrophe e.g .
  - GIT bleeding in a jaundiced child (ie, coagulation factors already depleted)
  - Bleeding in someone who is already anemic

History

Present history

- Age of onset: early in hereditary coagulaopathy hemophilia
- Site of bleeding: skin, mucosal or internal
- Duration of bleeding
- Outcome of bleeding
- Relation to trauma or viral respiratory infection
- Secondary cause eg drugs (warfarin), alcohol, liver disease, sepsis
- Severity of bleeding: is it spontaneous or traumatic

Past history: Drugs/ operations / transfusions

Family history: For similar cases or consanguinity

Menstrual history: For adolescent girls

Examination

* Purpura
  - Multiple, spontaneous hemorrhages in the skin & mucous membranes.
  - Range from pin point (petechiae) to several centimeters (ecchymosis)

* Coagulopathy
  - Hematomas in deep structures e.g. muscle hematomas
  - Extensive ecchymosis
  - Hemarthrosis

⌖ Associations
  - Organ system failures
  - Other systems involvement
  - Underlying systemic disease

Workup

1. Complete blood count / blood film
2. For purpura (See before)
3. For coagulopathy: (See before)
4. For an etiology /association e.g. sepsis workup in DIC, renal function in HUS,....

Leukemia

Definition

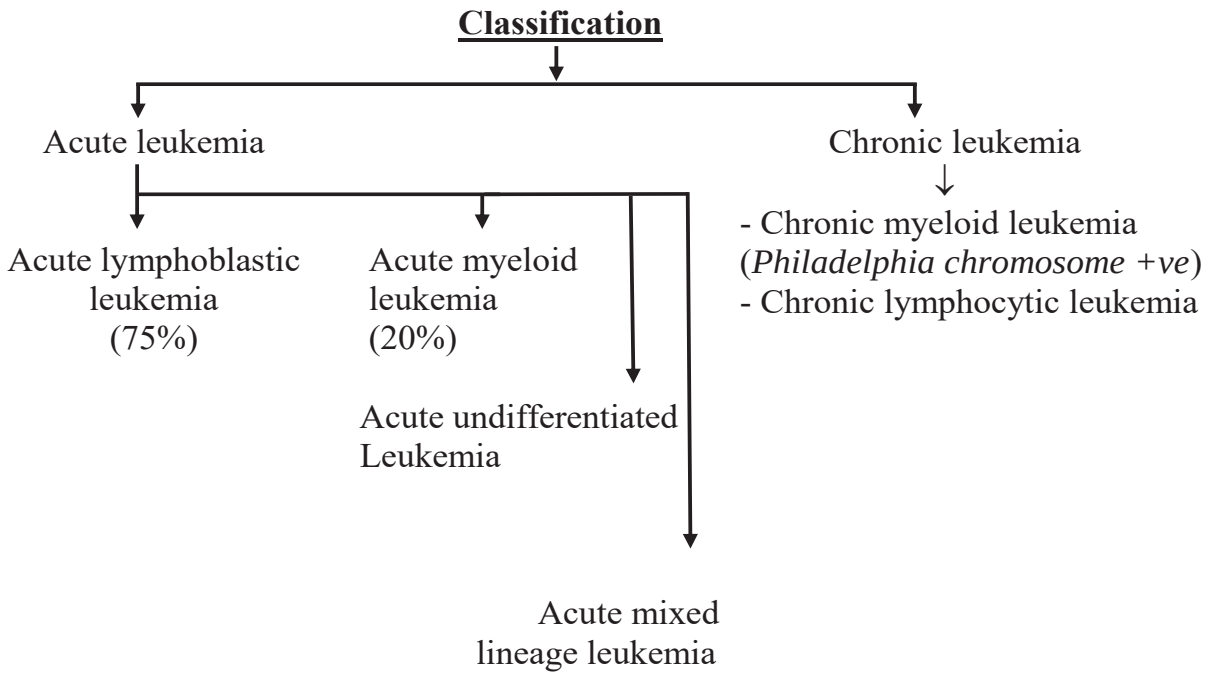
- Group of malignant diseases of hematopiotic cells in bone marrow
- Giving rise to uncontrolled clonal proliferation of cells
- With arrest of maturation at different stages
- With subsequent marrow failure.

Risk factors

The etiology of ALL is unknown, although several genetic and environmental factors are associated with childhood leukemia

1- Genetic predisposition
2- Chromosomal anomalies e.g. Down
3- Chromosomal breakage disorders e.g. Fanconi anemia, Bloom syndrome, ataxia telangiectasia
4- Immunodeficiency states
5- Ionizing irradiation which either diagnostic irradiation or therapeutic
6- Chemical carcinogens: Benzene, Pesticide, Alkylating agents.
7- Viral infections

Classification



Acute Lymphoblastic Leukemia (ALL)

Incidence

- Peak age = 2-5 years, more in boys than in girls in all ages.
- Among identical twins, the risk to the second twin if one develops leukemia is greater than that in the general population

Classifications

1. Morphologic: (French – American – British) FAB classification.

L1	L2	L3
85 %	14 %	1%
– Small cell – Small cytoplasm – Best prognosis	– Larger cell – Larger cytoplasm – Prominent nucleoli – Poor prognosis	– Large cell – Vacuolated cytoplasm – Worst prognosis.

2. Immunophenotyping

Classify ALL according to blast cell membrane & cytoplasmic markers.

1. Precursor B-cell ALL (CD10+ or common acute lymphoblastic leukemia antigen [CALLA] positive)
   - The commonest ALL: 85 %
   - Best prognosis
2. T cell type
   - Less common : 15 %
   - Poor prognosis
3. B cell type(Burkitt type)
   - least common : < 1%
   - Worst prognosis

Clinical picture

I. The initial presentation of ALL

Usually is nonspecific and relatively brief.

- Anorexia, fatigue, malaise, and irritability
- Intermittent, low-grade fever.
- Bone or, less often, joint pain, particularly in the lower extremities, may be present

II. Manifestations of bone marrow failure

- Anemia (erythriod cell infiltration, bleeding) → pallor
- Thrombocytopenia → purpura , ecchymosis , orificial bleeding.
- Granulocytopenia & granulocytic dysfunction → persistent infections (bacterial, viral, fungal and protozoal) and persisting pyrexia

III. Manifestations of organ infiltration

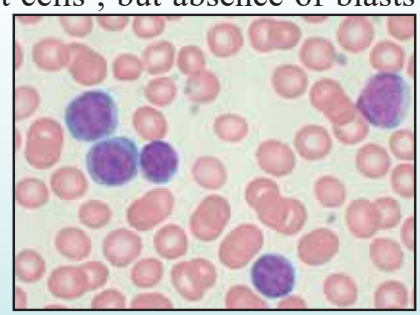
- Generalized lymphadenopathy
- Hepatosplenomegaly (HSM)
- Respiratory distress can occur in patients with an obstructive airway problem (wheezing) due to a large anterior mediastinal mass (e.g., in the thymus or nodes). This problem is most typically seen in adolescent boys with T-cell ALL.
- CNS leukemia: may manifest with:
  - Raised intracranial tension→ headache, vomiting, coma or
  - Focal lesion → fits, paresis, cranial nerve paralysis.
- Testis → painless swelling
- Kidneys → hematuria, renal failure
- Bone swellings (infiltration) / Arthritis→limping

Investigations

A. For diagnosis

1. CBC/Blood smear

- WBCs: → Count may be normal, low or high.
  → Blood smear show leukemic blast cells , but absence of blasts doesn't exclude leukemia.
- Platelets: →Thrombocytopenia
- RBCs: → Normocytic anemia.



2. Bone Marrow examination

- Increased cellularity; marrow is replaced by ≥ 25% leukemic blasts cells ; all are having malignant features
- Reduced erythroid and megakaryocytic cells

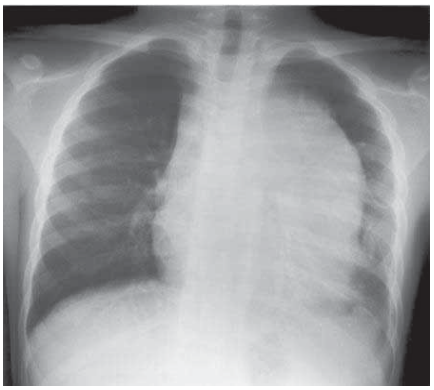


Blast cells must be subjected to:

1. Cytochemistry of blast cells show:
   - Absent peroxidase positive granules
   - Positive periodic acid schiff in clumps.
2. Immunophenotyping.
3. Cytogenetic studies
4. Immunoglobulin and T-cell receptor (TCR) gene re arrangement studies

B. To detect infilterations

- Lumbar puncture is done during workup ;approximately 5-10% show leukemic spread to the CSF
- Brain MRI scan
- Chest X-ray:
  Mediastinal mass often present in T -ALL



- Abdominal sonogram
- Bone survey for bone infiltration.

C. For planning therapy

- Biochemistry, serum uric acid, renal and liver biochemistry.
- Cardiac function; ECG and Echocardiogram

Differential diagnosis

1. Infections

Disease	Similarity to leukemia	Exclusion from leukemia
– **Typhoid** – **Brucellosis**	– Prolonged unexplained fever	– Blood culture /Serology – Absent blast cells
– **Infectious mononucleosis**	– Fever – Organomegaly – Purpura	– Absent blast cells – Monospot test – Positive IgM anti EBV
– **Perussis**	– Fever – Leukemoid reaction (WBCs count > 50.000/mm^3).	– No organomegaly – WBCs are mature lymphocytes – Normal blood smear & BM

2. Hematologic /oncologic disorders

	ITP	BM Aplasia	BM infiltration e.g. Neuroblastoma	Hypersplenism
Clinical	– Isolated purpura in clinically well child	– Purpura – Pallor – Pyrexia – No organomegaly	– Purpura – Pallor – Pyrexia – Organomegaly (Abdominal mass)	– Purpura – Pallor – Pyrexia – Splenomegaly
Blood film	– ↓Platelets	– Pancytopenia	– Pancytopenia	– Pancytopenia
Bone Marrow	– Megakaryocyte hyperplasia – No blasts	– Hypoplastic – No blasts	– Infiltrated – Abnormal cells	– Hyperplastic – No blasts

4. **Rheumatologic disease e.g.**
   Rheumatic fever/Juvenile rheumatoid arthritis/SLE
   - Cause fever and arthritis/limping
   - Other specific clinical features & Lab workup for lupus
   - Blood film and bone marrow exclude leukemia
5. **Acute myeloid leukemia**
   - Myeloblasts have: - Peroxidase +ve granules.
   - Positive PAS (diffuse reaction).
   - Immunophenotyping.
   - Cytogenetic studies

Prognosis

Prognostic factors

Clinical

- Age
- Sex
- Leukemic burden
- Response to therapy

Laboratory

- Initial WBC count
- Immunophenotyping
- Cytogentics

Risk group	Clinical features	Molecular/genetic features
Low risk	- Rapid response to induction therapy - Age 2-10 years - WCC < 50 x 10^9/L - Precursor B cell phenotype - No central nervous disease - No testicular disease - C-ALL or L1	- DNA index >1.6 - Polypliody - t(12;21) or some trisomies
High risk	- Induction failure ; > 4weeks to remission - Age > 10 years - High minimal residual disease	- Hypopliody - Presence of t(9;22) or t(4;11)

N.B: The single most important prognostic factor in ALL is the response to treatment

Treatment of ALL



A. Supportive

1. Psychological & nutritional support

2. Control infections by

- Education of patients, relatives and staff about hand washing and isolation facilities and mouth care
- Anti-microbial protocols in case of fever
- Granulocyte colony stimulating factor (G-CSF)

  Or
- Granulocyte Monocyte Colony Stimulating Factor (GM-CSF) for granulocytopenia; < 500 cell /mm^3
- Prophylactic treatment for *Pneumocystis jiroveci* pneumonia

3. Control bleeding by: Platelet transfusion if < 10-20 .000 / mm3

4. Control anemia by: Packed red cells if hemoglobin falls below 7 gm/dl.

5. Avoid tumor lysis syndrome

- High-count ALL (especially T-cell) and B-cell NHL (specifically Burkitt lymphoma) have the potential for bulky disease – a high cell mass, which will undergo lysis with treatment, resulting in the intracellular contents of potassium, phosphate and nuclear debris being released into the circulation
- Lymphoblast cells have four times the amount of phosphate compared to normal white cells
- Uric acid crystals and phosphate (precipitating out with calcium) crystals may cause acute renal failure and the following:
  - Rise in urate
  - Within 1-2 days; rise in phosphate with concomitant hypocalcaemia
  - Then rapid development of hyperkalemia

Prevention

- Treatment of hyperkalaemia
- Hyperhydration: Intravenous fluids without added potassium
- Alkalinaization prior to chemotherapy
- Uric acid-lowering agents : Urate oxidase (Rasburicase) is the drug of choice or Allopurinol
- Dialysis or hemofiltration

B. Specific treatment

1. Induction of remission

Aim	→ Eradicate malignant cells in bone marrow (i.e. remission).
Duration	→ 4 weeks
Drugs	– Vincristine weekly (IV) – Prednisone daily (Oral) – L-asparaginase (IM) – CNS therapy ; see below. – Daunomycin at weekly intervals for patients at higher risk
Criteria of remission: With this approach, 98% of patients are in remission	
* Clinical	→ No organomegaly nor detectable extramedullary disease
* CBC	→ No blast cells in peripheral blood nor in the CSF
* BM	→ Balst cells < 5%; none are having frank malignant features

2. CNS prophylaxis (CNS therapy)

Target patients	Patients who, at the time of diagnosis, have lymphoblasts in the CSF and either an elevated CSF leukocyte count or physical signs of CNS leukemia, such as cranial nerve palsy.
Aim	Prevent later CNS relapses→The likelihood of later CNS relapse is reduced to <5%.
Duration	→ 4 weeks
Drugs	– Intrathecal methotrexate. – A small percentage of patients with features that predict a high risk of CNS relapse may receive irradiation to the brain

3. Consolidation and intensification phase

- Aim: Maintain remission and avoid relapse
- Many regimens provide 14-28 wk of multiagent therapy
- Drugs and schedules used vary depending on the risk group of the patient.

4. Maintenance phase of therapy

Aim	→Maintain remission
Duration	→ Lasts for 2-3 yr, depending on the protocol used.
Drugs	– Oral 6 mercaptopurine daily . – Intravenous methotrexate weekly. – Usually with intermittent doses of vincristine and corticosteroid.

C. Bone marrow transplantation

Done in the first remission for high-risk patients such as those with the t(9;22) translocation known as the Philadelphia chromosome or extreme hypodiploidy

Complications

A. Complications of chemotherapy

- Bone marrow depression → Pancytopenia.
- Vincristine → Neuritis.
- Methotrexate → Renal toxicity.
- Adriamycin → Cardiomyopathy.

B. Relapse

Defined by any of the following

a- More than 25% lymphoblasts in bone marrow

or

b- Leukemic cell infiltration in the CNS or the testis.

Possible causes

- Persistence of leukemic cells in hidden sites (CNS, testis)
- Drug resistance

Decision

- Intensive chemotherapy and bone marrow transplantation
- Local irradiation

Minimal Residual Disease (MRD)

- Patients in clinical remission can have minimal residual disease (MRD) that can only be detected with specific molecular probes to translocations and other DNA markers contained in leukemic cells or specialized flow cytometry.
- MRD can be quantitative and can provide an estimate of the burden of leukemic cells present in the marrow.
- Higher levels of MRD present at the end of induction suggest a poorer prognosis and higher risk of subsequent relapse.
- MRD of 0.01-0.1% on the marrow on day 29 of induction is a significant risk factor for shorter event-free survival for all risk categories, when compared with patients with no MRD

Acute Myeloid Leukemia (AML)

Other names: Acute Myeloblastic Leukemia, Acute non-lymphoblastic Leukemia

Risk factors

- Chromosomal anomalies e.g. Down, Fanconi anemia, Bloom syndrome.
- Anti neoplastic drugs.

FAB classification

M0 : Acute myeloblastic leukemia without differentiation
M1 : Acute myeloblastic leukemia with minimal maturation
M2 : Acute myeloblastic leukemia with maturation
M3 : Acute Pro myeloblastic leukemia
M4 : Acute Myelomonocytic leukemia
M5 : Acute Monocytic leukemia
M6 : Acute Erythroleukemia
M7 : Acute Megakaryocytic leukemia

Clinical picture

- As ALL
- Signs and symptoms that are uncommon in ALL, including

1. Subcutaneous nodules or "blueberry muffin" lesions (especially in infants)
2. Chloromas (discrete granulocytic masses):
   - Typically are associated with the M2 with a t(8;21) translocation.
   - Common sites : retro orbital (→ proptosis),skin and epidural space.
3. Signs and laboratory findings of disseminated intravascular coagulation (especially indicative of M3).
4. Infiltration of the gingiva (especially in M4 and M5 subtypes)



"Blueberry muffin" lesions



Chloromas



Gingival hypertrophy

Investigation:

- As for ALL
- Bone marrow show myeloblasts which have:
  - Peroxidase positive granules.
  - Positive PAS (diffuse reaction)
  - Some have Auer Rods
- Immunophenotyping and cytogenetic studies of blast cells

Treatment

- **Unlike ALL ; therapy for AML is**
  - Of shorter duration
  - More intensive
  - Profound bone marrow aplasia (except for M3 subtype) is usually necessary.
  - Require more aggressive supportive care during the lengthy periods of marrow suppression
  - Overall outlook is less optimistic with survival rates reported of 50-70%
- **Induction chemotherapy includes**
  - Commonly used regimens use cytarabine and an anthracycline
  - In combination with other agents such as etoposide and/or thioguanine
- **Hematopoietic stem cell transplantation**
  - The treatment of choice for:
    - High risk cases
    - Refractory cases
    - Relapsed case
  - Matched-sibling bone marrow or stem cell transplantation *after remission* has been shown to achieve long-term disease-free survival in 60-70% of patients
- **Acute promyelocytic leukemia (FAB-M3)**

  Characterized by
  - A fusion gene involving the retinoic acid receptor t(15;17)
  - Very responsive to All-Trans-Retinoic Acid ; ATRA (Tretinoin) combined with anthracyclines and cytarabine.
  - ATRA allows maturation of the accumulated promyelocytes
  - The success of this therapy makes marrow transplantation in first remission unnecessary for patients with this disease.

Lymphoma

Definition

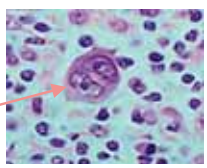
Malignant tumors of lymph nodes (LN) and extra nodal lymphoid tissue.

1- Hodgkin disease(HD): Mainly nodal disease

2- Non Hodgkin lymphoma(NHL): Mainly extra nodal disease

Incidence: 13 : million children

Hodgkin Disease



- B-lymphocyte malignancy.
- With characteristic *Reed-Sternberg cells*.
- Peak age → Bimodal; 15-30 and > 60 years ; rare before 5 years.

Histologic classification

Histologic type	Incidence	Criteria
1. Nodular sclerosing	50%	- Good prognosis - Females > males - Mediastinal mass common
2. Mixed cellularity	40%	- Present with more advanced disease
3. Lymphocyte predominance	10%	- Best prognosis
4. Lymphocyte depletion type	Very rare	- Present with disseminated disease - Poor prognosis

Clinical picture

1. Lymphadenopathy (Variable size)

Sites	Criteria	Effects
- Cervical (75 %)	- Painless	- Mediastinal syndrome: cough, dyspnea, dysphagia, & facial edema
- Supraclavicular	- Rubbery	
- Mediastinal	- Discrete	- Mesenteric: - Abdominal mass.
- Mesentric	- Later become matted and fixed	- May be intestinal obstruction





2. B Symptoms

- Unexplained intermittent fever (*Pel-Ebstein* fever).
- Night sweating requiring change of clothes.
- Unexplained loss of body weight > 10% in the last 6 months.

3. Other rare constitutional symptoms and extra nodal involvement

- Fatigue, pruritus, anorexia
- Splenomegaly with or without hepatomegaly.
- Bone marrow failure.

"Modified Ann – Arbor staging"

Based on radiograph and CT of chest, abdomen, and pelvis with or without bone marrow biopsy.

I	Confined to single lymph node region
II	Involvement of two or more nodal areas on the same side of the diaphragm
III	Involvement of nodes on both sides of the diaphragm
IV	Spread beyond the lymph nodes eg liver or bone marrow

Each stage is either

A : Absent of B symptoms

B : Presence of B symptoms

Investigations

A. For definitive diagnosis: Lymph node biopsy

B. For staging

- Chest and abdominal CT scans is the investigation of choice for staging
- Positron emission tomography (PET): Is increasingly being used for staging, assessment of response and direction of therapy

C. Indicators of disease activity

- ESR: is usually raised
- Serum lactate dehydrogenase; raised level is adverse prognostic factor

D. Others

- Chest X-ray may show mediastinal widening, with or without lung involvement
- Liver biochemistry is often abnormal, with or without liver involvement.
- Bone marrow aspirate and trephine biopsy seldom done but show involvement in patients with advanced disease; this is unusual at initial presentation (5%).

Management

- Treatment is determined largely by disease stage, presence or absence of B symptoms, and the presence of bulky nodal disease.
- Generally; treatment involves combined chemotherapy with or without low-dose involved-field radiation therapy
- The combination chemotherapy regimens in current use are based on:
  - **COPP** (cyclophosphamide, vincristine [Oncovin], procarbazine, and prednisone) **Or ABVD** (doxorubicin [Adriamycin], bleomycin, vinblastine, and dacarbazine),
  - For intermediate- and high-risk ; with the addition of prednisone, cyclophosphamide, and etoposide (**ABVE-PC** and **BEACOPP**) or **BAVD** (brentuximab vedotin, doxorubicin [Adriamycin], vincristine, dacarbazine) in various combinations
- Novel promising agents that target RS tumor cells:
  - Anti- CD20 antibody (rituximab)
  - Anti-CD30 agents (Brentuximab)
  - Anti-CD30 antibody linked to the antimitotic agent monomethyl auristatin E Brentuximab vedotin
  - Generated EBV-specific cytotoxic T lymphocytes (CTLs)

Non-Hodgkin Lymphoma

Incidence

- 3 times common than Hodgkin in children
- Peak age around 3 years
- ♂:♀ ratio = 3:1

Aetiology

The cause is unknown.

- Infective agents associated with development of NHL e.g. Ebstein Barr virus (Burkitt's lymphoma)
- Diseases known to predispose to NHLs
  - Congenital immunodeficiency states.
  - Acquired immunodeficiency e.g. HIV infection
  - Autoimmune disorders
  - Familial cancer syndromes

WHO Classification

1. B cell lymphoma e.g.
   - Burkitt's lymphoma/leukaemia
   - Precursor B lymphoblastic lymphoma/leukaemia
2. T/NK cell lymphomas e.g.
   - Precursor T cell lymphoblastic leukaemia/lymphoma
   - Diffuse large B cell lymphoma
   - Anaplastic lymphoma

Clinical features

i- Abdominal lymphoma (35%)

- Mainly B cell type; start in payer's patches, and mesenteric lymph nodes.
- Presentation:
  1- Rapidly enlarging abdominal mass with abdominal pain.
  2- May be: - Ascites.
  - Hepatosplenomegaly
  - Intussusception.

ii- Anterior mediastinal mass: (25%)

- Mainly T. cell type; starts in thymus.
- Presentation:
  1- Mediastinal syndrome (cough, dyspnea, dysphagia, face edema).
  2- May be: - Pleural effusion.
  - Pericardial effusion.

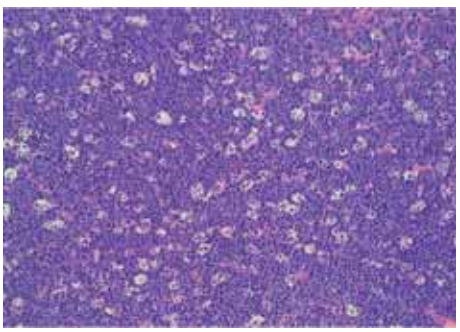
iii- Others

- Painless lymphadenopathy
- Bone marrow infiltration→ pancytopenia (occur in advanced lymphoma).
- CNS infiltration, oropharyngeal involvement, weight loss, bone pain,fever

Investigations

1. Biopsy & immunophenotyping & cytogenitic studies.
2. Cytological examination of ascitic fluid, pleural fluid, B.M
3. Chest X-ray, CT scans of chest, abdomen and pelvis.
4. PET and gallium scans help in staging.
5. Bone marrow aspirate and trephine biopsy are always performed
6. Other investigations as Hodgkin disease

N.B: Burkitt's lymphoma



Nature: B-cell type.
Histology: *Starry sky appearance*
Cytogenetics: May be (t 8; 14).

Types	Endemic	Sporadic
Distribution	African	World wide
Age	Children	Young adults
Site	Jaw, Ovary	Abdomen, Marrow
Association with Ebstein Barr virus	In > 97%.	In < 30%.

Treatment

1. Supportive; especially for life threatening complications:

Complication	Action
1. Upper airway obstruction by mediastinal mass	- Corticosteriods - Local radiation
2. Tumor lysis syndrome	- Rasburicase - Superhydration. - Na bicarbonate

2. Surgery: Only for small, easily, totally resectable tumors.

3. Chemotherapy:

- Protocols combining chemotherapy, Local radiation & CNS prophylaxis
- Protocols differ according to staging (localized or advanced) and Immunophenotyping. It includes:
  - High dose methotrexate (2-3 gm/m^2) I.V along with folonic acid
  - Rituximab
  - Targeted irradiation with Rituximab as a carrier
  - Allogeneic hematopoietic stem cell transplantation (HSCT)

Wilms' Tumor

(Nephroblastoma)

- Embryonic tumor of the developing kidney
- Very good overall prognosis – in excess of 90% 5-year survival rate
- 2nd common abdominal tumor
- 6% of malignancies in children around 3 years.
- Age: usually occurs in children < 5 ys .

Types

1. **Sporadic form** (common): Usually unilateral.
2. **Familial form**: Usually bilateral.
3. **Associations:**

A. Congenital anomalies: in 15 %;
- Genito-urinary anomalies.
- Congenital aniridia .
- Hemihypertrophy e.g. Beckwith–Wiedemann syndrome.

B. Syndromes:
- WAGR syndrome : Wilms', aniridia, genito-urinary anomalies, retardation.
- Denys-Drash syndrome: Wilms'tumor, renal disease, genital anomalies

Clinical picture

Presents in a well child with a painless (or minimal discomfort) abdominal mass, and/or haematuria and/or hypertension (independently or collectively)

1. **Abdominal mass** (the commonest presentation)**:**
   - Never cross midline (enlarged vertically)
   - Usually unilateral; Bilateral in 5-10%
   - Association: Microscopic hematuria.
2. **Hypertension :**
   - In 60 % of cases
   - Due to Renin-producing tumor or renal ischemia.
3. **Others**: - Polycythemia: occasional .
   - Metastasis e.g. Lungs (commonest) → cannon-ball lesions.

Investigations

i. Detect the tumor:

1. Abdominal ultrasonography.
2. Abdominal CT: - Exclude neuroblastoma.
   - Evaluate contralateral kidney.
   - Evaluate metastasis
   - CT guided biopsy.
3. Others: - Urine analysis for hematuria
   - Renal functions tests

Staging: Based on operative findings

Stage	Tumor extent
I	– Limited to kidney & is completely excised
II	– Extends beyond the kidney but is completely excised
III	– Residual abdominal extension after surgery
IV	– Hematogenous spread to lungs, liver, bone,…
V	– Bilateral renal involvement at time of diagnosis

Differential diagnosis: Causes of abdominal mass

1. Neuroblastoma.
2. Hydronephrosis.
3. Renal cyst.
4. Others: Hypernephroma, Clear cell sarcoma
5. Mesoblastic nephroma

Treatment

1. Surgical: Radical nephrectomy
2. Chemotherapy
3. Radiotherapy

Neuroblastoma

Characters

- Aggressive embryonal tumor of the autonomic system, originating from neural crest-derived sympathetic nerve cells in the adrenal gland & sympathetic ganglia (Abdomen // Thoracic // Cervical)
- Most common cancer diagnosed at age <1 year
- It accounts 8-10% of all childhood malignancies.
- Has the biological potential to involute and resolve spontaneously or behave aggressively with widespread metastases/organ invasion

Staging

Stage	
I	- The tumor is confined to site of origin.
II	- The tumor extend beyond site of origin but not cross midline
III	- Localized tumor with contralateral regional lymph node involvement.
IV	- Metastatic disease .
IVs	- Stage I or II + involvement of skin, liver and/or BM .

Clinical picture

Presenting symptoms are extremely variable

1. Abdominal mass (commonest)

- Origin → adrenal medulla or abdominal sympathetic chain.
- Hard with irregular surface
- Located in upper quadrant of abdomen
- May cross midline as it enlarges horizontally
- May be ascitis , hepatomegaly

2. Mediastinal or cervical mass

* Origin: Thoracic or cervical sympathetic chain.
* Manifestations:
  - Mediastinal syndrome: see before
  - Horner syndrome (unilateral ptosis, enophthalmos, meiosis & anhydrosis).

3. Spinal cord compression

* Origin → sympathetic chain.
* Manifestations: - Localized back pain
  - Paraplegia.
  - Sphincteric dysfunction.

4. Metastatic neuroblastoma

- BM → pancytopenia.
- Orbit→ Proptosis with ecchymotic eye lids → *Raccoon like appearance*.



5. Syndromes associated with Neuroblastoma

Eponym	Features
Excessive catecholamine	– Intermittent attacks of sweating, palpitation, hypertension , flushing, polyuria & polydipsia
Horner syndrome	– Unilateral ptosis, myosis, and anhidrosis associated with a thoracic or cervical primary tumor. – Symptoms do not resolve with tumor resection.
Opsoclonus-myoclonus-ataxia syndrome	– Myoclonic jerking and random eye movement with or without cerebellar ataxia. – Often associated with a biologically favorable and differentiated tumor. – The condition is likely immune mediated, may not resolve with tumor removal – Often exhibits progressive neuropsychological sequelae.
Kerner-Morrison syndrome	– Intractable secretory diarrhea and hypokalemia due to tumor secretion of vasointestinal peptides. – Tumors are generally biologically favorable.

Diagnosis

1. Detect origin of 1^{ry} site (adrenal, sympathetic chain)
   - Abdominal ultrasonography & CT.
   - Chest x ray & CT.
   - Biopsy
2. Urinary catecholamines: Vallinyl mandelic acid & Homovanillic acid

Treatment

- The usual treatment for low-risk neuroblastoma is surgery for stages 1 and 2 and observation for stage 4S with cure rates generally >90%
- Stage 3 : Surgery , radiotherapy and chemotherapy .
- Stage 4 : Radiotherapy or chemotherapy
- Therapies currently under investigation : radiolabeled targeted agents (e.g. MIBG), monoclonal antibodies (anti–tumor-associated GD2) combined with growth factors (GM-CSF), and antitumor vaccines

Lymphadenopathy

Causes

1. Nonspecific reactive hyperplasia

2. Infection

A. Bacterial: Staphylococcus, streptococcus, tuberculosis, atypical mycobacteria, Bartonella henselae(cat scratch disease) brucellosis

B. Viral: Epstein–Barr virus, cytomegalovirus, rubella, rubeola

C. Protozoal: Toxoplasmosis, malaria

D. Spirochetal: Syphilis, rickettsia

E. Fungal: Histoplasmosis, cryptococcus, aspergillosis

3. Connective tissue disorders

A. Rheumatoid arthritis

B. Systemic lupus erythematosus

4. Hypersensitivity states e.g. Serum sickness

5. Lymphoproliferative disorders

6. Neoplastic diseases

A. Hodgkin and non-Hodgkin lymphomas

B. Leukemia

C. Metastatic disease from solid tumors: neuroblastoma, nasopharyngeal carcinoma,

D. Histiocytosis

7. Storage diseases

A. Niemann–Pick disease

B. Gaucher disease

8. Immunodeficiency states

9. Miscellaneous causes

A. Kawasaki disease (mucocutaneous lymph node syndrome)

C. Sarcoidosis

E. Hyperthyroidism

Diagnosis

1. Thorough history of

- Infection
- Contact with rodents or cats
- Systemic complaints

2. Careful examination of the lymphadenopathy including

* All the lymph-node-bearing areas should be carefully examined
* Size, mobility, warmth, erythema, fluctuation & location.

* Texture of lymph nodes:
    - Consistency: soft, firm, rubbery, hard
    - Discrete or matted
    - Tender (in inflammation or rapid malignancy) or not

3. Physical examination for

* Draining area in localized adenopathy e.g.
    - Ear, nose and throat in cervical adenopathy
    - Occipital adenopathy in infections of scalp
    - Preauricular adenopathy in conjunctivitis and cat scratch disease
* Evidence of hematologic disease, such as hepatosplenomegaly and petechiae

4. Workup

* Blood count
* Erythrocyte sedimentation rate (ESR)
* Skin testing for tuberculosis
* Bacteriologic culture of regional lesions (e.g., throat)
* Specific serologic tests for e.g. Epstein–Barr virus (EBV)
* Chest radiograph ± CT scan
* Abdominal sonogram ± CT
* LN Ultrasonography
* Lymph node aspiration and culture
* Bone marrow examination if leukemia or lymphoma is suspected
* Lymph node biopsy

Indications

- Initial physical examination and history suggest malignancy
- Lymph node size is greater than 2.5 cm in absence of signs of infection
- Lymph node persists or enlarges
- Appropriate antibiotics fail to shrink node within 2 weeks
- Supraclavicular adenopathy.

Precautions

- Upper cervical and inguinal areas should be avoided
- The largest node should be biopsied, not the most accessible one.
- The node should be removed intact with the capsule, not piecemeal

Splenomegaly

* Spleen may normally be felt in children up to 3 or 4 years of age
* At an older age, the spleen tip is generally not palpable below the costal margin and a palpable spleen usually indicates splenic enlargement two to three times its normal size.

Causes of splenomegaly

I. Infectious splenomegaly

A. Bacterial: acute and chronic systemic infection, subacute bacterial endocarditis, typhoid fever,and miliary tuberculosis

B. Viral: infectious mononucleosis (Epstein–Barr virus), cytomegalovirus, hepatitis viruses

C. Spirochetal: Syphilis

E. Protozoal: malaria, toxoplasmosis, leishmaniasis, schistosomiasis, trypanosomiasis

F. Fungal infections

II. Hematologic disorders

A. Hemolytic anemias

B. Extramedullary hematopoiesis as in osteopetrosis and myelofibrosis

C. Myeloproliferative disorders (e.g., polycythemia vera)

III. Infiltrative splenomegaly

A. Nonmalignant

1. Langerhans cell histiocytosis
2. Storage diseases such as Gaucher disease, Niemann–Pick disease, amyloidosis and sarcoidosis

B. Malignant

1. Leukemia
2. Lymphoma

IV. Congestive splenomegaly: portal hypertension

B. Connective tissue disorders e.g.,

Systemic lupus erythematosus

Rheumatoid arthritis

VI. Primary splenic disorders

A. Cysts

B. Benign tumors (e.g., hemangioma, lymphangioma)

C. Hemorrhage in spleen (e.g., subcapsular hematoma)



Pulmonology

Acute Pharyngitis

It include acute tonsillitis, pharyngitis or tonsillopharyngitis

Causes

Viral or Bacterial (group A β hemolytic streptococci is the commonest.).

Complaint
- Fever, anorexia and malaise
- Sore throat
- Dysphagia



- Red , congested throat
- Inflamed tonsils with white or yellow exudates
- Enlarged tender lymph nodes on the front of the neck
- Associations
  - Conjunctivitis (Adeno virus)
  - Minute vesicles and ulcers (Coxachie virus)

Complications

- As that of scarlet fever +
- Mesenteric adenitis (→ abdominal pain).

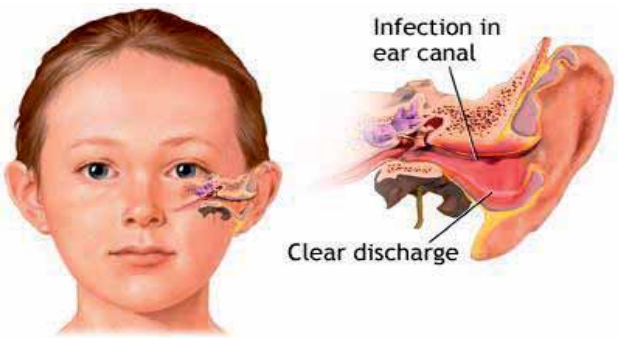
Treatment

* **S**ymptomatic for fever.
* **S**pecific: e.g. **10** days course of antibiotic (**5** days for Zithromax)
  - Penicillin V
  - Amoxicillin
  - Cephalosporins
  - Zithromax
  - Clarithromycin

* **S**urgical:
- A tonsillectomy, with or without adenoidectomy
- Indications:
  1. The most common indication for adenotonsillectomy is adenotonsillar hypertrophy associated with obstructive sleep apnea
  2. Recurrent tonsillitis defined as :
     - Seven or more documented infections in 1 year
     - Five per year for 2 years
     - Or three per year for 3 years
  3. Recurrent peritonsillar abscess
  4. Multiple antibiotic allergies.

Otitis Externa
"Swimmer's ear"

- Cellulitis of the soft tissues of the external auditory canal
- Risk factors : trauma ,humidity, heat, and moisture in the ear



Essentials to diagnosis

- Edema and erythema of the external auditory canal with debris or thick, purulent discharge.
- Severe ear pain, worsened by manipulation of the pinna.
- Periauricular and cervical lymphadenopathy may be present



Management

- Pain control
- Removal of debris from the canal
- Topical antimicrobial therapy, Fluoroquinolone eardrops are the first-line
- Avoidance of causative factors

Acute Otitis Media (AOM)

Risk factors

Eustachian tube obstruction by adenoids or edema in upper respiratory infection
Others: Impaired host immune defenses, bottle feeding, genetic susceptibility

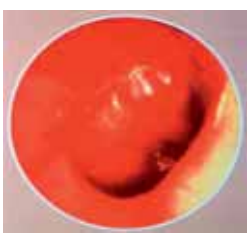
Causes

1. Viral; AOM is a known complication of bronchiolitis
2. Bacterial: mainly H. influenza , pneumococci, moraxilla catarrhalis

Clinical picture



- Fever
- Earache (irritability , rubbing the ears in infants)
- Otoscopic examination:
  - Drum is congested, bulging
  - Middle ear effusion
  - Drum may be perforated ± discharge.



Acute OM



Severe acute OM



Draining acute OM



Resolving acute OM

- Complications:
  - Mastioditis: tender swelling behind the ear
  - Chronic ear infection: draining ears for 14 days or more

Treatment

* **S**ymptomatic for pain & fever: ibuprofen or acetaminophen
* **S**pecific
  - Antibiotic or observation?
    - For infants younger than 6 months→ antibiotics are always recommended on the first visit, regardless of diagnostic certainty
    - For children ≥ 2 years with uncomplicated otitis media without otorrhea→ optional 48 hours of observation
  - Antibiotics
    - Amoxicillin-clavulanate enhanced strength ;ES (14:1 ratio of amoxicillin: clavulanate), with amoxicillin dose 90 mg/kg/d for 10 days
    - Alternatives: Ceftriaxone (injections for 3 days), Cefdinir, or Cefpodoxime →Then, according to culture and sensitivity
  - Surgical :Tympanocentesis & drainage ± Tympanostomy tubes
  - Patients with tympanostomy tubes with acute otorrhea → ototopical antibiotics (fluoroquinolone eardrops) are first-line therapy

Acute Sinusitis

- The maxillary and ethmoid sinuses are most commonly involved. These sinuses are present at birth.
- Major risk factors: Upper respiratory tract infections and immunodeficiency

Causes

- As in otitis media
- Mixed infections

Clinical picture

* Fever and headache
* Purulent or mucopurulent nasal discharge & post nansal discharge⇒ cough
* Others:
  - Nasal obstruction
  - Halitosis (fetid breath odor)
  - Diminished smell
  - Periorbital edema



Investigations

- Culture and sensitivity of sinus aspirate
- Trans illumination test ⇒ opaque sinus
- Plain X ray skull
- CT skull

Treatment

- **S**ymptomatic for pain & fever (paracetamol)
- **S**pecific:

1. Antibiotics for a minimum of 10 days or 7 days after resolution of symptoms:
   - High dose amoxicillin or amoxicillin/clavulanate
   - Alternatives
     - Ceftriaxone
     - Cefdinir , Cefpodoxime, Cefixime, Cefuroxime
   - Neither Zithromax nor Cotrimoxazole are recommended
2. Saline nasal washes or nasal sprays can help to liquefy secretions and act as a mild vasoconstrictor
3. The use of decongestants, antihistamines, mucolytics, and intranasal corticosteroids has not been adequately studied in children and is not recommended for the treatment of acute uncomplicated bacterial sinusitis

- **S**urgical = Sinuscopic sinus surgery for chronic cases

(Nelson Textbook of Pediatric)

Stridor

Definition

Harsh, continuous inspiratory sound due to variable obstruction in upper airways (larynx and trachea); may be associated with hoarseness of voice and respiratory distress

Causes

Acute

Infectious	Non infectious
* Viral: – Laryngeotracheobronchitis – Acute laryngitis. – Spasmodic laryngitis. * Bacterial: – Acute epiglottitis. – Acute tracheitis (staph. aureus). – Diphteritic laryngitis.	– Laryngeal foreign body. – Laryngeospasm(e.g. tetany). – Laryngeal edema(e.g. allergic) – Laryngeal compression.

Chronic

Congenital	Acquired
▪ Laryngeomalacia ▪ Laryngeal web or cyst ▪ Tracheomalacia ▪ Congenital vascular ring	▪ Laryngeal – Stenosis – Tumors – Paralysis ▪ Tracheal stenosis

Severity of Stridor / Croup

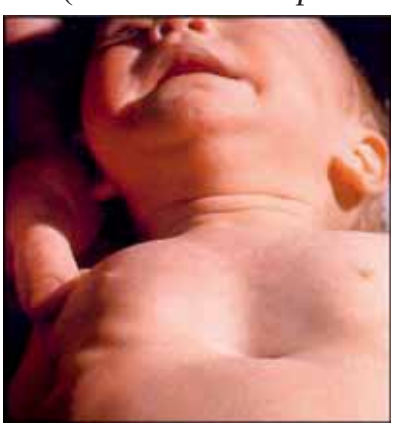
	Mild	Moderate	Severe
Stridor	±	+	++
Sternal tug	-	+	++
Recession	-	+	++
Accessory muscles	-	+	++
Nasal flare	-	+	++
Cyanosis	-	-	+
Drooling	-	-	+
Air entry	Normal	Reduced	Poor
Hydration	Normal	Normal/reduced	Reduced
If the child does not object:			
Saturation	Normal	Normal/reduced	Reduced
Heart rate	Normal	Raised	Raised (bradycardia is pre-terminal event)

(Oxford Paediatric Emergency Medicine)

Acute Infectious Stridor

Laryngeotracheobronchitis (Croup)

- Affects children 6 months - 6 years in the fall and early winter
- Cause: Viral
  - Para-influenza types 1, 3
  - Others: RSV, influenza, adenovirus, corona virus
- Presentation
  - Upper respiratory catarrh (Rhinitis, low grade fever)
  - Croupy ,barking, cough
  - Hoarseness of voice
  - Absence of drooling and toxic appearance
- Croup severity:
  Can be severe with inspiratory and expiratory stridor and respiratory distress (*substernal & suprasternal* retractions)





- Neck X ray
  Steeple sign: Sub glottic narrowing in antero posterior view
- Complication : Rarely; secondary bacterial infection → Bacterial tracheitis

Differential diagnosis

- **Acute laryngitis**
  - Less severe croup (inspiratory stridor)
  - No respiratory distress
- **Spasmodic laryngitis**
  - Viral but may be allergy or psychogenic (afebrile illness)
  - Occurs at midnight
  - Less severe
  - Recurrence is common
- **Acute epiglottitis and acute tracheitis** : see later

Management

- Most cases respond to home treatment
- **Indication for hospitalization**
  - Progressive stridor
  - Severe stridor at rest
  - Respiratory distress; hypoxia, cyanosis, depressed mental status
  - Poor oral intake
- **Mild croup** (barking cough and no stridor at rest)
  - Oral hydration
  - Minimal handling
  - Mist therapy (no evidence to support its use)
- **Moderate to severe stridor at rest**

1. Oxygen for patients with oxygen desaturation
2. Nebulized epinephrine
   - Racemic adrenaline nebulizer (0.25-0.5 ml in 3ml saline),
   - L- adernaline 5 ml 1:1000 solution is equally effective

   Value

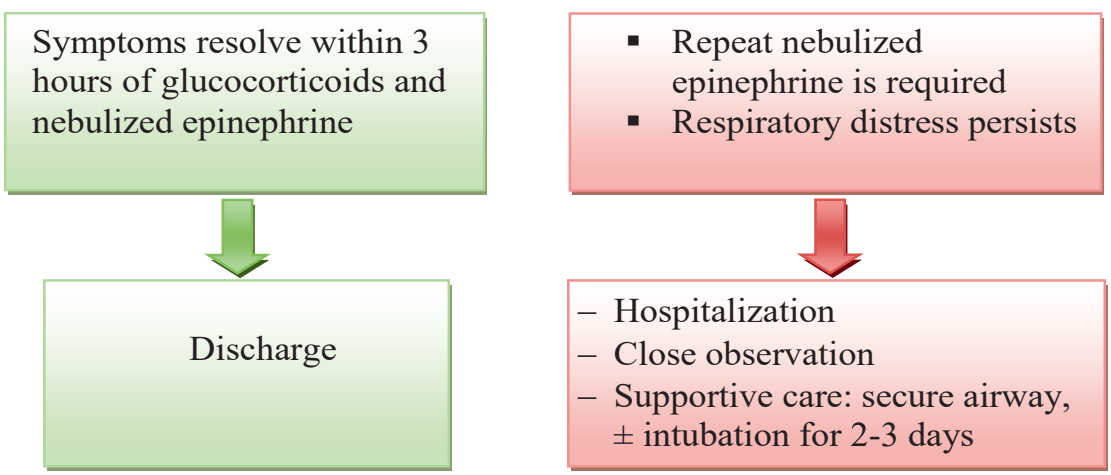
   Reduce need for intubation for moderate to severe stridor at rest
3. Oral corticosteroids
   - Dexamethasone: 0.6 mg/kg Oral or intramuscular as one dose. Lower dexamethasone dose (0.15 mg/kg) is equally effective
   - Inhaled budesonide (2–4 mg)

   Value

   Improves symptoms even in mild stridor

   Reduce need for intubation for moderate to severe stridor at rest

Outcome



(Nelson textbook of pediatrics)

Acute Epiglottitis (Supraglottitis)

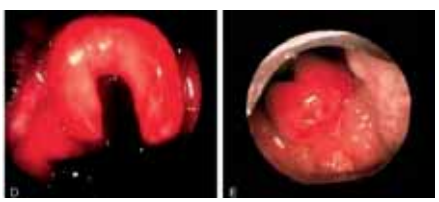
- Infection of the epiglottis by *Hemophilus influenza type B(in pre vaccine era)*
- *Strept pyogenes, Strept pneumoniae,* nontypeable *H. influenzae*, & *Staph aureus,* represent a larger portion of pediatric cases of epiglottitis in vaccinated children

Clinical picture

- Peak age = 2-7 years (now commoner in adults with sore throat)
- Toxic child with high fever
- Drooling of saliva (severe dysphagia)
- The child is severely exhausted :
  - Voice is muffled.
  - Stridor is mild.
  - Little or no cough
  - The child prefer upright posture and neck is hyperextended in an attempt to maintain the airway



- Laryngeoscopic examination shows large "*cherry red*" swollen epiglottis but this procedure and any minor procedure may precipitate complete airway obstruction.



Management

Medical emergency, once suspected, the patient must be admitted to the PICU

- Secure the Airway before any maneuver:
  - Endotracheal tube (or less often tracheostomy) is indicated ,regardless degree of respiratory distress , placed either in an operating room or ICU
  - The artificial airway is kept in place for 2-3 days
  - O_2 inhalation as needed



- Blood culture and, if possible, epiglottic surface culture should be done.
- Antibiotics:
  - Start parenteral Ceftriaxone or Cefotaxime or Meropenem pending result of culture & sensitivity
  - Continue antibiotics for at least 10 days.
- Lateral *X-ray* of the neck if done (after securing airway) may show swollen epiglottis (*Thumb sign*)



N.B: Household contacts < 4 years with incomplete HiB immunization or immunocompromised require Rifampin prophylaxis (20 mg/kg orally once a day maximum dose 600 mg for 4 days) **(Nelson textbook of pediatrics)**

Bacterial Tracheitis

- Acute bacterial infection of the upper airway that is potentially life threatening.
- *Staphylococcus aureus* is the commonest cause
- Often follows viral laryngotracheobronchitis
- Commoner than acute epiglottitis basically because of introduction of *Hib* vaccine in most vaccinations protocols √√

Clinical picture

- The early clinical picture is similar to that of viral croup
- Instead of gradual improvement, patients develop higher fever, toxicity, and progressive or intermittent severe upper airway obstruction that is unresponsive to standard croup therapy
- Differentiated from epiglottitis by:
  1. Preceded by viral prodrome
  2. No posture preference; the patient can lie flat
  3. No dysphagia or drooling!!
  4. Lateral neck radiographs show a normal epiglottis but severe subglottic narrowing; irregular tracheal border (absent thumb sign)
  5. During endotracheal intubation/ Bronchoscopy: Normal epiglottis and the presence of deep red mucosa and copious purulent tracheal secretions below the cords confirm the diagnosis
  6. Although cultures of the tracheal secretions are frequently positive, blood cultures are almost always negative
  7. Despite the severity of this illness, the reported mortality rate is very low if it is recognized and treated promptly

Treatment

- Patients with suspected bacterial tracheitis will require
  - Direct visualization of the airway in a controlled environment
  - Debridement of the airway
  - Most patients will be intubated because the incidence of respiratory arrest or progressive respiratory failure and respiratory arrest is high
  - Thick secretions persist for several days, usually resulting in longer periods of intubation for bacterial tracheitis than for epiglottitis or croup
- Antistaphylococcal agents: vancomycin or naficillin or oxacillin
- Supportive care in ICU including supplemental oxygen ,suctioning

Lower Respiratory Diseases

Chest Examination

	Pneumonia	Bronchopneumonia	Pleural effusion	Pneumothorax	Hydropneumothorax	Collapse
Inspection						
- Movement	Decreased	Decreased bilateral	Decreased			Decreased
- Shape	Normal	Normal	Bulge			Retraction
Palpation						
- Tracheal shift	Central	Central	Shifted to opposite side			To same side
- Tactile vocal fremitus	Increased	? Normal	Decreased			Decreased
Percussion						
- Note	Impaired note	? Impaired note	Stony dull	Hyperresonance	Shifting dullness	Dull
- Topography	Lobar	Bilateral	Rising to axilla	Allover the side	Transverse upper border	Lobar
Auscultation						
- Breath sounds	Diminished bronchial	? Normal vesicular	Markedly diminished vesicular			
- Adventitious sounds	Crepitations	Bilateral wheezes , Crepitations				
- Vocal resonance	Increased Bronchophony	May be normal	Decreased			
Special signs	--	--	Aegophony	Coin test	Succussion splash	--

Pneumonia

- Pneumonia is an infection of the lower respiratory tract that involves the airways and parenchyma with consolidation of the alveolar spaces
- Pneumonitis is a general term for lung inflammation that may or may not be associated with consolidation

Anatomic classification

- Lobar pneumonia: pneumonia of one or more lobes
- Bronchopneumonia: scattered bilateral inflammation both lungs
- Interstitial pneumonia: bilateral perihilar pulmonary inflammation

Etiologic classification

Category	Etiologic agents
Bacterial	– Gram-positive: e.g. Strept Pneumonae, group B and A streptococci ,Staphylococcus aureus – Gram-negative: e.g. H.influenzae , Legionella, Klebseilla
Viral	– Respiratory syncytial virus (RSV) parainflenza , influenza , adenovirus, Human metapneumovirus, Corona virus
Atypical	– Mycoplasma pneumoniae , *Chlamydophila pneumoniae* – Chlamydia trachomatis (in infants)
Mycobacterial	– Tuberculosis and atypical mycobacteria
Aspiration	– Oral anaerobic flora, with or without aerobes
Allergic	– Esinophilic pneumonia (Loffler's syndrome)
Rickettsial	– Coxiella Burnetii
Opportunistics in immunocompromised	– Fungal e.g. Aspergillus , histoplasma , cryptococcus, candida – Protozoal ; Pneumocystis jiroveci (carinii) – Bacterila; Klebsiella, and proteus

Symptoms

Onset is variable from acute, sub-acute or gradual

- General
  - Fever, malaise , toxemia (worst in bronchopneumonia)
  - May be abdominal pain: Referred from lower lobe pneumonia
- Chest
  - Cough (dry then productive)
  - Dyspnea and grunting

Signs

Respiratory distress

- Tachypnea is the most consistent clinical manifestation of pneumonia, nasal flaring, retractions and grunting
- Cyanosis and lethargy in severe infection specially in infants

Chest examination

- Pneumonia (See table previous page for pneumonia ± effusion)
- Bronchopneumonia (See table previous page)
- Interstitial pneumonia:→ Minimal chest findings.
  → Prolonged expiration & wheezes are common

Viral or bacterial pneumonia?

1. Clinical
   Large pleural effusion, lobar consolidation, and a high fever at the onset of the illness are suggestive of a bacterial etiology
2. Investigations : See later

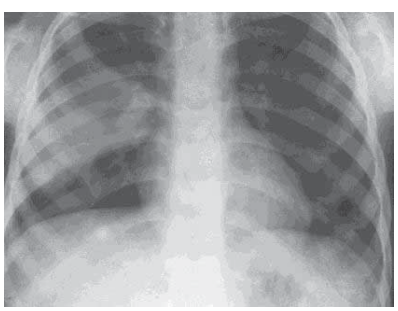
Investigations

A. Radiological

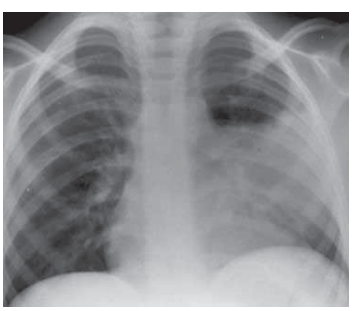
1. Chest X-ray findings:

A. Lobar pneumonia

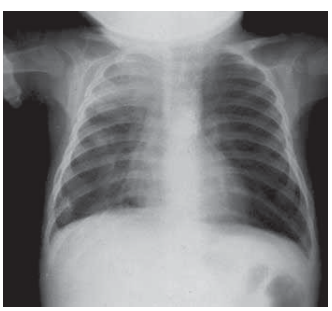
- Homogenous opacity in one or more lobes
- With clear costopherinic angle (differentiate it from effusion)
- Usually bacterial



Right sided middle lobe pneumonia



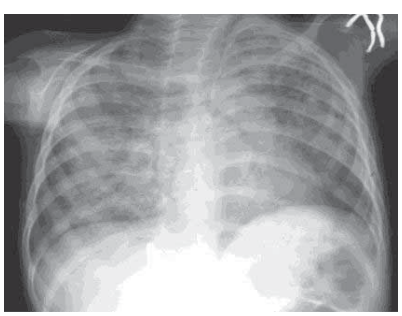
Left sided lower lobe pneumonia



Right sided upper lobe pneumonia

B. Bronchopneumonia

- Scattered opacities in both lungs
- Viral or bacterial

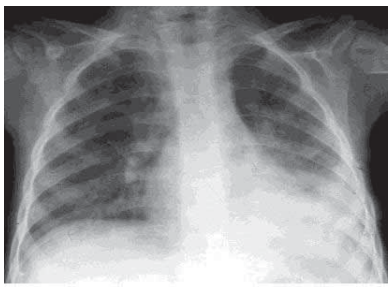


C. Interstitial pneumonia

- Scattered bilateral interstitial infiltrates and peribronchial cuffing
- Hyperinflation, and atelectasis
- Seen in viral bronchopneumonia and atypical pneumonia

D. Complications

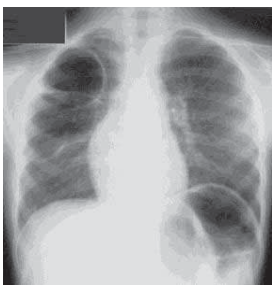
Effusion, abscess, or pneumatoceles (single or multiple, thin-walled, air-filled, cystlike cavities) may indicate S. aureus, gram-negative, or complicated pneumococcal pneumonia.



Effusion (Left)



Lung abscess (Right)



Pneumatoceles (Right)

Meningitis, suppurative arthritis, and osteomyelitis are rare complications of hematologic spread of pneumococcal or H. influenzae type b infection

2. Ultrasonography:

- Highly sensitive and specific in diagnosing pneumonia by determining lung consolidations and air bronchograms or effusions
- Differentiate simple effusion and empyema
- Guide thoracentesis of a loculated effusion



3. Contrast CT scan, CT or ultrasonography guided lung biopsy:

Reserved for complicated cases/ rare pneumonias

B. <u>Laboratory</u>

1- **WBC count**

- In viral pneumonia usually not higher than 20,000/mm^3, with a lymphocyte predominance
- In bacterial pneumonia, in the range of 15,000-40,000/mm^3, and a predominance of granulocytes
- Mild eosinophilia is characteristic of infant C. trachomatis pneumonia

2- Acute phase reactants: High ESR, positive C-reactive protein and Procalcitonin *usually* suggest bacterial rather than viral pneumonia

3- **Isolation of an organism**

<u>Indicated for</u>

- Ill cases that require hospitalization
- Immunocompromised patients
- Patients with recurrent pneumonia
- Pneumonia unresponsive to empirical therapy

Workup

- Blood cultures are positive in 10% to 20% of bacterial pneumonia
- Pleural fluid culture
- Lung tracheobronchial secretions culture
- Invasive: Bronchoscopy with bronchoalveolar lavage ,brush mucosal biopsy, needle aspiration of the lung, and open lung biopsy
- Specific testing e.g.
  - M. tuberculosis : tuberculin skin test, serum interferon-gamma release assay, or analysis of sputum or gastric aspirates by culture, antigen detection, or PCR
  - Detect the virus or viral antigens by DNA or RNA tests

4- Serology: for rising antibody titers:

- **C**old agglutinins in 50% of mycoplasma pneumonia (non specific test).
- ASO titer in streptococcal pneumonia

Complications

Respiratory	Systemic
– Pleural effusion – Empyema with or without bronchopleural fistula and pyopneumothorax – Lung abscess – Pneumatoceles – Unresolved pneumonia *These complications are more common with Staph and Klebseilla pneumonia*	– Meningismus especially with right upper lobe *pneumococcal* pneumonia – Heart failure – Distant infections e.g. Septicemia, meningitis, pericarditis – Paralytic ileus

Differential Diagnosis of pneumonia

1. Viral pneumonia

The commonest cause in pre-school children with peak at 2- 3 years

Causes: RSV, parainfluenza (1, 2, and 3) viruses, influenza (A and B) viruses, human metapneumovirus, and adenovirus

Clinically: - Preceding upper respiratory tract infection for several days.
- Fever & respiratory distress ⇒ milder than bacterial pneumonia.
- May be widespread wheezes and crepitations.

Diagnosis

- CXR: Bilateral peri hilar infiltrates (bronchopneumonia **or** interstitial pneumonia) ± Hyperinflation.
  Pleural effusions, pneumatoceles, abscesses, lobar consolidation, and "round" pneumonias are generally inconsistent with viral disease
- CBC : normal or mildly elevated WBCs with predominant lymphocytes
- Detect the virus or viral antigens by DNA or RNA tests

2. Bacterial pneumonia

	Pneumococcal*	Streptococcal	Staphylococcal	H. Influenzae*	Klebsiella
Age	Commonest bacterial. pneumonia in children	Peak age 3-5 yr.	Peak age below 1 year History : staph skin infection	Peak age below 3 yr	more in immune deficient - Has high mortality
C/P	- Moderate - Moderate fever - Usually lobar - Bronchopneumia is commoner in young infants	- Severe with extreme prostration - High fever - Bronchopneumonia with large pleural effusion	- Severe - High fever - May be bronchopneumonia or lobar or hemithorax - Complications (abscess, empyema , pneumatoceles, and pneumothorax)	- Insidious onset - Prolonged course over weeks - Usually lobar; involving two or more lobes	- Severe , fulminant - High fever with copious purulent secretions - Usually lobar - Complications as Staph.

* High incidence of penicillin resistance so treat with high doses of amoxicillin (80-90 mg/kg/24 hr) or cefuroxime or 3rd generation cephalosporin

3. Mycoplasma pneumonia (Primary Atypical Pneumonia): Common in school age (5-15 yr)

Clinically

- Severe nonproductive **cough** without significant respiratory distress
- Pharyngitis is common
- Minimal physical signs (walking pneumonia)
- May be chest wheezes and inspiratory crepitations

Diagnosis is mainly clinical.

- Blood: CBC is usually normal, Cold agglutinins may be detected
- Chest X-ray show:
  - Scattered bilarteral perihilar pulmonary infiltrates
  - Rarely: Lobar pneumonia ± effusion.

Treatment of pneumonia

i. Supportive

- Bed rest, humidified O_2 inhalation ± restricted I.V. fluids
- Symptomatic treatment e.g. antipyretics for fever
- Treatment of complications e.g. Heart failure.
- Aspiration /drainage for effusion or empyema
- Oral zinc (10- 20 mg/day) is recommended *add-on* in developing countries

ii. Specific treatment

1. Suspected bacterial pneumonia: *Antibiotics*

- As suggested by clinical picture & chest X-ray
- Based on the presumptive cause and the age
- Antibiotic combination if the cause cannot be detected

Duration: For 10-14 days, 5 days if azithromycin is used

Empirical therapy

Milder cases	– Amoxicillin (50–90 mg/kg/dose) or Cefuroxime or Amoxicillin clavulanate
Hospitalized cases	
▪ Children less than **4** weeks	– IV Ampicillin and an Aminoglycoside
▪ Infants **4–12** weeks of age	– IV Ampicillin for 7–10 days
▪ Older child fully immunized against H. influenzae type B and S. pneumoniae	Yes → Ampicillin or penicillin G. No → Parenteral cefotaxime or ceftriaxone
▪ Suspected Staph	– Add vancomycin or clindamycin
▪ Suspected Klebsiella	– Add aminoglycoside
▪ Mycoplasma pneumonia	– Erythromycin or azithromycin or clarithromycin
▪ In adolescents	– Fluoroquinolones may be considered

(Nelson textbook of Pediatrics, 2016)

2. Viral pneumonia

- Antibiotics may be considered as a coexisting bacterial infection exists in 30% of cases
- An appropriate antiviral (e.g. Amantadine, Rimantidine, Osetamivir, Zanamivir) should be considered for the child with pneumonia due to Influenza

Prognosis

- Uncomplicated community-acquired bacterial pneumonia show improvement in clinical symptoms within 48-96 hr of initiation of antibiotics.
- Radiographic evidence of improvement lags behind clinical improvement.
- Causes of none improvement with appropriate antibiotic therapy:

- **C**omplications, such as empyema
- **O**bstruction from endobronchial lesions or foreign body
- **B**acterial resistance
- Pre-existing **D**iseases: as causes of recurrent pneumonia***
- Nonbacterial **E**tiologies such as viruses and aspiration

Recurrent pneumonia

- Defined **as 2 or more** episodes in a single year **or 3 or more** episodes ever, with radiographic clearing between occurrences
- **Underlying disorder**

1. Hereditary disorders
- Cystic fibrosis
- Sickle cell disease

2. Immunodeficiency: Primary or secondary

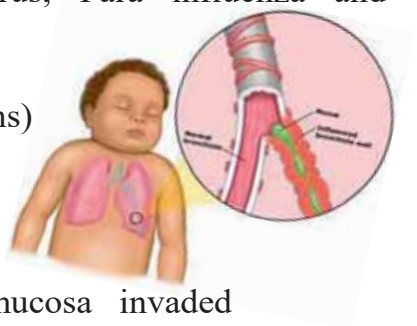
3. Disorders of Cilia
- Immotile cilia syndrome
- Kartagener syndrome

4. Anatomic disorders
- Aspiration (oropharyngeal incoordination)
- Gastroesophageal reflux
- Tracheoesophageal fistula (H type)
- Foreign body
- Bronchiectasis
- Pulmonary sequestration
- Lobar emphysema

(Nelson textbook of pediatrics)

Acute Bronchiolitis

- Acute inflammation of the bronchioles
- Usually viral infection
  - Respiratory syncytial virus (RSV) in 50% of cases
  - Others: human metapneumovirus, Adenovirus, Para influenza and Mycoplasma
- Incidence
  - Age: The 1st 2 years of life (peak age ~ 6 months)
  - Season: more in winter and spring
  - More in boys who are not breast fed



Pathogenesis

- Viral invasion of small bronchioles mucosa & submucosa invaded → acute inflammation → bronchiolar obstruction by edema, mucus and cellular debris
- Impaired pulmonary gas exchange (hypoxemia , hypercapnia) may occur with severe disease

Clinical picture

Symptoms

- Mild upper respiratory catarrh (rhinitis , mild fever) for few days then
- Gradually occurring dyspnea, cough and wheezy chest
- Along with irritability, difficult feeding, air hunger
- Apnea may be more prominent in very young infants (<2 mo old) or former premature infants

Signs

1. Respiratory distress
   - Tachypnea, retractions, grunting ± cyanosis
   - Degree of tachypnea does not always correlate with the degree of hypoxemia or hypercarbia, *so pulse oximetry and noninvasive determination of carbon dioxide are essential*
2. Hyperinflation → Ptosed liver and spleen
3. Chest examination:

Inspection	→ Hyperinflated chest , prolonged expiration
Palpation	→ May be palpable wheezes and decreased TVF
Percussion	→ Bilateral hyper resonance
Auscultation	→ Diminished vesicular breath sounds. → Prolonged expiration. → Bilateral expiratory wheezes → Bilateral fine crackles

Complications

- Dehydration → due to tachypnea & anorexia
- Lung collapse or pneumothorax → sudden deterioration
- Respiratory failure
- Heart failure

Investigations: Diagnosis of acute bronchiolitis is mainly clinical

- Chest X-ray
  - Indicated only for severe illness or bacterial superinfection suspected
  - Shows:
    - Hyperinflation (horizontal ribs , flat diaphragm)
    - Bilateral perihilar infiltrates ± areas of atelectasis





- Blood tests:
  - ESR, CRP and white blood cell count → are usually normal
  - Arterial blood gases for severe disease
- Detect the virus by cell culture or viral antigen /RNA by PCR using nasopharyngeal aspirate

Differential diagnosis: From other causes of **wheezy** infants e.g.:

- Bronchial asthma: suggested by
  - Recurrent attacks of wheezy chest ± viral prodrome
  - Related to certain allergens or exercise
  - Respond to anti-asthma therapy(bronchodilator trial)
  - Relatives with atopy or asthma/presence of atopy or dermatitis
- Congestive heart failure
- Aspiration syndromes /Foreign body inhalation.
- Cystic fibrosis
- Infections e.g. Pulmonary TB, Pertussis

Treatment

Treat at home or hospital?

Hospitalize if risk factors for severe disease exist e.g.

- Infants younger than 3 months
- Severe respiratory distress or apnea, oral feedings intolerance
- Preterm birth
- Underlying comorbidity

The mainstay of treatment is:

"Supportive care"

- Nurse sitting with head and chest elevated at a 30-degree angle with neck extended
- Humidified cool oxygen inhalation with high-flow nasal cannula
- Frequent suctioning of nasal and oral secretions often provides relief of distress
- Care of feeding (more calories are required)
- Parenteral fluids if risk of aspiration exists with respiratory distress
- Treat complications e.g.
  - Antibiotic therapy if secondary bacterial pneumonia suspected
  - CPAP or intubation and mechanical ventilation if deterioration with exhaustion or persistent apnea

Non evidence based and controversial strategies

1. Inhaled bronchodilator*	– Don't modify disease course
2. Steroids*	– Don't modify disease course – Prolong virus shedding
3. Combined nebulized epinephrine* & oral dexamethasone 4. Nebulized hypertonic saline 5. Heliox	– Under ongoing studies – Short term relief in severe cases
6. Chest physiotherapy 7. Cough sedatives	– Should be avoided

* Frequently used

Home oxygen therapy

- Low risk cases that require oxygen can be discharged ER on home oxygen
- Criteria for home oxygen therapy includes
  1. Mild illness as evident by feeding well ,alert and active; minimal retractions; respiratory rate <50 breaths/min, no apnea
  2. Age: 2 mo-2 yr of age with first episode of wheezing during RSV season
  3. Reliable family: good access to healthcare, can manage secretions by bulb suctioning
  4. Absent:
     - Toxic appearance or proven bacterial disease
     - Comorbidities: Cardiac, pulmonary, immunodeficiency, or neuromuscular

(Nelson textbook of Pediatrics)

Antiviral

Ribavirin aerosol
Indications: risky infants (see later)
Side effect: controversial benefits and very costly

Prevention

- Meticulous hand hygiene is the best measure to prevent nosocomial transmission.
- Monoclonal antibody to RSV F protein (Palivizumab) I.M is given before and during RSV season for *risky infants < 2 yr of age* with:
  - Chronic lung disease
  - Gestational age is less than 35 weeks
  - Comorbidities e.g. congenital heart disease, immunodeficiency, neuromuscular disorders

Prognosis

- The median duration of symptoms is approximately 14 days; the first 2-3 days are the most critical
- Mortality rate ≈ 1% due to: apnea , respiratory failure, dehydration
- There is higher incidence of wheezing and asthma in children with a history of bronchiolitis

.

Bronchial Asthma

Definition

Asthma is a chronic inflammatory condition of the lung airways resulting in airways hyper responsive to various stimuli and episodic airflow obstruction with high degree of reversibility

Incidence

- Boys/ girls: 2:1 before puberty and 1:1 after puberty.
- Most asthmatic children become symptomatic before 5th year

Risk factors/associations

- Parental asthma
- Other allergies e.g. eczema, allergic rhinitis, food allergies
- Rhinitis, sinusitis & gastro esophageal reflux disease (GERD).
- Early weaning from breast milk before 4 months.

Pathogenesis

- Genetic predisposition
- Imbalance between T Helper 1 lymphocytes (Th1) & and T Helper 2 (Th2) with raised Th2 → excessive release of proinflamatory cytokines (IL4, IL5, IL13)
- Exposure to asthma triggers

↓

- Accumulation of IgE in airways and blood (Type 1 hypersensitivity; atopy)
- Increased activated mast cells, eosinophils and chronic inflammatory cells in airways

↓

Bronchoconstriction and airways inflammation with edema, ↑mucus, ↑chronic inflammatory cells

↓

Airways narrowing especially in expiration

- Persistent airway inflammation leads to

↓

- Collagen deposition beneath basement membrane.
- Hypertrophy of muscles & glands.

↓

Airway remodeling and persistent narrowing (Chronic obstructive airway disease (*Cor Pulmonale*)

- Asthma triggers includes
  - Respiratory viral infections
  - Animals with fur, dust mites, cockroaches
  - Aerosol chemicals
  - Changes in temperature e.g. early morning
  - Drugs (aspirin, beta blockers)

- Exercise (? Air drafts moving in and out)
- Pollens
- Smoke, tobacco smoke
- Strong emotional expression

Clinical picture

Asthma is strongly suggested with

1. History of any of the following:
   - Cough, worse particularly at night
   - Recurrent wheeze, difficult breathing or chest tightness
2. Symptoms occur or worsen
   - At night, awakening the patient
   - In a seasonal pattern
   - In the presence of a trigger
3. Symptoms respond to short-acting inhaled β-agonist
4. History of other allergies eczema, hay fever,…
5. History of asthma or atopy in other family member
6. History of Patient's colds often "go to the chest" or take longer to clear up

During an attack (Exacerbation)

- Irritability, restlessness.
- Respiratory distress (tachypnea, retractions…)
- **Chest signs**

  Chest wheezing (a normal chest examination does not exclude asthma)

Inspection	→	Hyper inflated chest, prolonged expiration and ↓movement Intercostals and subcostal retractions
Palpation	→	Decreased TVF and may be palpable wheezes
Percussion	→	Bilateral hyperresonance with ↓ hepatic & cardiac dullness
Auscultation	→	Diminished vesicular breath sounds with prolonged expiration
	→	Bilateral expiratory wheezes

Types of asthma

1. Transient non atopic wheezing
   - Triggered by common respiratory viral infections
   - Usually resolves during childhood
2. Persistent atopy-associated asthma
   - Associated with atopy
     - Clinical e.g., atopic dermatitis in infancy, allergic rhinitis, food allergy
     - Allergen sensitization, ↑ Ig E and blood eosinophils
   - Tend to persist into later childhood with lung function abnormalities

Workups in asthma

"Diagnosis of bronchial asthma is mainly clinical"

1. Lung function tests

Usually feasible in children > 5 yr of age

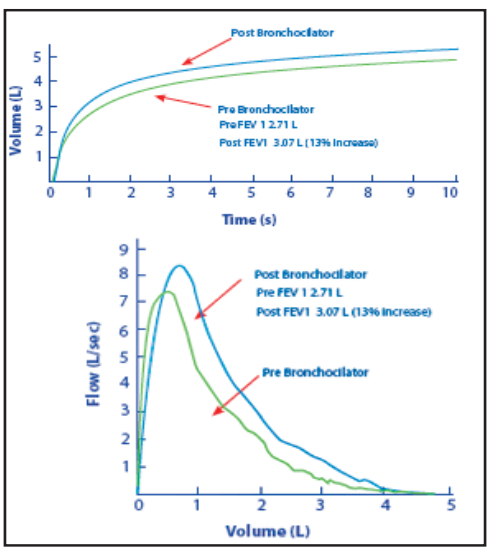
A. Spirometry (in clinic)

- FEV1 Forced expiratory volume in the 1st second
- FEV1:FVC ratio Forced expiratory volume in the 1st second/ Forced vital capacity

Findings in asthma

- Low (relative to percentage of predicted norms or previous best)
- Improve after inhaled bronchodilator
- Worsen after exercise challenge





Spirometry is recommended at least annually and more often if asthma is poorly controlled or abnormal lung functions detected

B. Peak Expiratory Flow (PEF)

- Used for home monitoring
- Less sensitive than spirometry





2. Immunologic

- High IgE and eosinophils in the blood and sputum
- Allergen sensitization : Skin testing with suspected allergens

3. Chest X-ray (During exacerbation) may show

- Hyperinflation.
- May detect complications e.g. collapse, pneumothorax

Treatment

Medications used in bronchial asthma

Reliever medicines

1. Short acting β2 Agonists (SABA)

▪ Preparations	– Salbutamol – Albuterol – Levalbuterol
▪ Action	– Selective β2 agonists; induce quick and short lived bronchodilatation
▪ Duration	– Short acting ; 4- 6hrs
▪ Indication	– Drugs of choice for acute asthma symptoms ("rescue" medication) and for preventing exercise-induced bronchospasm (2.5 – 5 mg by inhalation; dilute with saline to 3 mL)
▪ Side effects	– Tachycardia and tremors (less with Levalbuterol) – Hypokalemia – Overuse of SABAs as a "quick fix" for asthma, rather than using controller medications is associated with an increased risk of death from asthma

2. Ipratropium bromide

- Parasympatholytic
- Used primarily as add on to SABA in treatment of acute severe asthma
- 125 – 250 microgram by inhalation; dilute with saline to 3 mL
- Useful in wheezing due to bronchmalacia
- Side effects : Mild atropine like / less potent than the β-agonists

Controller medicines

1. Steroids

a. Inhaled corticosteroids (ICS)

▪ Preparations	– Beclomethasone (Qvar) – Budesonide (Pulmicort) – Fluticasone (Flixotide) – Ciclesonide (Alvesco)
▪ Action	– Potent anti-inflammatory→ reduce airway chronic inflammation and remodeling – ↑↑ expression of β-receptors in bronchial muscles
▪ Indication	– First-line treatment for persistent asthma – Available in metered-dose inhalers (MDIs), dry powder inhalers (DPIs), or suspension for nebulization

- Side effects
  a. Oral candidiasis (thrush)
  b. Dysphonia (hoarse voice) due to vocal cord myopathy
     Both are minimized by:
     – Using a spacer with MDI(reduce oropharyngeal deposition of the drug)
     – Mouth rinsing after ICS use
  c. ? Steroids systemic effects with high dose, long term ICS

b. Systemic corticosteriods

- Preparations
  – Oral: Prednisolone
  – Parenteral: Methyleprednisolone, Hydrocortisone
- Indication
  – Short courses to treat asthma exacerbations
  – Rarely, long courses in patients with severe disease who remain symptomatic despite optimal treatment
- Precaution
  – Children who require routine or frequent short courses of oral corticosteroids, especially with concurrent high-dose ICSs, should receive corticosteroid adverse effects screening and osteoporosis preventive measures

2. Long Acting β2 Agonists (LABA)

- Preparations
  – Salmeterol (onset after 1hour)
  – Formoterol (onset after 5- 10min→Useful reliever √)
- Action
  – Selective β2 agonist→ prolonged bronchodilatation (12hr)
- Indication
  – add-on agent in asthma suboptimally controlled with ICS therapy alone
- Precaution
  – *LABA should not be initiated as first-line or sole asthma therapy without the concomitant use of an ICS*
  – Should be stopped once asthma control is achieved, and maintain on ICS
- Risks
  – An increase in severe asthma episodes
  – Reported higher number of asthma-related deaths √√

Controller formulations that combine an ICS with a LABA (fluticasone/salmeterol, budesonide/formoterol, mometasone/ formoterol) are available and recommended, in lieu of separate inhaler delivery devices

3. Leukotriene receptor antagonist (LTRAs)

- Preparations
  – Montelukast (Singulaire); chewable tablets or sachets, licensed above 6 months
  – Zafirlukast, licensed above 5 years

▪ Action	–	Leukotriene receptor antagonist with anti-inflammatory and a bronchodilator effect
▪ Indication	–	Alternative treatment for mild persistent asthma
	–	Add-on medication with ICS for moderate persistent asthma
	–	Reduce exercise-, aspirin-, and allergen-induced bronchoconstriction.
▪ Precaution	–	Less effective than ICSs in patients with mild persistent asthma
▪ Risks	–	Montelukast has rarely been associated with mood changes and suicidality

4. Theophylline

▪ Action	–	Phosphodiesterase inhibitor → bronchodilatation and anti-inflammatory
▪ Use	–	Alternative monotherapy controller agent for older children and adults with mild persistent asthma
	–	It is **no longer** considered a first-line agent for young children
▪ Precaution	–	Narrow therapeutic window; therefore, when it is used, serum theophylline levels need to be routinely monitored
▪ Overdose	–	Headaches, vomiting, cardiac arrhythmias, seizures, and death

5. Sodium cromoglycate

- Mast cell stabilizer
- Inhibit exercise-induced bronchospasm, they can be used in place of SABAs, especially in children who develop unwanted adverse effects with β-agonist therapy (tremor and elevated heart rate).
- Not a preferred controller ; must be administered frequently (2-4 times/day) and are not nearly as effective daily controller medications

Asthma medicines delivery systems



Metered dose inhaler with a spacer



Dry powder inhaler with metered dose turbohaler



Solution for nebulization

Management of acute asthma exacerbation

A. At Home

- Inhaled SABA (**Salbutamol**): 2.5 - 5 mg with saline by nebulizer up to 3 treatments in 1 hr **OR**
- 2-6 puffs SABA by MDI (puff=100 mcg) up to 3 treatments in 1 hr

– Resolution of symptoms – No symptoms over the next 4 hr – Improved PEF ≥ 80% of personal best	– Incomplete response to initial treatment – Persistent PEF < 80% of personal best – Deterioration
– Continue SABA at 3-4 hr intervals for 1-2 days – Contact your physician for advice	Seek urgent medical advice

B. Emergency department treatment

Assessment of severity

	Mild	Moderate	Severe
▪ **Altered consciousness**	Absent		Agitated, confused
▪ **Cyanosis**	Absent		Likely present
▪ **Dyspnea**	On walking	On talking	At rest
▪ **Speaks**	In sentences	Phrases	In words
▪ **Pulse**	Normal	Mild tachycardia	Marked tachycardia (> 180bpm in young)
▪ **Pulsus paradoxus**	Normal	Less than 20 mmHg	20-40 mmHg
▪ **Wheezes**	End expiratory	Holo expiratory	Exp and inspiratory May be quiet
▪ **Retractions**	Absent	Common	Usual
▪ **Peak expiratory flow**	≥70%	40-69%*	<40%*
▪ **Oximetry in air**	> 95 %	90 –95 %	< 90%
▪ **$PaCO_2$**	< 42 mmHg*		> 42 mmHg
▪ **PaO_2***	Normal*		< 60 mmHg

* Means test not usually necessary

Signs of acute severe asthma with imminent respiratory arrest

- Drowsy or confused
- Paradoxical thoracoabdominal movement
- Absence of wheeze
- Bradycardia, Absent pulsus paradoxus (due to respiratory muscle fatigue)

Status asthamticus *: A severe asthma exacerbation that does not improve with standard therapy*

Action plan

- Pulse oximetry in air
- High flow oxygen to keep O_2 saturation > 92%
- SABA **(Salbutamol):** 2.5 - 5 mg with saline by nebulizer or 2 to 6 puffs by MDI plus spacer
  (Repeat every 20 minutes for the 1st hour)
- **Ipratropium bromide:** 125 - 250 mcg by nebulizer; if no adequate response to the first salbutamol nebulizer
  (Repeat every 20 minutes for the first hour only)
- Oral **Corticosteroids** (1-2mg/kg in divided doses) in moderate to severe asthma exacerbations to hasten recovery and prevent recurrence of symptoms
- **Epinephrine(1:1000) :**0.3-0.5 mg IM or SC may be given in severe cases

⇩

Outcome at 1 hour

Good outcome:
- Normal physical findings
- PEF >70% of predicted or personal best
- Oxygen saturation >92% in room air for 4 hr

↓

1. Wean gradually → SABA every 3-4 hr
2. If on a controller drug (ICS) → continue it during and after exacerbation
3. Continue oral steroids for 3-7 days

- Moderate to severe exacerbations that do not adequately improve within **1-2** hr of intensive treatment
- High risk patients*

↓

Admission

* High risk patients include:
- Previous severe asthma exacerbation (intensive care unit admission)
- Two or more hospitalizations for asthma in past year
- Three or more emergency department visits for asthma in past year
- Low birthweight
- Poverty

(Nelson Text Book of Pediatrics, 2016)

C. Admission to hospital /PICU

- Admission to an PICU is indicated for patients with
  - Severe respiratory distress
  - Concern for potential respiratory failure and arrest

Action plan

- Cardio-pulmonary monitoring
- Oxygen
- **Salbutamol**
  - Nebulizer every 20 min as needed, then every 1-4 hr as needed
    Or
  - Continuous nebulization with oxygen: 5-15 mg/hr
- **Corticosteroids**: Short course 3-7 days
  - Oral: Prednisone 1-2 mg/kg
  - Parenteral: Methyleprednisolone or Hydrocortisone

Ipratropium bromide nebulizer (poor evidence; no longer recommended)

↓

Persistent severe dyspnea and high-flow oxygen requirements

↓

- Obtain IV access, take electrolytes , glucose and ABG
- CXR in life threatening attack or suspected pneumothorax (deteriorate after a period of improvement)
- Start IV maintenance fluids
  - Dextrose 5% in 0.45% saline with 20-mmol/l kcl
  - At 70-80% of maintenance
  - Correct any dehydration
- Replace nebulizer by IV **Salbutamol** ; Watch for low potassium and ECG monitoring for arrhythmias

↓

Critically ill or at risk for respiratory failure

Available options

- **Epinephrine** IM or SC
- **Magnesium sulphate** (25-75 mg/kg, maximum dose 2.5 g, given intravenously over 20 min)
- **Aminophylline** ; loading dose 5-10 mg/kg over 1 hour followed by maintenance 1 mg /kg/hour (0.7 mg /kg/hour if >10 years)
- Inhaled heliox (helium and oxygen mixture)
- Assisted /mechanical ventilation

After exacerbation resolution:

1. Space SABA gradually
2. Continue steroids for full 3-7 days
3. Start controller therapy
4. Families of all children with asthma should have a written action plan to guide their recognition and management of exacerbations
5. With history of life-threatening episodes, especially if abrupt-onset in nature, providing an epinephrine autoinjector and, possibly, portable oxygen at home should be considered

(Nelson Text Book of Pediatrics, 2016)

Long term asthma management

I. Assessing asthma severity and initiating treatment (For patients who are not currently taking long-term control medications)

		Persistent		
	Intermittent	**Mild**	**Moderate**	**Severe**
1. Daytime symptoms (Wheezing, cough, breathless)	≤ 2 days/wk	> 2 days/wk but not daily	Daily	Throughout the day
2. Nocturnal symptoms /awakening (Nocturnal cough, wheezing, breathless)				
Age < 5 yr	0	1-2 /mo	3-4 /mo	>1 /wk
Age ≥ 5 yr	≤ 2 /mo	3-4 /mo	>1 /wk	Often 7 /wk
3. Need for reliever	≤ 2 days/wk	> 2 days/wk	Daily	Several times per day
4. Limitation of activities (Cough ,wheeze, or breathless on exercise, play or laugh)	None	Minor limitation	Some limitation	Extreme limitation
5. Lung function (FEV1); age ≥5 yr	> 80% predicted	≥ 80% predicted	60-80% predicted	< 60% predicted
Recommended step for initiating therapy	Step 1	Step 2	Step 3	Step 3 or Step 4

II. Stepwise Approach for Managing Asthma in Children

Step 1	Step 2	Step 3	Step 4	Step 5	Step 6
Rescue treatment for all steps: As needed inhaled short acting β2 agonist (SABA) ± Short course of oral steroids if exacerbation is severe or history of previous severe exacerbations					
Move to step 2 if : Rescue treatment is needed more than twice a week or If night-time symptoms at least once a week or If exacerbation in the last 2 years	Low dose ICS **Or** Montelukast	Medium dose ICS OR Low dose ICS + Montelukast* OR Low dose ICS + LABA	Medium dose ICS + Montelukast* OR Medium dose ICS + LABA	High dose ICS + Montelukast* OR High dose ICS + LABA	High dose ICS + Montelukast* OR High dose ICS + LABA + Oral glucocorticoids lowest dose
	Modified-release oral theophylline may substitute Montelukast				
				Consider Anti IgE(Omalizumab) for patients above 12 years with Allergies	
	Consider subcutaneous allergen immunotherapy for patients who have allergic asthma				
Patient education, environmental control, and management of comorbidities					

* For those less than 4 years



Step Down or Step Up gradually
According to assessment of current clinical control (See next table)



III. Assessment of current clinical control (preferably over 4 weeks)

Characteristic	Well controlled (All of the following)	Not well controlled (Any measure in any week)	Very poorly controlled
Daytime symptoms	None or < 2 /week; very short	>2 days/wk	Throughout the day
Nocturnal symptoms /awakening			
Age < 5 yr	≤ 1 /mo	> 1 /mo	>1 /wk
Age ≥ 5 yr	≤ 1 /mo	≥ 2 /mo	≥ 2 /wk
Limitation of activities	None	Some limitation	Extreme limitation
Need for reliever	≤ 2 days/wk	> 2 days/wk	Several times per day
Lung function (FEV1 or PEF)	> 80% predicted	60-80% predicted	< 60% predicted
Exacerbations requiring systemic steroids courses	0-1/yr	≥ 2/yr	> 3/yr

Action	***Step Down*** gradually to the least medication necessary to maintain control if asthma is well controlled at least 3 months	***Step Up*** gradually after checking inhaler technique, adherence, environmental control, and comorbid condition Improvement should be seen within 4-6 weeks

IV. Avoid exposure to triggering agents

- Eliminate or reduce problematic environmental exposures
  - Avoid drugs, foods, and additives known to cause symptoms.
  - Avoid allergens as suggested by skin testing
- Treat co-morbid conditions: sinusitis, GERD and rhinitis.
- Give annual influenza vaccine unless egg allergic
- In exercise induced asthma give:

	SABA inhalation	→ 10 minutes before exercise
or	Montelukast oral	→ 1 hour before exercise

V. Patient education

- Explain basic facts about asthma
- Written asthma management plan
- Demonstrate optimal technique of use of asthma devices
- Insist on adherence to medications
- Ensure avoidance of risk factors
- *Two to four asthma checkups* per year for:
  1. Frequency of asthma symptoms during the day, at night, and with physical Exercise
  2. Frequency of "rescue" SABA medication use and refills
  3. Lung function measurements (spirometry) for older children at least annually
  4. Number and severity of asthma exacerbations
  5. Presence of medication adverse effects since the last visit

Differential diagnosis of asthma (Wheezy Chest)

Definition of wheeze

- Expiratory musical continuous sound
- Due to partial obstruction of small bronchi & bronchioles
- Can be sibilant or sonorous
- May be also inspiratory in severe obstruction

Possible mechanisms

- Bronchoconstriction (spasm of airways smooth muscles)
- Bronchial mucosal edema.
- Excessive, viscid secretions inside airways lumens

Causes

Acute	Chronic /recurrent
▪ **A**cute bronchiolitis. ▪ **B**ronchial asthma exacerbation ▪ Foreign **B**ody inhalation ▪ **C**ongestive Heart failure(e.g. congenital heart diseases or cardiomyopathy) ▪ Aspiration e.g. GER**D**	▪ Bronchial asthma ▪ Congestive heart failure ▪ Cystic fibrosis ▪ Dynamic airway collapse: e.g. bronchomalacia &tracheomalacia ▪ Recurrent aspiration e.g. – GERD – Tracheo esophageal fistula – Neuromuscular disorders – Foreign body ▪ Pulmonary tuberculosis(LN+) ▪ Airway compression by: lymph nodes, vascular ring or tumor

Foreign body aspiration

Clinical picture

Squeals

1. Immediate expelling by cough reflex
2. Retained foreign body: manifestations differs according to site

Type of obstruction	Laryngeal	Tracheal	Bronchial
Partial	- Respiratory distress - Stridor - Hoarsness of voice - Aphonia	- Metallic cough	- Chest wheezes
Complete	- Suffocation	- Respiratory distress - Cyanosis	- Collapse - Abscess - Pneumonia

History

- Commonly reported in children 3months to 6 years
- History of sudden chocking or frank history of foreign body aspiration
- Triphasic history may be obtained:
  - Initial phase: cough, chocking, stridor or gagging
  - Silent phase: if foreign body pass and impact in smaller airways
  - Phase of complications: recurrent pneumonia , abscess, bronchiactasis

Signs

- Fixed localized wheeze; unresponsive to treatment.
- Unexplained lung collapse
- Diminished breath sound over one lung, one lobe **or** one segment
- Mediastinal shift (unilateral collapse or emphysema).
- "Same site" recurrent pneumonia, abscess, bronchiectasis

Chest X-ray

- Positive only in about **50%** of cases
- May show obstructive collapse **or** obstructive emphysema in expiratory film.

Treatment

A. Without respiratory distress→ bronchoscopic extraction

B. With respiratory distress:

1. If the child is breathing well:

- Encourage cough to clear the foreign body
- Be vigilant for any deterioration

2. If cough becomes ineffective:

- Try to assist expulsion of the foreign body
- Provide rescue breathing in between trials.



- Use the Alert, Verbal, Pain, Unresponsive pediatric scoring system(AVPU) to determine both a child's level of consciousness and cerebral cortex function
- If trials fail and infant becomes unconscious, attempt to visualize foreign body and remove manually.



i. First aid for the choking infant < 1 year of age

- Hold infant prone with the head down.
- Give 5 interscapular back blows, using heel of hand.
- Turn the infant supine, with head dependent and perform 5 quick downward chest thrusts .

ii. First aid for the choking child older than 1 year of age

A) In conscious patient ⇒ abdominal thrust in sitting or standing (Heimlich maneuver):



- Encircle the child chest with arms from behind.
- Place one fist against patient's abdomen in midline just below tip of xiphoid.
- Grasp fist with other hand and exert 5 quick, upward thrusts.

B) In unconscious patient ⇒abdominal thrust in lying down:



- Place the patient supine.
- Open patient airway using chin lift or jaw thrust.
- Place heel of one hand on child's abdomen just below costal margins.
- Place the other hand on top of the first hand.
- Press both hands into abdomen with quick, upward thrusts in midline.

iii. Further interventions

- Laryngoscopic removal.
- If failed; push foreign body more distally.
- If failed, perform immediate cricothrotomy

Prevention

Avoid chocking materials in infants and young children e.g. small toys, nuts, popcorn

Dry Pleurisy

Definition: Fibrinous inflammation of the pleura.

Causes

- Infections: Viral pneumonia, bacterial pneumonia, tuberculosis
- Chest wall trauma
- Collagen diseases e.g. Rheumatic fever, systemic lupus erythematosus

Clinical picture

- Manifestations of **the cause**
- Chest pain: Stitching, ↑ with deep respiration, cough & sneezing
- Patients may prefer to lie on same side.
- Auscultation: *Pleural rub*: - Scratchy sound.
  - Decrease by holding breathing.

Treatment: - Treat the cause.
- Analgesics.

Serofibrinous Pleurisy (Pleural Effusion)

- Normally, only 4-12 mL of fluid is present in the pleural space, but if formation exceeds clearance, fluid accumulates.
- Definition: Serofibrinous inflammation of the pleura.

Types of effusion

Transudate	Exudate	Bloody	Cheylous
Characters - Clear; straw colored - Proteins < 3gm/dl - ↓ Cells (mesenchymal) - ↓ Specific gravity - Sterile - ↓Lactate dehydrogenase	- Turbid ; opaque - > 3 gm/dl. - ↑ Cells (PMNLs) - ↑ Specific gravity (>1015) - May reveal organisms - Lactate dehydrogenase >200 iu /l	Bloody with RBCs on mic. examination	- Milky white - Dissolved with ether - Spread on filter paper
Mechanisms - Increased hydrostatic capillary pressure - Decreased plasma osmotic pressure	Increased capillary permeability due to inflammation , malignancy, mediastinal or chest wall diseases		Impaired lymphatic drainage
Causes Passive transudation in renal, cardiac & hepatic causes of generalized edema	- Pneumonia. - T.B. - Ruptured Lung abscess - Mediastinitis - SLE, uremia, metastasis - T cell lymphoma.	- Tumors - Trauma - Hemorrhagic blood Diseases	Thoracic duct obstruction **or** trauma

Clinical picture

Symptoms

- Manifestations of **the underlying cause** (e.g. fever, dyspnea,…..)
- Respiratory distress
- Chest pain: dull aching pain; patient prefers to lie on the affected side

Chest examination

Small effusion:

Clinical picture of an underlying cause e.g. pneumonia → bronchophony, bronchial breathing and crepitation

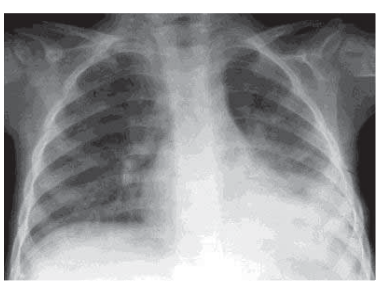
Massive effusion

- Inspection → Unilateral bulge, full intercostal spaces with diminished movement
- Palpation → Decreased TVF & trachea shifted to opposite side
- Percussion → Stony dullness, rising to axilla
- Auscultation → Marked diminished breath sounds (or absent).
  → Aegophony (nasal tone of voice) may present at the top of effusion due to kinked bronchi

Investigations

1. Chest X-ray in supine and upright positions:

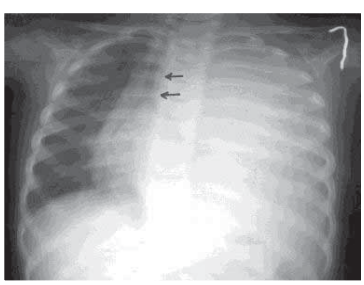
- In small effusion: homogenous opacity just obliterating costophernic angle
- In moderate to large (Massive) effusion: homogenous opacity
  - Filling the costophernic angle
  - Rising to the axilla.
  - With shift of the mediastinum to the opposite side



Mild left sided effusion



Moderate right sided effusion



Massive left sided effusion

2. Chest ultrasonography

- Diagnostic for pleural fluid
- Guide thoracentesis/ Chest tube insertion



3. Thoracentesis:

a. Inspect the fluid:
- Straw colored → Transudate
- Turbid → Exudates
- Milky white → Chylous
- Fetid odor → Anaerobic infection , empyema

b. Cytology:
- Polymorph → Infection e.g. Pneumonia , early TB
- Lymphocytes → TB, chylous , malignancy ; lymphoma
- Esinophils → Parasitism, emboli
- Red cells → Trauma, tumors,……

c. Order culture & sensitivity

d. Biochemical examination: Mention from previous table

4. Tests for TB: tuberculin test, sputum analysis and culture

5. Pleural biopsy, **thoracoscopy and/ or broncoscopy**: if TB or malignancy is likely

Outcome of effusion

- Massive effusion may impair cardiac function
- Secondary pyogenic infection→ empyema
- Organization of unresolving exudates may lead to fibrothorax

Treatment

1. Treat the cause
2. Thoracocentesis is both diagnostic and therapeutic
3. **Thoracostomy tube drainage**
   - Closed drainage using intercostal tube with underwater seal
   - Indicated for:
     a. Massive effusion
     b. Marked respiratory distress
     c. Effusion not resolved with medical treatment
     d. Empyema
   - Site of aspiration → 5th space mid axillary line
   - Precaution: Rapid removal of ≥1 L of pleural fluid may be associated with the development of re-expansion pulmonary edema

Purulent Pleurisy (Empyema)

Definition

- Exudative pleural effusion with marked ↑↑ pus cells
- "Effusion is empyema if bacteria are present on Gram staining, pH is < 7.20, and there are >100,000 neutrophils/μL"

Causes

- Pneumonia (Pneumococci, Staph, H. influenza and klebseilla).
- Rupture lung abscess.
- Rupture abdominal abscess or subphernic abscess
- Rupture of chest wall abscess
- Secondary contaminated chest trauma or surgery
- Secondary infection of an effusion
- Secondary to infection from suppurated lower cervical lymph nodes

Clinical picture

1. Acute empyema: Same as pleural effusion with:
   - High fever, toxic patient.
   - Same side chest wall edema
   - High incidence of complications.
2. Chronic empyema; empyema lasting for 3 months or more:

Clinical	Laboratory
– Pale clubbing – Pallor – Low grade fever (Pyrexia) – Eventual fibrothorax ,collapse, with same side mediastinal shift , scoliosos and narrow ribs – Risk of amyliodosis	– Anemia of chronic illness ; normocytic normochromic – Elevated ESR – Poly morph nuclear leucocytosis

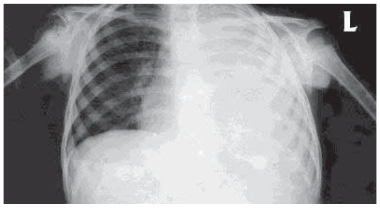
Complications

1. Local spread to:
   - Lung → Bronchopleural fistula.
   - Abdomen → peritonitis.
   - Chest wall → empyema necessitatis.
   - Pericardium → purulent pericarditis.
2. Distant spread e.g. Meningitis, septicemia,…..

Investigations

1. Chest X-ray: as effusion but;

- Opacity is denser.
- Ribs crowding
- May be lung collapse in chronic cases



2. Thoracocentesis

- For character of the fluid (exudate with ↑↑ pus cells).
- For culture & sensitivity.

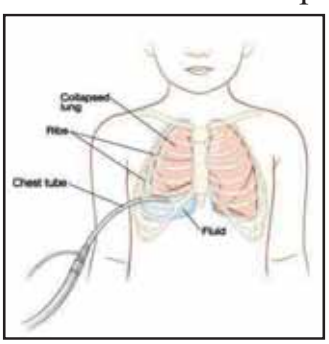
3. Ultrasonography or CT chest: Detect pleural fluid *septa* and loculated empyema

4. Blood cultures: have a higher yield than cultures of the pleural fluid.

Treatment

1. Thoracostomy tube drainage

- Closed drainage using intercostal tube with underwater seal(Open drainage may be necessary in chronic cases)
- For about 1 week
- More than one tube may be needed to drain pockets of pus.
- Use intra pleural fibrinolytic agents (Streptokinase or Urokinase):
  - For 3-5 days
  - Promote drainage, decrease fever, and shorten hospitalization
  - Precaution: risk of anaphylaxis, and hemorrhage





2. Antibiotics: According to culture and sensitivity for 2- 4 weeks

3. Surgical decortications

- Via video-assisted thoracoscopic surgery (VATS) or open thoracotomy
- Indicated for child who remains febrile and dyspneic >72 hr after initiation of therapy with intravenous antibiotics and thoracostomy tube drainage



N.B: Pseudochylous effusion: Chronic serous effusion with cellular degeneration:

Criteria: - High cholesterol /Low triglycerides level.
- Doesn't clear with ether or alkali.
- Doesn't spread on filter paper

Hydropneumothorax

Definition: Presence of both fluid & air in the pleural cavity.

Causes

- Thoracocentesis for pleural effusion → hydropeumothorax.
- Thoracocentesis for hemothorax → hemopneumothorax.
- Empyema with bronchopleural fistula → pyopneumothorax

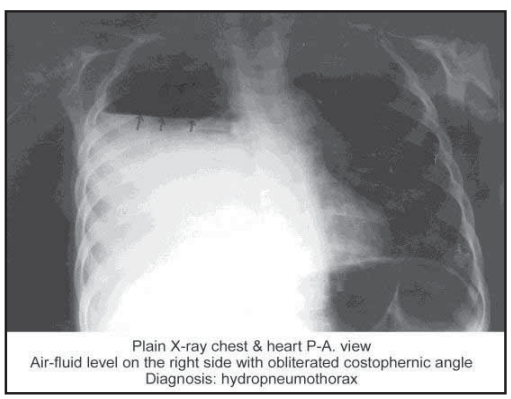
Clinical picture

Chest examination

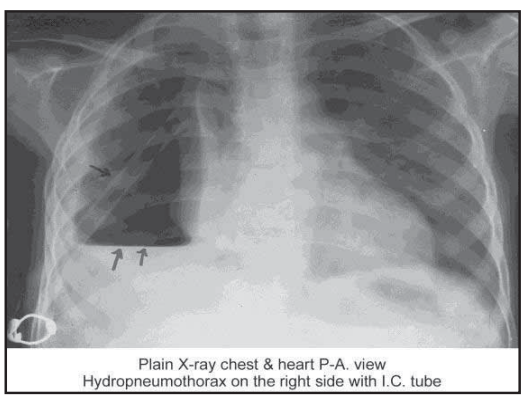
- Inspection → Unilateral bulge.
- Palpation → Decreased TVF & trachea shifted to opposite side.
- Percussion → Shifting dullness.
- Auscultation → Marked diminished breath sounds.
  → Succession splash

Investigations

- As pleural effusion;
- Chest X-ray → air- fluid level



Plain X-ray chest & heart P-A. view
Air-fluid level on the right side with obliterated costophernic angle
Diagnosis: hydropneumothorax



Plain X-ray chest & heart P-A. view
Hydropneumothorax on the right side with I.C. tube

Treatment

1- Antibiotics according to culture and sensitivity.

2- Closed drainage with underwater seal⇒ If failed → surgical closure of the fistula.

Pneumothorax

Definition: Presence of air in the pleural cavity

Causes

- Rupture preumatoceles
- Rupture tuberculous cavity
- Rupture lung abscess.
- Rupture surface alveoli in air trapping
- Vigorous resuscitation
- Chest wall trauma

Clinical picture

Symptoms

- Asymptomatic (in small pneumothorax) → discovered accidentally
- Symptomatic: → Respiratory distress (↑↑ with tension pneumothorax).
  → Symptoms of the cause

Chest examination

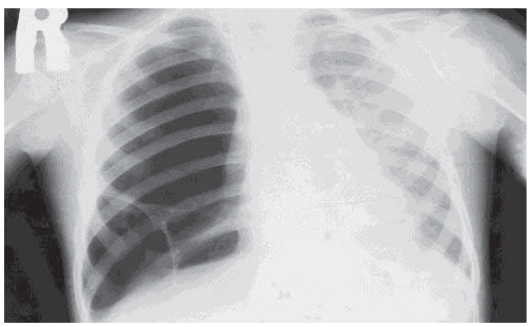
- Inspection → Unilateral decreased movement & unilateral bulge.
- Palpation → Decreased TVF & trachea shifted to opposite side.
- Percussion → Hyper resonance.
- Auscultation → Marked diminished breath sounds.
  → Coin test

Evidence of tension

- Mediastinal shift
- Circulatory compromise
- Hearing a "hiss" of rapid exit of air under tension with the insertion of the thoracostomy tube

Investigations

- Chest X-ray/CT → jet black opacity ± mediastinal shift to the opposite side
- CT chest: may identify underlying pathology such as blebs
- For the cause



Treatment

1- Small pneumothorax: usually resolve within 1 week.

2- Symptomatic:

- Closed drainage with underwater seal.
- Tube is inserted in the 2nd space mid clavicular line.

3- Treat the underlying cause.

Tuberculosis

Definition

Chronic infectious disease caused by Mycobacterium TB bacilli (human and bovine types) which is alcohol and acid fast aerobic intracellular bacilli.

Modes of transmission

- Inhalation → pulmonary tuberculosis
- Ingestion(with milk) → intestinal T.B(& tonsillar tuberculosis)
- Wound contamination → cutaneous tuberculosis
- Hematological spread form primary T.B. focus

Risk factors

- Children exposed to high-risk adults
- Low Socioeconomic standard (Homeless persons)
- Suppressed immunity e.g. HIV, malnutrition & immunosuppressive therapy
- Susceptible age: disease is more severe in infants and young child
- Susceptible Race: More in Negroes

Pathogenesis

Primary exposure to T.B bacilli result in formation of primary complex at the site of entry of the bacilli (the commonest form in children).

1. Primary pulmonary complex:

Composed of
- Primary focus (Ghon's focus)
- Lymphangitis
- Hilar lymphadenitis

2. Primary cervical complex (tonsillar T.B)

Composed of
- Primary focus in tonsils
- Lymphangitis
- Cervical lymphadenitis

3. Primary intestinal complex

Composed of
- Primary focus in pyere's patches
- Lymphangitis
- Mesenteric lymphadenitis

Each primary focus is formed of tubercles each tubercle is formed of :

- Central caseation
- Epitheloid cells
- Macrophages and lymphocytes
- Langerhans giant cells



Fate of primary pulmonary complex

With good immunity ⇓ **Regression (in 90%)**

- Small focus → complete fibrosis
- Large focus → capsulation & calcification → In which T.B bacilli may remain viable for years ;Latent TB Infection (LTBI)

With poor immunity ⇓ **Progression**

N.B: Risk for progression of latent TB: Infants and children ≤4 yr of age, especially those <2 yr of age, Adolescents and young adults, immunocompromised, and infection with measles and pertussis.

Primary Cavitation

Direct spread ⇓

- T.B pneumonia
- Tuberculous effusion

Bronchial spread (Endobronchial T.B) ⇓

- Incomplete obstruction → emphysema
- Complete obstruction → collapse

Hematologic spread ⇓

- One organ T.B
- Miliary T.B

Clinical stages of tuberculosis

There are 3 major clinical stages of tuberculosis

Exposure

- The child has had significant contact ("shared the air") with an adult with infectious tuberculosis but lacks proof of infection : →Negative tuberculin skin test (TST) or interferon- γ release assay (IGRA)
- Normal physical examination
- Normal chest radiograph
- These cases have risk of developing tuberculosis disease rapidly

Infection

- The child inhales droplet nuclei containing *M. tuberculosis,* which survive intracellularly within the lung and associated lymphoid tissue
- Positive TST or IGRA are the hallmark of tuberculosis infection
- Normal physical examination
- The chest radiograph is either normal or reveals only granuloma or calcifications in the lung parenchyma.

Disease

Occurs when signs or symptoms or radiographic manifestations caused by *M. tuberculosis* become apparent

(Nelson textbook of pediatrics, 2016)

Clinical Picture

1.Pulmonary TB

Common

- ✓ Asymptomatic in up to 50%; may be mild fatigue & poor appetite
- ✓ Nonproductive cough and mild dyspnea are the most common symptoms
- ✓ Some infants have difficulty gaining weight or frank failure-to-thrive

May be

- Hilar lymphadenopathy may present with:
  - Obstructive emphysema/wheezing: due to partial bronchial obstruction
  - Lung collapse: due to complete bronchial obstruction
  - Positive D'Espine sign (Bronchial breathing below level of tracheal bifurcation)
- Wheezy chest : due to endobronchial TB with partial bronchial obstruction
- Allergic manifestations:



o Erythema nodosum



o Phlyctenular keratoconjunctivitis

- Toxic manifestations (uncommon) → night fever & sweating.
- Manifestations of extension; usually with toxic manifestations and hectic fever e.g. Bronchopneumonia, tuberculous effusion, miliary tuberculosis

N.B. *Cough with sputum is rare, seen literally in progressive primary pulmonary TB with formation of T.B cavity.*

2. Extra pulmonary tuberculosis

A. Tuberculous lymphadenopathy

Common sites: Cervical, Mediastinal, Mesenteric

Criteria:

- Firm
- Non-tender
- Early discrete then matted after caseation

Complications:

- Cold abscess
- Draining sinus

Diagnosis: Biopsy and histologic examination



B. Miliary tuberculosis

- Hemtogenous spread of tubercle bacilli from any focus (usually pulmonary) → causing disease in 2 or more organs; lung, kidneys, liver, spleen, bone marrow, meninges.
- Usually complicates the primary infection, occurring within 2-6 mo of the initial infection

Common in

Infants, malnourished, immunosuppressed, with measles or pertussis

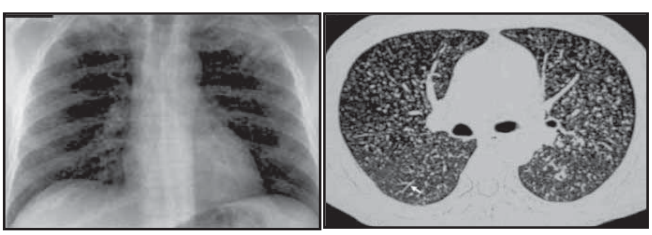
Clinical picture

- Often, the onset is insidious, with anorexia, weight loss, and low-grade fever
- Weeks later:
  - Generalized lymphadenopathy and hepatosplenomegaly
  - Fever higher and more sustained
- Weeks later:
  - The lungs become filled with tubercles→ dyspnea, cough, rales, or wheezing. May be respiratory distress, hypoxia, and pneumothorax
  - Meningitis (recurrent headache) or peritonitis (abdominal pain) are found in 20-40%
  - Cutaneous lesions include papulonecrotic tuberculids, nodules, or purpura

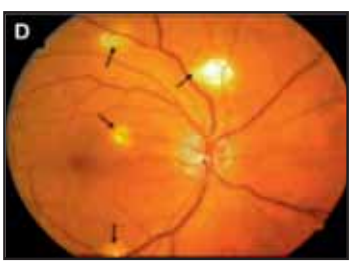


Diagnosis

1. **History** of recent exposure to an adult with infectious TB→ The most important clue
2. **Biopsy** of the liver or bone marrow with appropriate bacteriologic and histologic examinations more often yields an *early diagnosis*
3. **Chest x ray/CT**:
   - Small miliary shadows; < 2-3 mm (*Snow storm opacities*)



4. **Fundus** examination:
   - Show choriod tubercles in 13-87 %
   - Highly specific



5. **TST** is non-reactive in 40 %

C. Tuberculous meningitis

- Complicates about 0.3% of untreated tuberculosis infections in children
- Due to hematogenous spread either isolated or as a part of miliary TB
- Tubercle bacilli spreading into the subarachnoid space form a gelatinous exudate that infiltrates the corticomeningeal blood vessels, producing inflammation, obstruction, and subsequent infarction of cerebral cortex

Clinical picture

- In infancy and early childhood
- Insidious onset
- Pass in **3** stages (each lasts 1-2 weeks)

1st stage (Nonspecific)	2nd stage (Meningitis)	3rd stage (Terminal stage)
– Fever – Headache, irritability – Drowsiness, and malaise	– Meningeal irritation – ↑ Intra cranial tension – Cranial nerve palsies	– Hemiplegia or paraplegia – Coma – Eventually death

D. Intestinal tuberculosis

Occur secondary to

- Ingested tubercle bacilli in milk
- Swallowed sputum from tuberculous lesions in the lungs

Clinical picture

- Tabes mesentrica; enlarged mesenteric lymph nodes.
- Tuberculous enteritis:
  - Chronic diarrhea → failure to thrive
  - Chronic abdominal pain

E. Tuberculous peritonitis

Occur 2ry to: Spread from intestinal or genitourinary T.B lesions

Clinical picture

- Ascites
- May be adhesions.

F. Pott's disease

Common sites: mainly affect lower dorsal spine.

Clinically

- Back pain and stiffness
- Cold abscess formation with persistent angular kyphosis

X ray spine: Diagnostic

Diagnosis of tuberculosis

1. History of recent exposure to an adult with infectious TB→ The most important clue

2. Tuberculin Skin Test (TST):

- Detects delayed hypersensitivity reaction to tuberculoprotein
- Mantoux test: intradermal injection of 0.1 ml containing 5 tuberculin units of purified protein derivative (PPD).
- Interpretation: measure the induration after **48 -72** hours
- Indications : see later

A. Positive test (= TB infection or disease)



1. Induration ≥ 5 mm^2 in high risk patients;
- Close contact with active tuberculosis patient
- Immunodeficiency
- Child having clinical or chest x ray compatible with tuberculosis

2. Induration ≥ 10 mm^2 in moderate risk patients;
- Child < 4 years
- Child from endemic area or exposed to people from endemic area
- Chronic diseases with increased risk e.g. diabetes , renal diseases

3. Induration ≥ 15 mm^2 in any child above 4 years without risk factors

B. False positive test; usually less than 10 mm induration, consider:
- Recent BCG vaccination; reactivity is lost by 5-10 years after vaccine
- Non tuberculous mycobacteria

C. Negative test: induration less than 5 mm^2
- True negative test → no T.B infection
- False negative test → in
  - Technical error
  - Transient suppression of tuberculin reactivity with viral infections e.g. measles, mumps or live virus immunization
  - Early in the disease
  - Miliary TB.
  - Immunodeficiency

3. Interferon-γ Release Assays (IGRA)

- Two blood tests available (T-SPOT.*TB* and QuantiFERON-TB)
- Detect IFN-γ generation by the patient's T cells in response to specific human *M. tuberculosis* antigens
- Like the TST, IGRAs cannot differentiate between tuberculosis infection and disease.

Advantages

- Lack of cross-reaction with BCG vaccination and most other mycobacteria
- One spot test; unlike TST

Indications:

- CDC recommends using IGRA interchangeably with TST

Indications of TST or IGRA

1. Contacts of people with confirmed or suspected contagious TB
2. Children with radiographic or clinical findings suggesting tuberculosis
3. Children immigrating from countries with endemic infection (e.g. Asia, Middle East)
4. Children with travel histories to countries with endemic infection
5. Children infected with HIV should have annual TST or IGRA
6. Before initiation of immunosuppressive therapy e.g. Prolonged steroid

- Specific indications of IGRA:
  - Child ≥5 yr who have a positive TST and have received the BCG vaccine
  - If nontuberculous mycobacterial disease is suspected
  - Child ≥5 yr who is unlikely to return for TST reading
  - Supportive to positive TST before starting therapy

(Nelson textbook of pediatrics, 2016)

4. Specific:

A. Pulmonary tuberculosis

1. Isolate M. tuberculosis:

Sampling

- Expectorated sputum in older children
- Induced sputum with a jet nebulizer and chest percussion followed by nasopharyngeal suctioning is effective in children as young as 1 year
- 3 consecutive early morning gastric aspirate before the infant has arisen

Workup

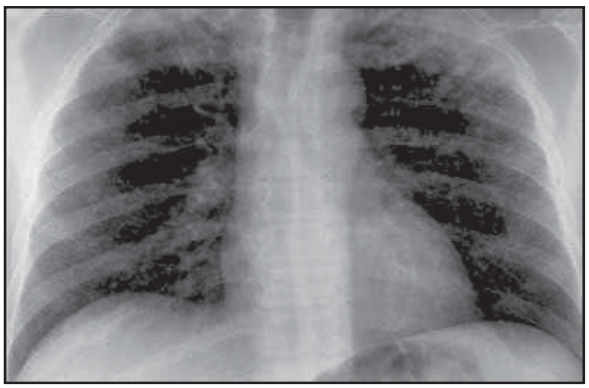
- Acid-fast bacilli staining (Zehl Nelsen stain and light microscopy)
- Culture
- Polymerase chain reaction ;PCR(of limited value)
- Recently , Gene Xpert MTB/RIF is a real-time PCR assay for *M. tuberculosis* that simultaneously detects rifampin resistance

N.B For many forms of tuberculosis, the culture yield is only 25-50%. So, negative cultures never exclude the diagnosis of tuberculosis in a child.

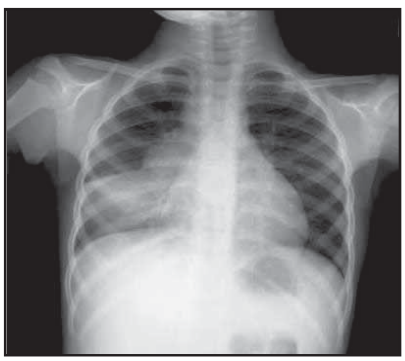
2. Chest X-ray: May reveal



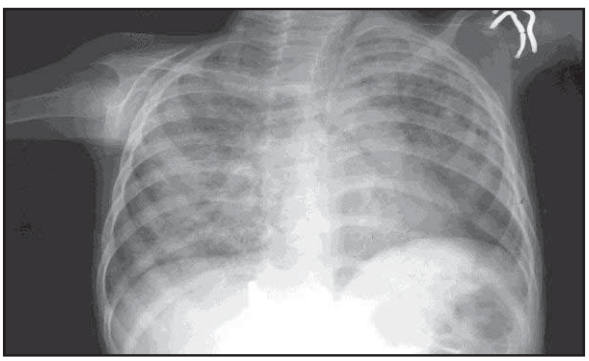
Enlarged hilar lymph nodes
Localized emphysema



Miliary TB; small miliary shadows 1-2 mm (snow flake opacities).



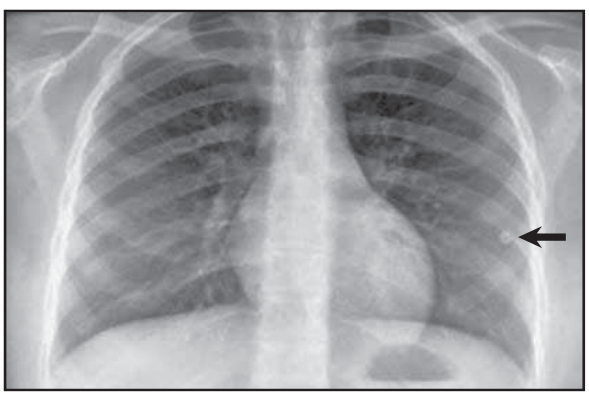
Enlarged hilar lymph nodes
Localized collapse



T.B bronchopneumonia ;fluffy cotton appearance



Pleural effusion



Calcified granuloma (primary focus)

3. Detect the pathology:
   - Pleural biopsy
   - Lymph nodes biopsy
   - Bronchoscopy (for suspected endobronchial TB) and biopsy



4. Pleural fluid examination:
   - Color : Yellow with blood tinge
   - Characters: Usually unilateral, massive; recollect after aspiration
   - Cells : ↑ lymphocytes but it is very rare to discover T.B bacilli
   - Cultures of the fluid are positive in <30% of cases.
5. Blood: Elevated ESR

B. Tuberculous meningitis

- Lumbar puncture and CSF analysis, culture and PCR(See neurology)
- CT, MRI may detect *tuberculoma*; a tumor-like mass resulting from aggregation of caseous tubercles



MRI of brain of a 3 yr old child showing multiple pontine tuberculomas **(Nelson 2016)**

N.B The TST is nonreactive in up to 50% of cases, and 20-50% of children have a normal chest radiograph.

C. Intestinal tuberculosis:

- Mesentric lymph node biopsy
- Ascitic fluid analysis

The presence of a positive TST or IGRA, an abnormal chest radiograph consistent with tuberculosis, and history of exposure to an adult with infectious tuberculosis is adequate for the probable diagnosis of tuberculosis disease

Treatment

Prevention

* BCG vaccine (see before)
* Milk sanitation (boil milk for10- 15 minutes before use)
* Isolate and treat infective cases with open pulmonary TB.
* Avoid contact with cases.
* **Window prophylaxis:**
  - For children who:
    - Have unavoidable close contact to an adult with potentially contagious tuberculosis disease and
    - Have a negative TST or IGRA result
  - Break the contact with the source case for tuberculosis (i.e. physical separation or adequate initial treatment of the source case) and Give INH 10 mg/kg/d for **3** months (the time delayed hypersensitivity develops)
  - Perform TST or IGRA at **3** months
    - If positive result (≥ 5 mm^2) → continue INH for **9** months
    - If negative result→ stop INH and INH resistant BCG can be given
  - Trace the possible adult source and treat adequately to prevent other secondary cases.

Curative

A. General lines

- Good nutrition, fresh air
- Follow up carefully to promote adherence to therapy, and to monitor for toxic reactions to medications

B. Anti-Tuberclous drugs

First line drugs

Drug	Daily dose*	Twice weekly dose *	Side effects
▪ Isoniazide (INH)	10-15	Double the dose	- Hepatotoxic - Peripheral neuritis (?? add vit B_6)
▪ Rifampicin	10-20	Same	- Hepatotoxic - Red staining of secretions
▪ Pyrazinamide	20-40	50	- Hepatotoxic - Hyperuricaemia
▪ Ethambutol	20	50	- Optic neuritis (usually reversible) - Color blindness (green,red)

* mg/kg

Alternative drugs

Used as additive drugs in

- Multiple drug resistant tuberculosis
- Life threatening tuberculosis e.g. T.B. meningitis.

Drug	Dose (mg/kg/d)	Side effects
▪ Streptomycin	20-40 (I.M)	Ototoxic & nephrotoxic
▪ Ethionamide	15-20 (oral)	Hepatotoxic (similar to INH).
▪ Amikacin	15-30 (IM)	As streptomycin

Regimens for treatment

The specific treatment plan must be individualized for each patient according to the results of susceptibility testing on the isolates from the child or the adult source case

1. Six months regimen

- Standard therapy for intrathoracic tuberculosis and cervical lymphadenopathy
- Rifampicin and INH (for **6** months) + Pyrazinamide and Ethambutol (in the1st **2** months)

1. Rifampicin
2. INH
3. Pyrazinamide
4. Ethambutol
For **2** months

Rifampicin + INH
For **4** months

- When **directly observed therapy** is used: initial period as short as 2 wk of daily therapy followed by intermittent (twice weekly) therapy is as effective as daily therapy for the entire course.

2. Nine month regimen

- Using only isoniazid and rifampin
- Highly effective for drug-susceptible tuberculosis
- Carry risk of initial drug resistance and poor compliance

3. In miliary T.B, meningitis and bone T.B

- Extend treatment period for **9-12** months.

4. In drug resistance

- Treatment is undertaken by a clinician with specific expertise
- Initial treatment
  - Isoniazid resistance ⇒9 mo with rifampin, pyrazinamide, and ethambutol
  - Isoniazid and rifampin resistance ⇒ extend total duration of therapy to 12-24mo, and avoid twice-a-week regimens

5. Latent TB infection: isoniazid for 6-9 months

Steroids in T.B

Used in

1- Miliary tuberculosis → to improve the general condition
2- Endobronchial tuberculosis with localized emphysema.
3- Enlarged hilar lymph nodes with airway obstruction.
4- Tuberculosis of serous cavities e.g Pleurisy , Pericarditis , Meningitis
5- Adrenal tuberculosis

Precautions

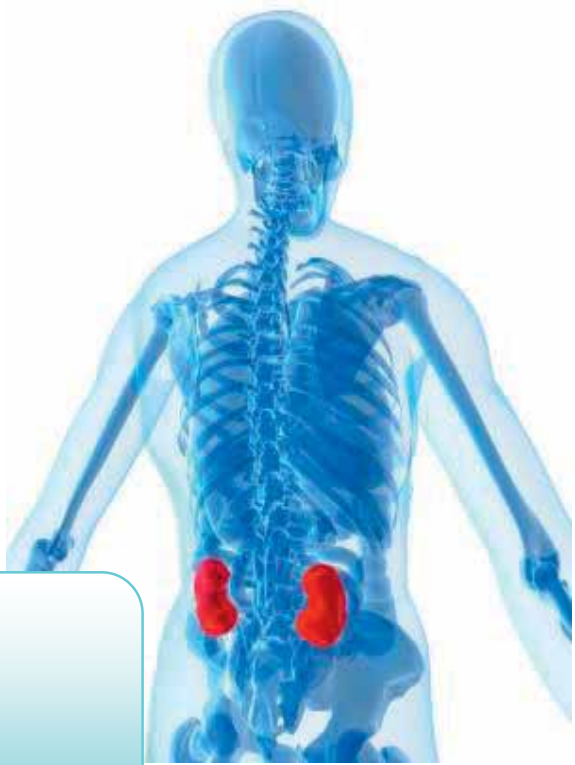
1- Under umbrella of antituberculous drugs.
2- Dose 2 mg/kg/d for 4-6 weeks followed by gradual tapering.

Respiratory failure

Definition: Failure of the lungs to keep normal level of arterial blood gases(O_2 & CO_2)

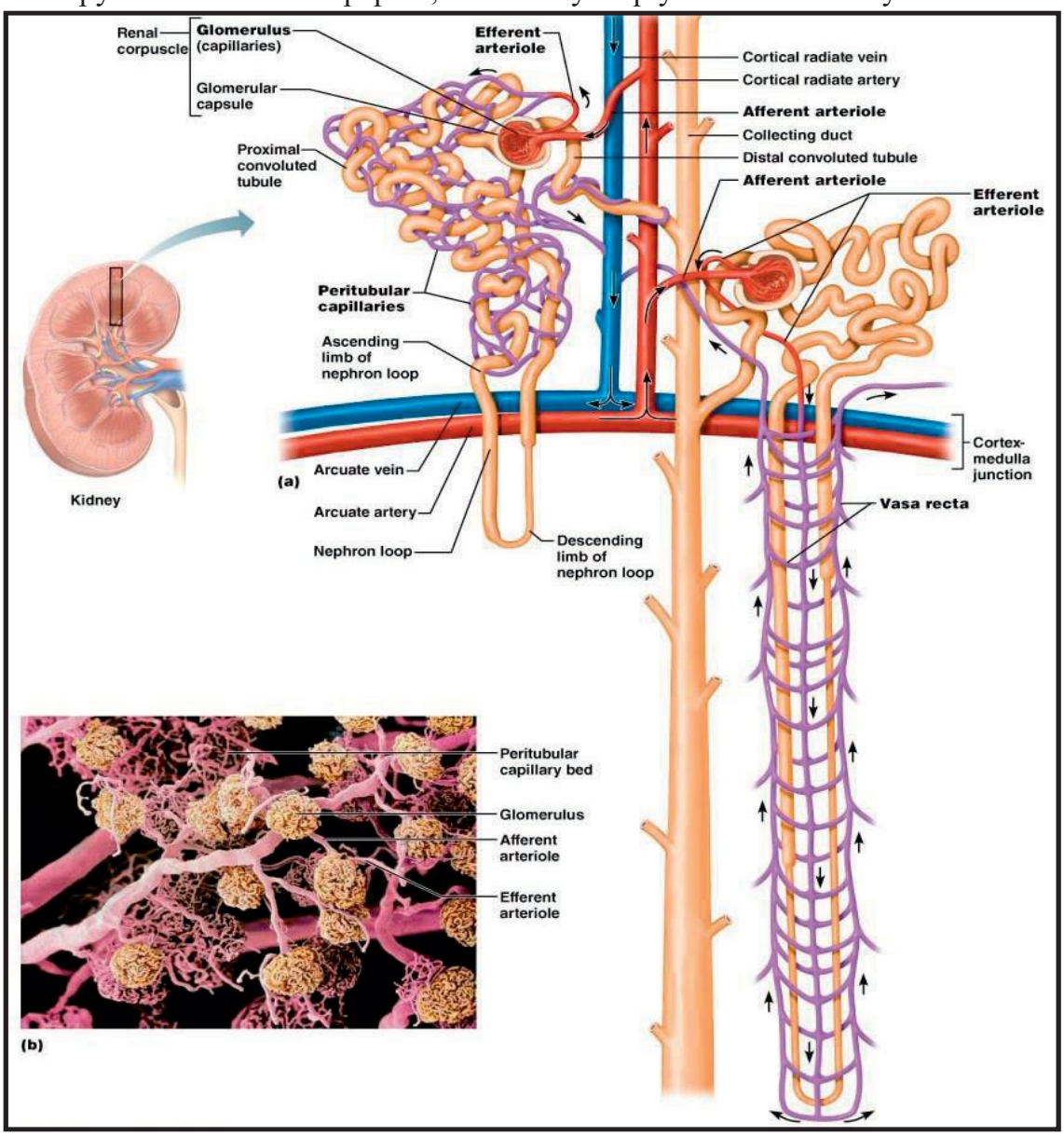
	Peripheral(type I)	**Central (type II)**
Causes	- Airway obstruction e.g asthma - Pneumonia - Pneumothorax - Massive effusion	- Brain : hemorrhage,drugs - Neuromuscular: spinal muscle atrophy, Guillian Barre syndrome - Skeletal: severe kyphosis, scoliosis
Clinically	* Manifestations of the cause * Respiratory distress * Mainly hypoxemia: irritability restless, dizziness , cold pale extremities	* Manifestations of the cause * Irregular , shallow respiration * Mainly hypercapnia: cyanosis, lethargy, headache and impaired consciousness
ABGs	$\downarrow PaO_2$ – $\uparrow PaCO_2$ – $\downarrow pH$	
Treatment	Treat the cause Oxygen therapy(See neonates)	Treat the cause Ventilation

Nephrology



Structure of the Nephron

- Nephrons are the structural and functional units of the kidneys that carry out processes that form urine
- Each nephron consists of a renal corpuscle composed of a tuft of capillaries (the glomerulus), surrounded by a glomerular capsule (Bowman's capsule) and a renal tubule. The renal tubule begins at the glomerular capsule as the proximal convoluted tubule, the loop of Henle, and turns into a distal convoluted tubule before emptying into a collecting duct.
- The collecting ducts collect filtrate from many nephrons, and extend through the renal pyramid to the renal papilla, where they empty into a minor calyx



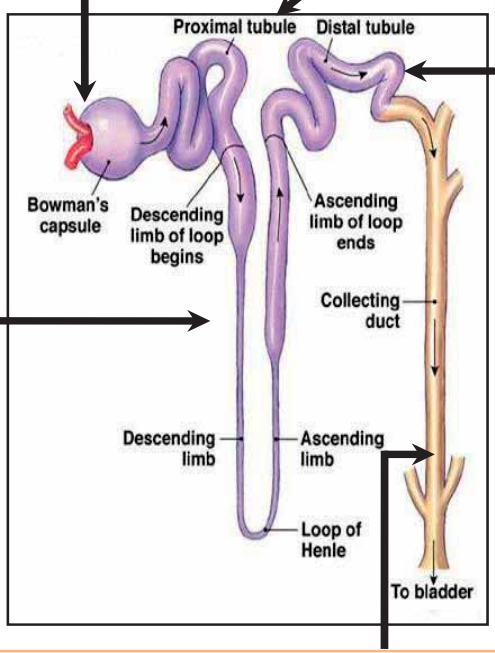
Note: - Efferent arterioles provide blood supply to the whole renal tubules
- Erythropoietin is produced by peritubular capillary endothelium cells

Functions of the Nephron

Glomerular filtration:
- Plasma is passively **filtered** through the glomerular capillary walls.
- The ultrafiltrate, which is cell free, contains all of the substances in plasma (water,electrolytes, glucose, phosphate, urea, creatinine, peptides, low molecular weight proteins) except proteins having a molecular weight of ≥68 kd (e.g. albumin and globulins).

Reabsorption at the PCT (Main bulk)
- Sodium: 65-70% of filtered by active transport by Na-K pump
- Water :By osmosis (65-70% of filtered water) as obligatory water reabsorption
- HCO3: Linked to sodium transport
- Nutrients :100% of glucose, amino acids, vitamins by secondary active transport
- Phosphate (80% reabsorbed)
- Ions: Ca,Mg,K by passive diffusion

Loop of Henle:
- Na,K,Cl: active transport by Na-K-2Cl cotransporter in ascending limb(These transporters are the targets for loop diuretics e.g **Furosemide**)
- Water :By osmosis (10% of filtered water) in descending limb helped by ADH



Distal tubules :(DT)
- Early DT has Na+-Cl- cotransporters (inhibited by **Thiazide** diuretics)
- Late DT has Na+ (and K+) channels that are increased by aldosterone hormone (inhibited by K sparing diuretics e.g. **Ameloride**).Net result is Na reabsorption and secretion of either K+ or H+ and
- Water reabsorption follows Na
- Reabsorption of Ca controlled by parathyroid hormone

Collecting ducts:
- Like late DT, the collecting tubule has Na+ (and K+) channels induced by Aldosterone
- Water reabsorption secondary to Na reabsorption and passive osmosis 2ry to interstitial hypertonicity with help of Antidiuretic hormone

Summary of renal functions:

1. **Control body water**
2. **Get rid of waste e.g. Blood urea nitrogen , creatinine, H^+, excess K^+, Drugs**
3. **Regulation of acid –base balance by:**
   - About 90% of filtered bicarbonate is absorbed in the **proximal tubule** aided by carbonic anhydrase (CA)
   - Secretion of hydrogen ion as titratable acid or as NH4+ at the **distal tubule**
4. **Hormonal role e.g. Release of Erythropoietin, Activation of vit D**

Estimation of Glomerular Filteration Rate (GFR)

- Formation of nephrons is complete at 36-40 wk of gestation, but functional maturation with tubular growth and elongation continues during the 1st decade of life
- Because new nephrons cannot be formed after birth, any disease that results in progressive loss of nephrons can lead to renal insufficiency

Serum creatinine:

- Estimates the GFR in the steady state
- Insensitive measure of decreased renal function because its level does not rise above normal until the GFR falls by 30-40%

Clearance tests (require timed urine collection)

- The clearance is represented by the following formula:
  Cs (mL min) = *U*s (mg mL) × *V* (mL min) *P*s (mg mL)
  Where *Cs* equals the clearance of substance s, *Us* reflects the urinary concentration of s, *V* represents the urinary flow rate, and *Ps* equals the plasma concentration of s.
- To correct the clearance for body surface area, the formula is
  Corrected clearance (mL/ m/1.73m^2) = C (ml/min) ×1.73/Surface area (m^2)
- Clearance substances
  - Exogenous: Inulin clearance (Gold standard – Difficult)
  - Endogenous: Creatinine clearance

Formula in pediatrics:

1. Haycock-Schwartz formula

eGFR = k × ht / *P*cr

eGFR	Estimated glomerular filtration rate(in ml/min/1.73m2)
k	Empirical value relating height to muscle mass = **0.33** for LBW infant = **0.45** for Full term infant = **0.55** for child or adolescent girl = **0.7** for adolescent boy (13-18 years old)
ht	Height in centimeters
*P*cr	Plasma concentration of creatinine in mg/dl

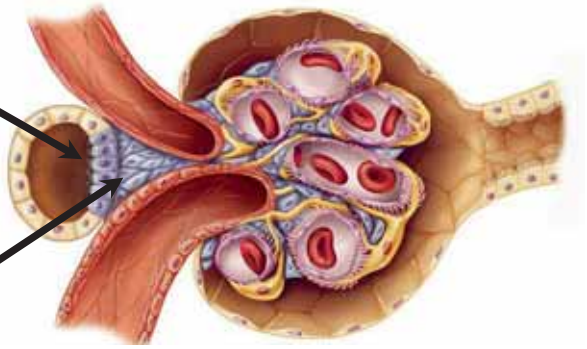
When timed urine collection is not possible, simply Haycock-Schwartz formula is the best, easiest and cheapest way to assess GFR.

2. DTPA scan: Can assess split functions of both kidneys

Control of renal hemodynamics occurs through the following mechanisms:

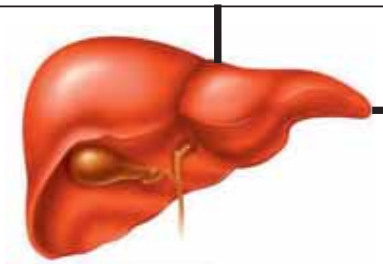
1. **Renin-Angiotensin-Aldosterone System (RAAS)** primed by Juxtaglomerular apparatus (JGA): composed of two cell types, macula densa cells(modified distal tubular cells) and juxtaglomerular cells, located at the junction of the afferent and efferent arterioles

- Low renal vascular flow
- Low sodium delivery at the macula densa cells



Stimulate release of **Renin** from the juxta glomerular cells

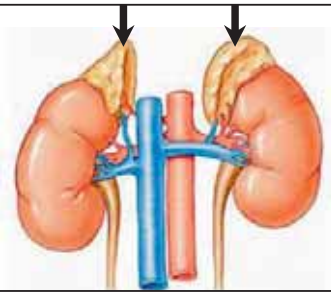
Renin converts angiotensinogen released by the liver to **Angiotensin I**





Angiotensin I is converted to Angiotensin II in the lungs by **Angiotensin Converting Enzyme (ACE)**

Angiotensin II
- A very potent systemic vasoconstrictor
- Stimulates release of Aldosterone by supra renal gland



Increase renal blood flow

Aldosterone is secreted by supra renal gland zona glomeruloza. It enhances distal tubular salt and water retention in exchange with H^+ or K^+

2. Intra renal prostaglandins

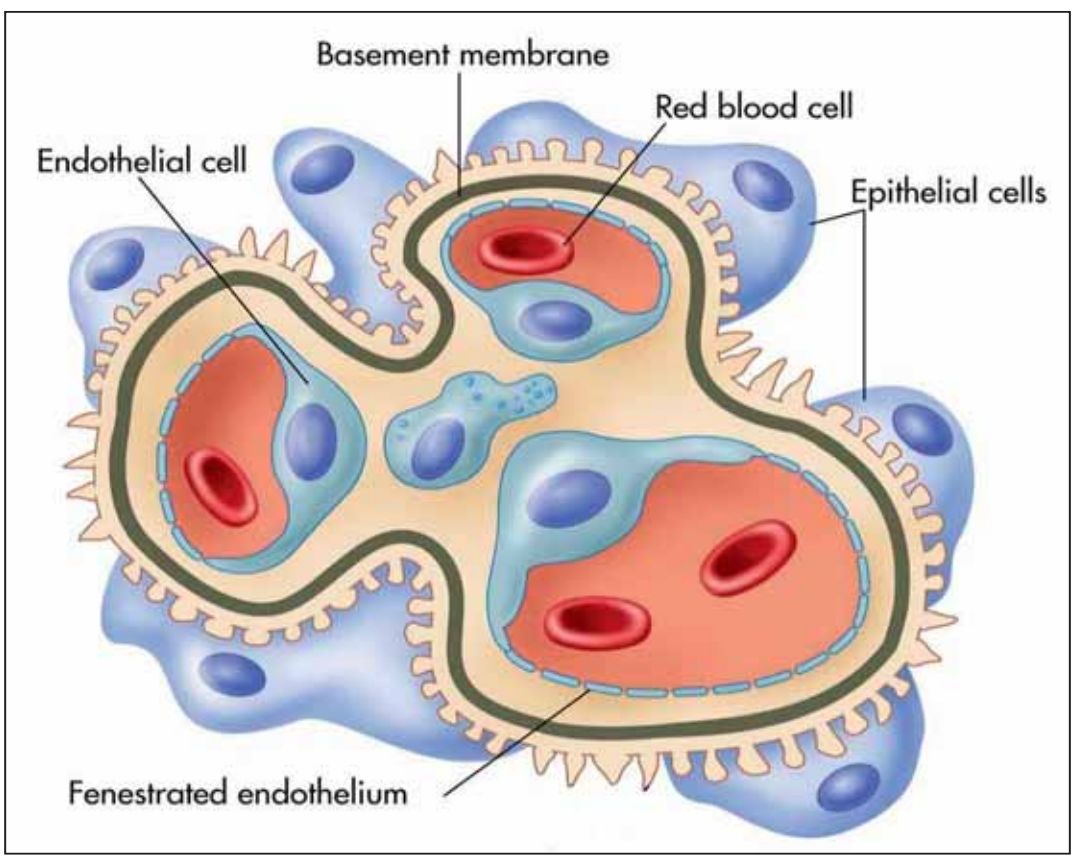
- PGE2 and prostacyclin [PGI2] are vasodilators
- They counteract primarily angiotensin II–mediated vasoconstriction
- Nonsteroidal anti-inflammatory drugs (NSAIDs) such as aspirin will block prostaglandin synthesis and restrict the compensatory renal vasodilation

3. Atrial natriuretic peptide (ANP)

- Is released from the right cardiac atrial myocytes in response to stretch (at high blood volume)
- ANP dilates the afferent arteriole, and constricts the efferent arteriole, increasing glomerular capillaries hydrostatic pressure, and thus, GFR.
- The enhanced flow increases sodium and water excretion, reducing blood volume.

Blood filtrate barrier ultrastructure

1. Epithelial cells (Podocytes)
2. Glomerular basement membrane
3. Endothelial cells



Hematuria

Definition

- **Macroscopic (Gross) hematuria:** Blood in the urine visible to the naked eye
- **Microscopic hematuria:**
  - More than **5** red blood cells (RBCs) per high-power field on freshly voided and centrifuged urine
  - Isolated asymptomatic microscopic hematuria is found in up to 4% of healthy children.

Presentation

- Episode of macroscopic hematuria (causes alarm to child/family).
- Incidental finding of microscopic hematuria.
- Family screening and routine urinalysis.

Other causes of 'red urine'

The following can usually be distinguished from hematuria by taking a careful history, and with urine dipstick testing and microscopy:

A. Pathologic
   - Hemoglobinuria
   - Myoglobinuria

B. Non pathologic
   - Foods coloring (e.g. beetroot).
   - Drugs (e.g. rifampicin).
   - Urate crystals (in young infants, usually 'pink' nappies).
   - External source (e.g. menstrual blood losses).
   - Fictitious: consider if no cause found

Causes of haematuria

A. Glomerular

1. Immunologic injury :Glomerulonephritis (GN)
2. Structural disorder (Alport syndrome, thin basement membrane disease)
3. Toxin-mediated injury (HUS)

B. Extra Glomerular

- **Tubulo interstitial/Parenchymal**
  a. Inflammation (interstitial nephritis, pyelonephritis)
  b. Vascular (sickle cell trait/disease, renal vein thrombosis)
  c. Structural (cyst rupture, Wilms tumor, urinary tract obstruction, trauma)
- **Lower urinary tract**
  a. Inflammation (cystitis, hemorrhagic cystitis, urethritis)
  b. Injury (trauma, kidney stone)
  c. Hypercalciuria

(Essential Nelson)

Glomerulonephritis (GN)

Definition: Group of diseases with acute glomerular injury. Initiated in most cases by immunologic mechanism. With variable presentation of:

- Acute kidney injury (AKI) (oliguria, uremia, elevated creatinine)
- Hematuria
- Hypertension
- Peripheral edema
- Proteinuria

Classification

With low serum complement (C3,CH50)	With normal serum complement
Renal disease	**Renal disease**
▪ Acute post infectious GN (> 90%) - Post streptococcal GN - Other infections: e.g. staph, pneumococci, HBV, …. ▪ Membranoproliferative GN type 1	▪ IgA nephropathy
Systemic disease	**Systemic disease**
▪ Chronic Post infectious GN: - Infected shunt - Infective endocarditis ▪ Lupus nephritis	▪ Vasculitis ▪ Henoch-Schönlein purpura ▪ Goodpasture's syndrome

Acute Post Streptococcal Glomerulonephritis (APSGN)

Definition and Etiology

- Acute Nephritic syndrome which follow infection with nephritogenic strain of group A-β hemolytic streptococci causing throat infection or skin infection
- Streptococcal pyogenic exotoxin (SPEB) mimic glomerular basement membrane

	Post pharyngitis GN	Post skin infection GN
Stains of strept.	- Serotypes :1,3, 4,12,18	- Serotypes :2,49,55,57
Season	- Cold(winter)	- Tropical(summer)
Sex incidence (M: F ratio)	- 2:1	- 1:1
Susceptible age	- School age	- Pre School age
Latent period	- 1-2 weeks	- 2-6 weeks; typically in child with eczema
Serology: ASO titer	- Elevated	- May be absent
Anti DNAse B	- Elevated	- Elevated

Pathogenesis

Streptococcal infection → (Latent- period) → Antibodies

↓

Antigen + Antibody + Complement (C_3) immune complexes

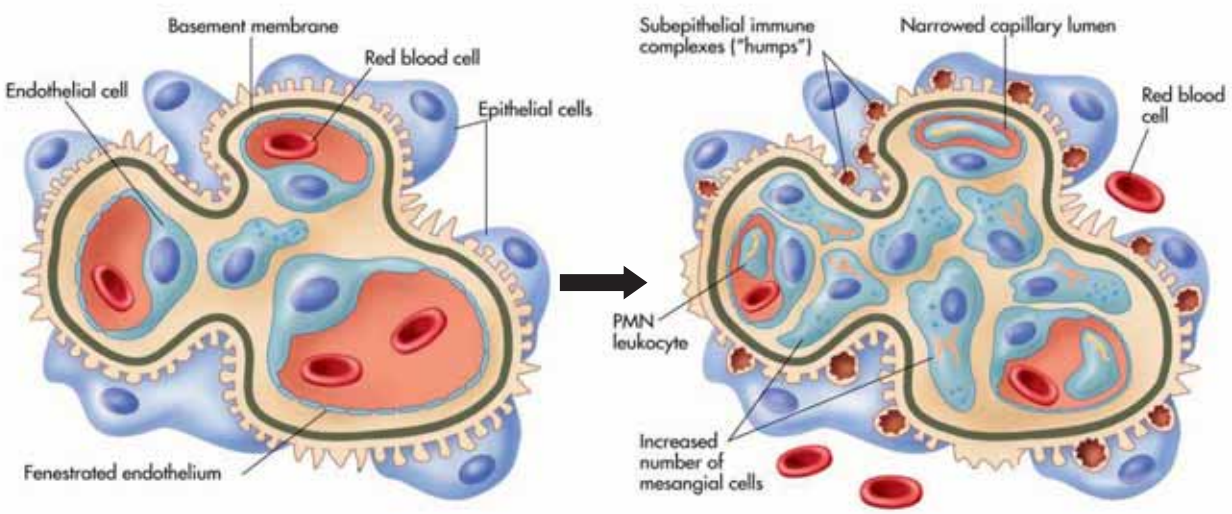
↓

Deposited in glomerular basement membrane (subepithelial humps)

↓

Acute inflammation

- Proliferation of mesangeal and endothelial cells.
  - ↓
  - **Glomerular capillaries narrowing**.
  - ↓
  - Impaired Glomerular blood flow
    - ↓ Glomerular filtration rate.
      - **Oliguria**
      - Fluid retention
        - ↓
        - Hypervolemia
          - ↓
          - **Edema**
          - → **Hypertension**
    - + + **J**uxta-**g**lomerular **a**pparatus (JGA)
      - ↓
      - + + Renin- Angiotensin Aldosteron system
      - ↓
      - **Hypertension**
- **Glomerular endothelial damage**
  - ↓
  - **Hematuria ± proteinuria**



Clinical picture

Mean age: 7 years.

History of a preceding skin or throat infection 1-3 weeks ago may be obtained

Gradual or sudden onset of:



1. Hematuria : - The 1st presenting feature
- Painless
- Cola colored (smoky) urine or bloody with gross hematuria
- Urine rarely appears normal with microscopic hematuria

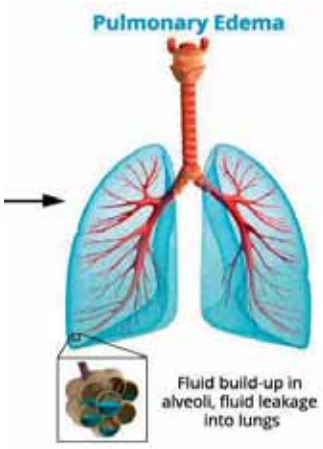
2. Edema : - 2nd presenting feature
- Mild, morning periorbital puffiness & pretibial edema
- Mainly due to reduced GFR and hypervolemia

3. Oliguria : - Urine output (UOP) < 1 ml/kg/hr or < 400 ml/m^2/day
- Some others have anuria or rarely normal urine volume

4. Hypertension : - Transient ; mostly resolves by the end of 1st week
- Mild to severe
- Seen in 60-70% at any time in the acute phase
- Principally due to *salt and water retention*

5. Nonspecific: - Headache, anorexia, vomiting, abdominal pain, mild fever.
6. Some cases are asymptomatic and discovered accidentally on routine urinalysis

Complications: May be the presenting event



1. Heart Failure and acute pulmonary edema

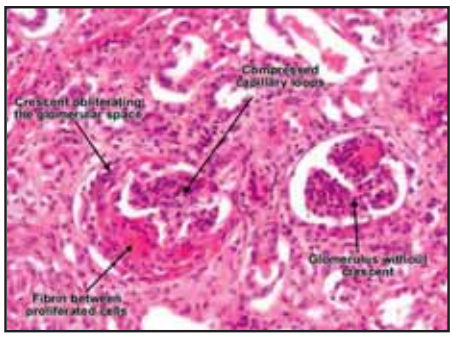
- Due to hypertension or hypervolemia
- Clinically
  - Respiratory distress and orthopnea
  - Tachycardia, tachypnea, tender liver
  - Acute pulmonary edema
    - Severe respiratory distress
    - Cyanosis
    - Wide spread crepitations

2. Hypertensive encephalopathy

- Due to acute hypertension → cerebral edema ± hemorrhages
- Clinically
  - Severe headache
  - Blurred vision
  - Vomiting → convulsions → coma

3. Acute renal failure (ARF)

- Due to rapidly progressive (*crescentic*) GN
- Usually transient
- Clinically
  - Marked oliguria or anuria
  - Acidotic breathing
  - Uremic encephalopathy



Investigations

A. For diagnosis → Urinalysis

- Color: Smoky **or** gross hematuria.
- Specific gravity: High
- Proteinuria: Usually < 1gm/dl (nephrotic range is seen in 10-20%)
- Microscopy: Dysmorphic RBCs / RBCs casts which is pathognomonic to glomerular bleeding
- *Timed urine output collection or 24 hours collection can prove oliguria/anuria*

B. For effect

- Electrolytes → may be hyperkalemia & dilutional hyponatremia
- Renal function tests → may be impaired.
- Anemia → due to hemodilution

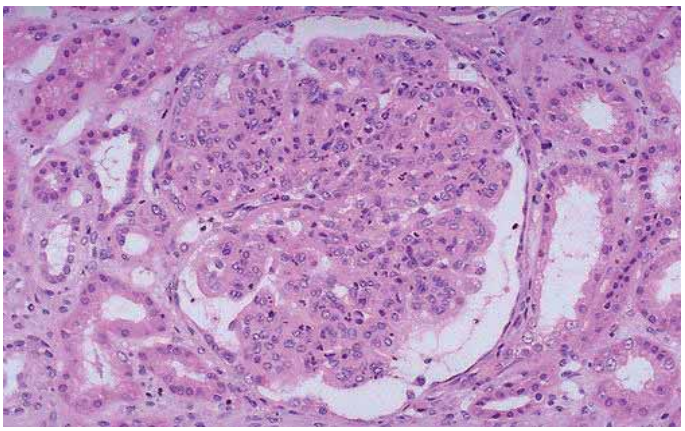
C. For etiology

- Low complement component C3 (Normal C4)
- Evidence of recent streptococcal infection:-
  - Throat or skin lesion swab culture
  - Anti- streptolysin O (ASO) titer →> 1/200 todd unit; may be negative after skin infection. **(** *ASO titer is raised in up to 20% of healthy children*)
  - Anti Deoxyribonuclase B titre (Anti- DNase B).
- **Renal biopsy** is indicated for :

  a. In acute phase:
  - Nephrotic syndrome (nephritic nephrosis)
  - Rapidly rising creatinine suggesting rapidly progressive glomerulonephritis

  b. For prognosis in atypical course
  - Abnormal creatinine at 6 weeks
  - Low C3 > 3 months
  - Proteinuria > 6 months

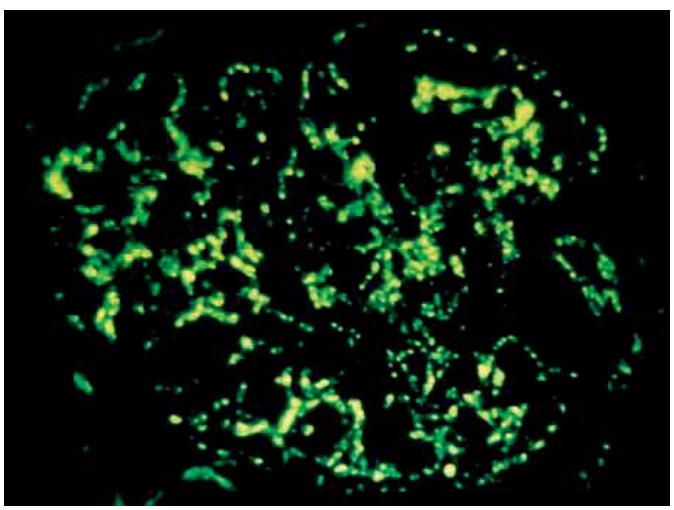
Light microscopy



Glomerulus is

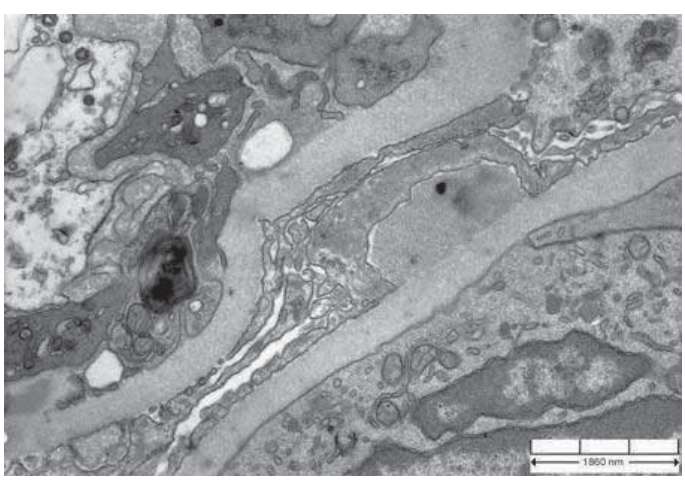
- Hypercellular
- Capillary loops are poorly defined.
- Massive influx of neutrophils

Immunofluorescence microscopy



Reveals:
"*lumpy-bumpy*" deposits of immunoglobulin and complement on the glomerular basement membrane (GBM) and in the mesangium

Electron microscopy



Reveals:
Glomerular subepithelial cone-shaped electron-dense deposits, referred to as *"humps."*

Course

- Nephritis manifestations usually resolve by 2-3 weeks
- C3 normalize by 6-8 weeks of onset
- Abnormal urinalysis improve within 4-8 weeks of onset
- Microscopic hematuria may persist for 1-2 years with *no long term relevance*

Differential diagnosis:

From other causes of Hematuria (see later)

Treatment

1. **B**ed rest as needed by the patient or with complications
2. **C**ourse of penicillin for 10 days is necessary to prevent spread to contacts but will not help the nephritis.
3. **D**iet
   - Assess fluid balance
   - Fluid restriction:
     - In oligo-anuria to avoid hypervolemia.
     - Intake = urine output plus insensible loss (200-400ml/m^2/d or 20-40 ml/kg/day) plus any additional losses (vomitus or diarrhea)
     - Given as 0.45% saline /2.5 % dextrose or 0.45% saline /5-10 % dextrose in infants
   - Salt restriction
   - Potassium and protein restriction→ only with renal failure
     - Provide calories (reduce catabolism) by giving enteral *Maxijul* 10-20% via naso-gastric tube
4. Hypertension (**E**levated blood pressure):

Mild to moderate hypertension	Severe hypertension
- Fluid restriction - Furosemide 2-5 mg/kg IV - ACE inhibitor e.g Captopril - Angiotensin Receptor Blockers - Nifidipine or Amelodipine.	- Furosemide - Na nitroprusside (infusion) - Hydralazine - Diazoxide (i.v. push).

5. Treat Complications (**F**ailures):
   - ✧ Heart **f**ailure: Treatment depends on the underlying cause:
     - ○ Hypertensive heart failure→ treat hypertension
     - ○ Hypervolemic heart failure → diuretics ± dialysis.
   - ✧ Acute renal **f**ailure:
     - ○ Conservative treatment with or without peritoneal dialysis(see later)
     - ○ Cresentic nephritis on biopsy may benefit from immunosuppression e.g. pulsed IV methyleprednisolone

(Oxford Pediatric Nephrology)

6. Discharge from hospital (**G**o home) if there is:

- No gross hematuria
- Normal renal functions
- No or controlled hypertension

Prognosis

- Over 95% of post streptococcal glomerulonephritis recover completely
- Less than 5% go into rapidly progressive glomerulonephritis may end in chronic renal failure.
- Mortality is due to heart failure, hypertension and renal failure
- Recurrence is extremely rare

IgA Nephropathy (Berger Nephropathy)

- IgA nephropathy is an immune complex disease
- It is characterized by a predominance of IgA within mesangial deposits of the glomerulus in the absence of systemic diseases such as systemic lupus erythematosus or Henoch-Schönlein purpura.
- The most common chronic glomerular disease worldwide.

Clinical

- Recurrent gross (macroscopic)hematuria*
  - Commonest presentation
  - Occur 1-2 days following upper respiratory tract infection
- Asymptomatic microscopic hematuria ± proteinuria.
- Nephritic syndrome
- Nephrotic syndrome
- Nephritic-nephrotic syndrome.
- Progressive renal dysfunction in adulthood.

Diagnosis

- Normal serum levels of C3
- Serum IgA levels have *no diagnostic* value
- Renal biopsy : mesangeal deposits of IgA

Treatment

There is no accepted treatment regimen; most are controversial.

A. Microscopic hematuria /recurrent gross hematuria:

- Of no clinical significance
- Tonsillectomy is advised by some investigators
- Some reports suggest vit E

B. Proteinuria & hypertension

- ACE inhibitors & angiotensin II receptor antagonists

C. Patients with adverse risk factors:

- Treat proteinuria and hypertension
- Immunosuppression including: steroids azathioprine, cyclophosphamide, mycophenolate mofenil (MMF)
- Fish oil (contains anti-inflammatory omega-3 fatty acids)
  - May decrease the rate of renal progression in patients with progressive disease
  - Preparation : Omacor capsule and Maxepa syrup
  - Main side effect is halitosis and none adherence

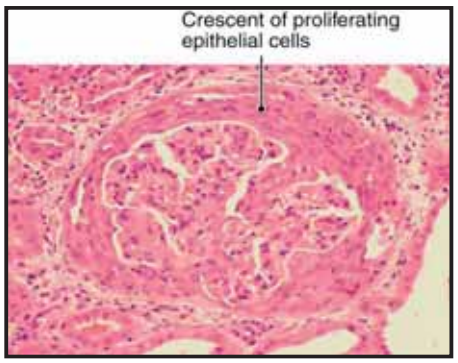
D. Successful renal transplantation

Recur in only 15–30% of patients

Rapidly Progressive (Crescentic) Glomerulonpehritis

Crescents in the majority of glomeruli progress rapidly to end-stage renal failure.

Causes



1- Immune complex glomerulonephritis:
   - Post-streptococcal glomerulonephritis.
   - Henoch Shönlein purpura
   - IgA nephropathy
   - Lupus nephritis
   - Membranoproliferative GN (MPGN)

2- Anti Glomerular Basement Membrane GN (Good Pasture Syndrome):
   - Pulmonary hemorrhage and glomerulonephritis associated with antibodies against lung and against glomerular basement membrane (anti-GBM)
   - Hemoptysis: is usually the presenting complaint (that may lead to death).
   - Hematuria, proteinuria, and progressive renal failure
   - Anti-GBM antibodies confirm the diagnosis

3- Idiopathic.

4- ANCA mediated glomerulonephritis (Wegner granulomatosis)

Pathology

- Characteristic Crescents inside Bowman capsule composed of:
   - Proliferating epithelial cells
   - Fibrin
   - Basement membrane like material
- Idiopathic form: absent deposits; only crescents.

Clinical picture:

Suspected in acute nephritic episode leading to acute renal failure

Diagnosis

1- Serologic studies e.g. C3, anti-Dnase-B,….

2- Renal biopsy

Prognosis and Treatment

☆ Spontaneous recovery may occur with post streptococcal type

☆ Treatment options (choice is cause related):
- Pulsed methyle prednisolone / oral prednisone
- Plasma exchange
- Cyclophosphamide
- Biological agents: Rituximab, Anti-TNF therapy
- Supportive care of ARF e.g. Dialysis

Hemolytic Uremic Syndrome (HUS)

Definition

- One of the most common causes of community-acquired acute kidney failure in young children
- It is characterized by the triad of : Microangiopathic hemolytic anemia thrombocytopenia, and renal insufficiency

Causes

1. Infection induced

- Often preceded by acute bloody diarrhea (*Typical HUS*) caused by:
  - Vero toxin -producing *E. coli* (Enterohemorrhagic E coli); mainly O157:H7 (Europe , America)
  - Shiga toxin producing *Shigella dysenteriae type 1* (Asia-Africa)
  - ✓ Transmitted by undercooked meat or unpasteurized milk
  - ✓ Epidemics have followed ingestion of undercooked, contaminated hamburger at fast food restaurants
- Neuraminidase (T antigen) -producing *Strept.pneumonia* is a rare cause.

2. Other less common causes

Genetic (inherited HUS)

- Defective Von Willebrand factor–cleaving protease (ADAMTS 13)
- Defective Complement factor H, I, or B
- Defects in vitamin B12 metabolism

Secondary HUS

- Antiphospholipid syndrome
- Systemic lupus.
- Drugs (Cyclosporin ,chemotherapy)
- Bone marrow transplantation

Pathogenesis

Toxins (bacterial & others) → + + Leucocytes → ↑ tumor necrosis factor & interleukin 2

↑ O_2 Free radicles. → endothelial damage

↑ Procoagulants

→ Intra vascular thrombi

→ Damaged circulating RBCs & platelets → Micro angiopathic hemolytic Anemia / Thrombocytopenia.

→ Glomerular obstruction → Acute renal failure /Hematuria

Clinical picture

- Common age: most common in preschool and school-aged children
- Acute diarrhea (often bloody) or respiratory infection is followed few days later by :
  - Acute hemolysis → Acute pallor, jaundice and purpura
  - Acute renal failure → Oliguria, edema, hypertension, acidotic breathing.
  - Hematuria and may be hemoglobinuria
- Genetic forms
  - Onset is insidious
  - Triggered by a variety of illnesses, including mild, nonspecific gastroenteritis or respiratory tract infections
- Complications:
  - Acute renal failure
  - Acute heart failure (hypervolemia, hypertension, anemia)
  - Acute neurologic dysfunction ; encephalopathy in ≤ 20 %)

Workup

For diagnosis

For ARF	- High creatinine and blood urea nitrogen - High potassium - Metabolic acidosis (↓ pH, ↓ $PaCO_2$, ↓ HCO_3)
For MAHA	- Anemia - Reticulocytosis - Negative Coombs test (except pneumococci-induced HUS) - Blood Film: fragmented RBCs (Helmet's cells, shistocytes)
For bleeding	- Thrombocytopenia counts usually 20,000-100,000/mm3. - Coagulation profile → Normal (unlike DIC)
Urinalysis	- Hematuria and may be hemoglobinuria

For etiology

- History of diarrheal prodrome or pneumococcal infection
- If history is negative, consider evaluation for genetic causes
- Stool culture is often negative in diarrhea-associated HUS (even, risk of developing HUS with intestinal E.Coli 0157:H7 is only 10%)

Differential diagnosis

1. **From other causes of intrinsic renal failure**
2. **From other causes of microangiopathic hemolytic anemia: e.g.**
   - Bilateral renal vein thrombosis (Marked renal enlargement , Doppler)
   - DIC

Treatment

i. Prevention

- Adequate cooking of meat (especially Hamburger)
- Isolation of cases to avoid cross infection with E.coli

ii. Curative

- Rules of management
  - Early recognition of the disease
  - Monitoring for potential complications
  - Meticulous supportive care
- **ARF**
  - Meticulous fluid and electrolyte management
  - Early dialysis or hemofiltration is usually indicated
  - Treatment as for intrinsic acute renal failure
- **Packed RBCs**
  - May be repeated in active phase as hemolysis take up to 2 weeks
  - For hemoglobin < 6 gram/dl
  - In pneumococci-associated HUS use washed red cells (remove residual plasma containing IgM directed against T antigen)
- **Platelets**
  - Should generally not be administered(consumed by the active coagulation and can theoretically worsen the clinical course)
  - Considered only if elective surgery is indicated or with ICH
- **Others**
  - Antibiotics can result in increased toxin release, potentially exacerbating the disease. It is not recommended except for any underlying pneumococcal infection
  - Plasma infusion or plasmapheresis is considered for serious CNS involvement and in atypical HUS
  - ***Eculizumab***; Mono clonal antibody against C5 is promising in atypical (genetic) HUS, even in patients resistant to plasma therapy.

Prognosis

- ✓ Course is unpredictable
- ✓ HUS can be relatively mild or can progress to a severe, and even fatal, multisystem disease
- ✓ More than 50% require dialysis ; 5% remain dependent on dialysis, and up to 20-30% are left with some level of chronic renal insufficiency.
- ✓ Patients who have recovered completely, with no residual urinary abnormalities after a year, are unlikely to manifest long-term sequelae

Alport Syndrome

Definition

- Multisystem disorder : Hereditary nephritis, sensineural deafness and often eye abnormalities
- Classical Alport syndrome is due to mutations in α chain of type IV collagen (COL4); a major component of basement membranes.

Genetics

- Mainly X- Linked disease (85%) caused by a mutation in the COL4 α 5 gene; some are associated with diffuse leiomyomatosis.
- Autosomal recessive or autosomal dominant form (less common).

Clinical manifestations

i. Renal manifestations

1. Hematuria

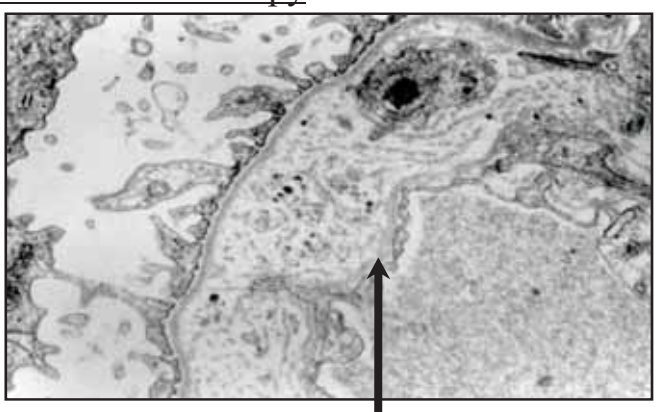
- Affects 100 % of males and 95% carrier females
- Asymptomatic microscopic hematuria, which may be intermittent
- Recurrent gross hematuria*

2. Proteinuria

- Common in males; may be absent, mild, or intermittent in females.
- Progressive by the second decade of life

Renal pathology

- Light : non specific
- Electron microscopy



Reveals:

- Diffuse thickening, thinning, splitting of the basement membranes of the glomeruli &tubules
- Lamellation of the basement membrane (basket wave pattern)

Prognosis

- ESRD occurs before age 30 years in approximately 75% of hemizygotes with X-linked AS.
- Risk factors for progression
  - Gross hematuria during childhood.
  - Nephrotic syndrome.

ii. Extra Renal manifestations

Hearing deficits

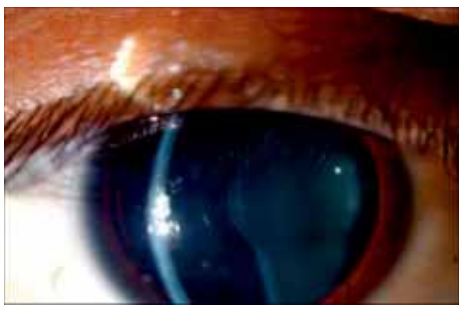
- Affects 90% of males with X-linked AS, less common in cases with autosomal recessive AS, and only 10% of heterozygous females with X-linked AS.



  - Bilateral sensorineural hearing loss (Never congenital in onset)
  - Starts for the high-frequency range but progresses to involve conversational speech
  - Hearing aids are eventual.

Ocular abnormalities

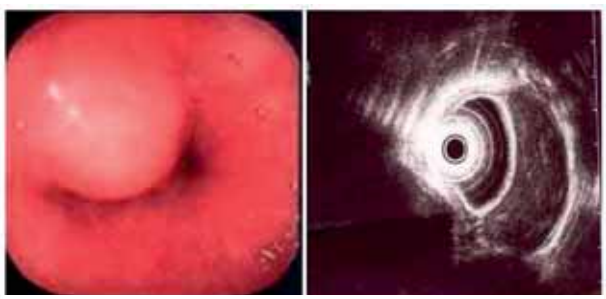
- Mainly affects patients with X-linked



  - Anterior *lenticonus* (extrusion of the central portion of the lens into the anterior chamber) is pathognomonic
  - Corneal erosions.
  - Macular flecks.

Rarely

- Leiomyomatosis of the esophagus, tracheobronchial tree and female genitalia
- Platelet abnormalities



Endoscopic image of eosophageal submucosal leiomyoma and endoscopic US

Diagnosis

Requirements

- Family history
- Screening urinalysis of first-degree relatives
- Audiogram
- Ophthalmology exam

Diagnositic

- Hematuria *with* anterior lenticonus is pathognomonic
- Hematuria with at least **two** of the following characteristics:
  - Sensorineural deafness
  - GBM thickening and thinning (by EM)
  - Macular flecks
  - Recurrent corneal erosions
- Basement membrane collagen studies.

Note

- Mutation screening or linkage analysis is not clinical use.
- Prenatal diagnosis is available for familial sex linked disease.

Treatment

- ✓ No specific therapy for Alport syndrome.
- ✓ Angiotensin-converting enzyme inhibitors may slow the rate of renal progression.
- ✓ Supportive treatment for renal failure (conservative, dialysis , kidney transplantation); 5% of renal transplant recipients develop anti-GBM nephritis.

Approach to Hematuria

We should exclude other causes of red urine without RBCs by urine analysis (Dipstick) which includes:-

A. Heme positive

- Hemoglobinuria in case of acute hemolytic anemia.
  - CBC shows fragmented RBCs & reticulocytosis
  - Hemoglobin in urine
- Myoglobinuria in case of rhabdomyolysis (myositis, crush)
  - High serum creatine kinase.

B. Heme negative

- Foods e.g. Beet roots, black berries.
- Drugs e.g. Rifamipicin, Desferal, Nitrofurantoin.
- Urate crystals (red diaper).

History

- Glomerulonephritis: sore throat/rashes/body swelling
- UTI: fever/frequency/dysuria.
- Renal stones: colicky abdominal pain/family history.
- Coagulopathy: easy bruising.
- Trauma
- Family history: hematuria, deafness (Alport's), sickle cell disease.

Examination

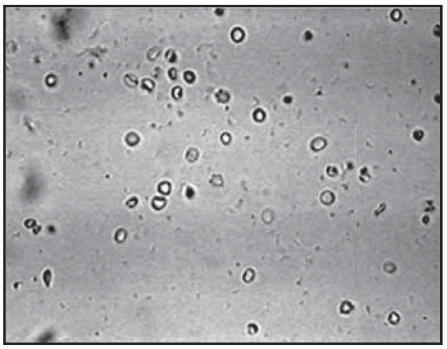
- Blood Pressure (use age,sex and height appropriate blood pressure centiles)
- Abdomen: palpable masses (polycystic kidneys, tumors, hydronephrosis).
- Skin: rashes.
- Joints: pain/swelling.

Investigations

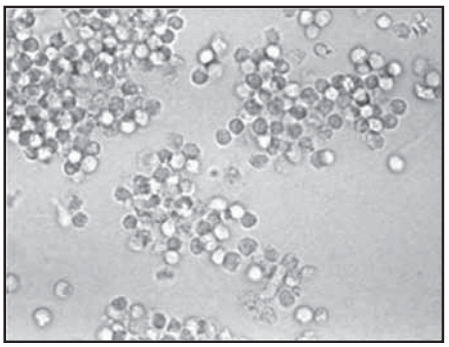
- It is important to identify serious, treatable, and progressive conditions.
- During an acute illness, exclude UTI by urine culture.
- Asymptomatic or 'benign haematuria' in children without growth failure, hypertension, oedema, proteinuria, urinary casts, or renal impairment is a frequent finding.

1. Localize hematuria

	Glomerular	Extra glomerular
Acute nephritic syndrome	Present	Absent
Color	Cola or tea colored	Bright red
Clots	Absent	May present
RBCs Shape	Dysmorphic (distorted)	Normal
RBCs casts	Present	Absent
Proteinuria	> 30 mg / dL.	< 30 mg / dL.



Typical appearance of RBCs in glomerular hematuria: RBCs are small and vary in size, shape, and hemoglobin content



Typical appearance in non-glomerular hematuria: RBCs are uniform in size and shape, along with numerous polymorphs

(*Consensus Statement on Evaluation of Hematuria, Indian Pediatrics 2006*)

2. For Glomerular hemturia:

- **Hematology**
  - CBC with differential
- **Chemistry**
  - Electrolytes, Ca
  - BUN/ Creatinine /Creatinine clearance
  - Serum protein/Albumin /Cholesterol
  - Urine protein
- **Immunology**
  - C3/C4
  - ASO/Anti-DNase B
  - ANA
  - Antineutrophil antibody

Reduced C_3 in	- Post infectious glomerulonephritis - Systemic lupus nephritis (and low C4) - Nephritis with chronic infection - Membrano proliferative glomerulonephritis

3. Renal Biopsy

- Unexplained persistent or recurrent gross hematuria
- Lupus nephritis
- Glomerulonephritis with: → nephritic nephrosis
  → Absent low C_3
- Unexplained acute renal

4. For extra glomerular hematuria

Step 1

- Urine culture

Step 2

- Urine calcium/creatinine ratio
- Sickle prep (African American)
- Renal/bladder ultrasound

Step 3

- Urinalysis: siblings, parents
- Serum electrolytes, Cr, Ca
- If crystalluria, urolithiasis, or nephrocalcinosis: 24-hour urine for Ca, creatinine, uric acid, oxalate
- If hydronephrosis/pyelocaliectasis: Cystogram, renal scan

(Nelson textbook of pediatrics)

Treatment

- If obvious cause (e.g. UTI), treat.
- If complex diagnosis (impaired renal function, proteinuria, or family history) refer to paediatric nephrology unit.
- If no cause found and normal renal function, BP, and no proteinuria, monitor until resolves.
- If no resolution after 6mths or change in any of above parameters refer to paediatric nephrology unit.

Proteinuria

Normal values

- Most of the proteins filtered by the *glomeruli* are reabsorbed by the *proximal convoluted tubules*.
- The normal daily urinary protein loss is < 4 mg/m^2/hr or < 150 mg/24hr.

Detection

1. Urine dipstick analysis (Albustix)

- Less sensitive; affected by urine specific gravity and pH.
- Primarily detects albuminuria ; less sensitive for other forms of proteinuria
- Reported as:
  - Negative
  - Trace (10-20mg/dl)
  - 1+ (30 mg/dl)
  - 2+ (100mg/dl)
  - 3+ (300 mg/dl)
  - 4+ (1000–2000mg/dl)



2. Timed (24-hr) urine collections

- Quantitative for proteinuria (not practical)
- Normal value < 4 mg/m^2/hr.
- Abnormal 4 – 40mg/m^2/hr.
- Nephrotic range > 40 mg/m^2/hr or > 50 mg/kg

3. Spot urine protein: creatinine ratio (UPr: UCr)

- Quantitative for proteinuria and more practical timed urine collections
- Use early morning urine sample
- Normal value < 0.5 (<0.2 in children ≥2 yr)
- Nephrotic-range proteinuria > 2

Causes of proteinuria

i. Transient proteinuria (Never exceed 2+):

- Postural or orthostatic : Proteinuria in upright posture only.
- Non postural: - Fever, vigorous exercise, seizures

ii. Persistent proteinuria

	Tubular	Glomerular
Due to	Decreased reabsorption of filtered proteins.	Increased glomorular basement membrane (GBM) permeability.
Level	Usually < 1gm/24 hours	Can exceed 1gm/24 hours
Type	Low molecular weight proteins	Low and high molecular weight proteins
Albuminuria	Absent.	Present.
Associations	Other proximal tubular defects e.g glucosuria, phosphaturia	Edema. May be hypertension, hematuria.
Causes	Fanconi syndromes (Cystinosis, Lignac,…… …)	- Damage to GMB, e.g. GN - Dysfunction of GBM e.g.: Minimal change nephrotic syn.

Nephrotic Syndrome

Definition: Clinico-laboratory condition characterized by:-

- Nephrotic range proteinuria defined as urinary proteins > **40** mg/m^2/hr or a first morning protein : creatinine ratio of > **2-3 : 1**
- Hypoalbuminemia
- Generalized edema

Hyperlipidemia

Incidence: 15 times commoner in children than adults

Causes of Nephrotic Syndrome

1. Idiopathic (? Lymphocyte dysfunction → altered GBM permeability)

- 90% of cases
- Histologic types
  - Minimal change disease (85%)
  - Focal segmental glomerulosclerosis
  - Membranous nephropathy

2. Genetic nephrotic syndrome

- Finnish-type congenital nephrotic syndrome (absence of nephrin)
- Focal segmental glomerulosclerosis (podocin, actinin mutations)
- Diffuse mesangial sclerosis (laminin mutations)
- Denys-Drash syndrome (mutations in WT1 transcription factor)

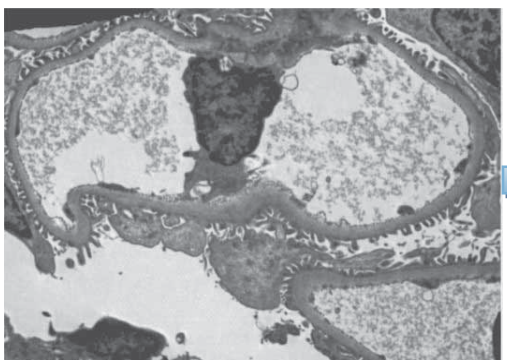
3. Secondary nephrotic syndrome

Due to	Examples
Glomerulonephritis with heavy proteinuria	- Systemic lupus nephritis - Henoch Schonlein purpura
Infection	- Hepatitis B, C - HIV-1 - Malaria - Syphilis - Toxoplasmosis
Allergy	- Serum sickness - Bee sting
Drugs	- Penicillamine - Gold salts - Interferon - Nonsteroidal anti-inflammatory drugs
Diseases	- Sickle cell disease - Amyliodosis - Thrombosis of renal veins
Tumors	- Hodgkin lymphoma /Leukemia

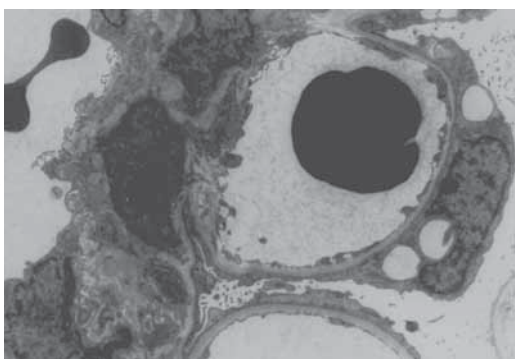
Histological classification

1. Minimal change nephrotic syndrome (MCNS)

- Light microscopy and immunoflorescence → normal.
- Electron microscopy → loss of podocytes foot processes.
- Proteinuria is selective
- Steroid responsive in > 95%.
- In prolonged cases as in relapses or resistance to treatment→ more notable changes are seen as focal glomerulosclerosis



Electron microscopy photo of a normal glomerulus with epithelial cells with normal loose foot processes



Electron microscopy photo of a glomerulus in MCNS showing diffuse effacement of epithelial foot processes

2. Focal segmental glomerulosclerosis (FSGS):

- Light and electron microscopy → Segmental sclerosis
  → Focal sclerosis
- Steroid responsive in < 20%.

3. Mesangeal proliferative glomerulonephritis

- Light and electron microscopy → increase mesangeal cells & matrix.
- Steroid responsive in < 50%.

4. Membranous

- Light & electron microscopy:
  - Uniform thickening of glomerular basement membrane
  - Focal sclerosis as the disease progress

Idiopathic Nephrotic Syndrome

Pathogenesis

1. Proteinuria

* Due to increased glomerular basement membrane (GBM) permeability
(Due to defect in size **or** altered negative charges of GBM barriers)

⇓

Proteinuria is either

- Selective = escape of low molecular weight proteins **as** albumin.
- Non selective = escape of both low & high molecular weight proteins

⇓

2. Hypoproteinemia

Decreased total serum proteins → basically hypoalbuminemia

3. Hyperlipidemia

⇓

Mainly hypercholestrolemia.

Due to

- Hypoproteinaemia
  ↓
  Stimulate the liver
  ↓
  ↑ Protein synthesis including lipoproteins
- ↓ lipoprotein Lipase
  ↓
  ↓ Lipid catabolism

4. Generalized oedema

⇓

Due to ↓ plasma osmotic pressure
(↓ proteins & albumin)
↓
fluid shift to interstitial tissue
(mainly in lax & gravity dependent)
↓
↓ intra vascular volume

- ↑ADH
  ↓
  Water retention
- ↓ Renal blood flow
  ↓
  ++ JGA
  ↓
  ↑ Aldosterone
  ↓
  Salt and water retention

Water retention + Salt and water retention → More oedema.

Clinical picture

- ✓ Peak age in MCNS =2-7 years/ Boys: Girls 2:1
- ✓ The initial episode and relapses may follow viral upper respiratory tract infection.

1. Generalized edema

- Start as morning periorbital puffiness then progress to involve lower limbs,genitalia and abdominal wall with stretched skin and skin breaks
- Oedema is very soft, pitting,
- Ascites and pleural effusion are very common→ may be respiratory distress.
- With developing edema and increasing weight → urine volume is declining





2. Gastro intestinal mucosal oedema → anorexia, abdominal pain & diarrhea
3. Hypertension may occur in only 5-10%.

Complications

1. Hypovolemia

- **Precipitated by**
  - Sepsis
  - Diarrhea
  - Use of diuretics



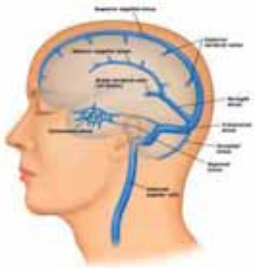
- **Presentation**
  - Abdominal pain
  - Hypotension
  - Poor perfusion
  - Hemoconcetration

2. Acute renal failure (Nephrotic crisis)

- **Precipitated by**
  - Severe hypovolemia → ↓↓ renal blood flow (pre renal failure).
  - Renal vein thrombosis.
  - Sepsis

3. Intra vascular thrombosis

- **Precipitated by**
  - Hypovolemia → hemoconcetration→ sluggish circulation
  - Increased platelet adhesiveness and certain coagulation factors.
  - Decreased natural anti-coagulants e.g. Anti thormbin III and protein C.
- **Common sites**



Cerebral cortical veins



Renal veins



Deep venous thrombosis

4. Infections

- **Precipitated by**
  - Loss of immunoglobulins and complement factor B
  - Edema or ascites acting as a potential culture medium
  - Immuno suppressive therapies
- **Common infections**: basically by capsulated bacteria and viruses
  - Spontaneous bacterial peritonitis (Strept. pneumoniae, E.Coli)
    - ✓ Child looks ill, feverish with persistent abdominal pain
    - ✓ Prompt evaluation (including cultures of blood and peritoneal fluid), and early initiation of antibiotic therapy are critical.
  - Others: urinary tract infections (*E.coli*) ,pneumonia (*H. influenza*), cellulitis (*Staph aureus*), and sepsis

5. Relapse

- Recurrence of significant proteinuria ;urine Albustix ++ or more for **3** consecutive days
- Frequent Relapsing Nephrotic Syndrome (FRNS): Relapses 4 or more within 12 months period.
- Steroid Dependent Nephrotic Syndrome (SDNS): nephrotic syndrome that relapses during steroid tapering or immediate after steroid withdrawal.

6. Complications in Resistant and FRNS

- Protein depletion: muscle wasting, osteoporosis, short stature
- Chronic renal failure → in non- MCNS
- Drugs complications: ***See later***

Investigations

1. For diagnosis

A. Urine analysis

- Heavy Proteinura:
  First morning protein: creatinine ratio of >2-3: 1 (Timed urine protein > 40 mg/m^2/hr)

- Color: Yellowish, frothy
- Specific gravity: High
- Waxy (lipoid) & hyaline casts.
- Microscopic hematuria in about 10% of MCNS.

B. Biochemical

- Low serum albumin < **2.5** gm/dl (normal: 3.5 - 4.5 gm/dl), Low total proteins < **4.5** gm/dl. (normal :6.5 - 8 gm/dl)
- Increased serum cholesterol > **220** mg/dl

- Complement C3 and C4 levels (Normal in MCNS)

2. Workup before starting therapy

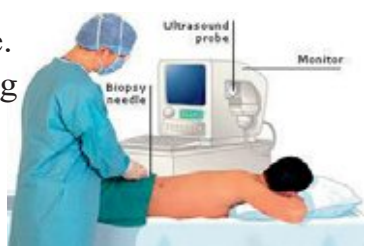
- CBC.
- Urine microscopy and culture as there is an increased rate of UTI.
- Infectious disease workup including
  - PPD (Mantoux) skin test and chest X-ray in TB endemic areas.
  - Varicella zoster serology to determine immune status.
  - Hepatitis B and C serology.
- Lupus antibody serology (ANA, extractable nuclear antibodies (ENA), and ds-DNA) in older children, and those with atypical presenting features

3. Renal biopsy

- Not indicated if MCNS is suggested.
- Indications.

Before treatment
- Age < 1 **or** >12 years.
- Gross hematuria.
- Renal failure
- Low C3

After treatment
- Steroid resistance.
- Frequent relapsing



Differential Diagnosis

1. Form other causes of generalized edema (See Clinical)

2. From causes of secondary nephrotic syndrome:

A diagnosis other than MCNS should be considered in

- Children <1 yr of age
- A positive family history of nephrotic syndrome
- Presence of :
  - Extrarenal findings (e.g., arthritis, rash, anemia)
  - Hypertension or pulmonary edema
  - Acute or chronic renal insufficiency
  - Gross hematuria

Treatment of Nephrotic Syndrome

Supportive care

1. Fluid balance, hypovolemia, and blood pressure

- Mild peripheral edema:
  - Do not require fluid restriction
- Significant edema
  - Mild fluid restriction (70 % of maintenance requirements)
  - A low salt diet control thirst and minimizes edema
- Regular assessment of temperature, weight, BP, hydration state

2. Avoid infections

- **A**void contact with infectious patients
- Broad spectrum **A**ntibiotics for any current infections pending bacterial cultures
- Nonimmune child, if exposed to **V**aricella, should receive varicella-zoster immunoglobulin dose within 4days after significant exposure.
- Edematous child should receive prophylactic penicillin **V** 12.5mg/kg bd
- **V**accines :
  - ✓ Routine vaccines should be given during remission
  - ✓ Live vaccines should only be administered when the child is off all immunosuppressive therapy (*1 month following discontinuation of steroids and 3-6 months following discontinuation of cyclosporine and cyclophosphamide*).
  - ✓ Other recommended vaccines: Pneumococcal(both types) , H. influenza vaccines and annual influenza/H1N1 vaccine

3. Avoid thrombosis

- Low dose aspirin
- Treat hypovolemia either by:
  - Plasma 20 ml /kg or
  - 4.5 % albumin solution 10 – 20ml/kg
  - Diuretics should be stopped or avoided in this setting

4. Salt free albumin

Indications	**Massive generalized oedema (anasarca) with respiratory difficulty**
Precaution	May cause acute increase of intravascular volume *so* observe closely for developing hypertension or heart failure
Dosage	- Up to1g/kg (up to 5mL/kg 20 % albumin) - Should be infused slowly over 4h - Furosemide 1–2mg/kg being given IV during the second half of the infusion

Specific therapy

Induction of remission by **Prednisone** 60 mg/m^2/d or **2** mg/kg/d (maximum dose 80mg) for **4** weeks in a single daily dose

↓

Response

- Urine albustix negative or trace for 3 consecutive days (timed urine protein < **4**mg/m^2/hr)
- Serum albumin ≥ **3.5** gm/dl.

↓

Remission of nephrotic syndrome

↓

Steroid Responsive Nephrotic Syndrome

- Proteinuria (2+ or greater) persist:
  - ✓ After 8 weeks of steroids Or
  - ✓ After 4 weeks of steroids followed by 3 methyl prednisolone pulses.
- In absence of infection or non-adherence to medication

↓

Steroid Resistant Nephrotic Syndrome

1. Steroid Responsive Nephrotic Syndrome:

After the initial 4 weeks induction:

- Use alternate day prednisone **40** mg/m^2 as single morning dose for **4** weeks
- Followed by gradual withdrawal over 4 months with monitoring for proteinuria

Suggested plan:

- 40 mg/m^2 (−1.5 mg/kg) Q48 hours for 4 weeks, then
- 20 mg/m^2 (−0.75 mg/kg) Q48 hours for 4 weeks, then
- 10–15 mg/m^2 (0.5 mg/kg) Q48 hours for 4 weeks, then
- 5–7.5 mg/m^2 (0.25 mg/kg) Q48 hours for 2 months

(Manual Of Pediatric Nephrology)

Outcomes

- No relapses (30%) → Observe
- Relapse > 3 months after steroid withdrawal → Re treatment as above
- Relapse while on alternate day steroid therapy or < 28 days after steroid withdrawal are termed steroid dependent nephrotic syndrome ; 60%

↓

* Re induction followed by alternate day steriods
* Then to minimize relapse rate try the following:

1- Prolong alternate day steroid therapy for up to12-18 months

2- Steroid sparing therapies:

Levamizole	2.5 mg/kg Every 48H for 12 months
Cyclophosphamide	2 mg/kg/day for 12 week
Cyclosporine (CSA)	5−7 mg/kg/day
Mycophenolate Mofitel (MMF)	600–1200mg/m^2/day 12 months
Tacrolimus(Prograf)	0.1−0.2/kg/day or at least 12 months
Rituximab	Every 4−6 months 375 mg/m^2 per infusion

2. Steroid resistant nephrotic syndrome: may benefit from

- Methylprednisolone pulses with cyclophosphamide (Mendoza protocol). The recent guidelines advise against this protocol for SRNS.
- Steroid pulse therapy :
  - Methylprednisolone 10 mg/kg or 300 mg/m^2 IV infusion every other day for 3 doses followed by Prednisone 40 mg/m^2 every other day.
  - Kidney biopsy and genetic testing are essential to avoid excess glucocorticoid toxicity
- Steroid sparing medicines: Cyclosporine ,Tacrolimus, MMF
- Anti proteinuric agents: ACE inhibitors and Angiotensin II Blockers(ARBs)
- Supportive therapy
  - Lipid-lowering drugs
  - Thrombosis prophylaxis
  - Vitamin D
- Renal transplantation

(Oxford Pediatric Nephrology)

N.B Drugs Complications

Steroids



- **C**ataract
- **U**lcers (peptic) → add ranitidine or proton pump inhibitor
- **S**triae
- **H**ypertension
- **I**nfections due to immunosuppression
- **N**ecrosis of bone → may require calcium and vit D
- **G**rowth retardation →monitor height Q 3 months
- **O**steoporosis
- **I**ntracranial hypertension
- **D**iabetes Mellitus
- **M**yopathy (proximal)
- **A**dipose tissue hypertrophy (moon face, Bufflo hump, trunkal obesity)
- **P**ancreatitis

Cyclophosphamide

- **A**lopecia
- **B**one marrow suppression
- Hemorrhagic **C**ystitis (prevented by I.V. Mesna)
- **D**ecreased fertility

Cyclosporin A

- **H**ypertrichosis
- **H**ypertension
- **H**yperplasia of gums
- **N**ephrotoxicity

MMF

- Leucopenia /anemia
- GIT ulcerations

Tacrolimus

- Hyperglycemia
- Nephrotoxicity

Rituximab

- Immunosuppression
- Unclear long term safety profile

Causes of death in nephritic syndrome?

1. Renal failure
2. Overwhelming infections
3. Hypovolemic shock

Congenital and infantile nephrotic syndrome

- Congenital nephrotic syndrome (CNS):
  Presentation of nephrotic syndrome during the first **3** months of life (often present before or at birth)
- Infantile nephrotic syndrome :
  Presentation of nephrotic syndrome between **3** and **12** months of age

Primary causes

- Finnish-type CNS
- Diffuse mesangial sclerosis
- Focal segmental glomerulosclerosis
- Membranous nephropathy
- Minimal change disease

Secondary causes

- Congenital Infections
- Syndrome-associated:
  - Denys Drash syndrome: Wilm's tumor, genitourinary anomalies , nephrotic syndrome
  - Nail-patella syndrome
  - Lowe's syndrome
  - Frasier syndrome : Male pseudohermaphroditism and proteinuria

Other: e.g.

- HUS
- Nephroblastoma
- Drug reaction

Fennish type congenital nephrotic syndrome

- Autosomal recessive disorder
- Most patients have mutations in the NPHS1 gene which encodes for *Nephrin* protein in the glomerular basement membrane.

Presentation

- Low birth weight
- Large edematous placenta (> 25 % mass of newborn)
- Generalized edema
- Histology: early tubular dilatation (microcytic change), later on glomerular sclerosis and fibrosis ending in end stage renal failure

Treatment

- Bilateral nephrectomies and peritoneal dialysis + Transplantation once the child has reached approximately 9 –10kg
- Alternatively, perform early unilateral nephrectomy plus medical therapy + Second nephrectomy and subsequent transplantation at 3 – 4 years of age

Medical supportive therapy (as before)

- Regular albumin infusions via a central venous catheter until bilateral nephrectomy is performed
- ACEI/ARBs.
- NSAID e.g. Indomethacin up to 4mg/kg divided into three doses.
- Thyroxine is necessary from birth.
- Childhood vaccines should be completed after nephrectomy prior to transplantation
- Close attention to nutrition and growth — high calorie and high protein diet (4mg/kg/day).
- Anticoagulation with warfarin/aspirin may have a role.

(Oxford pediatric nephrology handbook)

Acute Renal Failure (ARF)

Definition: A sudden, potentially reversible inability of the kidney to maintain normal body chemistry and fluid balance. The term ARF has been replaced by the term Acute Kidney Injury (AKI).

Etiology

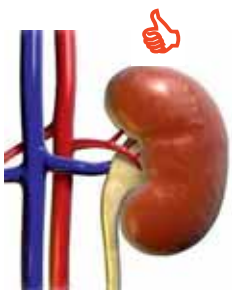
A. Pre renal (60%)

Due to: Marked reduction of renal blood flow



1. Excess losses
- Gastro-enteritis
- Polyuria
- Burns
- Hemorrhage

2. Hypo albuminenic states
- Nephrotic syndrome
- Liver failure (hepato-renal synd.)



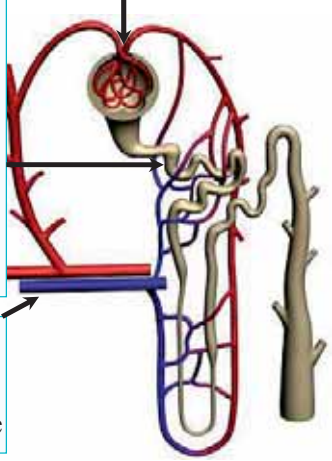
3. Impaired cardiac output
- Heart failure
- Shock

B. Intrinsic renal (30%)

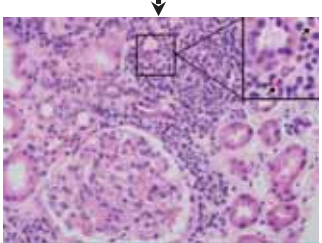
Due to: Renal parenchymal damage

1. Glomerular: Acute glomerulonephritis

2. Tubular necrosis
- Untreated pre-renal failure
- Nephrotoxins e.g. aminoglycosides
- Myoglobinuria
- Hemoglobinuria
- Crystal nephropathy



3. Interstitial
- Tubulointerstitial nephritis
- Pyelonephritis



4. Vascular
- Renal vein thrombosis.
- Hemolytic uremic syndrome

5. Acute on top of chronic: Decompensated CKD due to inter current illness

C. Post renal (10%):

Due to: urine outflow obstruction (Obstructive uropathy)
By stones, tumors, trauma

Clinical picture

1. Manifestations of the cause; basically in pre renal failure
2. **Oliguric phase:**

Mechanism	Clinical sign
Water retention	– Oliguria or anuria – Edema – Hypertension
Waste retention	– Acidotic breathing (rapid & deep) – Hyperkalemia → dysrhythmias – Urea → uremic encephalopathy (confusion → convulsions → coma)
Systemic upset	– Heart failure up to acute pulmonary edema may occur due to hypervolemia – GIT bleeding may occur due to gastric stress ulcers and thrombocytopathy – Convulsions may occur due to excessive salt loss(hyponatremia) or uremic encephalopathy

3. **Polyuric phase**:

It may occur indicating early recovery → new tubular cells can't retain fluid & electrolytes → polyuria & electrolyte loss.

Diagnosis

1. Is there an Acute Kidney Injury

▪ Urine volume	– Oliguria: ▪ Urine output < 0.5mL/kg/h (< **300** ml/m^2/day) or ▪ < 1mL/kg/h in an infant – Anuria : urine output < **30** ml/m^2/day
▪ Renal function tests	– Elevated serum creatinine – Elevated urea & blood urea nitrogen(BUN)
▪ Acid / base disturbance	– Metabolic acidosis (↓pH, ↓$PaCO_2$, ↓HCO_3).
▪ Electrolytes	– Hyperkalemia. – Hyponatremia. – Hypocalcemia, hyperphosphatemia

Creatinine is an insensitive and delayed measure of decreased kidney function. Other biomarkers under investigation include changes in plasma neutrophil gelatinase-associated lipocalin (NGAL) and cystatin C levels and urinary changes in NGAL

2. For effect/cause

a) ECG: for evidence of hyperklemia
- Tall peaked T wave; the earliest ECG sign of hyperkalemia
- Widening of QRS complex
- Arrhythmias

b) Ultrasonography
- Exclude obstructive uropathy
- Differentiate between acute (enlarged and echobright kidneys) and acute on top of chronic failure (shrunken kidneys)

c) Radiologic studies:for
- Heart failure: cardiomegaly with prominent pulmonary markings
- Renal stones

3. Diagnosis of the type

i. Post renal

1- Palpable bladder & kidneys.
2- Bladder catheterization
3- Abdominal ultrasound to exclude obstructive uropathy.

	ii. Pre renal	**iii. Intrinsic renal**
Clinical	Hypovolemia : – Tachycardia – Core-peripheral temperature gap > 2 °C – Prolonged capillary refill time – Dehydration – Hypotension	Euvolemia or Hypervolemia : – Hypertension – Raised jugular venous pressure – Tachycardia, gallop rhythm – Palpable liver
Laboratory ▪ Urine osmolality ▪ Urine specific gravity ▪ Urine sodium ▪ Fractional excretion of Na(FNa) ▪ BUN / creatinine ratio(mg/dl)	 High > 1020 < 20 meq/L < 1 > 20:1	 Low < 1010 > 30 meq/L > 1 10-20:1 (+ oliguria)
US	Normal	Kidneys are enlarged and echobright

Pediatric RIFLE (p*RIFLE*) criteria

Used for the detection and classification of AKI and for correlation with clinical outcomes

	Serum creatinine rise	GFR decline	Urine out put
***R*isk**	1.5-fold	25 %	<0.5 ml/kg/h × 8 h
***I*njury**	2-fold	50 %	<0.5 ml/kg/h × 16 h
***F*ailure**	3-fold	75% or < 35 ml/min/1.73 m^2	<0.3 ml/kg/h × 24 h or anuric for 12 h
***L*oss**	Persistent failure > 4 weeks		
***E*nd stage**	ESRD (persistent failure >3 months)		

Treatment of ARF

I. Hospitalization and Monitoring

- Weight → Twice daily
- Fluid input-output recording → Hourly
- Vitals; including BP → Hourly
- Blood sugar → 6-hourly
- Neurological observations → Hourly
- Renal parameters (BUN/Creatinine) Blood gases, Electrolytes → May be appropriate to perform up to every 6hours

II. Correct Post-renal causes

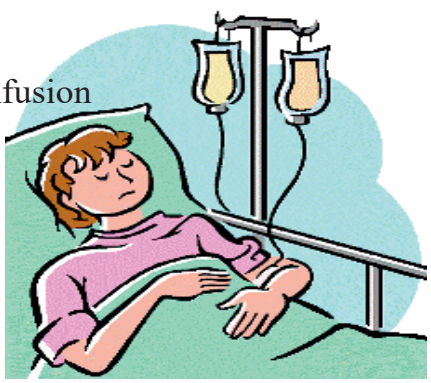
→ Remove obstruction (by catheterization ± surgical)

III. Correct pre-renal causes

- Fluid resuscitation → Restores renal blood flow

↓

- Fluid loss→ saline infusion 10 ml/kg over 30 minutes
- Blood loss→ fresh blood transfusion
- Plasma loss → plasma transfusion
- Protein loss (Nephrotic) → albumin infusion



↓

- Repeat if necessary
- With good correction → the patient should void within 1-2 hours

IV. Treatment of intrinsic renal failure

1. **Monitoring**
2. **Maintain fluid balance:**

Euvolemic	– Fluid challenge 10–20mL/kg normal saline over 1h with – Furosemide 2–4mg/kg IV
Hypervolemic	– Furosemide 2–4mg/kg IV

- Maximum dose of diuretic = 12mg/kg/day
- Value : Reduce volume overload & enhance potassium excretion
- If there is no response to a diuretic challenge, diuretics should be discontinued and start fluid restriction / Dialysis

Fluids intake

- Daily intake = insensible loss (400 ml/m^2 or 30mL/kg/day) + plus urine output and other ongoing losses
- Type: feeds if tolerated or IV D5 / Half normal saline
- Replace 100 % of urine output if euvolemic.
- Restrict to 50–75 % of urine output if overloaded

Nutrition

- Consult renal dietician
- Maximize caloric intake
- Restrict sodium, potassium and phosphorus and proteins

3. **Management of electrolyte and acid base abnormalities**

a. Hyperkalemia (potassium level ≥ 6 meq/L):

- Restrict potassium intake
- Enhance GIT excretion by kayexalate (polyestrene resin) oral or enema.
- With potassium levels above 7 meq/l or ECG changes appear
  - Stabilize cardiac membrane by 1 ml/kg calcium gluconate 10% slow I.V
  - Shift potassium intracellular by:
    - Salbutamol nebulizer
    - Sodium bicarbonate 1-2 mEq/kg over 10 minutes i.v
    - Regular insulin(0.1 u/kg) + glucose 50% solution(1 ml /kg) over 1 hour
- Persistent hyperkalemia should be managed by dialysis

b. Hyponatremia (sodium level < 130 meq/L):

- Dilutional hyponatremia usually respond to fluid restriction
- Hyponatremia < 120 meq/l with seizures is treated with Na CL 3% (Target serum sodium 125 mEq/l)

c. Hypocalcemia and hyperphosphatemia

- Tetany is rare as acidosis increases ionized calcium.
- Restrict phosphate intake & give phosphate binders e.g. sevelamer (Renagel), calcium carbonate or calcium acetate (PhosLo).
- Calcium gluconate 10 % slow iv only for cases with tetany

d. Metabolic acidosis

- Slow partial correction with IV bicarbonate 1–2mmol/kg
- Monitor serum calcium

4. Treatment of complications

- Anemia → Fresh packed RBCs transfusion over 4 - 6 hours
- Hypertension → Diuretics, Isradipine , amelodipine etc

5. Dialysis

Indications

Clinical

1. Volume overload with evidence of hypertension and/or pulmonary edema refractory to diuretic therapy
2. Neurologic symptoms (altered mental status, seizures)
3. Inability to provide adequate nutritional intake because of the need for severe fluid restriction(dialysis provides a space for nutrition)

Laboratory

1. Persistent hyperkalemia
2. Severe metabolic acidosis unresponsive to medical management
3. Blood urea nitrogen >100-150 mg/dL (or lower if rapidly rising)
4. Calcium: phosphorus imbalance, with hypocalcemic tetany

Modes of dialysis

1) Peritoneal dialysis : most commonly employed in neonates and infants with ARF
2) Continual Renal Replacement Therapy (CRRT): useful in patients with unstable hemodynamic status, concomitant sepsis, or multiorgan failure in the intensive care setting
3) Intermittent hemodialysis: useful in patients with relatively stable hemodynamic status

Prognosis

- The mortality rate depends entirely on the nature of the underlying disease
- Recovery of renal function is likely after ARF resulting from prerenal causes, HUS, Acute tubular necrosis, or acute interstitial nephritis.
- Poor outcome with AKI following rapidly progressive glomerulonephritis, bilateral renal vein thrombosis, or bilateral cortical necrosis

Chronic kidney disease (CKD)
Chronic Renal Failure (CRF)

Definition

- Irrecoverable, bilateral abnormalities of the renal parenchyma → permanent reduction of glomerular filtration rate (GFR)
- CKD has replaced "chronic renal failure" and "chronic renal insufficiency" as the globally accepted terminology for persistent renal dysfunction

Stages of CKD

CKD stage	Description	GFR (ml/min/1.73 m^2)
1	Kidney damage with normal or increased GFR	≥ 90
2	Kidney damage with mild decrease in GFR	60–89
3	Moderate decrease in GFR	30–59
4	Severe decrease in GFR	15–29
5	Kidney failure	<15 (or dialysis)

(Manual of pediatric nephrology)

Causes

In < 5 years of age

- Congenital malformations:
  - Hypoplastic/dysplastic kidneys
  - Reflux nephropathy(VUR+UTI)
  - Obstructive uropathy
- Metabolic/genetic disorders:
  - Oxalosis
  - Polycystic kidney disease
  - Congenital nephrotic syndrome
  - Wilms' tumor

In > 5 years of age

- Glomerular disease:
  - Focal segmental glomerulosclerosis
  - Hemolytic uremic syndrome
  - Chronic glomerulonephritis
  - Alport's syndrome
- Tubulointerstitial disease:
  - Chronic tubulointerstitial nephritis
  - Cystinosis
  - Nephrotoxic drugs

Clinical picture

Non specific so, need high index of suspicion

Clinical feature	Mechanism
o Growth retardation (Short stature)	- Resistance to growth hormone - Anemia - Anorexia (metabolic acidosis) - Renal osteodystrophy
o Unexplained anemia	- ↓ Erythropiotine - ↓ intake of iron, B12, folic acid. - Bone marrow depression by uremic toxins - Defective iron utilization.
o Polyuria / Polydipsia/ Nocturia	- Renal tubular concentration defect
o Hypertension	- Increased renin & fluid retention
o Renal osteodystrophy (ROD)	- Hyperphosphatemia - Decreased 1,25 $(OH)_2D_3$ - Secondary hyperparathyriodism
o Bleeding tendency	- Platelet dysfunction
o Infection	- Defective granulocyte function
o Neurologic (fatigue , drowsiness,polyneuropathy)	- Uremic toxins
o Pericarditis, cardiomyopathy	- Uremic toxins - Hypertension
o Hyperlipidemia	- ↓lipoprotein lipase activity
o Hyperkalemia and hyponatremia	

Diagnosis

1. Assessment of renal function:
   - Serial measurements of creatinine: An abnormal serum creatinine value persisting for more than 3 months confirms CKD.
   - Estimated GFR by Schwartz's equation & DTPA scan
   - Proteinuria: Persistent proteinuria is a marker of ongoing renal disease.
2. Renal ultrasound & DMSA scan show shrunken kidneys(Kidneys may appear large e.g. Polycystic kidney disease , Obstructive uropathy)
3. For complications:
   - CBC for anemia
   - Lipid profile
   - Echocardiography for pericardial effusion & cardiomyopathy.
   - In ROD → High phosphate - low calcium - high intact PTH
     → Bone X-ray → subperiosteal erosions ± bone cysts.
4. For the cause
5. Follow up investigations (see later)

Treatment

Stages of treatment:

- Stage 1: diagnosis and treatment of primary disease
- Stage 2- 3: retard progression, treat complications/comorbidities
- Stage 4: prepare for renal replacement therapy
- Stage 5: renal replacement

Outpatient checks in the child with CKD

- Height, weight and head circumference
- Pubertal stage
- BP

Investigations at each clinic visit

1. Full blood count (FBC; and ferritin if needing an erythropoiesis stimulating agent).
2. Urea & electrolytes (U&Es), bicarbonate, and creatinine
3. Calcium, phosphate, albumin, alkaline phosphatase, intact PTH
4. Urine protein or albumin to creatinine ratio
5. Fasting HDL and LDL, total cholesterol and triglycerides 6 monthly.

Management

A. Slowing the progression of chronic kidney disease

1. Control proteinuria
   - Rate of decline of kidney function is closely related to the quantity of proteinuria
   - Use:
     - ✓ Angiotensin Converting Enzyme Inhibitors; enalapril or captopril and/or
     - ✓ Angiotensin Receptor Blockers(Irbesartan)
2. Control hypertension
   - BP should be maintained within the normal range for age and height
   - There is some evidence that BP <50th centile may be beneficial
3. Control Dyslipidaemia by
   - Dietary intervention
   - Statins
4. Treatment of anemia

B. Conservative treatment

Nutrition

- Consult expert dietician
- Treat vomiting (ranitidine)
- High caloric diet
- Dietary supplements, either orally or enterally, via nasogastric tube or gastrostomy, if anorexic
- Protein and potassium restriction may be needed in stage 4-5 CKD!
- Feeding infants with CKD:
  - Breast milk
  - Special formula e.g. Low salt formula (*SMA*), high energy formula (*Maxijul*), low potassium , low phosphate formula (*Renastart*),….

Fluid and electrolyte balance

- If polyuria→ free access to water
- If salt and bicarbonate loser→ salt and bicarbonate supplementation

Growth

- Correct acidosis, hyperparathyroidism and anemia
- Optimize nutrition
- Recombinant human growth hormone (rhGH)

Anemia

- Target Hb should be in the range of 11.0–12.0 g/dl
- Iron supplement
- Erythropoietin 50-100 U/kg/week or Darbepoetin α (longer half-life)
- RBCs Transfusion for symptomatic severe anemia

Mineral and bone disorder (ROD)

- **The target range of serum PTH is closely related to stage of CKD**
- Treatment
  1. Control hyperphosphatemia
     - Dietary phosphorus restriction
     - Phosphate binders;administered with meals
       - Calcium based e.g. Calcium carbonate, calcium acetate
       - Non calcium based: Sevelamer (Renagel)is now more popular
  2. Maintain serum calcium
  3. Vitamin D therapy: One Alpha = 1 α (OH) D3 and Calcitriol
  4. Ca sensing receptor blockers, *Cinacalcet*, blocks calcium sensing receptors (CaSR) in parathyroids →Reduce PTH.
  5. Partial parathyroidectomy for treatment - resistant tertiary hyperparathyroidism with persistent hypercalcemia

Hypertension (HTN)

- Target blood pressure should be lower than the 90th percentile for normal values adjusted for age, gender, and height
- Maintaining BP below the 50th percentile may be effective in delaying progression of CKD
- Use
  - Diuretics
  - ACEI/ARBs
  - Calcium channel blockers
  - Beta blockers

Immunizations

- All children must complete all routine childhood vaccines.
- BCG, Varicella, pneumococcal polysaccharide (PPV) and hepatitis B vaccines must be added in children approaching RRT
- Annual influenza vaccine
- Transplantation can be delayed till vaccination schedule is completed

C. Renal replacement therapy (RRT)

Dialysis

Indications:

Laboratory criteria:	– GFR <15 ml/min/m^2 – Refractory hyperkalemia, hyperphosphatemia, and metabolic acidosis
Clinical criteria	Children with symptoms of – Nausea, vomiting – Malnutrition, growth retardation – Fluid overload, hypertension – Uremia Despite optimal medical management

Types

- Hemodialysis
- Chronic peritoneal dialysis

Renal transplantation

- The renal replacement therapy of choice in children
- Offering a near normal life to a child with end-stage renal disease

Urinary Tract Infections

Urinary tract infections can present as

- Upper urinary tract infections → acute & chronic pyelonephritis
- Lower urinary tract infections → acute & chronic cystitis & urethritis
- Asymptomatic bacteruria
- Septicemia

Causes

Risk factors	Common organisms
– Females (short urethra) – Uncircumcised boys – Vesico ureteric reflux(VUR) – Obstructive uropathy – Constipation – Instrumentation	- G -ve → - Escherichia coli (80%) - Proteus (more in boys) - Pseudomonas - G +ve → staph, strept. fecalis

Causes of recurrent UTI

1. Obstructive uropathy:
   - VUR either congenital or secondary to UTI
   - Congenital anomalies e.g. phimosis
   - Renal calculi
   - Neurogenic bladder
2. Catheters or foreign bodies
3. Constipation; elimination disorders
4. Mal treatment or inadequate treatment of an acute attack

Clinical picture: (presentation differs according to age)

Asymptomatic bacteruria: - positive urine culture without manifestations

Newborn	– Sepsis (Jaundice,↓ feeding,……)
Infant	– Fever (UTI is the most common bacterial cause of fever without focus in infants) – Screaming during micturition – Failure to thrive(vomiting→ weight loss)
Child	▪ Lower UTI - Urgency, dysuria, frequency - Suprapubic pain - 2^{ry} nocturnal enuresis - May be hematuria ▪ Upper UTI - Acute → Fever, rigors & loin pain - Chronic → Prolonged fever → May be hypertension

Investigations

1. Diagnosis of urinary tract infections: by **Urinalysis and culture**

Urine sample obtained by:

- *For toilet-trained children*
  - Mid-stream voided urine
- *For non-toilet-trained children*
  - Clean catch into a waiting sterile pot when the nappy is removed
  - Urine bag (high a rate of contamination→ unreliable culture results)

    ↓

    If UTI is suggested; order a urine culture using a sample by either

    ↓
  - In and out catheter sample; reliable and more practical than SPA
  - Suparpubic aspiration; SPA (Reserved for sick patients)

Findings

- Pyuria:
  ≥ 5WBCs/hpf Or ≥ 10 WBCs/ml
  [Absence of pyuria is rare if UTI is present]
- Leukocyte esterase and Nitrite tests are usually positive with UTI
- Gram stained films: For bacteruria
- *Urine culture*: necessary for confirmation and appropriate therapy

  Minimum colony counts indicative of a urinary tract infection:

Specimen	(CFU/mL)
– Clean catch (midstream)	$\geq 10^5$
– Catheter	$\geq 5\times10^4$
– Suprapubic aspiration	Any growth

2. Indicators of upper UTI (Acute pyelonephritis):

- Leukocytosis, neutrophilia
- Elevated serum erythrocyte sedimentation rate and C-reactive protein
- Blood cultures, particularly in infants and in obstructive uropathy
- DMSA scan

3. Imaging studies:

Imaging tools	Yield
Ultrasound of kidneys and bladder	– Assess kidney size , anatomy and position – Evaluate bladder wall thickness and emptying – Detect ureteral dilation – Detect hydronephrosis

MCUG (Micturating Cyst Urethro Gram)	– The gold standard investigation for the diagnosis of urethral abnormalities and VUR – Evaluate bladder anatomy and emptying
Plain abdomen	– Fecal masses – Spinal dysraphism
DMSA scan	– The gold standard investigation for the diagnosis of renal parenchymal damage ▪ Acute : diagnose acute pyelonephritis ▪ Chronic : detect renal scarring

NICE Guideline for imaging in UTI

Ultrasound

- Indicated during the acute infection for:
  - Infants below 6 months
  - Atypical UTI
  - Recurrent UTI

MCUG

- Done at 2-4 weeks post UTI allowing bladder inflammation to resolve
- Indications:
  1. After a second UTI is diagnosed
  2. After a first UTI with:
     - Abnormal ultrasound findings e.g. hydrnephrosis, scarring, obstructive uropathy or VUR
     - Poor urine flow
     - Non E.coli UTI
     - Family history of vesico ureteral reflux VUR

DMSA scan

- Done at 4-6 months in order to avoid a false positive result due to renal parenchymal inflammation that may resolve
- Indications:
  - Child > 3 years of age with clinical pyelonephritis
  - Atypical UTI
  - Recurrent UTI

- In children with infection of the lower urinary tract, imaging is usually unnecessary.
- Instead, assessment and treatment of bladder and bowel dysfunction is important.
- If there are numerous lower urinary tract infections, then a renal sonogram is appropriate

Treatment

NICE guidelines regarding antibiotic treatment

Age	Antibiotic Plan
Age < 3 months	– IV antibiotics for 2–4 days then – Switch to oral antibiotics if clinically improved
Age > 3 months with upper tract UTI	– Oral antibiotic for 7–10 days – IV antibiotics for 2–4 days if vomiting, then oral antibiotics for a total of 10 days
Age > 3 months with lower urinary tract symptoms	– Oral antibiotics for 3 days
If a child is on prophylaxis	– Change the antibiotic

(Oxford pediatric nephrology)

Empirical antibiotics

Oral antibiotics e.g.	Parenteral antibiotics e.g.
– Cotrimoxazole – Amoxicillin – Cefixime – Ciprofloxacin for resistant cases	– Ceftriaxone – Cefotaxime – Ampicillin plus an aminoglycoside e.g. gentamicin

Follow on

- Change antibiotics according to culture and sensitivity
- Urine culture after 1 week to ensure recovery
- Continue antibiotics till urine culture turns sterile, pyuria disappears, afebrile without clinical evidence of UTI
- Urine culture after 3 months → detect recurrence

Supportive care

– Adequate hydration
– Diet, rest and symptomatic treatment e.g. antipyretics
– Admit to hospital sick children (dehydrated, vomiting, ≤1mo of age)

Prevention: For recurrent urinary tract infections

- Treat risk factor e.g. stones, constipation, voiding dysfunction
- Adequate hydration
- Frequent bladder emptying
- Antibiotic prophylaxis
  - e.g. Low dose cotrimoxazole or nitrofurantion
  - For high risk conditions e.g. neurogenic bladder, and VUR
  - Usually for patients below 3 years
- Probiotic and cranberry juice may have added benefit

Enuresis

Definitions

- Most children achieve night-time dryness by 5 years of age, when
- bladder volume exceeds nocturnal urine production
- Nocturnal enuresis is involuntary loss of urine at night, in the absence of physical disease, at an age when the child could reasonably be expected to be dry (developmental age of 5 years by consensus)
- Nocturnal enuresis is common, affecting 15–20 % of 5-year-olds, 5 % of
- 10-year-olds and 1–2 % of 15-year-olds
- Enuresis differs from urinary incontinence that in enuresis the child can control bladder while incontinence means loss of bladder control

Clinical types:

- Primary: the child has never been dry
- Secondary: the child has previously been dry at night for 6 months or more after the age of 5 years;
- Monosymptomatic (uncomplicated): not associated with other urinary tracts symptoms
- Polysymptomatic (complicated) :associated with symptoms suggestive of lower urinary tract dysfunction

Etiology:

Poorly understood

- Developmental disorder: Delay in maturation of bladder control
- Strong genetic component: a family history is found in most children
- Psychological disturbances e.g. emotional deprivation
- Organic causes e.g. mental retardation, urinary tract infections, polyuria as in diabetes mellitus or insipidus.

Management:

1. Proper history taking.
2. Investigations to exclude organic causes.
3. Psychotherapy if emotional factors are present.

General measures

- Adequate daytime fluid intake to develop bladder capacity and reduce evening fluid intake
- Avoid caffeine-based drinks
- Exclude/treat constipation if present
- Consider correction of airway obstruction in heavy snorers
- Pass urine regularly during the daytime and before sleep
- Waking for toileting is used only as short-term management

Reward systems

- Star charts for dry nights
- Rewards should be given for agreed behavior e.g toileting before sleep,….
- Where these measures alone are unsuccessful, consider the use of an alarm system or drug treatment

Alarm systems



- With alarm system around 50 % of children achieving long-term dryness
- Alarm systems are superior to behavioral techniques and drug therapy
- Use until either 14 consecutive dry nights or 3 months
- Alarms and desmopressin equally effective but alarm has more prolonged effect
- Drawbacks: requires time, motivation, and hard work

Desmopressin





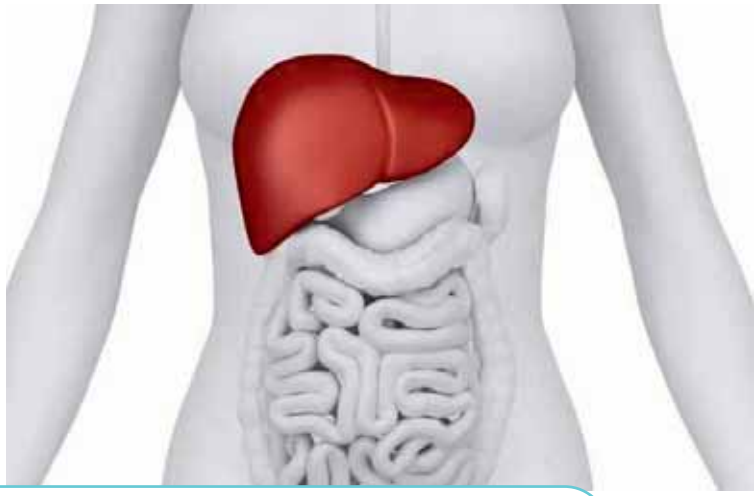
- Has more immediate effect than alarm
- May be used “on important nights only”
- Where successful, treatment should be withdrawn every 3 months to assess response
- Dosage: oral tablets 0.2–0.4 mg or 120–240 micrograms sublingually at bed time
- Keep evening fluid intake below 200 ml and no nighttime drinking
- Stop treatment if no effect within 2 weeks
- Drawbacks: high relapse rate upon stopping
- Can be combined with alarm

Anticholinergics e.g. Oxybutynin

- May be considered for cases unresponsive to the above therapy
- With or without desmopressin/alarm
- Never combined with tricyclic antidepressants
- Exclude residual urine and/or constipation
- Useful for cases with day time wetting

Tricyclic antidepressants (Imipramine)

- Restricted to those who have not responded to alarm and/or desmopressin
- Have significant adverse-effects ; cardio toxic ,anticholinergic effects
- If used, imipramine should be withdrawn gradually



Gastroenterology

Causes of vomiting in infants and children

1. Acute infections	2. Metabolic	3. Acute intestinal obstruction
- CNS infections	- **D**rug poisoning	* Functional: Paralytic ileus
- Pulmonary infections	- **R**ye's syndrome	* Organic:
- Gastroenteritis	- **D**iabetic keto acidosis	- Intussusception
- Acute pyelonephritis	- **R**enal failure	- Volvulous
- Sepsis	- **D**rugs: e.g. aspirin	

4. Chronic vomiting

- Over feeding
- Gastro-Esophageal reflux
- Congenital pyloric stenosis
- Inborn errors of metabolism/ adrenal insufficiency
- Psychogenic

Causes of Abdominal pain

i- Acute abdominal pain

Acute infections	Acute medical conditions	Acute intestinal obstruction
- Strept. Pharyngitis (mesenteric adenitis)	- Pneumonia(lower lobe)	
- Acute hepatitis.	- Rheumatic fever (peritonitis)	
- Acute pancreatitis	- Henoch Schonlein purpura.	
- Acute pyelonephritis	- Familial mediterranean fever.	
- Acute appendicitis.	- Diabetic keto acidosis	
- Acute peritonitis.		

ii- Chronic (recurrent) abdominal pain

Functional	Organic
- Irritable bowel syndrome (in 90%; psychic related)	- Intestinal parasites e.g. Giardiasis
- School phobia	- Chronic diarrhea (and Malabo sorption)
	- Chronic constipation
	- Inflammatory bowel disease
	- H. Pylori infection
	- Chronic hepatitis
	- Stones (urinary, biliary)

Q. Causes of constipation?

- Anal fissure
- Spina bifida
- Cretinism
- Intestinal obstruction
- Habitual constipation
- Medications (narcotics)

Stomatitis

Definition: Inflammation of the oral mucosa

Clinical types

1. Catarrhal stomatitis:

Mild generalized inflammation of the oral mucosa

Causes

- It accompanies acute infections as measles
- Local factors (chemical, traumatic or thermal).

Treatment: may use gentian violet 1 % or local oral anesthetics

2. Herpetic gingivo-stomatitis:

Caused by herpes simplex virus

Clinical picture

a. Acute onset of fever, anorexia; few days later

b. Herpetic vesicles appear → minute ulcers

c. Associations:

- Pain, excessive salivation
- Bad mouth odor
- May be bleeding, swollen gums.



(Nelson 2016)

Treatment

- Care of feeding: cold fluids ± nasogastric tube feeding in severe cases.
- Mouth paint with gentian violet 2% or silver nitrate 2%.
- Antiviral agents as Acyclovir (Zovirax).

3. Thrush stomatitis (Moniliais):

Caused by candida albicans in:

- New born: infected from mothers with genital moniliasis or infected teats
- Malnourished infants due decreased cell mediated immunity
- As a side effect of excess use of antibiotics.

Clinical picture

- Numerous small white flakes
- On the dorsum of tongue, inner side of cheeks & palate.
- In extensive cases a thick white membrane is formed



Prevention

The nipple and areola of the nursing mother should be cleaned before feeding and painted with nystatin ointment in between meals.

Treatment

4. Gangrenous stomatitis:

- Rare; Seen in severe debilitated children with diseases as measles
- A deep large ulcer on the inn surface of the cheeks that spread slowly by necrosis of adjacent tissues
- It may lead to perforation of the cheek

Treatment: antibiotics plus surgical excision of necrotic areas.

Differential diagnosis of oral ulcers

Common	Criteria
▪ Aphthous (canker sore)	– Painful – Circumscribed lesions – Last 10-14 days – Recurrences ▪ Treatment: benzocaine and topical lidocaine, are effective, as are topical steroids
▪ Traumatic	– Accidents – Chronic cheek biter
▪ Hand, foot, mouth disease	– Painful – Lesions on tongue, anterior oral cavity, hands, and feet
▪ Herpangina	– Painful – Lesions confined to soft palate and oropharynx
▪ Herpetic gingivostomatitis	– Vesicles on mucocutaneous borders – Painful – Febrile
Uncommon	
o Neutrophil defects	– Agranulocytosis, leukemia, cyclic neutropenia – Painful
o Systemic lupus erythematosus	– Recurrent – May be painless
o Behçet syndrome	– Resembles aphthous lesions – Associated with genital ulcers, uveitis
o Oral Crohn's disease	– Aphthous-like – Painful
o Stevens-Johnson syndrome	– May be isolated or appear initially in the oral cavity

Achalasia

Definition

- Esophageal motility disorder characterized by:
  - Failure of relaxation of lower esophageal sphincter (LES)
  - Due to lack of ganglionic cells in the lower esophageal sphincter
  - With subsequent increased lower esophageal sphincter pressure
- Incidence: most cases occur after age of 15 years

Clinical picture

- Dysphagia mainly to liquids rather than to solids that may open LES by weight
- Regurgitation of food
- Cough due to overflow of fluids into trachea (frequent aspiration)
- Recurrent chest infections
- Failure to gain weight in progressive cases

Diagnosis

- Upright chest x ray show air fluid level in dilated esophagus
- Barium swallow **fluoroscopy** shows massive dilatation of the esophagus with tapered lower end



- Esophageal manometry is *diagnostic*; it reveals:
  - ✓ Aperistalsis in the distal esophageal body and
  - ✓ Incomplete or absent LES relaxation, often accompanied by
  - ✓ High pressure LES

Treatment

- Pneumatic dilation is the initial treatment of choice, and does not preclude a future myotomy
- Temporary relief of dysphagia can be offered by calcium channel blockers (nifedipine) and phosphodiesterase inhibitors or endoscopic intra sphincteric injection of botulism toxin
- Heller myotomy: surgical division of muscles at gastroesophgeal junction.

Gastro- Esophageal Reflux Disease (GERD)

Definition: Abnormal retrograde of gastric contents into oesphagus due to persistent relaxation of lower oesphageal sphincter (LES)

Incidence: - Mainly in neonates & young infants
- 60% improve with age (resolve by 6 mo-2years).

Clinical picture

A. Uncomplicated cases

1. Vomiting
   - At the end of the feed
   - From the 1st week of life
   - Increase with lying flat
   - May be bile stained.
2. Sandifer syndrome: abnormal head posture and opisthotonus to protect airways.
3. Substernal pain and dysphagia in older child

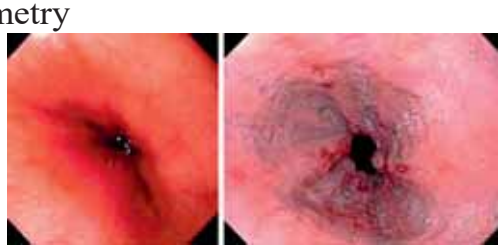
B. Complicated cases

- Oesphagitis → GIT bleeding
- Recurrent aspirations → recurrent aspiration pneumonia
- Chronic cough & chest wheezes
- Growth retardation
- May be sudden infant death syndrome(SIDS) due to laryngospasm and apnea

Investigations

1. Diagnostic:

- **Radiologic**
  - Barium swallow under screen → retrograde of the dye
  - Doesn't assesses mucosal inflammation nor the severity of GERD
- **Extended esophageal pH monitoring**: PH< 4 in 5-8% of the monitoring time
- **Endoscopy**
  - Detects low LES pressure by manometry
  - Detect acidic esophageal pH (< 4)
  - Visualize erosive esophagitis and complications such as strictures or Barrett's esophagus



Normal esophagus versus Erosive esophagitis

2. For complications:

Check stool for occult blood

Treatment

Medical

- Feeding : Thickening feedings with rice or oat cereal (1tablespoon/ounce)
- Position
  - Head up for 30 min after feeding. However, the "infant seat" may worsen reflux by increasing intraabdominal pressure
  - Supine position reduces SIDS but worsen GERD
  - When the infant is awake and observed, prone position and upright carried position can be used to minimize reflux.
- Pharmacotherapy
  - Ameliorate acidity by:
    - Anti-acids
    - Histamin-2 (H2) receptor blockers e.g. Ranitidine
    - Proton pump inhibitors(PPIs) e.g. omeprazole, esomeprazole
  - Prokinetics: controversial and not recommended
    - Domperidone (Risk of cardiac dysrhythmias)
    - Metoclopramide (Risk of tardive dyskinesia)
    - Baclofen ; a centrally acting γ-aminobutyric acid agonist
    - Under research: Metabotropic glutamate receptor 5 antagonists

  Avoid methylexanthines → it lowers LES tone

Surgical

Operation

- Fundo plication

Indications

- Failed medical treatment
- Complications. e.g. Growth retardation

Congenital Hypertrophic Pyloric Stenosis

Definition

Progressive hypertrophy of circular muscles fibers of the pylorus with subsequent pyloric narrowing due to ganglionic cells immaturity

Pathogenesis

Progressive vomiting results in:

- Loss of HCL→ metabolic alkalosis
- Loss of potassium and chloride → hypokalemia and hypochloremia
- Loss of water and nutrients→ dehydration and failure to thrive

Clinical picture

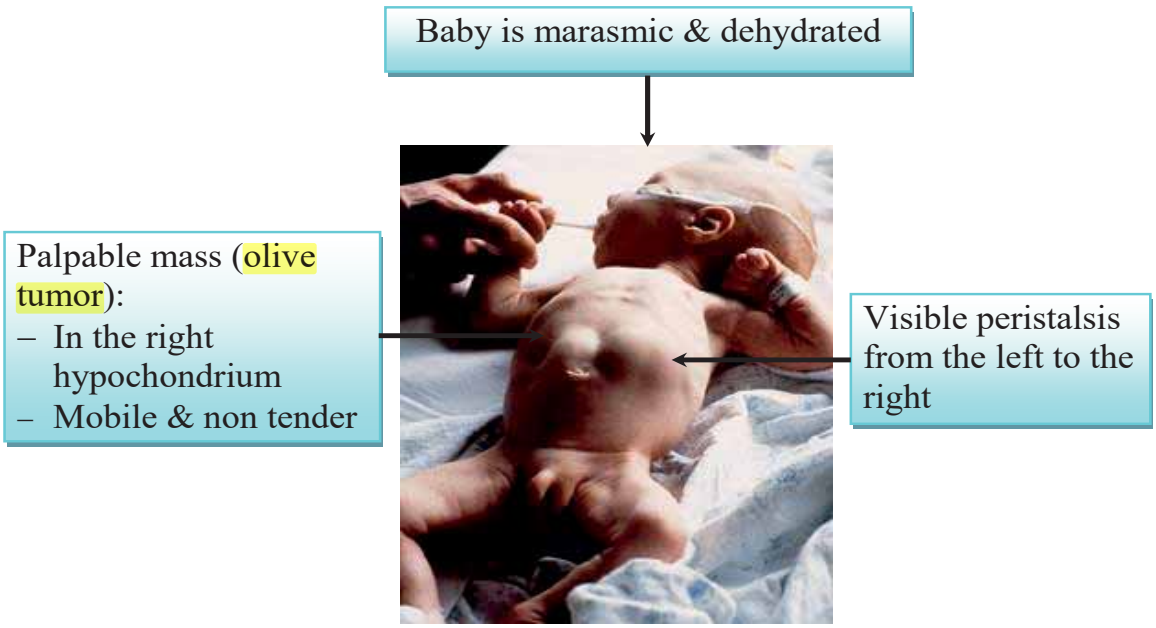
Incidence

- Males (especially first born) affected than females.
- Positive family history may exist.

Symptoms

1. Vomiting:
   - Occurs shortly after feeding.
   - Usually starts after the 2nd – 3th weeks of life (Rarely before or after)
   - Initially, non-projectile then projectile
   - Non bile stained.
   - Baby is often hungry after vomiting.
2. Constipation; passage of small, infrequent stools

Examination



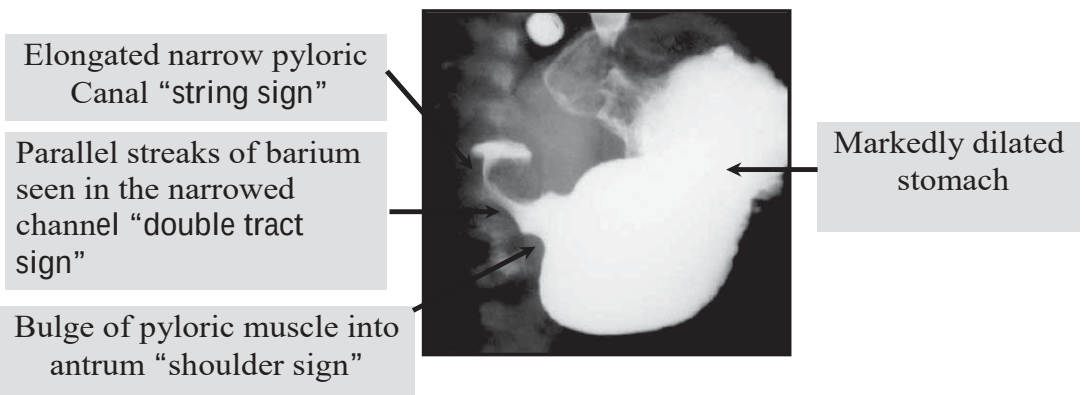
Differential diagnosis

1. Gastroesophageal reflux disease
2. Adrenal insufficiency (metabolic acidosis and elevated serum potassium and urinary sodium concentration)
3. Inborn errors of metabolism can produce recurrent emesis with alkalosis (urea cycle) or acidosis (organic acidemia) and lethargy, coma, or seizures.
4. Gastro enteritis
5. Other anomalies: pyloric membrane or pyloric duplication, Duodenal stenosis

Investigations

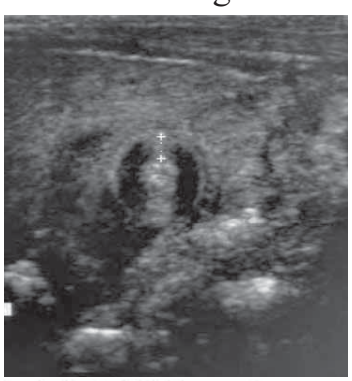
1. For diagnosis

A. Barium meal demonstrates:

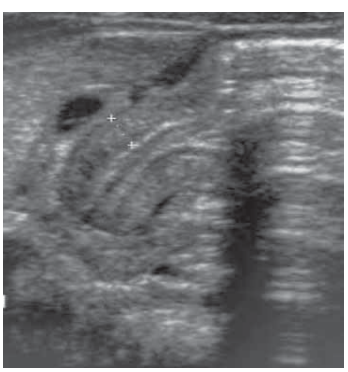


B. Abdominal ultrasound: Confirms diagnosis with 95% sensitivity

Criteria for diagnosis:



1. Pyloric thickness 3-4 mm



2. Pyloric length 15-19 mm
3. Pyloric diameter of 10-14 mm

2. For complications

* Hypochloremic metabolic alkalosis (↑pH, ↓CL)
* Hyponatremia & hypokalemia

Treatment

1. **Surgical:** *Ramstedt's pyloromyotomy*
   - Pre-operative → correct electrolytes disturbance and dehydration
   - Post-operative → start small feeds → gradually increasing
   - Efficiency :100% curative



2. **Medical:** Not efficient, it includes:
   - Antispasmodic before feeds
   - Small, thick, frequent feeds
   - Keep upright for 1 hour after feeding

Congenital Aganglionic Megacolon

(Hirsch sprung Disease)

Definition

– Functional obstruction of the colon due to absence of ganglion cells in bowel wall
– Commonest site: rectosegmoid in 75%

Incidence

– 1 / 5000
– Male: Females = 4: 1 (positive family history may be present).

Clinical picture

1. Presentation may be:

Neonatal (80%):
– Delayed passage of meconium beyond 48 hours.
– May be acute obstruction

In older child:
– Chronic constipation and abdominal distension.
– Large fecal mass felt in left lower abdomen with empty rectum

2. Complicated cases:

- Enterocolitis:
  – Infection with clostridia difficile, staph aureus and anaerobes
  – Presented with bloody diarrhea & toxemia
- Intermittent attacks of intestinal obstruction.
- Failure to thrive due to protein losing enteropathy

Investigations

1. **Rectal suction biopsy** (from narrow segment)
   – The gold standard for diagnosing Hirschsprung disease
   – Reveals absent ganglia
2. **Barium enema:**
   – Unprepared contrast enema aid in the diagnosis in children older than 1 month of age
   – Classic findings: an abrupt narrow transition zone between the normal dilated proximal colon and a smaller caliber obstructed distal aganglionic segment



Treatment

1. Surgical repair.
2. Preparation before surgery:
   - Regular evacuation of rectum.
   - Antibiotics.

Hepatology

Functions of the liver

- Synthesis of all proteins (except gamma globulin and F VIII)
- Synthesis and excretion of Bile.
- Synthesis and excretion of cholestrol.
- Carbohydrate: - Post prandial → convert glucose to glycogen.
  - Fasting → convert glycogen to glucose.
- Detoxication e.g. converts ammonia to urea.

N.B Hepatic enzymes

❶ Intra cellular; (Markedly raised in hepatitis):
- Alanine aminotransferase (ALT); more specific to the liver.
- Aspartate aminotransferase (AST).

❷ Intra canalicular → Alkaline phosphatase, Gamma glutamyle transferase, 5 Nucleotidase ⇒ markedly raised in cholestaisis.

Causes

Hepatitis

Metabolic
With some inborn errors of metabolism e.g.
1- Glactosemia
2- Wilson disease
3- α1 antitrypsin deficiency.
4- Tyrosinemia.

Toxic
- Drug e.g.: - INH, rifampicin
  - Paracetamol overdose
- Toxins e.g.: Mush-Room.

Infections

Bacterial ⇓
- Septicaemia
- Enteric fever
- Brucellosis

Protozoa ⇓
- Malaria
- Toxoplasma

Parasitic ⇓
- Bilharsiasis
- Hydatid cyst.

Viral

Hepatotropic
Viruses attack the liver mainly

Enterally transmitted ⇓
- Hepatitis A virus (HAV)
- Hepatitis E virus (HEV)

Parenterally transmitted ⇓
- Hepatitis B virus (HBV)
- Hepatitis C virus (HCV)
- Hepatitis D virus (HDV)

Non hepatotropic
Viruses may cause hepatitis during its course e.g.
- EBV
- CMV
- Herpes viruses
- Rubella

Acute Viral Hepatitis

Enteral viruses

Include hepatitis A (HAV) and Hepatitis E viruses (HEV)

Criteria

- Excreted only in stool (not in body fluids) **so** infection occur by faeco-oral route via contaminated water & food.
- None enveloped RNA viruses.
- Incubation period is short (2-6 weeks).
- **Epidemiology**:-
  - Occur in epidemics (highly contagious) or sporadic.
  - Mainly in low socioeconomics
  - HAV is the commonest cause of acute viral hepatitis in school age.
  - HEV is rare in children.
- **Outcome**
  - Complete recovery is the rule; no carrier state nor chronic hepatitis.
  - Fulminant hepatic failure may rarely occur.
  - HEV has high fatality in pregnant women.

Hepatitis A Virus (HAV)
(Infective Hepatitis)

HAV is the commonest cause of acute viral hepatitis in school age.

Pathology of acute hepatitis

- Hepatocyte injury is due to cytopathic effects
- Cholestatic jaundice with elevated both direct and indirect bilirubin

Clinical picture of acute hepatitis

- Many cases of viral hepatitis pass asymptomatic.
- Symptomatic cases pass in following phases:

I. Pre-icteric phase (2 – 4 weeks)

- Fever, malaise
- Anorexia, nausea, vomiting
- Abdominal pain
- Diarrhea is frequent in children while constipation predominates in adults

II. Icteric phase (2-4 weeks)

- Improved previous symptoms with appearance of:
  - Jaundice
  - Tender hepatomegaly
  - Dark urine + pale stool



III.Convalescent phase

- Most patients achieve full recovery
- 2 distinct complications can occur

Complications

1. Acute liver failure (ALF)

Incidence: Very rare

- Risk factors: adolescents and adults, immunocompromised patients and those with underlying liver disorders
- In the USA, HAV represents < 0.5% of pediatric-age ALF
- In endemic areas of the world, HAV represents up to 40% of all cases of pediatric ALF

Clinically:

- Deep progressive jaundice
- Bleeding tendency (↓ coagulation factors)
- Generalized edema with ascites
- Disturbed sleep rhythm, tremors, stupor & coma

Fate:

- High mortality rate
- Definitive treatment is liver transplantation.

2. Prolonged cholestatic syndrome

- With problematic pruritus and fat malabsorption
- Waxes and wanes over several months

Investigations

A. To prove acute hepatitis:

1. Liver function tests:
   - Aminotransferase levels (ALT & AST)
     - Markedly elevated; in thousands
     - The elevation level does not correlate with the extent of hepatocellular necrosis nor to the prognosis
   - Serum bilirubin
     - Moderately elevated ;mainly conjugated
     - The first marker to normalize with recovery
   - Alkaline phosphatase → mild↑
   - Albumin and prothrombin time → usually normal
2. Urine: Dark color due to ↑ cholebilirubin and bile salts
3. Stool: Pale color due to ↓ stercobilinogen.

In ALF:

1. Aminotransferases rise initially then rapidly decline with rising bilirubin
2. Altered synthetic function:
   - ✓ The most important marker of liver injury and defining severity
   - – Abnormal protein synthesis (prolonged prothrombin time, low serum albumin levels)
   - – Metabolic disturbances (hypoglycemia, lactic acidosis, hyperammonemia)

B. For the cause → viral serology; HAV markers

Marker	Significance
Anti – HAV (IgM)	Recent HAV infection.
Anti – HAV (IgG)	Previous HAV infection or HAV vaccination; confers long-term protection

Management

General

- – Exclusion of school or child care till one week after clinical jaundice
- – Hand washing after defecation or dealing with infected child
- – Sterilization of toilet after use

HAV prophylaxis

Indications:

- ▪ Pre exposure for susceptible travelers to HAV endemic countries
- ▪ Household post exposure as soon as possible
  - • *HAV Ig alone* intramuscular (0.02 mL/kg) *if*
    - – *Younger than 12 mo old*
    - – *Vaccine is contraindicated*
  - • Both *HAV* Ig and the HAV vaccine if
    - – Travel is planned in <2 wk
    - – *Older than 12 mo old*
    - – Immunocompromised
    - – Those with chronic liver disease

HAV vaccine (e.g. *Havrix*):

- – Inactivated vaccine, approved for children older than 12 mo
- – Dose: 2 doses 6 months apart, IM.

Treatment

- – Serial monitoring for signs of ALF and, if ALF is diagnosed, a prompt referral to a transplantation center can be lifesaving (**See Later**)
- – Supportive: bed rest, hydration, high carbohydrate diet, and multivitamins.

- Avoid hepatotoxic drugs
- Treat complications e.g. antipruritic for prolonged cholestatic syndrome

Hepatitis E virus

- Share criteria of enteral viruses (see before)
- The epidemic form of what was formerly called non-A, non-B hepatitis.
- Transmission is fecal–oral (often waterborne)
- The prevalence appears to be increasing, so that it is postulated to be the most common cause of acute hepatitis and jaundice in the world

Clinically

- The clinical illness associated with HEV infection is similar to that of HAV but Often more severe especially in pregnant women, in whom it causes ALF with a high fatality incidence
- Tends to affect older patients, with a peak age between 15 and 34 yr

Prevention

- A recombinant hepatitis E vaccine is highly effective in adults

Diagnosis

- Antibodies to HEV particles; IgM and IgG assays are available to distinguish between acute and resolved infections

Parenteral viruses

Include Hepatitis B virus (HBV), Hepatitis C virus (HCV) and Hepatitis D virus (HDV)

Criteria:

Acute hepatitis due to parenteral viruses occur sporadically **not** in epidemics

	HBV	HCV	HDV (Delta virus)
Nature	Enveloped DNA virus.	Non enveloped RNA virus.	Non enveloped RNA virus → need HBV coat to be infective (defective virus; dependent on HBs Ag).
Risk factors	▪ Perinatal exposure to an infected mother (The most important risk factor for acquisition of HBV in children) ▪ Parenteral via: – Post transfusion ; via contaminated blood products (The most important risk factor for acquisition of HCV) – Hemodialysis patients – Drug abusers ▪ Sexual ▪ No risk factors identified in approximately 40% of cases		
Incubation period	2-6 months	2-4 months	

Pathology of acute hepatitis

- Hepatocyte injury can be due to:
  - Cytopathic effects (by all except HBV)
  - Immune mediated cell lysis (by HBV & HCV)
- Cholestatic jaundice with elevated both direct and indirect bilirubin
- In perinatal HBV infection : markers of infection and antigenemia appear 1-3 mo after birth

Clinical picture of acute hepatitis

- Many cases of acute hepatitis pass *asymptomatic*
- Symptomatic cases are similar to that of **HAV**
- Acute HCV infection tends to be mild and insidious in onset unlike HBV
- Extrahepatic manifestations due to circulating immune complexes; mainly in HBV & HCV:
  - Serum sickness like prodrome marked by arthralgia or skin lesions e.g. urticarial, purpuric, macular, or maculopapular rashes
  - Aplastic anemia
  - Glomerulonephritis
  - Vasculitis

Complications

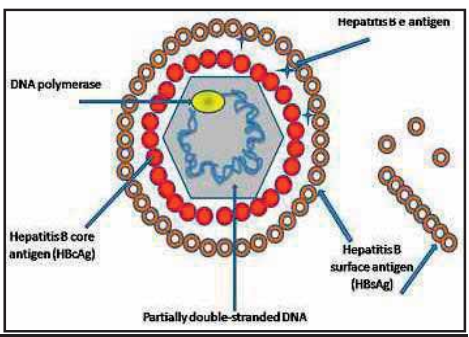
1. Chronic **c**arrier state
2. **C**hronic hepatitis
   Risk:
   - Risk of chronic HBV infection is 90% in children younger than 1 yr; 30% for those 1-5 yr and 2% for adults
   - HCV is the commonest cause ever of chronic viral hepatitis

   Complications
   - **C**irrhosis & liver cell failure.
   - Hepatocellular Carcinoma
3. Acute Liver Failure (ALF); risk increases when there is coinfection or superinfection with HDV and in an immunosuppressed host
4. Reactivation of chronic infection has been reported in immunosuppressed children

Investigations

I. To prove acute hepatitis: As in HAV

II. For Etiology: Viral Markers



1. For HBV

Marker	Significance
▪ **HBs Ag**	– Acute infection; its rise closely coincides with the onset of symptoms – If persist > 6 months → indicate chronic hepatitis
▪ **Anti HBc Ag IgM**	– **Acute infection**; Reliable single marker later in the acute phase as HBs Ag fall before the symptoms disappear
▪ **Hbe Ag**	– **Acute infection** ; a marker of active viral replication and usually correlates with HBV DNA levels – **HBe Ag** positive mothers put very high risk of perinatal transmission to their babies
▪ **Anti HBc Ag IgG**	– Infection; recent **or** chronic
▪ **Anti HBs Ag**	– If present alone, it indicate previous vaccination. – If present with anti-HBc Ag → resolved infections.

N.B. HBc – Ag is present only inside the hepatocytes.
Hbe – Ag is not structural antigen but it is produced by self-cleavage of the core antigen

2. For HCV and HDV

- Detect specific RNA by PCR.
- Detect specific antibodies.

N.B: Children born to HCV-infected women→ test for qualitative PCR in infancy and anti-HCV after 12-18 mo of age

Prevention

1. Blood donation screening
2. HBV immunization

Combined: Both vaccine and immunoglobulin

A. Post exposure prophylaxis for infant born to HBs Ag + ve mothers

- HBV immunoglobulin 0.5 ml IM, within 12 hr after birth *plus* the first dose of HBV vaccine (which is given as 0, 1, 6)
- Protective value of this regime is > 95%
- Follow up: Post vaccination testing for HBsAg and anti-HBs should be done at 9-18 mo:

If
- Positive for anti-HBs → The child is immune
- Positive for HBs Ag only → The child is infected
- Negative for both HBsAg and anti-HBs→ Repeat the vaccine

B. Post exposure in older child:

HBV immunoglobulin: 0.06 ml/kg within 24 hrs *plus* HBV vaccine which given as 0, 1, 6

Active

- HBV vaccine: (Recombivax HB and Engerix-B)
- Nature: Recombinant DNA vaccine.
- Time: 3 doses, IM at 0, 1, 6 months

Treatment

- As for HAV; Treatment of acute infection is largely supportive.
- Close monitoring for liver failure and extrahepatic morbidities is key
- **Acute liver failure** management include:
  - Referral to PICU in a center expert in liver transplant
  - Monitor vitals , conscious state and hepatic / renal chemistry
  - Fluid ,glucose, ammonia and electrolytes monitoring and correction especially for hypophosphatemia (high phosphate is a bad sign)
  - Restrict or forbid proteins initially
  - Gastriprotection :proton pump inhibitors
  - Oral antibiotics (e.g. Neomycin) and Lactulose to sterilize the colon
  - Frequent enema preferably with lactulose
  - Correct coagulopathy by vit K, fresh frozen plasma or factor VIIa
  - In advanced cases with coma: Hepatic dialysis/transplantation

Chronic Hepatitis

Definition: An inflammatory process of the liver lasting longer than **6** months.
Recently → continuing hepatic inflammatory process manifested with severe liver disease **or** features of chronicity (shorter time can be employed)

Chronic persistent hepatitis	Chronic active hepatitis
Causes Viral → HBV, HCV	▪ Auto immune → the commonest ▪ Viral → HBV, HCV, Delta virus. ▪ Metabolic → e.g. Wilson disease.
Pathology - Inflammation limited to the portal zone - Little or no fibrosis. - No cirrhosis.	- Erosions of the limiting plate. - Piecemeal necrosis of hepatocytes. - If severe → bridging necrosis → fibrous septa
Clinical picture - Asymptomatic - May be non-specific: malaise, anorexia - May be tender hepatomegaly.	**1.** Most cases have: - Hepatosplenomegaly (HSM) - Liver cell failure (LCF) **2.** In auto immune; type there may be also: - Iridocyclitis - Thyroiditis - Vasculitis → nephritis - Serositis → arthritis , pleurisy - Immune hemolytic anemia - Clubbing
Complications Very uncommon	Common:-Cirrhosis→ portal hypertension -Fulminate hepatic failure.
Investigations **1. Is it hepatitis? Yes.** - ALT & AST → mild increase. - Bilirubin → No or slight increase. **2. Is there liver decompensation? No** - Albumin → Normal. - Prothrombine time → normal. **3. What is the cause?** - HBV & HCV markers.	**Yes** → High (not alkaline phosphatase) → High. (2-10 mg/dl – mainly direct). **Yes** → Low → Prolonged 1- Viral markers. 2- For auto immunity: - Anit nuclear antibody (ANA). - Anti – smooth muscle antibody. - Anti liver kidney microsomal antibody - Anti soluble liver antigen antibody. - Elevated γ-globulin levels 3- Metabolic assay.
- Liver biopsy → diagnostic	

Treatment

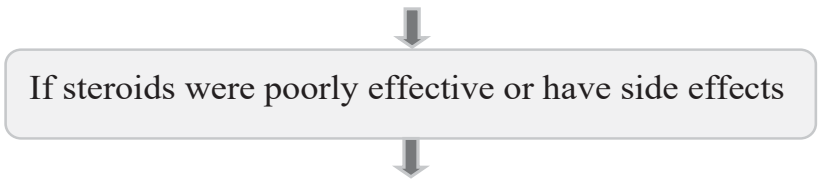
1. Supportive→ As in acute hepatitis

2. Follow up → Clinical (for signs of decompensation) **and** laboratory.

3. Specific:

Auto immune hepatitis:

- Steroids 1-2 mg / kg /day till ALT & AST less than twice high normal
- Then taper slowly over 2-4 months to reach maintenance dose of 0.1 – 0.3 mg/kg/day



- Azathioprine is added with frequent monitoring for bone marrow suppression
- Measurement of thiopurine methyltransferase activity before azathioprine therapy is a predictive of myelotoxicity

Post viral:

HBV

- Treatment is only indicated for patients in *the immune active* form of the disease, as evidenced by elevated ALT and/or AST, who have fibrosis on liver biopsy
- Medications
  - Interferon-α2b
  - Peginterferon-α2
  - Lamivudine
  - Adefovir

HCV

- Medications approved by the FDA for use in children older than 3 yr of age with HCV hepatitis
  - Interferon-α2b
  - Peginterferon
  - Ribavirin
  - New therapies: direct-acting antivirals: Sofosbuvir and Simeprevir

4. Treat complications e.g.

- Cirrhosis and fulminant hepatic failure → liver transplant

Reye's syndrome

Definition

Acute mitochondrial encephalopathy with hepatic fatty degeneration

Clinical picture

- Precipitated in a *genetically susceptible* person by the interaction of a viral infection (influenza, varicella) and salicylate and or antiemetic use
- Manifestations appear 5-7 days following viral infection



- Acute hepatomegaly
- The patient remains anicteric



- Acute rise of the intra cranial tension:
  - Severe profound vomiting
  - Delirium and stupor
  - Generalized fits → coma
  - Eventual herniation is the main cause of death

Diagnosis

- Coagulopathy and elevated aminotransferases, and ammonia and hypoglycemia
- Liver biopsy → Fatty infiltration and mitochondrial damage

- Cranial CT → Brain edema
- CSF is normal but with raised pressure (be cautious with lumbar puncture)

Differential diagnosis

Reye like conditions with some inborn errors of metabolism e.g. Fatty acid oxidation defects and valproate toxicity

Treatment

Largely supportive

- Supportive care for acute liver failure
- Coma care and reduction of raised intracranial tension

Wilson diseases

(Hepato lenticular degeneration)

AR defect in ceruloplasmin (Copper carrying protein) → Copper accumulate in:

▪ **Liver**	Forms of hepatic disease include – Asymptomatic hepatomegaly (± splenomegaly) – Subacute or chronic hepatitis – Acute hepatic failure – Cryptogenic cirrhosis, portal hypertension
▪ **Basal ganglia**	– Intention tremor, dysarthria, rigid dystonia, chorea – Deterioration in school performance – Behavioral changes/ Psychiatric manifestations
▪ **Anemia**	– Coombs-negative hemolytic anemia
▪ **Cornea**	– Kayser Flisher ring (*absent in young patients*) 
▪ **Renal tubules**	– Tubular defects (Fanconi like)

Diagnostic triad

- ↓ Serum ceruloplasmin (< 20 mg/dL).
- ↑ Urinary copper excretion after loading dose of D-penicillamine
- Liver biopsy → hepatic copper accumulation.

Screening

Family members of patients with proven cases require screening for presymptomatic Wilson disease. Screening include determination of the serum ceruloplasmin level and urinary copper excretion

Treatment

- Restrict dietary copper intake to <1 mg/day
- Copper chelating agent
  - D-penicillamine 20 mg/kg/day(penicillamine is an antimetabolite of vitamin B6, additional amounts of this vitamin are necessary)
  - Alternative : triethylene tetramine dihydrochloride (trientine)
- Adjuvant therapy: Oral zinc
- Liver transplant

Liver Cirrhosis

Definition: Chronic liver disease with triad of:
- Hepatocytes necrosis.
- Regeneration nodules.
- Lost hepatic architecture.

Causes:
- Post hepatitic.
- Metabolic: e.g. Wilson & hemochromatosis.
- Biliary: 1ry or 2ry to bile flow obstruction.
- Chronic hepatic congestion: cardiac cirrhosis.

Clinical picture

- **Compansated :** Clinical picture of the cause.
- **Decompansated: Features of liver cell failure**

1- Jaundice.
2- Bleeding tendency: - Skin bruises.
- GIT bleeding → Hematemesis & melena.
3- Ascites & generalized edema.
4- Hepatic encephalopathy:
- Due to increased ammonia and neurotoxins (false neurotransmitters).
- Manifested by: Disturbed sleep rhythm, flapping tremor, coma.
5- Hepato-renal syndrome: Functional renal failure in patients with end stage liver disease due to intense renal vasoconstriction with systemic vasodilatation → renal hypoperfusion→ pre renal failure
6- Hepato pulmonary syndrome: - Intrapulmonary vascular dilatation → right to left shunting of blood → hypoxemia, dyspnea, cyanosis & clubbing.
7- Others: - Feotor hepaticus.
- Palmar erythema.
- Spider nevi.
- Muscle wasting.

Diagnosis

1- To prove cirrhosis: - Abdominal ultrasound & MRI.
- Liver Biopsy → diagnostic (but avoided if decompensated)
2- For the cause: e.g. viral markers.
3- For complications:
- Liver functions tests → bilirubin, prothrombin time, albumin.
- Portal hypertension → **see later**.

Treatment

1- Supportive → Carbohydrates and vitamins rich diet
→ Low salt diet (for cases with edema)
→ Limit protein (for cases prone to encephalopathy)
2- Ant fibrotic: Colchicine.
3- Treatment of complications.
4- Liver transplant.

Ascites

Definition: Accumulation of fluid in peritoneal cavity.

Transudate	Exudate	Bloody	Chylous
- Clear - ↓proteins (<3gm/dl) - ↓ Cells - ↓Specific gravity. - **No** organisms.	- Turbid - > 3 gm / dl - ↑ cells (PMNL) - ↑ (> 1018) - may be organisms.	Bloody with RBCs on mic. ex.	Milky white
Causes: 1. Causes of generalized edema; cardiac, hepatic , renal 2. poly serositis	- Septic peritonitis - T.B. peritonitis - Non microbial: • systemic lupus • Metastasis • B-cell lymphoma	- Trauma - Tumors - Bleeding disorders - Acute hemorrhagic pancreatitis	Rupture thoracic duct due to trauma. **Or** obstruction.

Other causes: Bilious, urinary

Clinical diagnosis: see *SHOTS* clinical examination

Treatment of hepatic ascites:

1- Liver support (vitamins, avoids hepato toxic drugs, high carbohydrate diet).

2- Low salt and protein diet.

3- Diuretic: Aldactone.

4- Albumin or plasma infusion.

5- Therapeutic paracentesis provided:

- Tense ascites.
- Prothrombin concentration > 40%.
- Bilirubin < 10 mg/dl.
- Platelets > 40.000/mm^3.
- Creatinine < 3 mg/dl.
- Aspirated volume not more than 20ml/kg/setting



Portal Hypertension

Definition: an elevation of portal pressure >10-12 mm Hg (normal < **7** mmHg)

Causes:

Portal hypertension can result from obstruction to portal blood flow anywhere along the course of the portal venous system

i. Extra hepatic portal hypertension:

- Portal or splenic vein thrombosis due to:
  - Umbilical infection.
  - Neonatal sepsis.
  - Hypercoagulable states e.g. protein S & protein C deficiency.
  - Intra-abdominal infections
- Inferior vena cava obstruction with e.g. constrictive pericarditis/ thrombosis
- Congestive heart failure/tricuspid regurge

ii. Intra hepatic portal hypertension:

1. Pre sinusiodal	→	- Chronic hepatitis.
	→	- Schistosomiasis.
	→	- Portal tract infiltrations
2. Sinusiodal	→	- Cirrhosis (the commonest cause)
		- Veno oculsive disease
3. Post sinusoidal	→	- Budd-Chiari syndrome

(Nelson textbook of pediatrics 2016)

Clinical picture

- Opened collaterals
  - Esophageal varices → hematemesis & melena(the commonest presentation)
  - Caput medusa
  - Hemorrhoids
  - Venous hum
- Splenomegaly (the next most common presentation)
- Ascites
- Liver is
  - Shrunken in cirrhosis.
  - Enlarged tender in post sinusoidal causes
  - Normal in prehepatic causes.

Investigations

1- Abdominal ultrasound → for liver, spleen, ascites.
2- Measure portal vein pressure by ultrasound Doppler.
3- Upper GIT endoscopy for esophageal varices
4- Search for the cause.

Treatment

i. Emergency treatment (Control bleeding esophageal varices):

- ABC
- Take blood sample for investigation & ask for blood.
- Order fresh blood transfusion
- Fluid resuscitation followed by replacement of red blood cells
- Correct coagulopathy by:
  - Vitamin K I.V
  - Fresh frozen plasma or platelets
  - Place nasogastric tube → to monitor ongoing bleeding.
  - H2-receptor blocker or proton pump inhibitor should be given intravenously → reduce the risk of bleeding from gastric erosions

In most patients, particularly those with extrahepatic portal hypertension and with normal hepatic synthetic function, bleeding usually stops spontaneously



With continued bleeding

- Vasopression or Octreotide I.V infusion → ↓ splanchnic flow.
- Nitroglycerine skin patch decrease portal pressure and can ameliorate some of its untoward effects of vasopressin
- Endoscopic sclerotherapy with ethanol amine or much better band ligation of varices.



With continued bleeding

- Sengstaken – Blackmore tube → mechanical compression of esophageal & gastric varices (this device is rarely used now)

ii. Prophylactic (Prevent subsequent bleeding):

- Propranolol → ↓ portal pressure
- Porto-systemic Shunt operation
- Trans jugular intrahepatic porto systemic shunt (TIPS) by an interventional radiologist (problem→may precipitate hepatic encephalopathy and is prone to thrombosis).

iii. Orthotopic liver transplantation

Gastrointestinal Bleeding

Causes

- **Upper GIT bleeding**: Bleeding from above the ligament of Treitz:
  a. Esophageal:
    - GERD
    - Varices
    - Tumors.

  b. Gastric ulcers.
  c. Duodenal ulcers.
- **Lower GIT bleeding**: Bleeding below the ligament of Treitz:
    - Inflammatory bowel disease.
    - Intestinal obstruction (intussusception & volvulus
    - Meckel's diverticulum.
    - Gastroenteritis.
    - Anal fissure.
- **Hemorrhagic blood disease**:
  Result in either upper or lower GIT bleeding e.g. hemophilia, purpura, DIC.

Management

1. Emergency treatment as **(See bleeding esophageal varices)**.
2. Search **for the cause**:

a. History of: - Bleeding disorder// Liver disease.// Gastroenteritis.

b. Examination:

- Skin for → Signs of chronic liver disease.
  → Signs of coagulopathy (e.g. purpura & Bruises)
- Abdominal → Hepatosplenomegaly (in chronic liver disease & leukemia)
  → Distension (intestinal obstruction)
- P/R examination → For perianal ulcers & polyps.

c- Investigations:

- Rule out hemorrhagic blood diseases by → CBC
  → Coagulation profile
  → Liver function tests
- Abdominal X-ray and ultrasound → for obstructions & organomegaly.
- Stool analysis → For gastro enteritis & enterocolitis.
- Endoscopy → for varices, ulcers, polyps.

Treatment: of the cause

- **Medical** e.g. for Hemorrhagic blood diseases, Gastro Enteritis
- **Interventional / Surgical** e.g. for Varices, Polyps

Veno Occlusive Disease(VOD)

(Sinusoidal obstruction syndrome)

The most common cause of hepatic vein obstruction in children

Definition : Intrahepatic obstruction of hepatic veins.

Causes : After total body irradiation with or without cytotoxic drug
May be toxic injury by herbal teas or drugs

Clinical picture

Acute VOD	Subacute VOD	Chronic VOD
		
- Portal hypertension - Hepatomegaly. - No splenomegaly.	- Portal hypertension - Hepatomegaly - Splenomegaly	- Portal hypertension - Cirrhotic liver - Huge Splenomegaly

Diagnosis

1- As for portal hypertension
2- Liver biopsy → diagnostic.

Treatment

- Supportive
- In severe cases: liver transplantation

Budd Chiari Syndrome

Definition: Obstruction of the main hepatic veins.

Causes

- Hypercoagulable states e.g. Polycythemia ,
- Can complicate hepatic or metastatic neoplasms, collagen vascular disease, infection, and trauma

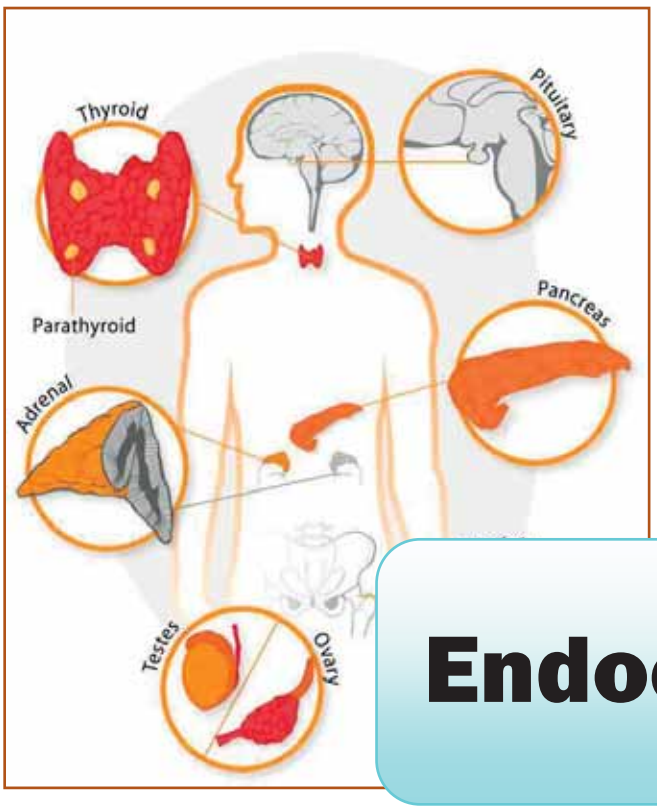
Clinical picture

Acute stage
- Acute hepatomegaly
- Acute abdominal pain & vomiting.
- Acute ascites

Chronic stage.
- Hepatomegaly.
- Portal hypertension.

Treatment

- Supportive
- In severe cases: liver transplantation



Endocrinology

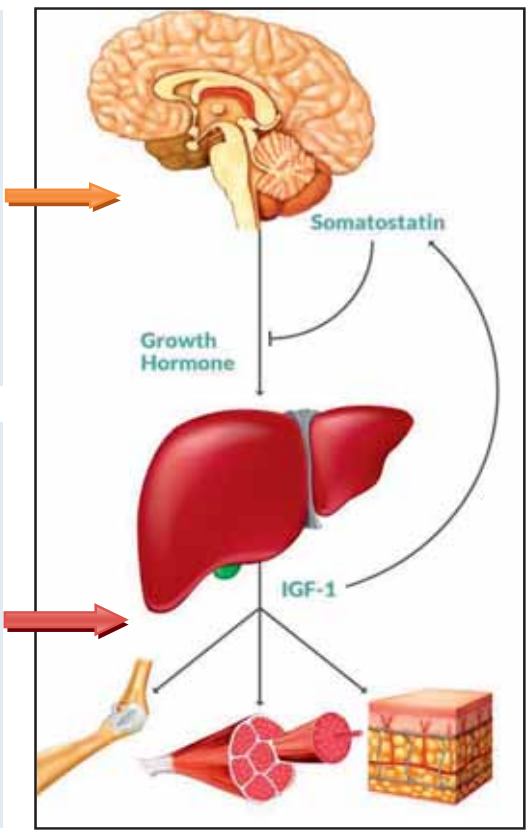
Hypopituitarism

Definition

Underproduction of growth hormone (GH) alone or in combination with deficiencies of other pituitary hormones

Physiology of growth hormone

- GH is secreted form the anterior pituitary in bursts (i.e. pulsatile pattern)under control of hypothalamic GH releasing hormone (GH. rH)
- GH release is:
  - Stimulated by sleep, exercise and hypoglycemia
  - Inhibited by somatostatin



Actions

- Anabolic hormone especially on long bones and muscles
- Action is mediated by **I**nsulin **G**rowth **F**actor **1** produced by the liver:
  - Increase protein synthesis
  - Anti-insulin effect → lipolysis & ↑ blood glucose

Causes

i. Isolated growth hormone deficiency:

A. Genetic: due to
- Mutation of growth hormone genes; AR, AD or XLR.
- End organ resistance:
  - Defective GH receptors e.g. Laron syndrome
  - Post receptor GH insensitivity e.g. abnormal IGF gene or receptor

B. Acquired:
- Idiopathic (The most common).
- Post cranial irradiation (e.g. for leukemia).

ii- Multiple pituitary hormones deficiency:

A. Genetic:
- Due to mutations of multiple pituitary hormones genes e.g. PROP1
- Associations: May be optic nerve dysplasia (septo optic dysplasia).

B. Congenital:
- Pituitary aplasia or hypoplasia.
- Association: May be mid facial anomalies e.g cleft palate, solitary maxillary central incisor

C. Acquired any lesion at hypothalamo- hypophyseal region:
- Tumors e.g. Craniopharyngioma
- Trauma
- Infiltration e.g. histoiocytosis

Clinical picture

1. At birth:
- Normal size (near normal weight and height)
- Micropenis is a diagnostic clue.
- May be neonatal emergency as apnea, cyanosis, hypoglycemia
- May be mid facial anomalies e.g. cleft lip & palate



2. Later on:

Severe growth failure
- Proportionate short stature
- Height below the 1st percentile for age and sex **or** height >2 SD below sex adjusted Mid-parent height
- Growth velocity < 5cm/year
- Appear by the end of the 2nd year



Normal intelligence

Childish facies
- Small face, nose & mandible
- Prominent forehead & depressed nasal bridge
- Wide anterior fontanel & fine hair
- Delayed teething

- Micropenis in childhood; normal for body size in adults
- Hypoglycemia: in infants and children; fasting symptoms in some adults

3. May be features of:
- Septo optic dysplasia : nystagmus & visual impairment in infancy
- Mid facial anomalies e.g. cleft lip / palate, solitary maxillary central incisor
- Increased intra cranial tension (in destructive lesions)
- Associated hormonal deficiency e.g. hypothyroidism, diabetes insipidus



Investigations

1. Screen for GH deficiency

- Low serum levels of IGF-1 and the GH-dependent IGF-Binding Protein 3 matched to normal values for skeletal age rather than chronological age
- Normal values IGF-1 and IGF-Binding Protein 3 indicates GH deficiency unlikely

2. Confirm GH deficiency

- Measure peak levels of GH after provocative agents; insulin, clonidine, or arginine or glucagon:
  - Normal peak levels of GH ⇒ > 10 ng/ml
  - Low peak levels of GH (<10 ng/mL) ⇒ GH deficiency
  - To confirm diagnosis 2 provocative tests should be done
- In prepubertal children: A 3 days of estrogen priming should be used before GH testing to achieve greater diagnostic specificity.

3. For associated deficiencies:

- Measure other anterior pituitary hormones.

4. For the cause:

- Skull CT & MRI for pituitary tumors, aplasia or hypoplasia
- TRH stimulation test (differentiate between hypothalamic and pituitary causes)
- Random prolactin level: high in hypothalamic defects

5. For effect: Radiograph for delayed bone age

Treatment

1. Recombinant GH:

- Dose: 0.18-0.3 mg/kg/week, divided into 6-7 daily subcutaneous injections should be used as soon as GH deficiency is diagnosed
- Criteria to stop therapy:
  - A decision by the patient that he or she is tall enough
  - A growth rate <1 inch/yr
  - And a bone age >14 yr in girls and >16 yr in boys
- Concurrent treatment with a gonadotropin-releasing hormone (GnRH) agonist to interrupt puberty will delay epiphyseal fusion and prolong growth
- Side effects of GH therapy:
  - A 6-fold increase in the risk for type 2 diabetes
  - Pseudotumor cerebri
  - Slipped femoral epiphysis

2. Recombinant IGF-1 for end organ unresponsiveness (e.g. Laron syndrome)

3. Treatment of other hormonal deficiency

4. For infants with micropenis:

One or two a 3-mo courses of monthly intramuscular injections of 25 mg of testosterone enanthate can bring the penis to normal size

Puberty

Definition

- It is a period of growth lasting 5 years, consisting of 3 stages and includes physical, sexual & psychological changes.
- Onset Girls: 8-13 years, Boys: 9-14 years
- Sequence :

Girls	Boys
- Breast development	- Testicular growth
- Pubic hair	- Pubic hair
- External genitalia maturation	- Penis and scrotal growth
- Feminine habitus	- Increased muscle bulk
- Axillary hair	- Body hair (beard, axillary)
- Oil secretion and acne	- Oil secretion and acne
- Menstruation	- First seminal discharge.

Thyroid Gland

Thyroid gland secrete

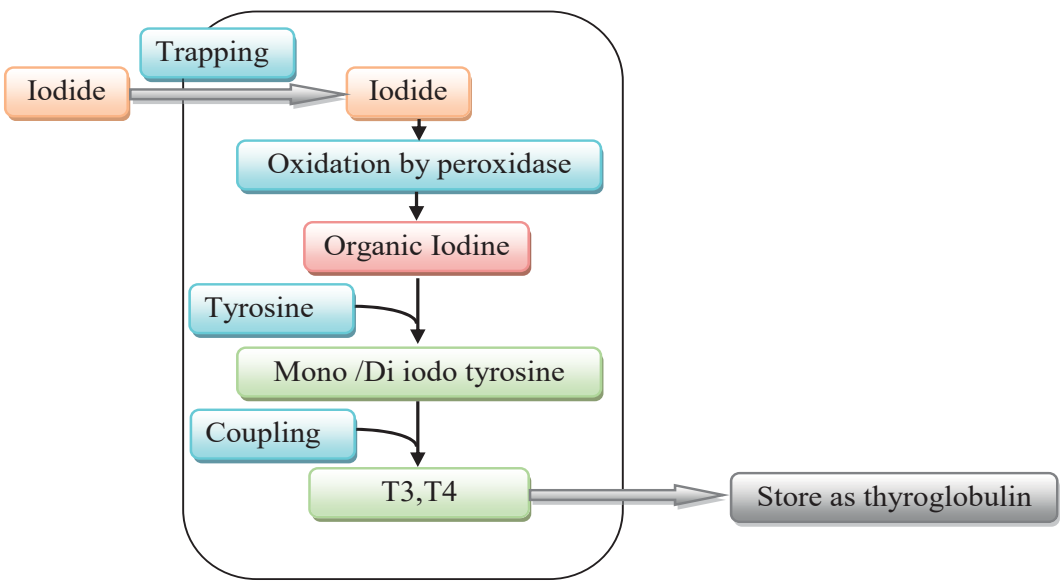
- Thyroid hormones:
  - Thyroxine (T4)
  - Triiodothyronine (T3) ; more potent than T4
- Calcitonin (which deposit calcium salts in bone).

Functions of thyroid hormones

1- Normal maturation of the growing brain in the 1st year of life.
2- Normal skeletal growth.
3- Oxidative metabolism & energy production in all cells

Thyroid hormones synthesis

- Iodide transport (Trapping).
- Iodide is oxidized to iodine by thyroid *peroxidase enzyme* (organification).
- Iodination of tyrosine to form Mono & Di iodo tyrosine.
- Coupling of:
  - 2 Di iodotyrosine → T4
  - Monoiodotyrosine & Di iodotyrosine → T3
- T3 & T4 are stored in thyroid gland as colloid (thyroglobulin).
- Only 20% of circulating T3 is produced by thyroid while 80% is produced by peripheral conversion of T4 by deiodinase



Control of thyroid function:

- Thyroid is regulated by pituitary thyroid stimulating hormone (TSH) in a feedback mechanism.
- TSH synthesis & release is controlled by hypothalamic **T**SH **r**eleasing **h**ormone (TRH).

Congenital hypothyroidism

Causes



A. Primary hypothyroidism:

1. Thyroid dysgenesis:

- The commonest cause (85%).
- Aplasia, hypoplasia or ectopic gland (may be lingual, sublingual or subhyoid).

2. Defective thyroid hormone synthesis (Dyshormonogenesis):

- The second common (15%)
- Autosomal recessive disorders
- Associated with goiter.
  - Examples:- Iodide transport defect.
    - Organification defect: defective thyroid peroxidase enzyme

3. Transient hypothyroidism:

- Trans placental passage of maternal anti thyroid drugs
- Neonatal iodine containing antiseptics

4. Maternal iodine deficiency → Endemic goiter

5. End organ unresponsiveness to: - TSH.
- T3 & T4 (Pseudohypothyroidism).

B. Secondary hypothyroidism:

Due to TSH deficiency either: - Isolated or.
- With multiple pituitary deficiencies.

C. Tertiary hypothyroidism:

Due to TSH releasing hormone deficiency

Incidence: 1:4000; Female: male = 2:1.

Clinical picture

A. In neonatal period: there is may be

- Prolonged physiologic jaundice
- Poor feeding→ chocking spells during feeding.
- Subnormal temperature

- Noisy breathing due to large tongue.
- Abdomen : constipation & umbilical hernia



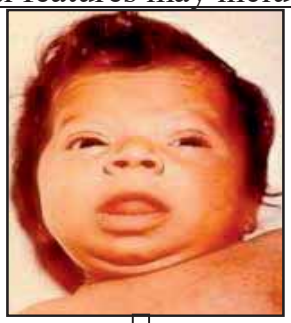
- **L**ethargy; cry little, sleep much.
- Widely open posterior and anterior fontanels (*Good initial clue*)

- May be heavier at Birth
- May be Limbs and genital edema

N.B: *Most infants with congenital hypothyroidism are asymptomatic (due to Trans placental maternal thyroxin), so neonatal screening is mandatory*

B. Full picture (by end of the 1st year): (cretinism)

* Delayed growth ⇒ Short stature with persistent infantile proportions
* Delayed mental milestones
* Delayed motor milestones
* Physical features may include:





Head

- Coarse, brittle hair with low anterior hair line
- Delayed closure of anterior fontanel
- Eyes are puffy, narrow palpebral fissure
- Broad nose & depressed bridge
- Delayed teething
- Thick large protruding tongue

Neck

- Short neck with supraclavicular pad of fat
- Thyroid is enlarged in:
  - Endemic goiter.
  - Dyshormonogenesis
  - Pseudohypothyroidism
- Hoarse cry

Cardiac

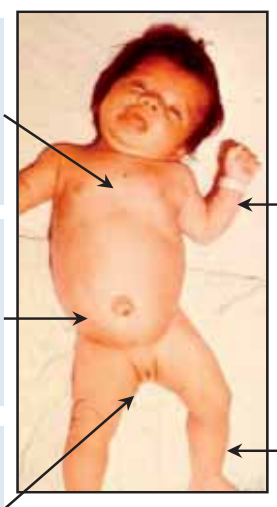
- Bradycardia
- Pericardial effusion
- Cardiomegaly

Abdomen

- Protuberant
- Umbilical hernia
- Constipation

Genitalia

- Delayed maturation
- Rarely precocious puberty



Limbs

- Short broad hands
- Generalized hypotonia
- Occasional reversible generalized pseudohypertrophy most prominent in calf (*Kocher Debre Semelaigne Syndrome*)

Skin

- Cold & pale (resistant anemia)
- Dry (↑myxematous tissue)
- May be yellow (↑ carotene)

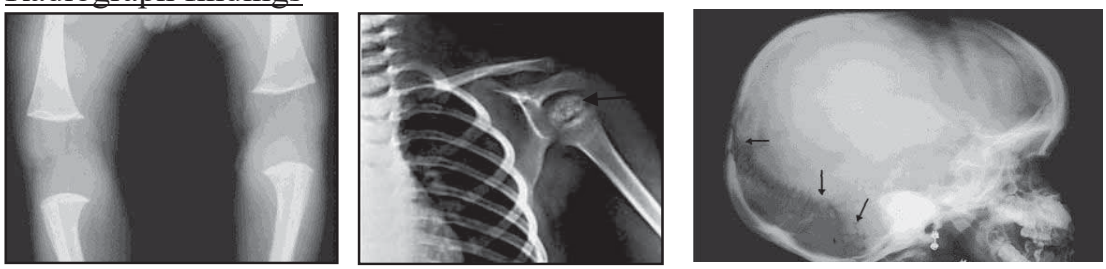
Investigations

1. Confirm diagnosis of hypothyroidism

- Low serum free T_4 (*In hypothyroidism there's compensatory increase in peripheral conversion of T_4 to T_3; so measuring of T_3 may be misleading*)
- Serum TSH
  - High in primary hypothyroidism
  - Low in secondary and tertiary hypothyroidism.
- In pseudohypothyroidism T_4, T_3 and TSH all are high

2. For effect

Radiograph findings



(a) (b) (c)

- Delayed bone age:
  - At birth → absent distal femoral epiphysis (in plane knee radiograph) in 60% of cases (a)
  - Later → delayed appearance of ossific centers (by wrist x-ray)
- Epiphyseal dysgenesis: multiple foci of ossification in heads of femur & humerus (b)
- Skull X-ray → Intrasutural (Wormian) bones (c) and large fontanels
- Beaking of anterior part of T_{12} & L_1 vertebrae.

Cardiac

- ECG shows bradycardia and low voltage.
- Echo / Chest x ray may show cardiac enlargement and effusion.

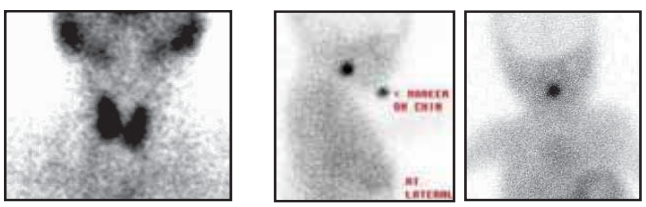
Others

- High serum cholesterol/ Macrocytic anemia

3. For the cause

a. Thyroid scintigraphy (using radioactive ^{123}I):

- Absent uptake in:
  - Aplasia
  - Iodide trapping defect
- Normal or increased uptake in dyshormonogenesis.
- Can localize ectopic thyroid.



Normal thyroid position **Lingual thyroid in an infant**

b. TRH stimulation test: (performed only with ↓ TSH)
- Differentiate between hypothalamic & pituitary defects
- Give an i.v. bolus of TRH:
  - If T4 increases → hypothalamic defect i.e. Tertiary hypothyroidism
  - If T4 does not increase → pituitary defect i.e. Secondary hypothyroidism

c. Skull CT & MRI for pituitary tumors.

Treatment

- Replacement therapy with sodium L-thyroxin (Eltroxin 50 µg tablet) for life
- Dose: **10-15** µg/kg/day in neonate (start with the higher dose if T4 < 5 µg/dl)
  6 - 8 µg/kg/d in infant
  4 µg/kg/d in child
- Follow up and monitoring
  - Clinical: for activity, milestones, growth and overtreatment
  - Lab: for T_4 and TSH (done monthly in the first 6 mo of life, then 3 monthly)
  - Radiologic: monitor bone age

NB

- Overtreatment carries risk of craniosynstosis and temperament disorders
- To rule out *transient hypothyroidism:*
  - Discontinue treatment at 3 years for 3-4 weeks and test for TSH; it will shoot in permanent hypothyroidism
  - This test is unnecessary in:
    1. Infants with proven thyroid ectopia
    2. In those who manifest elevated levels of TSH after 6-12 mo of therapy because of poor compliance or an inadequate dose of T4

Prognosis

- Diagnosis & treatment before 3 months → normal linear growth and intelligence
- Delay in diagnosis, failure to correct initial hypothyroxinemia rapidly, inadequate treatment, and poor compliance in the first 2-3 yr of life result in variable degrees of brain damage.
- Without treatment, affected infants are profoundly mentally deficient and growth retarded
- As diagnosis of hypothyroidism is difficult in the first 3 months screening for thyroid function (usually TSH) in all neonates is done in the first week of the life

Acquired Hypothyroidism

Definition: Juvenile hypothyroidism with manifestations appearing after the 1st year.(After a period of normal thyroid function).

Causes

1. Thyroiditis
   - Autoimmune thyroiditis(Hashimoto disease; chronic lymphocytic thyroiditis):
     - The most common cause of thyroid disease in children and adolescents. It is also the most common cause of acquired hypothyroidism, with or without goiter
     - May be part of polyglandular auto immune syndromes (associated with e.g. DM, Addison)
     - Children with Down, Turner, and Klinefelter syndromes and celiac disease or diabetes are at higher risk for associated autoimmune thyroid disease
   - Suppurative.
   - Viral e.g. mumps.
2. Injury to thyroid → trauma, surgery, irradiation, cystinosis
3. Iodine containing drugs e.g. cough mixtures.

Clinical picture

- Poor academic progress; but no mental retardation
- Unexplained short stature
- Skin: cold, pale(*refractory anemia*), excess myxoedematous tissue
- Cold intolerance
- Constipation
- Delayed puberty (may be precocious).

Investigations

As before but

a. Search for auto antibodies for Hashimoto thyroiditis e.g.
   - Thyroid antiperoxidase antibodies (TPO-Abs) and Antithyroglobulin antibodies (anti-Tg Abs)
   - TSH blocking antibody (may identify patients at future risk of having babies with transient congenital hypothyroidism).

b. Check for associated auto immune disorders e.g. auto immune hepatitis, diabetes mellitus, Celiac disease, Addison disease

Treatment:

- Overt hypothyroidism (elevated TSH, low T4 or free T4), require replacement treatment with levothyroxine
- For subclinical hypothyroidism (elevated TSH, normal T4 or free T4), many clinicians prefer to treat such children until growth and puberty are complete, and then reevaluate their thyroid function

Causes of deafness & hypothyroidism

1- Pendred syndrome

Mutation in the chloride-iodide transport protein common to the thyroid gland and the cochlea→ impaired iodide organification, & positive perchlorate discharge test → Sensorineural deafness and Goitrous hypothyroidism



2- Endemic goiter
3- Neglected hypothyroidism
4- Congenital rubella syndrome

Short Stature

- Growth is strongly related to the genetic potential. A child's target height (TH) is calculated by Mid Parent Height as follows:
  - Girl = (height of mother in inches + height of father in inches)/2 - 2.5inches
  - Boy = (height of mother in inches + height of father in inches)/2 + 2.5inches
- Short stature is height below 3rd percentile for age and sex
- Short stature is either:

Proportionate	Disproportionate
- Upper segment/lower segment is normal for age - Height equals span	- Upper segment/lower segment is abnormal for age. - Height does not equal span

I. Proportionate Short Stature

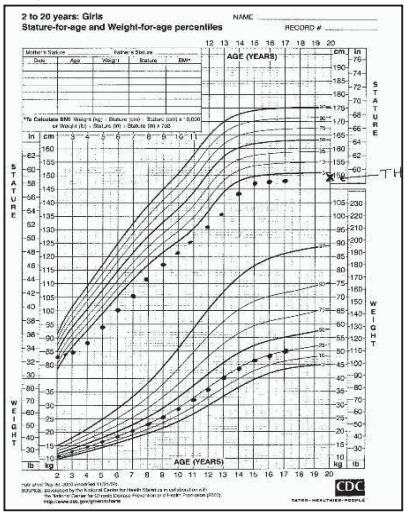
A. Normal types of short stature (about 90% of cases)

1. Familial (genetic) short stature

A short child who is growing close to his/her target height percentile

Clues:

- Small birth length (normal for the family)
- Normal bone age and age of onset of puberty
- Short parents (familial)
- Short target height like their parents

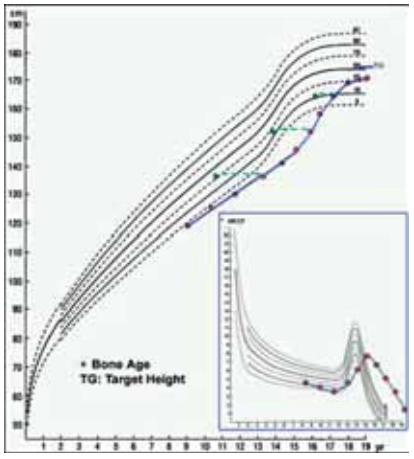


2. Constitutional Growth Delay (CGD)

Growth is normal for the 1st 4-12 mo of life. Height is sustained at a lower percentile during childhood with acceleration late in adolescence when their peers stop growing leading to a final height that is close to the target height

Clues:

- Delayed bone and age of onset of puberty
- Other family members (often one or both parents) with histories of short stature in childhood, delayed puberty, and eventual normal stature
- IGF-1 levels tend to be low for chronological age but normal for bone age



B. <u>Pathologic types</u>:

❶ Chronic undernutrition → marasmus & nutritional dwarfism

❷ Chronic systemic disease
- Malabsorption syndrome e. g celiac disease, inflammatory bowel disease
- Chronic hemolytic anemia e.g. thalassemia
- Chronic renal failure, renal tubular acidosis, urinary tract infections.
- Chronic chest diseases e.g. cystic fibrosis, asthma
- Chronic heart diseases: congenital, rheumatic, cardiomyopathy

❸ Endocrinal causes:
- Hypothyroidism.
- Hypopituitarism.
- Hypercortisolism (Cushing syndrome) and adrenal insufficiency
- Precocious puberty
- Diabetes mellitus.
- Diabetes insipidus.

❺ Psychosocial dwarfism:
- Due to maternal neglect <u>or</u> emotional deprivation
- Evidence of functional hypopituitarism is indicated by low levels of IGF-1 and suboptimal GH on provocation
- History and careful observations reveal disturbed mother-child or family relations
- Associations: perverted or voracious appetites, enuresis, encopresis, insomnia, crying spasms, and sudden tantrums

❹ Syndromes with short stature e.g.:
- Turner
- Down
- Prader Willi
- Noonan
- Other trisomis; e.g. 13 , 18
- Silver Russell

❻ Intra uterine growth retardation: 10 –15% will be short.

II. <u>Disproportionate Short Stature</u>

A. <u>With short limbs</u> e.g.
- Achondroplasia
- Rickets
- Osteogenesis imperfecta

B. <u>With short trunk</u> e.g.
- Skeletal dysplasia e.g. Ectodermal dysplasia and Morquio s syndrome
- Fanconi anemia

Approach to Diagnosis

I. History

* Perinatal for:
  - Exposures; infections, maternal drugs
  - Birth weight and length (differentiate prenatal and postnatal causes)
  - Problems e. g microphallus & hypoglycemia in hypopituitarism.
* Past history suggestive of:
  - Chronic systemic disease.
  - Endocrinal disorder
* Dietetic history for under nutrition or eating disorders.
* Family history for parent's height, other short siblings and social problems

II. Examination

Clinical tests

- Check parent's height → to rule out genetic causes.
- Determine type of short stature → proportionate or disproportionate
- Plot patient weight and height on growth charts:

1. Short stature following own growth curve:
   a. Familial short stature
   b. Constitutional growth delay
2. Short stature with decelerating growth pattern

Weight for age < height for age	Weight for age > height for age
- Chronic systemic disease - Chronic undernutrition - Cardiac disease	- Endocrinopathy e.g. Hypothyroidism Cushing , Hypopituitarism - Syndromes - Skeletal dysplasia

3. Short stature with normal weight for height: Emotional deprivation

Clinical examination

- Evaluate nutritional state: check for muscle wasting, subcutaneous fat loss and signs of vitamin deficiencies.
- Complete systemic examination: including cardiac, chest, abdomen, neurologic
- Check for features suggesting endocrinal disorders e.g. hypothyroidism.
- Check for dysmorphic features e.g. Down, Turner

III. Investigations

- Assess bone age by left wrist X-ray:
  - Normal in familial short stature
  - Delayed in most other causes

- Search for systemic diseases by:
  - Urine analysis → for glucosuria, UTI, osmolality
  - Stool analysis → for malabsorption
  - CBC → for anemia
  - ESR → increased in infection & inflammation.
  - Urea, creatinine → for renal failure
  - Chest X-ray → for suspected chest disease
  - Echocardiography → for suspected cardiac defect
  - Serum pH, calcium, phosphate for metabolic bone disease
  - Specific e.g. Celiac screen, malabsorption workup, chromosomal studies

3. Hormonal assay:
   - Free T4 & TSH for hypothyroidism (Because thyroid hormone is necessary for normal GH synthesis, it must always be assessed before GH evaluation)
   - Provocative growth hormone level & IGF-1 for hypopituitirsm
   - Night time blood or salivary cortisole level for Cushing
   - Blood glucose for diabetes
   - Serum &urine osmolality for diabetes insipidus

Treatment

1. Treat the cause e.g.
   - Gluten free diet for celiac
   - EL-troxin for hypothyroid
2. Recombinant growth hormone

   "FDA approved indications of GH therapy":
   - GH deficiency
   - Turner syndrome
   - Chronic renal failure before transplantation
   - Idiopathic short stature
   - Small-for-gestational age short stature
   - Prader-Willi syndrome
   - Noonan syndrome
3. Adequate balanced diet
4. Psychologic support

Diabetes Mellitus

Definition:

- Chronic, metabolic syndrome characterized by hyperglycemia as a cardinal biochemical feature
- Deficiency of insulin or its action with subsequent defect in metabolism of carbohydrates, protein & lipids

Actions of insulin

- ↓ Blood glucose by → ↑ Glucose uptake by cells
  → ↓ Gluconeogensis
  → ↓ Glycongenolysis
- ↑ Lipogenesis
- Anabolic effect

Types of diabetes mellitus

1. Insulin dependent (Type 1 DM)
2. Non-insulin dependent (Type 2 DM)
3. Secondary diabetes mellitus:
   - Endocrinopathies: Cushing disease, Hyperthyroidism, Acromegaly
   - Drug- or chemical-induced: Steroids, Thyroid hormone
   - Diseases of the exocrine pancreas: Hemochromatosis, Cystic fibrosis, Pancreatitis

Criteria for diagnosis

	Fasting plasma glucose	2-hr plasma glucose during the OGTT
• Diabetes mellitus	≥126 mg/dL (7.0 mmol/L)	≥ 200 mg/dL
• Impaired glucose tolerance	100-125 mg/dL	≥ 140 mg/dL, but < 200 mg/dL

OGTT: Oral Glucose Tolerance Test, Values in mg/dl

Insulin dependent diabetes mellitus

Etiology

β-cell destruction leading to absolute insulin deficiency, may be related to:

- Genetic predisposition: associated with HLA -DR_3, DR_4
- Auto immune response (Humoral & cell mediated response against islet cells):
  Evidence:
  - Presence of Islet cell autoantibodies
  - Association with other auto immune diseases e.g. thyroiditis
  - Environmental factors:
  - Viral infection e.g.: mumps, rubella, measles, EBV

Pathogenesis

Insulin deficiency results in disturbance of carbohydrate, fat & protein metabolism.

- Fat metabolism: ↓ Lipogenesis → ↑ free fatty acids (FFA) →↑ Ketone bodies
- Carbohydrate metabolism:
  - Hyperglycemia due to - ↓ glucose uptake & utilization by the cells
    - ↑ gluconeogenesis & glycogenolysis
  - Hyperglycemia leads to osmotic diuresis → polyuria & polydipsia
- Protein metabolism: ↑ proteolysis → ↓ body weight.

Clinical picture

- Polyuria, polydipsia, polyphagia & weight loss.
- Diabetic keto-acidosis (DKA) is the presenting feature in 25% of cases (10% in T2DM).
- Secondary nocturnal enuresis

Diagnosis of diabetes mellitus

- Symptoms of diabetes mellitus (include polyuria, polydipsia, and unexplained weight loss with glucosuria and ketonuria) **plus** random plasma glucose ≥ 200 mg/dL (11.1 mmol/L)

or

- Fasting (at least 8 hr) plasma glucose ≥126 mg/dL (7.0 mmol/L)

or

- 2-hr plasma glucose during the OGTT ≥ 200 mg/dL

or

- Hemoglobin A1C ≥ 6.5% (*Results should be confirmed by repeat testing if in absence of unequivocal hyperglycemia*)

(Nelson textbook of pediatrics 2016)

Diabetic Keto Acidosis (DKA)

Definition

- Metabolic disorders due to *acute* insulin insufficiency
- Risk Factors:
  - Omission of insulin dose by error or during inter current illness
  - Insulin pump failure
  - Adolescent girls
  - Psychiatric disorders including eating disorders
  - Previous DKA

Pathogenesis

History of exposure of diabetic patient to stresses (e.g. infection, trauma, and psychic).

⇓

Acute insulin insufficiency

Hyperglycemia

↑↑ FFA



↑↑ ketone bodies.

Ketonemia

Aceto acetic acid & B (OH) Bytric acid

⇓

Metabolic acidosis

acetone



acetone odor of breath

Osmotic diuresis

Polyuria

⇓

Dehydration

ketonuria

Acute abdominal pain & vomiting

Rapid & deep breathing (Kussmaul breathing)

Along with metabolic acidosis ⇒ coma

Clinical stages of DKA

Early	Then	Later
– Polyuria – Nausea and vomiting – Abdominal pain mimicking an acute abdomen	– Dehydration – Rapid, deep, sighing (Kussmaul)breathing	– Progressive obtundation – Impaired consciousness

Diagnosis

- Diagnostic criteria of DKA
  - Diabetic → Hyperglycemia ; blood glucose >200 mg/dL [≈ 11 mmol/L]
  - Keto → Ketonemia and ketonuria
  - Acidosis → Venous pH <7.3 and/or bicarbonate <15 mmol/L
- Severity of DKA

Mild	Venous pH 7.25-7.35 Or Bicarbonate 16-20 mmol/L Clinical: Oriented, alert but fatigued
Moderate	Venous pH 7.15-7.25 Or Bicarbonate 10-15 mmol/L Clinical: Kussmaul respirations; oriented but sleepy; arousable
Severe	Venous pH <7.15 Or Bicarbonate <10 mmol/L Clinical: Kussmaul or depressed respirations; depressed sensorium to coma

Management

- Assess
  - Severity of dehydration (most cases are considered 5- 8% dehydrated)
  - Level of consciousness using Glasgow coma scale
- Request
  - Plasma glucose, HbA1c, urinalysis for ketones, blood ß-hydroxybutyrate and venous pH
  - Electrolytes, blood urea nitrogen, creatinine, calcium, phosphorus, and magnesium concentrations
  - CBC and cultures (blood, urine) if infection is suggested
  - ECG for baseline evaluation of potassium status

A. Acidotic phase (in ICU)

DKA is a medical emergency that should follow ABC scheme

- Admit to the ICU

▪ Airway 	– Secure the airway – Insert airway in comatose – Insert nasogastric tube if conscious level depressed or the child is vomiting
▪ Breathing	– 100% O2 for hypoxia – ± intubation and ventilation
▪ Circulation 	– Site 3 IV lines one for I.V fluids , a 2nd for I.V insulin and a 3rd sampling cannula – Obtain blood sample for investigations – Give antibiotics to febrile patients

Monitoring using flow sheet

Hourly	**Two hourly lab tests for the first 12 hours**
– Vital signs – Neurologic state for warning signs of cerebral edema and Glasgow coma scale if comatose – Cardiac monitor for potassium status – Capillary blood glucose – Amount of insulin – Fluid input and output (catheter)	– Glucose – Blood gases – Blood ß-hydroxybutyrate – Serum electrolytes – Blood urea nitrogen – Calcium, magnesium, phosphorus
Urine ketones until cleared	**Weigh** the patient twice a day

1. Fluid therapy



Shock therapy

- 10 ml/kg over 1 hour can be repeated up to 30 ml /kg
- 0.9% saline or lactated ringer

In the 2nd hour after the shock therapy and until resolution of DKA:

- Fluid requirement = Fluid maintenance + Fluid deficit (**85** ml/kg) – Boluses fluids given over next 23 hour (36 hours in severe cases)

N.B

- Fluid deficit = % Dehydration × Body weight
- Fluid maintenance (24hr) = 100ml/kg(for the 1st 10 kg) + 50ml/kg (for the 2nd 10) + 25 ml/kg (for all remaining kg)
- Replace any ongoing fluid loss in vomiting or diarrhea or massive polyuria

Type of fluid

- 0.45% NaCl plus 20 mEq/L K phosphate and 20 mEq/L K acetate
- If K <3 mEq/L, give 0.5 to 1.0 mEq/kg as oral K solution OR increase IV K to 80 mEq/L
- Bicarbonate therapy is rarely necessary and may even increase the risk of hypokalemia and cerebral edema

2. Insulin therapy

- Use regular insulin (e.g. Actrapid)
- Slow infusion **0.1** unit/kg/hour without bolus must be given *at the beginning of therapy*
- Prepared by adding 50 units (0.5 ml) soluble insulin to 49.5 ml 0.9% saline in 50 ml syringe pump to provide 1unit/ml concentration and insulin started at **0.1ml/kg/hour**

- Precautions with insulin therapy

  If the blood glucose concentration drops too quickly before the DKA has resolved:
  - Serum glucose has decreased < 300 mg/dL → Glucose is added as a 5% solution
  - Serum glucose has decreased < 200 mg/dL→ Glucose is added as a 10% solution
  - If the serum glucose falls further despite these interventions, → The insulin infusion can also be lowered to 0.05 U/kg/hr

  ⇩

  Aim to keep blood glucose at about 11 mmol/L (200 mg/dL) until resolution of DKA
- Shift to to oral intake and subcutaneous insulin
  - ✓ When DKA has resolved indicated by:
    - Bicarbonate >15 mEq/L; pH >7.30
    - Sodium stable between 135 and 145 mEq/L
    - No emesis
  - ✓ Subcutaneous insulin
    - Dose: 0.2- 0.4 u/kg every 6-8 hr
    - To prevent rebound hyperglycemia the first injection should be given 30 minutes before stopping the insulin infusion

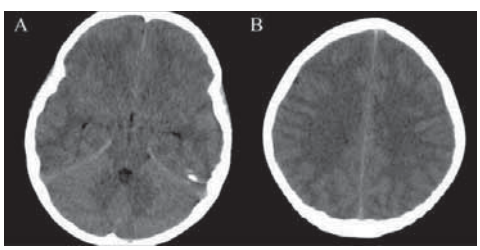
3. Treat precipitating factors e.g. Antibiotics for infections.

Hazards during treatment of DKA

1. Cerebral edema

Warning clue

Serum sodium should increase by about 1.6 mmol/L for each 100 mg/dL decline in the glucose; the sodium should steadily increase with therapy. *Declining sodium may indicate excessive free water accumulation and the risk of cerebral edema*



Signs of cerebral edema

- Headache & slowing of heart rate
- Change in neurological status (restlessness, irritability, increased drowsiness, incontinence)
- Specific neurological signs (e.g., cranial nerve palsies)
- Rising blood pressure
- Decreased O2 saturation

Treatment

- Initiate treatment as soon as the condition is suspected.
- Reduce the rate of fluid administration by one-third.
- Give mannitol 0.5–1 g/kg IV over 20 minutes and repeat if there is no initial response in 30 minutes to 2 hours
- Hypertonic saline (3%), 5–10 mL/kg over 30 minutes, may be an alternative to mannitol or a second line of therapy if there is no initial response to mannitol
- Elevate the head of the bed
- Intubation may be necessary for the patient with impending respiratory failure

1. **Hypokalemia.**
2. **Severe hypophosphatemia**
3. **Hypoglycemia**

*Hint: "I myself prefer adopting the British Society of Paediatric Endocrinology **and Diabetes protocol for DKA, 2015. It is very easy and less complicated.** Please google it"* ***Mohamed El Koumi***

Mortality rate of DKA: About 0.2% due to:

- Cerebral edema accounts for 60% to 90% of all DKA deaths
- Hypokalemia induced arrhythmias
- Sepsis / aspiration pneumonia

Differential diagnosis: From other causes of coma in diabetic child:

A. Hyperosmolar non ketotic coma (Hyperglycemic hyperosmolar state;HHS)

- Plasma glucose concentration >33.3 mmol/L (600 mg/dL)
- Arterial pH >7.30
- Serum bicarbonate >15 mmol/L
- Small ketonuria, absent to mild ketonemia
- Effective serum osmolality >320 mOsm/kg
- Clinically : Stupor or coma, severe dehydration

B. Hypoglycemic coma

- History :
  - Known diabetic with insulin overdose or exercise or delayed meals
- Clinically
  - Reactive sympathetic stimulation (↑catecholamines): Pallor, hunger pains, tachycardia, sweating, jitterness, tremor, irritability
  - Glucopenia of CNS: Lethargy, limpness, may be seizures.
- Blood glucose **< 50** mg/dl (< **2.6** m mol /l)
- Rapid response to I.V. glucose

B. Post acidotic phase

Regular Insulin	Diet
– Dose: 0.2-0.4 u/kg/ 6-8 hrs; SC – Given 0.5 -1 hr before meals – Increase **or** decrease insulin dose by 10% to keep blood glucose between 100-180 mg/dl	– Seek dietitian advice – Sips of water – Skimmed milk – Dietetic juice – Two days later give average diet

C. Lifelong management

1. Insulin

Starting Doses of Insulin (units/kg/day)

	No DKA	**Post DKA**
▪ Prepubertal	0.25-0.50	0.75-1.0
▪ Pubertal	0.50-0.75	1.0-1.2
▪ Postpubertal	0.25-0.50	0.75-1.0

Types

- Short acting
  - Regular insulin (Crystalline insulin, Humulin R)
  - Ultra short acting: Insulin Aspart (Novorapid) / Lispro (Humalog)
- Intermediate insulin (NPH, insulin Monotard , Humulin N)
- Long acting insulin (Glargine ; Lantus and Detemir)

Regimens

1. Multiple dose injections (recent, very effective):
   - Use Glargine or Detemir as basal insulin at bed time
   - The basal insulin should be 25-30% of the total dose in toddlers and 40-50% in older children
   - The remaining insulin is given as pre meals boluses of ultra-short acting analogs (usually Novorapid or Humalog) based on carbohydrate in meals
2. Two injections regimen:
   - 2/3 the total daily dose before breakfast
   - 1/3 the dose before evening meal
   - Each dose contain 1/3 short acting analog & 2/3 NPH
3. Three injection regimen:
   - NPH and short acting analog bolus at breakfast
   - Short acting analog bolus alone before afternoon snack or the main evening meal
   - NPH at bed time
4. Continuous subcutaneous insulin infusion using automated insulin pump

Adjustment:

Increase or decrease insulin dose by 10% to keep blood glucose:

- Pre meal 70-145 mg/dl
- Post prandial (1-2 hours) < 180 mg/dl
- Pre bed time 125-180 mg/dl

2. Instructions

- Diet → 3 main meals with 2 snacks (engage a dietician expert in diabetics diets).
- In infections increase rapid acting insulin by 10%
- Decrease insulin before exercise

3. Monitor

a. Daily blood glucose at least 4 times

- Before breakfast, lunch, supper& at bed time.
- Initially test blood glucose also between 12AM & 3AM.to exclude nocturnal hypoglycemia

b. Glycosylated hemoglobin (Hb A_{1C}).

- Fraction of hemoglobin to which glucose has been attached.
- Measured as a percent of total hemoglobin.
- Value: Reflect average blood glucose over previous 2-3 months:
  - Normal, non-diabetic → < 6%
  - American diabetes association recommends:
    Hb A_{1C} of <8.5 % in toddlers
    Hb A_{1C} of < 8 % in children
    Hb A_{1C} of < 7.5 % in teenagers

Honey moon period

* Due to residual β-cell function → release insulin **so** About 75% of new diabetics complain recurrent hypoglycemia which may recur for weeks to months.
* Advice: Never stop insulin but reduce the dose to avoid hypoglycemia.

Somogi phenomenon

* Due to large insulin dose > 2 u/kg/d → Late nocturnal hypoglycemia occur → ↑↑ anti insulin hormones → early morning hyperglycemia.
* Advice: Reduce the evening intermediate insulin by 10%

Dawn phenomenon

* Due to overnight growth hormone secretion → antagonise insulin action → early morning hyperglycamia.
* Advice: Increase the evening intermediate insulin by 10%

Polyuria

Definition: Passage of excessive urine output > 2 liters/m^2

Causes

Endocrinal	Renal	Psychogenic
* Diabetes mellitus * Diabetes inspidus	* Hypokalaemia (<2.5 meq/l). * Renal tubular acidosis * Barttar syndrome * Hypercalcemia (> 13 mg/dl). * Chronic renal failure.	* Compulsory water ingestion

Diabetes Insipidus

Definition

Inability to produce concentrated urine due to either

1. Decrease ADH production ⇒ Neurogenic diabetes insipidus.
2. Lack of response of renal tubules to ADH ⇒ nephrogenic diabetes insipidus.

Clinical picture

i- Polyuria = Urine output : 4-10 Liter/day.
- Polydipsia (Irritable infants)
- 2ry nocturnal enuresis.
- If water inaccessible → Dehydration
  → Electroyte disturbance.
  → Fever (no sweating).
  → Shock in severe cases.

ii- Growth retardation

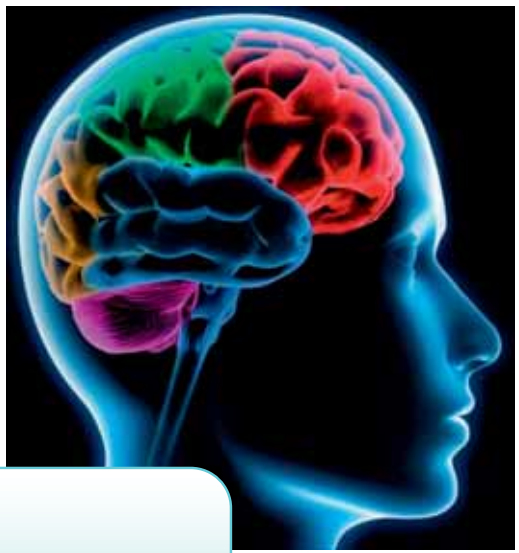
iii- May be features of the cause e.g ↑ ICT in craniopharyngioma

Investigations

1. Urine:
   - Specific gravity: 1002-1005 (diluted)
   - Osmolality: low (50-200 m.osmol/L)
   - No pathological constituents
2. Plasma osmolality: High (> 295 m.osmol /L)
3. Water deprivation test.
4. Vasopression stimulation test

Treatment

- ♦ Neurogenic → Desmopressin intranasal twice daily
- ♦ Nephrogenic → Adequate hydration.
  → Correct hypokalemia by:
  - Oral potassium.
  - Potassium sparing diuretics.
  → Indomethacin .



Neurology

Meningitis

Definition: Inflammation of the membranes covering the brain & spinal cord.

Types: - Bacterial
- Aspetic e.g. viral, fungal
- Tuberculous

Bacterial (Septic) meningitis

Causes

	G –ve bacteria	G +ve bacteria	
Cocci	• Nisseria meningitides type A, B, C, D, Y, W 135**	• Pneumococci • Staphylococci	++ in infants & children
		• Streptococci	++ in neonates
Bacilli	• E.coli	• Listeria monocytogenes	
	• Hemophilus influenza.		

N.B. ** Serogroup B is responsible for more than 50% of cases in children less than 1 year and has also been associated with outbreaks on college campuses

Transmission: - Droplet infection mostly (Blood borne in neonatal sepsis)

Clinical picture

1. **Non specific**
   - High fever (may be hypothermia in neonates).
   - Poor feeding
   - Rose spots may appear on the trunk & extremities in meningeococcal septicemia.
2. **Features of increased intracranial pressure (ICP)**
   - Before fontanel closure → tense, bulging anterior fontanel
   - After closure of fontanels:
     - Severe bursting headache (irritability)
     - Blur of vision
     - Projectile vomiting (in the morning, not preceded by nausea)
     - Cushing response (hypertension & bradycardia)
3. **Features of meningeal irritation:** (less sensitive in infants)
   - Neck rigidity (stiffness) → limited neck flexion
   - Opisthotonus → arched back
   - Kernig's sign → inability to extend the leg after the thigh is flexed to a right angle with the axis of the trunk.
   - Brudzinski leg sign: Passive flexion of one hip → flexion of the other hip and knee
   - Brudzinski neck sign: Passive flexion of the neck → flexion of the hip & knee.

4. Neurologic signs

- Stupor & drowsiness
- Convulsions → usually generalized
- Coma

Clinical types

1- Meningitic form → the classic presentation as before.

2- Fulminant meningitis.

- Abrupt fever.
- Severe headache and convulsions.
- Rapidly progress to coma.
- Fatal within 48 hrs.

3- Septicemic form (usually complicating meningeococcal form)

- Very bad general condition
- Shock
- Purpura & ecchymosis
- Meningitis develop within 1-2 days (**or** not at all)

Complications

1- Syndrome of inappropriate secretion of antidiuritic hormone (SIADH) → so, maintenance fluids must be at 2/3 normal to avoid brain edema.

2- Neurologic complications:

- Increased intracranial pressure (ICP) → May leads to cerebral or cerebellar herniation
- Subdural effusion
- Cranial nerve lesions (commonly oculomotor, 6th & 8th nerves).
- Hydrocephalus.

3- Peripheral circulatory complications

i- Waterhouse Friedrichson syndrome

- Septicemia
- Shock
- Extensive purpura
- Adrenal hemorrhage (acute adrenal failure).



ii- DIC: Gangrenous patches & extremities

4- Dissemination of infection: endocarditis, arthritis , osteomyelitis

Investigations

1. CBC → Leukocytosis (↑↑ PMNL)
2. Blood culture reveals the responsible bacteria in up to 80-90% of cases

3. C-reactive protein, ESR, and procalcitonin have been used to differentiate bacterial (usually elevated) from viral causes of meningitis
4. Lumbar puncture (LP) & CSF examinations:

Value

- Diagnostic (Discover organism, ↑PMNL, ↑Protein, ↓Glucose)
- Determine appropriate antibiotics by culture & sensitivity.
- Evaluate treatment: CSF become sterile within 24- 48 hours of appropriate antibiotics

Contraindications for an immediate LP

1. Evidence of increased ICP (other than a bulging fontanel), such as 3rd or 6th cranial nerve palsy with a depressed level of consciousness, or hypertension and bradycardia with respiratory abnormalities
2. Severe cardiopulmonary compromise requiring prompt resuscitative measures for shock
3. Infection of the skin overlying the site of the LP

What to do if an LP is delayed?

- Initiate empirical antibiotic therapy
- CT scanning for evidence of a brain abscess or increased ICP
- LP may be performed after increased ICP has been treated

Condition	Appearance	Pressure (mmH_2o)	Protein (mg/dl)	Glucose (mg/dl)	Leukocytes / ml	Organism
Normal CSF	Clear	50 - 80	5 – 20	40-80	0-5 (monocytes)	Nil
Bacterial meningitis	Turbid	↑↑	↑↑ (> 100)	↓↓	↑↑ (100-60.000) Mainly PMNLs	+ve Gram stain* +ve culture
TB meningitis	Web On stand	↑↑	↑↑ (> 100)	↓↓	↑ (10-500) Early PMNLs then lymphocytes	Acid fast bacilli by zehl nelsen stain.
Viral meningitis	Clear	Normal or slightly ↑	Mild ↑ (< 100)	Normal or ↓↓	↑ (10-500) Early PMNLs later mononuclear cells predominate	Viruses may be isolated

***** Gram stain is positive in 70-90% of patients with untreated bacterial meningitis

N.B: In *partially treated meningitis*

- ✓ Culture and gram stain are usually negative
- ✓ But Pleocytosis with a predominance of neutrophils, elevated protein level, and a reduced CSF glucose usually persist for several days

Differential diagnosis

1- From other causes of meningitis
2- Meningism:
- Noninfectious meningeal irritation due to extra cranial lesions
- Causes: Upper lobe pneumonia, otitis media, shigellosis
- CSF is normal

3- Brain abscess
4- Encephalitis

Management

A. Treatment

1. Antibiotic therapy

- Parenteral antibiotics according to culture and sensitivity for **2- 4** weeks
- Empirical therapy while waiting for culture results:
  - ✓ Third-generation cephalosporins (cefotaxime 300mg/kg/day divided 6 hourly **or** ceftriaxone 100mg/kg/day)
    Plus
  - ✓ Vancomycin (60 mg/kg/24 hr, given every 6 hr)
  - ✓ If a patient is immunocompromised and Gram-negative bacterial meningitis is suspected, initial therapy might include ceftazidime and an aminoglycoside or meropenem
  - ✓ If Listeria monocytogenes infection is suspected, as in young infants ,give ampicillin (200 mg/kg/24 hr, given every 6 hr)

(Nelson 2016)

2. Supportive therapy

Measures to ↓ ICP
- Mannitol 0.5 –1gm/kg iv
- Furosemide 1mg/kg iv

Corticosteroids

Indications:

a. H. influenza meningitis:
- Value: Reduce inflammatory response caused by cell lysis
- Use dexamethasone 0.15 mg/kg/dose every 6 hours for 2 days
- Maximum benefit if given 1-2 hr before antibiotics are initiated

b. Septic shock to improve general condition.

c. Adrenal failure

Treatment of complications e.g. convulsions
- Immediate relief by diazepam or lorazepam
- Then phenytoin loading and maintenance

B. Prevention

- Isolation of the case
- Vaccination against H.influenza, meningococci, pneumococci
- Chemoprophylaxis for contacts: e.g. rifampicin 10-20 mg/k/day ⇒ for 2-4 days.

Prognosis Depends on:

1- Age: the younger the age, the worse the prognosis.
2- Course: fulminant meningitis has worse prognosis.
3- Cause: - E.coli & staph →↑ fatality & ↑ long term sequalae.
- H.influenza & pneumococci → moderate prognosis.
- Meningococci → < 5% fatality & no residual disability.

Aseptic meningitis

Meningitis with no micro organisms detected in CSF by gram stain or bacterial culture.

Causes:
- Mostly viral → Herpes simplex virus
  → Enteroviruses (Echo & coxachie)
  → Mumps
  → Ebstein barr virus
- Protozoa → Malaria
  → Toxoplasma
- Non infectious → CNS leukemia
  → Intrathecal injection
  → Post vaccination.

Diagnosis: - CSF analysis
- Viral isolation

Treatment: - Supportive ± antiviral.

Seizures

Definition:

- **A seizure** is
  - Transient , paroxysmal, time limited, involuntary disturbance of brain function
  - Manifested by abnormal motor, sensory, behavioral or autonomic activities
  - With or without impaired consciousness

Causes

A. Acute Seizure

1. Febrile Seizure
2. First epileptic fit.
3. Symptomatic seizures
   - CNS causes:
     - Infection → meningitis, encephalitis, brain abscess.
     - Irritation → brain edema
     - Tumors of the brain
     - Toxic → tetanus, drug (e.g aminophylline), lead encephalopathy
     - Hemorrhage → trauma, hemorrhagic blood diseases.
     - Hypoxia → hypoxic ischaemic encephalopathy.
     - Hypertensive encephalopathy.
   - Metabolic causes:
     - Bilirubin encephalopathy
     - Uremic encephalopathy
     - Hepatic encephalopathy
     - Hypo (glycemia, calcemia, magnesemia)
     - Hypo **or** hypernatremia.
     - Pyridoxine (B_6) deficiency
     - Inborn errors of metabolism

B. Recurrent Seizures

1. Epilepsy
2. Symptomatic seizures
   - Tetany
   - Degenerative brain diseases
   - Chronic metabolic causes
     - Inborn errors of metabolism
     - Hepatic /Uremic encephalopathy

Febrile Seizures

Definition: Seizures in age vulnerable children due to:
- Rapid rise of body temperature.
- Due to extra cranial causes (mostly viral)

Incidence:
- Affect 4% of children.
- Family history in about 20 % of cases (genetic base do exist)
- Recurrent in 30-50% of cases specially in those with family history

Diagnostic criteria

1. Age: 6 - 60 months (convulsions below or above this age is not febrile)
2. Fits occur within 8-12hrs from onset of fever.
3. No evidence of CNS infection (e.g. meningitis), nor metabolic disease
4. Evidence of extra cranial infection (e.g. tonsillitis, otitis media, roseola)
5. Occur in the absence of a history of prior afebrile seizures
6. Type of convulsions:

Simple (Typical)	Complex
- Generalized tonic-clonic. - Last < 15 min. - One fit only in the same illness. - The commonest form	- Focal - Last > 15 min - Recurring within 24 hr - Uncommon.

Investigations

1. Lumbar puncture to rule out meningitis

Mandatory in
- Infants below 6 months presenting with fever and seizures
- Ill looking children
- Clinical suspicion of meningitis

Optional in
- Children 6-12 months not vaccinated for *Haemophilus influenzae type b & Streptococcus pneumonia* **or** immunization status is unknown
- Children who have been pretreated with antibiotics

2. EEG and Neuro imaging (CT, MRI)
- Only for cases with high risk of epilepsy ; usually not required for 1st simple febrile seizure
- EEG is done > 2 weeks of the attack

Risk of subsequent epilepsy is higher with:
- Neurodevelopmental abnormalities
- Complex febrile seizures (focal)
- Family history of epilepsy

3. Blood tests e.g. electrolytes, blood glucose if clinically indicated

N.B: Major risk factors predicting recurrence of febrile seizures:

- Age <1 yr
- Duration of fever < 24 hr
- Fever 38-39 °C

Differential diagnosis

1. Intracranial infections: Meningitis, meningeo encephalitis, and brain abscess
2. Epileptic fit precipitated by associated fever
3. Epilepsy syndromes typically start with febrile seizures e.g.
   - Generalized epilepsy with febrile seizures plus **(GEFS+)** : multiple febrile seizures and several types of afebrile generalized seizures
   - Severe myoclonic epilepsy of infancy **(Dravet syndrome):** seizures starts febrile then become afebrile with evolving developmental delay

Treatment

In general, antiepileptic therapy, continuous or intermittent, is not recommended for children with one or more simple febrile seizures

1. **Acute care of febrile seizure attack:**
   - Full history and thorough examination
   - Fever control by paracetamol and tepid sponges or cold bath.
   - Fit lasting more than **5** minutes→ Diazepam, lorazepam, or midazolam
   - Investigate and treat the underlying cause
   - Treatment of *febrile status epilepticus*
2. **Parent education about:**
   - Acute handling of seizures at home
   - Seizures lasting > 5 minutes → rectal diazepam **or** buccal or intranasal midazolam
   - Fever control→ reduce the discomfort not the seizure recurrence
   - During fever → intermittent oral diazepam 0.3 mg/kg q8 hours **or** rectal diazepam (0.5 mg/kg as a rectal suppository every 8 hr), reduce, but do not eliminate, the risks of recurrence of febrile seizures

(Nelson 2016)

Epilepsy

Epilepsy is

- A brain disorder with predisposition to generate seizures with neurobiologic, cognitive, psychologic, and social consequences of this condition
- It is considered to be present when
  - 2 or more unprovoked seizures occur in a time frame of longer than 24 hr
    Or
  - At least 1 unprovoked epileptic seizure with enough EEG and clinical information to demonstrate recurrences

(Nelson 2016)

Causes

1. Idiopathic (Now termed genetic) in 80% of cases
2. Organic (secondary) in less than 20% of cases
   - Congenital cerebral malformation.
   - Degenerative brain diseases.
   - Post-traumatic, post-hemorrhagic, post-infection, post-toxic, post-anoxic

Classification

A. Focal (partial) seizures

* Only one part of the body is involved i.e. focal.

* Types:

1. Focal seizures without impairment of consciousness (Simple partial seizures)	**2. Focal seizures with impairment of consciousness (Complex partial seizures)**
▪ No aura	▪ Often preceded by aura (e.g. visual hallucinations)
▪ Brief	▪ Last 1-2 min
▪ Motor (focal tonic, clonic or atonic) or sensory ▪ Often there is a motor (Jacksonian) march from face to arm to leg	▪ Only motor fits
▪ No automatism ▪ **DD:** Tics: Unlike tics, motor seizures are not under partial voluntary control	▪ **Automatism** may occur → automatic semi purposeful movements of the mouth (oral, chewing) or of the extremities (manipulating the sheets, shuffling, walking).
✓ Consciousness is intact.	✓ Consciousness is impaired with staring.
▪ Postictal (Todd's) paralysis or sleepiness last minutes or hours	

3. Focal seizures with secondary generalization

4. Focal seizures epileptic syndromes:

- ***Benign childhood epilepsy with centrotemporal spikes (BECTS)***
  - Starts during childhood (ages 3-10 yr) and is outgrown in adolescence
  - The child typically wakes up at night owing to a focal (simple partial) seizure causing buccal and throat tingling and tonic or clonic contractions of one side of the face, with drooling and inability to speak but with preserved consciousness and comprehension
  - EEG shows typical broad-based *centrotemporal spikes* that are markedly increased in frequency during drowsiness and sleep
  - Drug of choice: Carbamazepine, oxcarbazepine
- ***Landau-Kleffner epileptic aphasia syndrome***
  - Focal seizures + verbal auditory agnosia and loss of speech
  - Drug of choice: Valproate

EEG in focal seizures

- ✓ Shows focal spikes or sharp waves in the lobe where the seizure originates.
- ✓ A sleep-deprived and 24-hour video EEG EEG increase diagnostic yield

B. Generalized seizures: The whole body is affected.

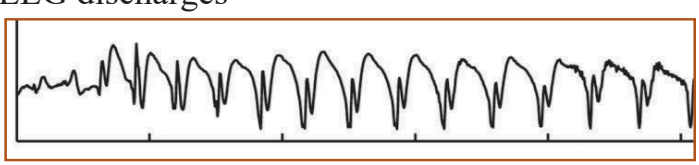
1. Absence seizures(Petit mal)

A. Typical Absence seizures

Incidence: More in girls. Usually start at 5-8 yr of age.

Description:

- Sudden cessation of all motor activities or speech with a blank facial expression; awareness of surroundings is cut off
- Accompanied by eye lid flutter or upward rolling of the eyes
- Last seconds; after seizure patient resume the pre seizure activity.
- Frequently recurrent; may occur countless daily
- No aura , loss of consciousness nor postictal phase
- EEG → typical 3 Hz spike–and–slow-wave discharges
- Hyperventilation for 3-5 min can precipitate the seizures and the typical EEG discharges



B. Atypical absence seizures

- Absences associated with myoclonic components **and** tone changes of the head (head drop) and body
- Precipitated by drowsiness
- Usually accompanied by 1-2 Hz spike–and–slow-wave discharges

2. Generalized motor seizures (Grand mal)

The commonest form; pass in 3 phases.

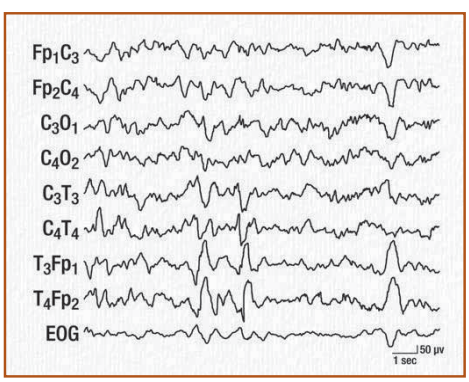
Aura (pre ictal phase)	Attack (ictal phase)	Post ictal phase
A warning signs before the attack may exist suggesting a focal origin of the epileptiform discharge: e.g. localized muscle spasm or paraesthesia.	– Sudden loss of consciousness – Tonic phase: tonic contraction of whole body → rigid posture, apnea, cyanosis, rolling of eyes & drolling of saliva. – Clonic phase: rhythmic contraction. & relaxation of all muscles groups → tongue biting & loss of sphincter control.	– Semiconscious for 30 min-2hr. – Headache – Sleepiness

3. Myoclonic epilepsies

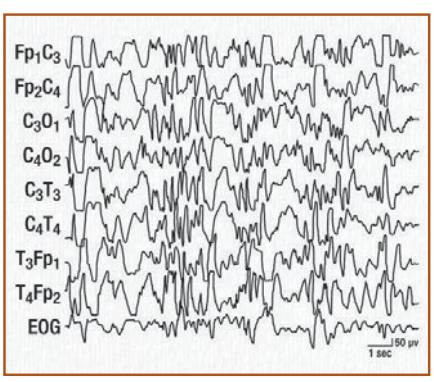
- Rapid shock like contractions, usually <50 msec in duration, that may be isolated or may repeat but usually are not rhythmic
- Intact consciousness.

4. Infantile spasms

- Starts in the 1st year of life
- Brief symmetric tonic contractions of the neck, extremities & trunk which may be flexor, extensor or mixed
- Repetitive; usually in the morning
- A cry may precede or follow the spasm; so may be confused with colic
- **West syndrome**: triad of infantile spasms, developmental regression, and a typical EEG ;hypsarrhythmia (high-voltage, slow, chaotic background with multifocal spikes)
- **EEG** → Hypsarrythmias



Normal EEG



Hypsarrythmias EEG

5. Atonic or Astatic seizures

- Often follow myoclonic seizures
- Cause a very momentary loss of tone with a sudden fall

Investigation

For the first unprovoked no febrile seizure

1. EEG (Electro Encephalogram); in awake and sleep state
2. Metabolic screen: Serum Na, Ca, Mg, glucose ± inborn errors of metabolism
3. CSF examination in suspected CNS infections.
4. MRI (preferable) / CT brain for:
   - Patients with focal seizures
   - Increased intra cranial pressure
   - Resistance to treatment
5. ECG to rule out long QT or other cardiac dysrhythmias
6. Genetic diagnosis is now available for a huge number of seizures disorder

Treatment of epilepsy

- **When to start anti-epileptic drugs (AEDs)?**

Low risk of recurrence	**High risk of recurrence**
Isolated first seizure with: – Normal neurodevelopmental status – Normal EEG – Normal MRI – Absent family history of epilepsy	Seizure with: – Abnormal neurodevelopmental status – Abnormal EEG – Abnormal MRI – Positive family history of epilepsy
⇓	⇓
✓ No long term AEDs ✓ Close observation ✓ Prescribe rescue medications (rectal diazepam) for seizures > 5 min	✓ Rule out Symptomatic Seizures ✓ Start AEDs even if the first seizure

- **For symptomatic seizures:**

  ⇓

  Treat the underlying cause (hypoglycemia, urea cycle abnormality, meningitis, temporal lobe tumor, etc.)

- **Educating the family and the child about**
  - The disease, and its management
  - How to handle seizures acutely and use of rescue medications
  - Watch the child during swimming in pools, passing traffic, ….
  - Never to stop the AEDs suddenly
  - Exercise: can share in hockey, baseball, basketball, and football

- **Anti-epileptic drugs (AEDs)**

How to start?

- Choice should be based on the type of seizure and the epilepsy syndrome
- Only **one** drug is used with small dose → if no response → gradually increase the dose.
- Control with 1 drug (monotherapy) should be the goal
- In resistant cases a **2nd** drug can be used alone or in combination.

When to stop?

- Children are free of seizures for at least 2 yr with normal EEG at discontinuation
- AED therapy should be discontinued gradually, over a period of 3-6 mo

Disorder	Drug of first choice
– Focal – Secondary generalized seizures	▪ Oxcarbazepine, levetiracetam, carbamazepine
– Absence seizures	▪ Ethosuximide (Zarontin) ▪ Treatment is guided by EEG
– Both absence & generalized motor seizures coexist – Generalized epilepsies	▪ Valproate /Lamotrigine
– Myoclonic epilepsy	▪ Valproate/ Lamotrigine
– Infantile spasms	▪ Adrenocorticotropic hormone (ACTH) – Suppresses the expression of corticotrophin-releasing hormone, a proconvulsant neuropeptide whose expression may be enhanced in patients with infantile spasms – Intramuscular **or** Gel in a tapering doses – Monitor the patient's response with serial EEG ▪ Vigabatrin (Sabril) ; retinal toxicity is a risk
– Migraine and epilepsy	Valproate or Topiramate (effective in both)

Other new FDA approved anti-epileptic medications e.g.

- Perampanel (Fycompa)
- Rufinamide (Banzel)
- Clobazam (Onfi)

	Na valproate	Carbamazepine	Levetiracetam	Lamotrigine
Dose	20-40 mg/kg/d	10-20 mg/kg/d	20-40 mg/kg/d	1-15 mg/kg/d
Side effect	– Sedation – Hepatotoxic – Alopecia – Weight gain	– Sedation – Hepatotoxic – Anemia – Leucopenia	– Somnolence – Asthenia – Behavioral disorders	– Dizziness – Headache, ataxia – Stevens-Johnson syndrome

- **Ketogenic diet:**
  - For infants < 2year with resistant myoclonic epilepsy
  - Most calories given form fat (never used with valproate)

Differential diagnosis: from conditions mimic epilepsy

1. Vagal syncope

- Triggered by sight of blood, pain, or sudden stress
- There is initially pallor and sweating followed by blurring of vision, dizziness, nausea, and then gradual collapse with loss of consciousness
- Rapid recovery with no postictal depression
- If prolonged ; lead to generalized convulsions, termed anoxic seizures

2. Cardiac syncope:

- Long QT syndromes
- Aortic stenosis

Syncope mostly predisposed by Exercise

3. Breath holding attacks

- Episode starts with a cry (often a "silent" cry and marked pallor in the case of the pallid type), and progresses to apnea and cyanosis.
- Spells usually begin between 6 and 18 mo of age.
- Syncope, tonic posturing, and even reflex anoxic seizures
- Association iron deficiency anemia is common

4. Psychogenic nonepileptic seizures

- Predisposed by stress
- Gradual onset
- Asynchronous flailing limb movements that vary between attacks
- No injury, closed eyelids
- May respond to suggestion during "loss of consciousness"
- Usually >2-3 min

Status Epilepticus

Definition

- Continuous seizure activity or recurrent seizure activity without regaining of consciousness lasting for >30 min
- Impending status epilepticus: seizures lasting between 5 and 30 min

Etiology

1. Prolonged febrile seizures (the commonest cause)
2. Sudden withdrawal of anticonvulsants in an epileptic patient
3. CNS anomalies or infections (e.g. encephalitis) or tumors.
4. Metabolic disorders e.g. hypoglycemia, inborn errors of metabolism

Clinical types

- Convulsive status epilepticus (generalized tonic, clonic, or tonic-clonic)
- Nonconvulsive status (complex partial, absence)
- Myoclonic status

Management

A. Initial assessment

- A brief physical examination should assess respiratory and circulatory status.
- A rapid neurologic examination provides a preliminary classification of the type of status epilepticus.
- A history from a parent or caregiver for possible cause of the seizures.

B. Initial intervention:

In the first 5 minutes of seizure activity

a) Airway	– Maintain airway. – Suction of secretions
b) Breathing	– O_2 inhalation – Assisted ventilation
c) Circulation	– Secure an I.V. line
d) Draw Samples for	– Electrolytes, Glucose, Calcium and magnesium – Basic metabolic panel for inborn errors of metabolism – Culture blood and CSF – Toxic screen – AEDs level in known epileptics
e) Continuous EEG	– Helps diagnosis – Monitor response to treatment
f) Glucose 10% 5ml/Kg For hypoglycemia	

C. Control convulsions

Emergent therapy (6-15 minutes)

IV line available	I.V slow **Lorazepam** (0.05mg/kg) May be repeat in 5-10 minutes **or** **Diazepam** (0.3mg/kg) May be repeat in 5-10 minutes **or** **Midazolam** (0.2 mg/kg) Followed by IV infusion *(With all benzodiazepines ; monitor and manage respiratory depression)*
IV line un available	Buccal or intranasal midazolam **or** Intranasal lorazepam **or** Rectal diazepam are effective options

Urgent therapy (16-35 minutes)

- Give immediately Phosphenytion or phenytion
  - Loading 15-20 mg/kg under ECG monitor
  - Not > 0.5-1 mg/kg/min
  - Take peak blood level 2 hr later
  - Maintain on 3-6 mg/kg/24 hr
- Phenobarbitone is often the next medication at a loading dose of 5-10 mg/kg
- IV Valproate (25mg/kg) is emerging as a strong evidence urgent therapy

If seizures controlled	**Refractory status epilepticus**
Use maintenance doses of phenytion and /or phenobarbitone	▪ Intubate and assist respiration ▪ Drug options: – Midazolam infusion – Propfol infusion – Barbiturate coma – General anesthesia ▪ Careful attention to blood pressure ▪ Monitor response with EEG

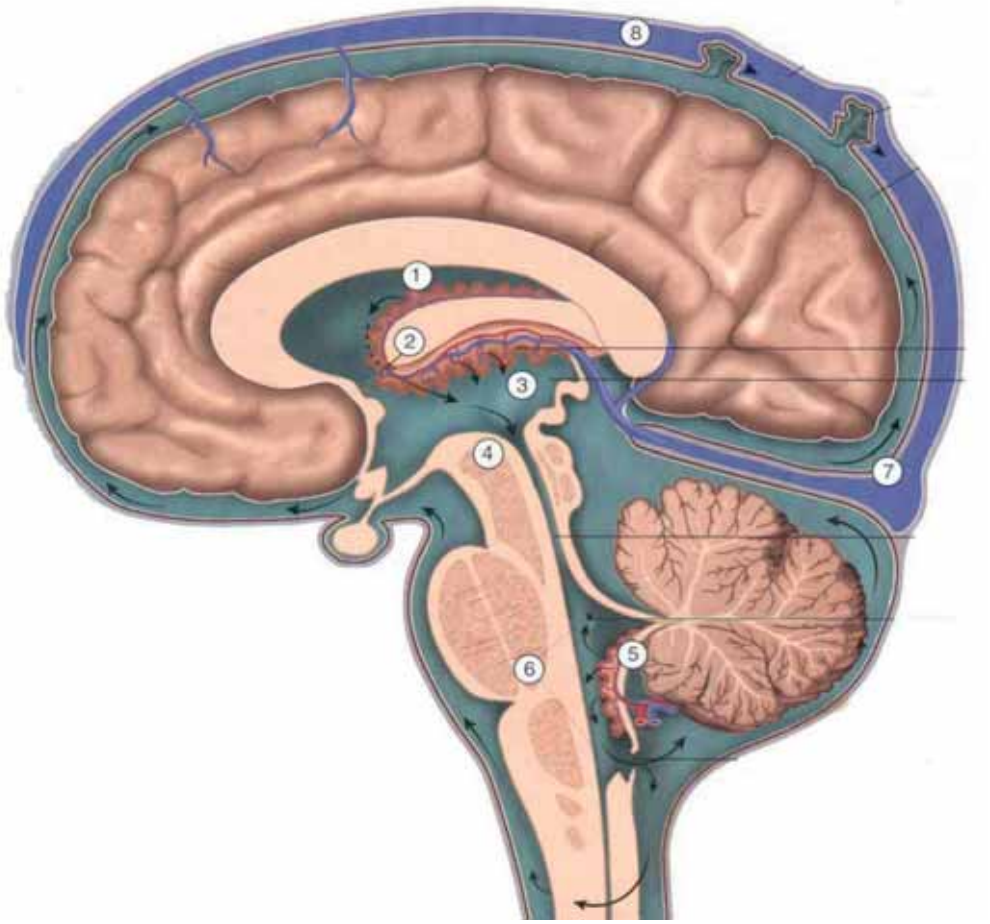
Hydrocephalus

Definition

- Excessive accumulation of CSF with enlargement of cerebral ventricles with or without increase of the intra cranial pressure; ICP
- Hydrocephalus is not a specific disease; it represents a diverse group of conditions that result mainly from impaired circulation and/or absorption of CSF .

Normal CSF circulation:

CSF amount in infant = **50** ml (150 in adult)



1. CSF is formed by active secretion by choroids plexus mainly in the lateral ventricles
2. CSF passes via foramen of Monro to the 3rd ventricle
4. Then via aqueduct of Sylvius to the 4th ventricle
5. Then via foramena of Lnuscka & Magendi to the subarachnoid space
7. CSF in subarachnoid space is absorbed by arachnoid villi to dural venous sinuses

Causes of hydrocephalus

I. Relative hydrocephalus: Normotensive hydrocephalus
- Apparent increase in CSF due to brain atrophy
- Not associated with raised ICP

II. Absolute hydrocephalus

A. Obstructive hydrocephalus

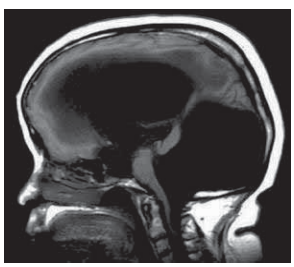
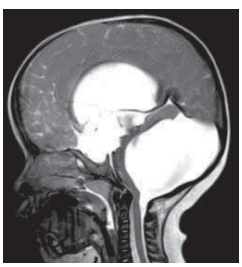
Obstructed CSF flow within the ventricular system (Non-communicating)

1. Obstruction of aqueduct of Silvius:

* Congenital atresia:
  - May be sex linked recessive.
  - May be associated with spina bifida occulta
* Obstruction from outside by:
  - Brain tumors.
  - Malformation of vein of Galen (listen for A cranial bruit).
* Obstruction from inside:
  - Post hemorrhagic (especially in premature).
  - Post meningitis (T.B., pneumocci, mumps)

2. Congenital atresia of:

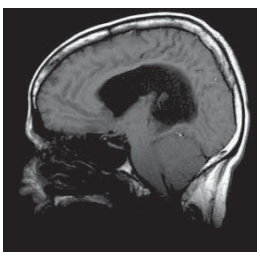
* Foramen of Monro.
* Foramina of Luscka & Magendi: Cystic dilatation of 4th ventricle usually with cerebellar vermis agenesis (**Dandy Walker malformation**)

3. Arnold Chiari malformation:

Congenital downward displacement of cerebellum, pons & medulla





4. Congenital infection especially toxoplasmosis

5. Brain tumors

B. Non obstructive hydrocephalus (Communicating) due to either:

1. Defective CSF absorption
   * Subarachnoid space adhesions: - Post hemorrhagic or post meningitic
   * Leukemic infiltration.
   * Dural sinus thrombosis
2. Excessive CSF secretion (Rare) due to:
   - Choroid plexus papilloma
   - Choriod plexus congestion as in meningitis

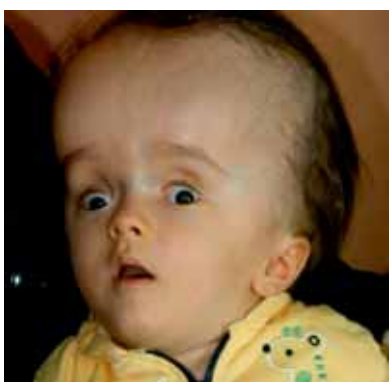
Clinical picture

In infant

1. **Head signs**

Marked; as the cranial sutures are still opened and yield under rising ICP





- Accelerated rate of enlargement of the head is the most prominent sign. (increasing head circumference on serial measurements)
- Fontanels are widely opened & bulging.
- Sutures are widely separated.
- Dilated scalp veins.
- Eyes deviated downwards → Sunset appearance
- Skull percussion → Cracked pot sound (Macewen sign).
- Craniotabes in all bones
- A foreshortened occiput suggests Chiari malformation, and a prominent occiput suggests the Dandy-Walker malformation.

2. **Neurologic sings**

Mild, as rapid ↑ in skull size protect against marked increase of ICP

- Mild vomiting
- Squint
- Delayed motor milestones
- Pyramidal tract lesion signs are common especially in lower limbs.
- In advanced cases: mental retardation & optic atrophy may occur.

In older child

Marked neurologic manifestations as the sutures are not easily separated with subsequent marked increase ICP

-Bursting headache; severe in the morning
-Blur of vision
-Projectile vomiting (unrelated to meals, not preceded by nausea)
-Bradycardia & hypertension (Cushing response)

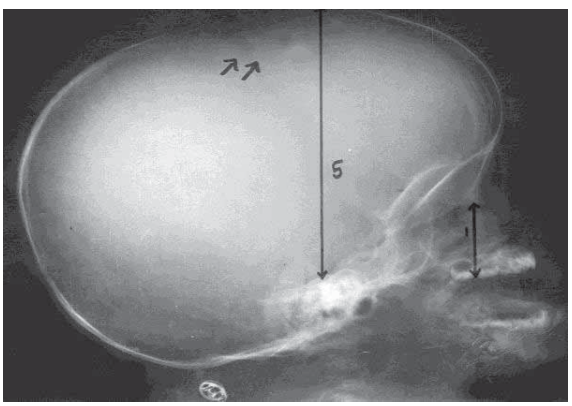
Diagnosis

A. Confirm hydrocephalus

1. Clinical picture: Progressive head enlargement on serial measurements

2. Cranial X-ray

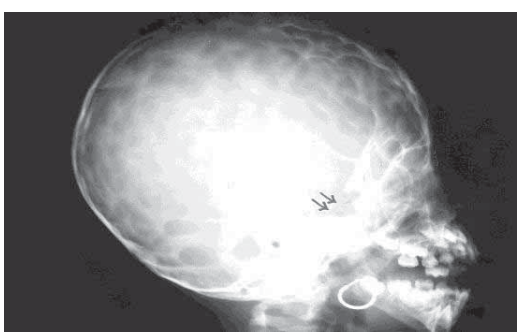
A. Before closure of sutures and fontanels:



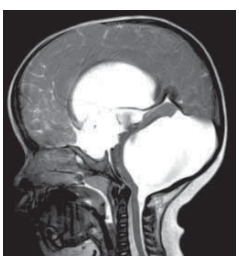


- Wide fontanels, wide separation of sutures.
- Craniofacial disproportion with large cranium.

B. After closure of sutures and fontanels:



- Increased ICP (beaten silver appearance, wide sella)



3. Trans fontanel cranial ultrasound

4. CT & MRI

- Diagnostic; can detect ventricular dilatation.
- Detect degree of cortical atrophy.
- May detect the cause

B. Type of hydrocephalus; obstructive or communicating?

1. CT & MRI: The most accurate and non-invasive √√
2. Simultaneous lumbar & ventricular manometry:
   - Normally, both are equal
   - Ventricular pressure > spinal pressure in obstructive hydrocephalus
3. CSF examination: xanthochromia & cytoalbuminous dissociation in obstructive type

N.B: Examination of fundus is mandatory for:

- Evidence of chorioretinitis in congenital infections e.g. toxoplasmosis
- Check for papilledema in older child

Treatment

Medical

Decrease CSF by:

- Carbonic anhydrase inhibitors; acetazolamide (Diamox tablets)
- Furosemide

For:

- Moderate or slowly progressive ventricular dilatation
- If response is not stable → proceed to shunt operation

Draw backs:

- Transient effect
- Electrolyte & pH disturbances

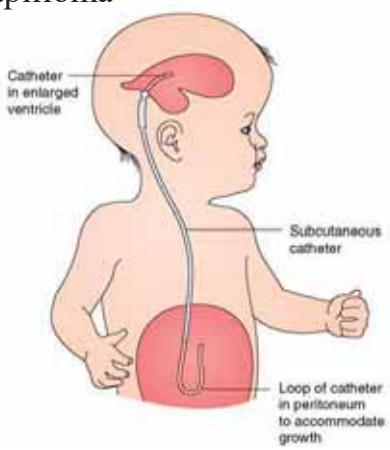
Surgical

1. Choroid plexectomy **or** diathermy for choroid papilloma
2. Extra cranial shunt operation

Types

- Ventriculoperitoneal ⟶
- Ventriculo artial (right)
- Ventriculopleural





Complications

- Shunt nephritis (immune complex mediated)
- Obstruction (headache, papilledema, emesis, mental status changes)
- Infection commonly with staph epidermidis (fever, headache, meningismus)
- Relative shortening as the child grow

Differential diagnosis macrocephaly

From causes of **macrocephaly** (H.C> **2** standard deviation above mean)

Cranial causes

- Constitutional
- Achondroplasia
- Familial
- Anemia (chronic hemolytic)
- Rickets

Intracranial causes

- Hydrocephalus
- Hydrancephaly
- Space occupying lesion e.g. tumor
- Megalencephaly which may be due to:
  - **C**retinism
  - **S**torage diseases (e.g. mucopolysacharidosis).
  - **F**amilial

Microcephaly

Definition: Head circumference measures > 3 SD below the mean for age and sex

Causes

1- True microcephaly due to small sized brain.

2- Craniosynostosis due to early fusion of sutures.

i. True microcephaly

Criteria

- Skull sutures & fontanelles → normal.
- No increase intra cranial tension.
- Skull X ray show small vault.
- CT scan shows brain atrophy.

Etiology

a. Genetic

- Familial → AR, (severe atrophy of frontal lobes → camel head)
- Chromosomal syndromes e.g. trisomy 21, 18, 13

b. Secondary (Non genetic)

* Prenatal:
  - TORCH infection.
  - Fetal irradiation; especially in the 2nd trimester.
  - Maternal diabetes or phenyle ketonuria.
  - Maternal drugs e.g. phenytoin, &alcohol.
* Natal: Hypoxic ischemic encephalopathy.
* Post-natal: Early meningitis& Encephalitis

ii. Craniosynostosis

Definition: early fusion of skull sutures;

1. Palpable ridge is felt at the affected suture.
2. If multiple sutures are affected:
   - Microcephaly → brain atrophy.
   - Increase intra cranial tension → hydrocephalus& beaten sliver appearance in skull X ray.
3. Skull examination → abnormal skull shape which may be:

a. Scaphocephaly (Dolicocephaly)

– Elongated due to premature closure of sagittal suture



b. Brachycephaly

- Short anteroposterior
- Due to bilateral closure of coronal sutures




c. Oxycephaly

- Conical head
- Due to multiple sutures closure



d. Trigonocephaly

- Triangular
- Due to closure of metopic suture



Treatment

Surgical separation of skull sutures is indicated in:

- Cases with hydrocephalus.
- Cases with progressively increase intra cranial tension.
- Cosmotic reasons.

Cerebral Palsy

(Little's Disease)

Definition

- A group of permanent disorders of movement and posture causing activity limitation
- Resulting from non-progressive lesions to the developing fetal or infant brain
- Affecting mainly the motor centers; cerebral cortex , cerebellum , and basal ganglia
- With frequent neurologic associations including:
  - Mental retardation
  - Epilepsy
  - Impaired hearing ;deafness
  - Impaired vision
  - Emotional disturbances
  - Behavioral disturbances

Causes

- **Pre-natal** (80%)
  - Antenatal Infections
  - Congenital malformations
  - Fetal asphyxia
- **Natal** (10%)
  - Birth asphyxia
  - Birth trauma
- **Post-natal** (10%)
  - VLBW with intracranial hemorrhage
  - Meningitis, encephalitis
  - Metabolic e.g. phenyle ketonuria
  - Hypoglycemia
  - Hyper bilirubinemia
  - Hydrocephalus.

Topographic classification: (distribution of motor defect)

1- Monoplegia → Only one limb is affected
2- Hemiplegia → Upper and lower limbs on one side are affected
3- Diplegia → All limbs are affected, the lower more affected than the upper limbs
4- Paraplagia → Only both lower limbs are affected
5- Quadriplegia → All the four limbs are affected

Clinical Types

1. Spastic cerebral palsy

Criteria

- The commonest type
- Pyramidal tract lesion (UMNL) signs:
  - Hypertonia
  - Hyper reflexia
  - Positive Babinski sign
  - May be clonus
- Persistence of primitive reflexes
- Pesudobulbar palsy → feeding disorder (poor suckling & swallowing), squint and speech disorders.

Types

1. Spastic diplegia: 35%

- Bilateral spasticity of the legs that is greater than in the arms
- More in premature with periventricular leucomalacia
  - Crawling is commando like rather than four limbed crawling.
  - Lower limbs scissoring (application of a diaper is difficult)
  - With paraspinal muscle involvement, the child may be unable to sit.
- MRI typically shows scarring and shrinkage in the periventricular white matter with compensatory enlargement of the cerebral ventricles

2. Spastic hemiplegia: 25%

- Due to in utero or neonatal stroke
  - Decreased spontaneous movements on the affected side
  - Shows hand preference at a very early age
  - Walking is delayed until 18-24 mo (tiptoe walking); gait is circumdactive
  - Examination of the extremities may show growth arrest, particularly in the hand and thumbnail
  - Upper extremity assumes a flexed posture when the child runs

3. Spastic quadriplegia: 20%

- More ischemia and infection
- The most severe type
- Marked motor impairment of all extremities and the high association with mental retardation and seizures

4. Spastic monoplegia

5. Spastic paraplegia

2. Ataxic cerebral palsy

Criteria

- Hypotonia and hyporeflexia
- Cerebellar ataxia→ incoordination of voluntary movements, nystagmus, staccato speech, intention tremors.

3. Extrapyramidal (dyskinetic, asthetoid) cerebral palsy

Commonest causes

- Asphyxia, kernictrus

Criteria

- Hypotonia (replaced with time with hypertonia & rigidity)
- Chorio asthetoid movements.
- Deafness.

3. Atonic cerebral palsy

- Profound hypotonia → floppy infant
- Preserved deep tendon reflexes

5. Mixed cerebral palsy

Diagnosis

1. Clinical: A thorough history and physical examination should rule out a progressive disorder of the CNS, including degenerative diseases, metabolic disorders, spinal cord tumor, or muscular dystrophy
2. Investigations:

Value: Exclude progressive brain insults and may detect a cause or association

a. CT & MRI
   - May detect the cause e.g. brain malformations and spinal cord lesions
   - Rule out brain tumors & degenerative brain disease.
   - CT scan may be useful for detecting calcifications associated with congenital infections

b. TORCH screen.

c. Genetic evaluation

d. Metabolic screen.

e. For associations: Test for Hearing, Visual function , EEG for seizures

N.B Conditions that can mimic cerebral palsy

- Spinal cord tumors
- Channelopathies
- Sandifer syndrome
- MECP2 duplication
- Congenital dopa-responsive disorders
- Genetic spastic paraplegia
- Some metabolic conditions (GLUT1 deficiency, glutaric aciduria type 1)
- Ataxia telangiectasia

Treatment

- Multidisciplinary approach is most helpful in the assessment and treatment of such children; A team of physicians from various specialties as:
  - Occupational and physical therapists
  - Speech pathologists
  - Social workers
  - Educators
  - Developmental psychologists
- **Assist:**
  - Feeding & defecation
  - Vision and hearing
  - Walking: Walkers, standing frames, motorized wheel chair
  - Communication by talking typewriters and special computers
  - Rehabilitation according to the degree of motor disability
- **Medications**

Anti spastic drugs

drug	Action	Side effect
Diazepam	– GABA agonist – Useful in short term relief of painful spasms	– Sedation – Dependency with long term use
Baclofen	– GABA agonist and inhibit spinal neuronal transmission – When used intrathecally, via a surgically implanted continuous-delivery pump → greater efficacy with fewer adverse effects	– Sedation
Tizanidine	– Alpha-2 adrenergic receptor agonist and inhibit spinal neuronal transmission – Useful in severely disabled by cerebral palsy and in those with night-time spasms	– Sedation
Dantrolene	– Block calcium intake by skeletal muscles→ ↓ free intracellular calcium	– Hepatic dysfunction – Blood dyscrasia

Botox A: injection in spastic muscles and salivary glands to reduce drooling. It stops the release of acetylcholine at the synapse and blocks neurotransmission. The effects gradually wear off (over about 3–6 months)

Levodopa: Small doses may be helpful for dystonia and rigidity

- **Surgery:**

For marked spasticity of the lower extremities → surgical soft tissue procedures that reduce muscle spasm e.g. adductor tenotomy or psoas transfer and rhizotomy (roots of the spinal nerves are divided)

Mental Retardation

Definition: Handicapping disorder with age of onset below 18 years characterized By subnormal I.Q. (< 70%).

$$\text{I.Q. (Intelligence Quotient)} = \frac{\textit{Mental}\ \text{age}}{\text{Chronological age}} \times 100$$

Diagnostic criteria

1- Subnormal intelligence quotient "IQ" (less than or equal to 70%)
2- Limitations exist in two or more of the adaptive skills e.g. communications, social skills, self care, safety, functional academics, work
3- Manifest before age of 18 years (if after 18 years, it is called dementia.)

Etiology

1.Physiologic (sub cultural)

- No demonstrable organic brain lesions
- Seen in children living in low socio economic standard with neglect and poverty

2.Genetic causes

- Chromosomal anomalies; e.g. Trisomy 21,18,13, klinefelter syndrome
- Genetic disorders e.g. Fragile-X syndrome , prader willi syndrome
- Developmental brain abnormalities e.g. hydrocephalus and familial microcephaly
- Degenerative brain diseases e.g. lipidosis and mucoploysacridosis
- Inborn errors of metabolism

2.Non genetic

- Cerebral palsy causes (**Mention**)
- Congenital hypothyroidism
- Severe hypernatremia or recurrent hypoglycemia

Presentations

Age	Manifestation
Infancy	– Delayed social development →Fail to interact with environment – Gross motor delay
Early childhood	– Language delay /difficulties – Behavior difficulties – Delayed fine motor
Late childhood	– Academic under achievement

Prevention

- Proper prenatal, natal and post-natal care
- Vaccination against rubella for females (not during pregnancy)

- Neonatal screening to identify preventable causes of MR (e.g. phenylketonuria).
- Treatment of neonatal jaundice, hypoglycemia, hypothyroidism ….

Evaluation:

1. Neuro imaging → CT , MRI
2. T4 ,TSH
3. Karyotyping
4. Fragile X screen
5. Metabolic → e.g. plasma amino acids , urine organic acids , …

Treatment

Only rehabilitation of the child depending on the degree of mental retardation:

- Mild (IQ **50-70**) → educable (may need special classes)
- Moderate (IQ **35-50**) → trainable (they are trained to care for themselves)
- Severe (IQ **20-35**) → ± trainable.
- Profound (IQ **0-20**) → non trainable (so, they need full time nursing care).

Acute post infectious polyneuropathy (Gillian Barre syndrome)

Etiology

- Auto immune, often postinfectious polyneuropathy involving mainly motor but also sensory and sometimes autonomic nerves
- Paralysis usually follows a nonspecific gastrointestinal (especially Campylobacter jejuni, or Helicobacter pylori) or respiratory infection (especially Mycoplasma pneumonia) or viral infections.

Clinical picture

1. Motor : *Acute ascending flaccid paralysis*:

Criteria:

- Bilateral & symmetric usually (asymmetric in 9%)
- Associated hyptonia & hyporeflexia.

Progress:

- Lower Limb (inability or refusal to walk)→ trunk → upper limb.
- Bulbar palsy (in 50%) → dysphonia, dysphagia & lost bulbar reflexes.
- Respiratory muscles → respiratory failure.
- ✓ **Miller-Fisher syndrome** consists of acute external ophthalmoplegia, ataxia, and areflexia

2. Sensory

- Mild
- Tender calf

3. Autonomic

- Labile blood pressure & heart rate
- Urinary incontinence or retention of urine in about 20%

Diagnosis

✓ **CSF**: Cyto albuminous dissociation :

- High CSF protein > twice the upper limit of normal and a lack of cellular response < 10 white blood cells/mm^3
- Negative bacterial culture

✓ **MRI of the spinal cord**

- Thickening of the cauda equina and intrathecal nerve roots in >90% of patients
- Rule out other spinal disorders

- Motor nerve conduction velocity are greatly reduced
- Electromyography shows evidence of acute denervation of muscle.
- Serum creatine kinase (CK) level may be mildly elevated or normal

Treatment

Patients in early stages of this acute disease should be admitted to the hospital for observation because the ascending paralysis can rapidly involve respiratory muscles during the next 24 hr

i. Supportive:

- Respiratory effort monitoring (spirometry) and support
- Cardiac monitoring
- Nasogastric feeding
- Care of bladder (catheterization & neostigmine).
- Physiotherapy

ii. Specific

- IVIG: A commonly recommended protocol is IVIG 0.4 g/kg/day for 5 consecutive days, but some studies suggest that larger doses are more effective (1 g/kg/day for 2 consecutive days)
- Alternatives: if IVIG is ineffective
  - Plasmapharesis is equally effective as IVIG.
  - Combined IVIG and interferon is effective in some patients
  - Steroids are not effective

Differential diagnosis: Other causes of acute flaccid paralysis

Prognosis

- The clinical course is usually benign, and spontaneous recovery begins within 2-3 wk.
- Most patients regain full muscular strength, although some are left with residual weakness.
- Improvement usually follows a gradient opposite the direction of involvement: bulbar function recovering first, and lower extremity weakness resolving last.

Differential diagnosis : Acute Flaccid Paralysis

A. Acute asymmetrical paralysis

- Cerebrovascular stroke e.g. acute hemiplegia
- Poliomyelitis
- Pseudo paralysis e.g. with osteomyelitis, trauma, scurvy

B. Acute symmetrical paralysis

* Spinal cord: Trauma, Compression by abscess or tumors, Transverse myelitis
* Infections: Botulism, Diphtheria, Rabies
* Post infections: Gillian Barre syndrome, Enterovirus Associated Post Infectious Myelitis (*Recently discovered in USA*)
* Tick-bite paralysis
* Myasthenia gravis
* Hypokalemic periodic paralysis

Inability To Walk

Normal walking require integration between CNS, muscles, skeleton and training

Causes of delayed walking

A. Central causes:

1-Brain	- Cerebral palsy - Mental retardation - Hydrocephalus - Congenital malformations - Brain damage with tumors or infections
2- Spinal cord	- Congenital→ Spina bifida. - Traumatic→ Spinal cord trauma. - Inflammatory→ Pott's disease of the spine. - Neoplastic→ Spinal cord tumors.
3- Anterior horn cells	- Poliomyelitis. - Spinal muscle atrophy.(Werding Hoffman disease)
4- Peripheral Nerve	- Guillian - Barre syndrome.. - Polyneuritis (Diphtheria, Drugs)
5- Neuro muscular junction	- Mysthenia gravis. - Botulism - Organophosphorus poisoning.

B. Muscular causes:

* Primary muscle disorders: Myopathies, Myositis and Metabolic.
* Secondary muscle disorders: Rickets and malnutrition.

C. Skeletal: (Bones, Joints)

- Rickets
- Inflammation (arthritis, osteomyelitis)
- Lower limb trauma

Differential diagnosis of inability to walk:

Causes	Primary (The child has not walked before)	Secondary (The child has walked before)
a.Paralytic	- Early poliomyelitis. - Early paralysis before walking. - Cerebral palsy	- Poliomyelitis. - Post diphtheric paralysis - Post encephalitic paralysis - Cerebro-vascular accidents
b.Non paralytic	- Rickets - Mental retardation - Simple delayed walking	- Rickets. - Malnutrition. - Fractures or osteomyelitis

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